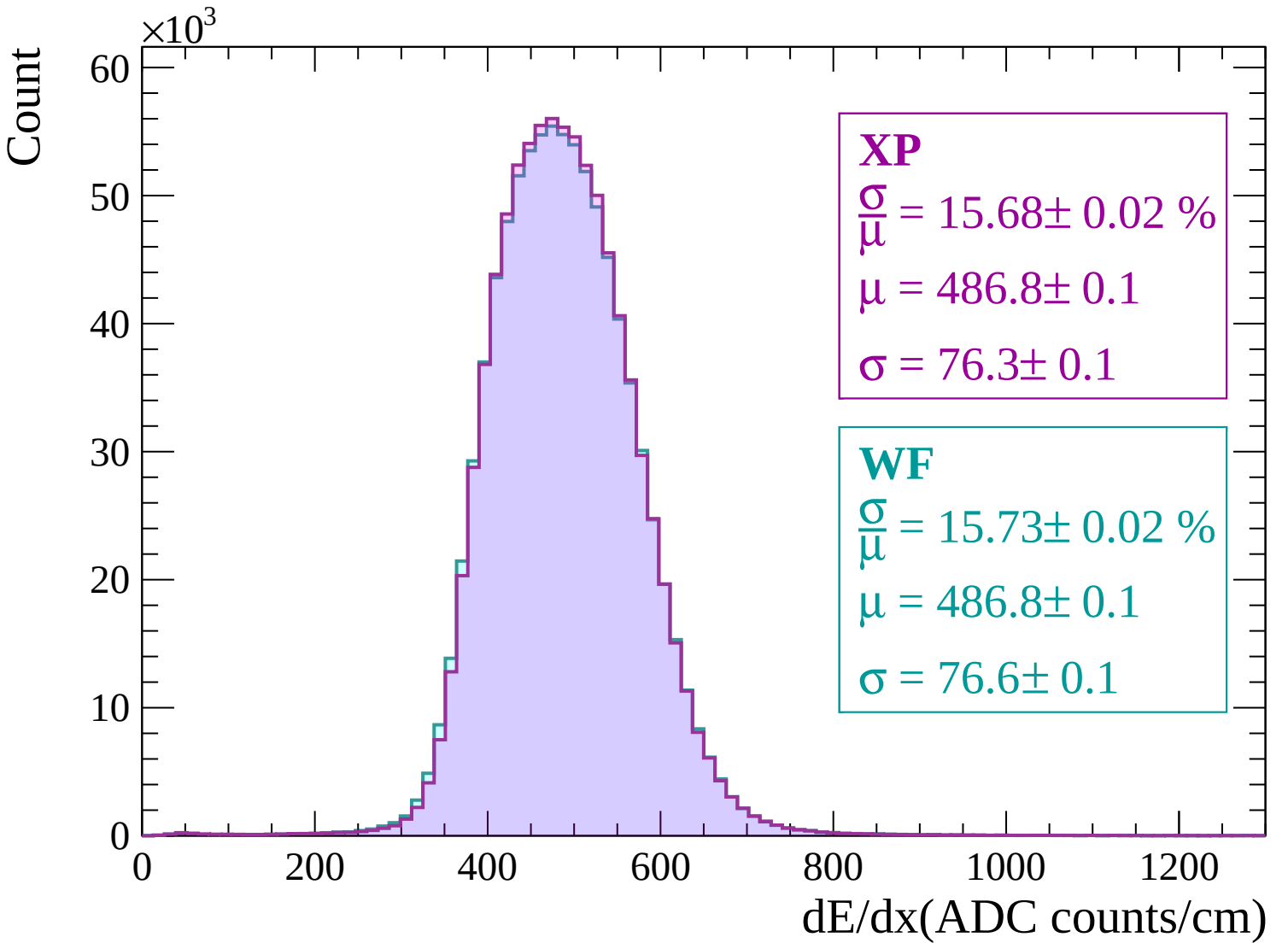
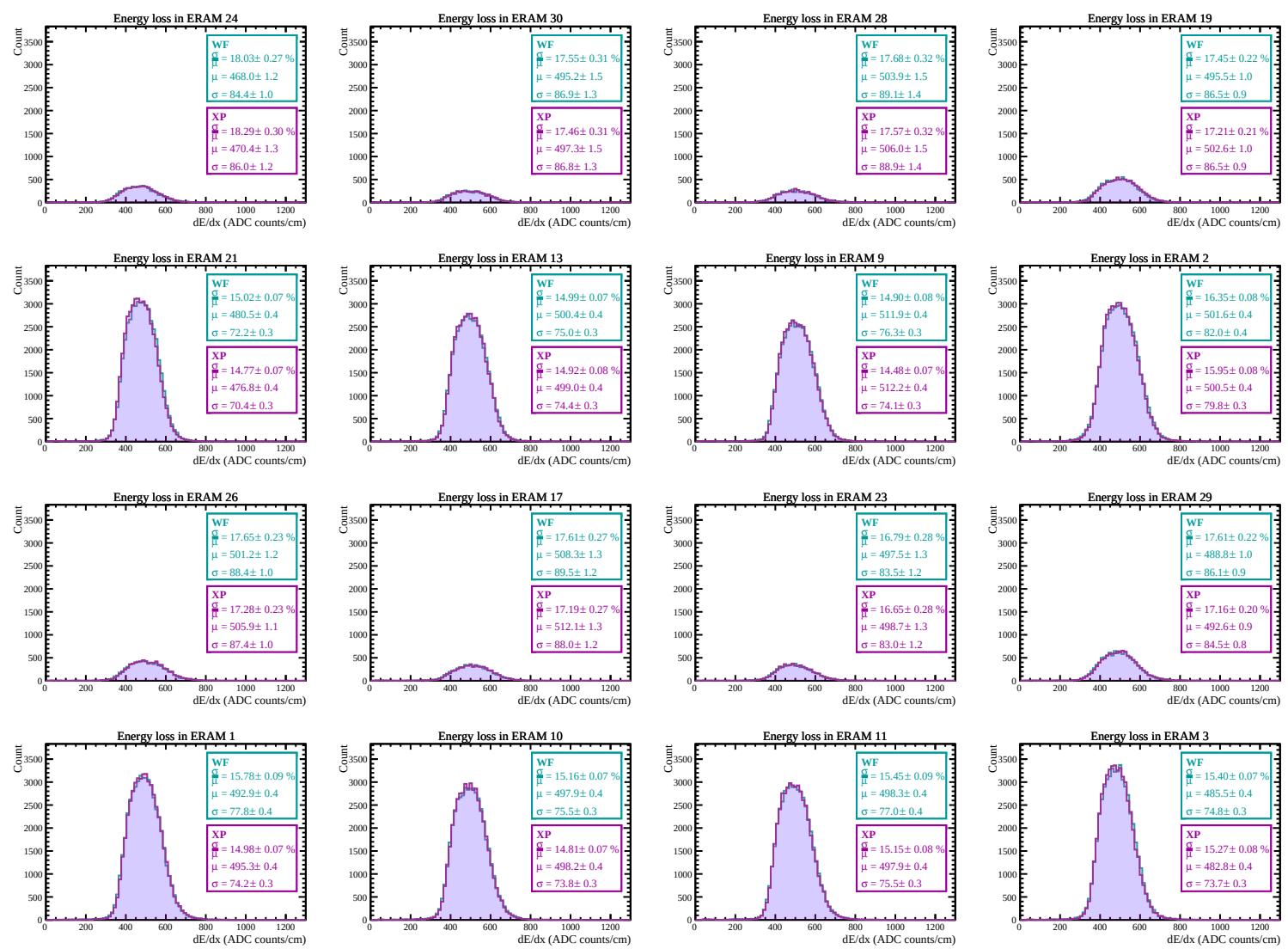
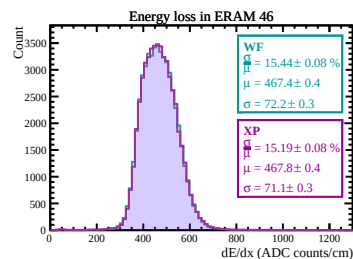
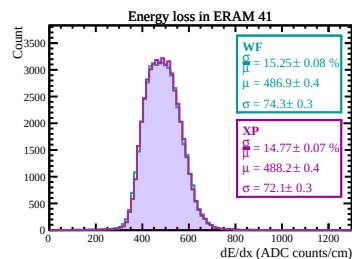
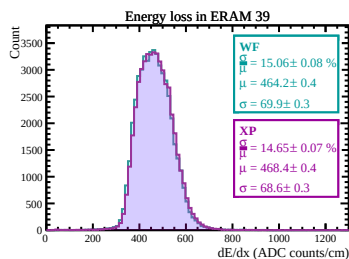
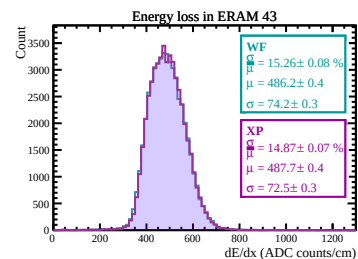
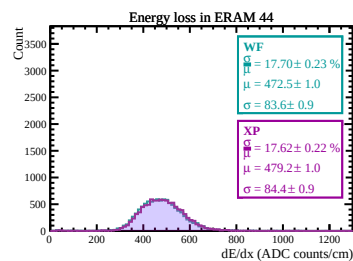
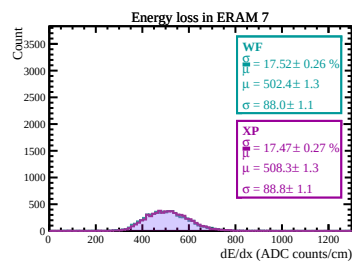
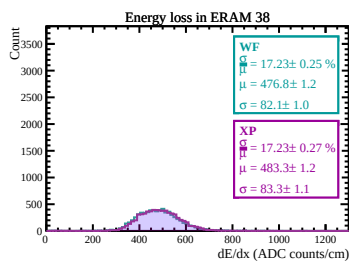
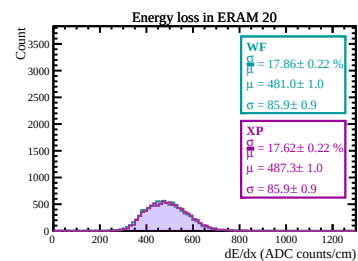
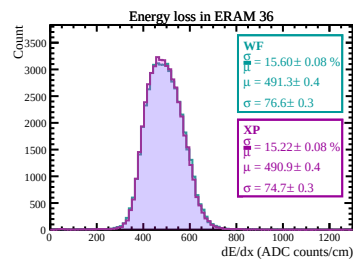
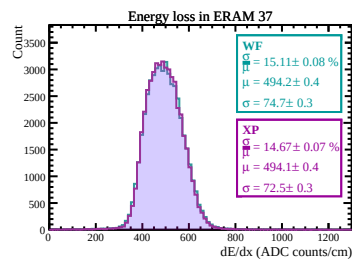
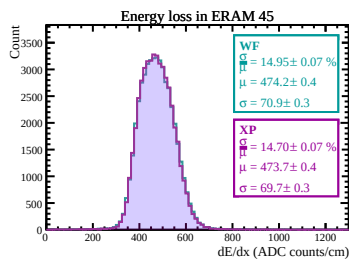
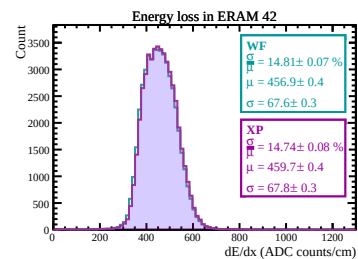
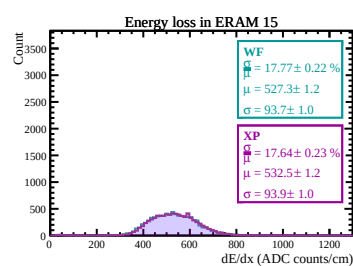
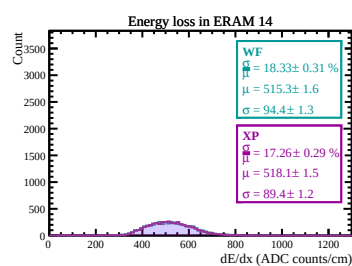
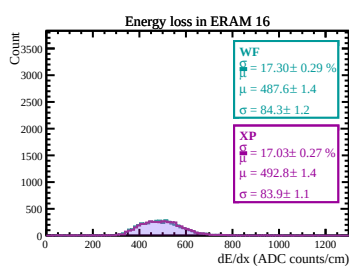
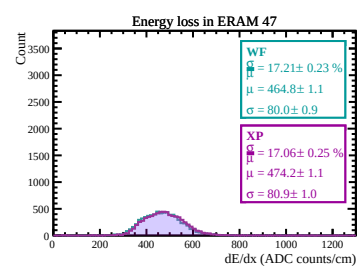
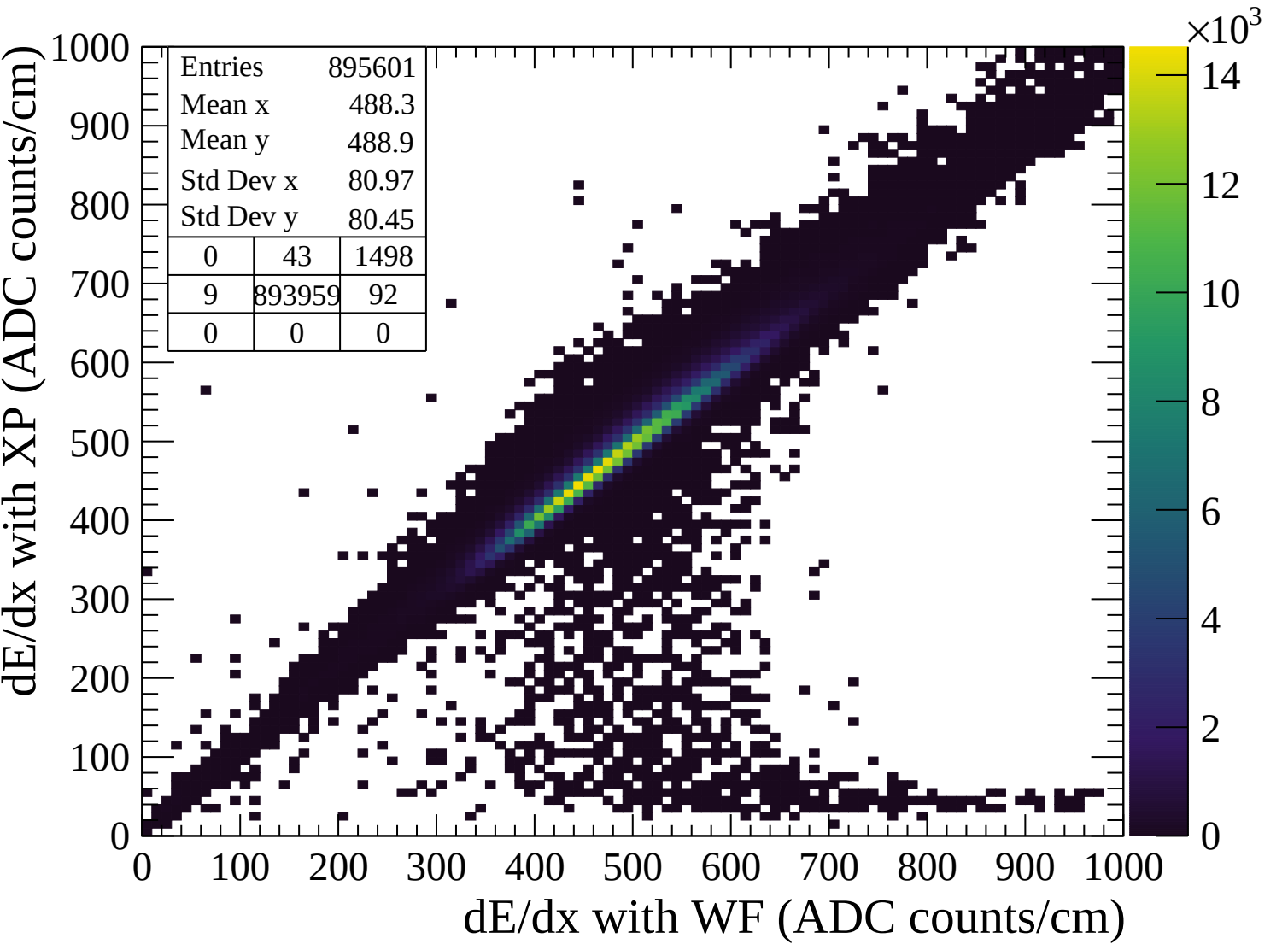


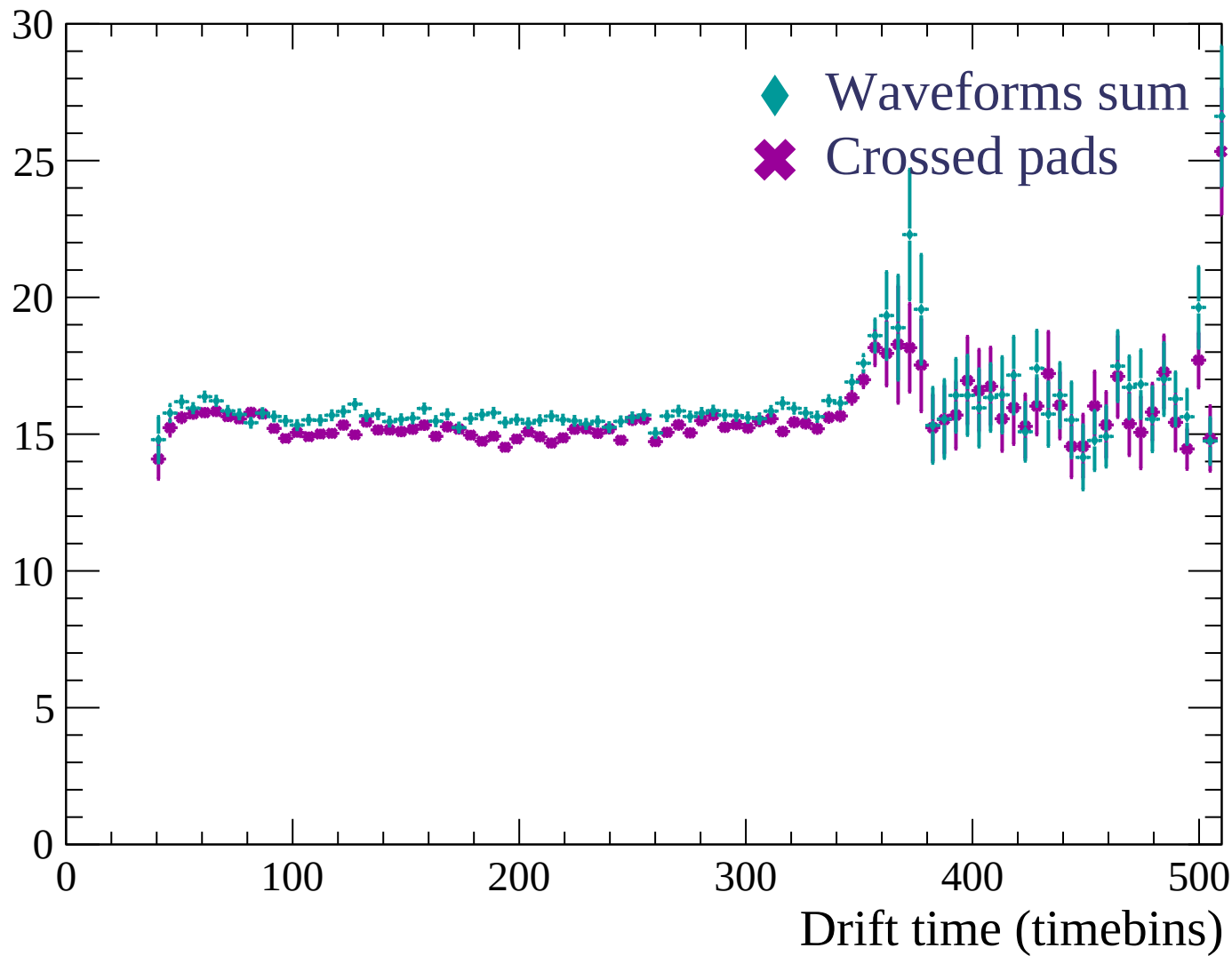


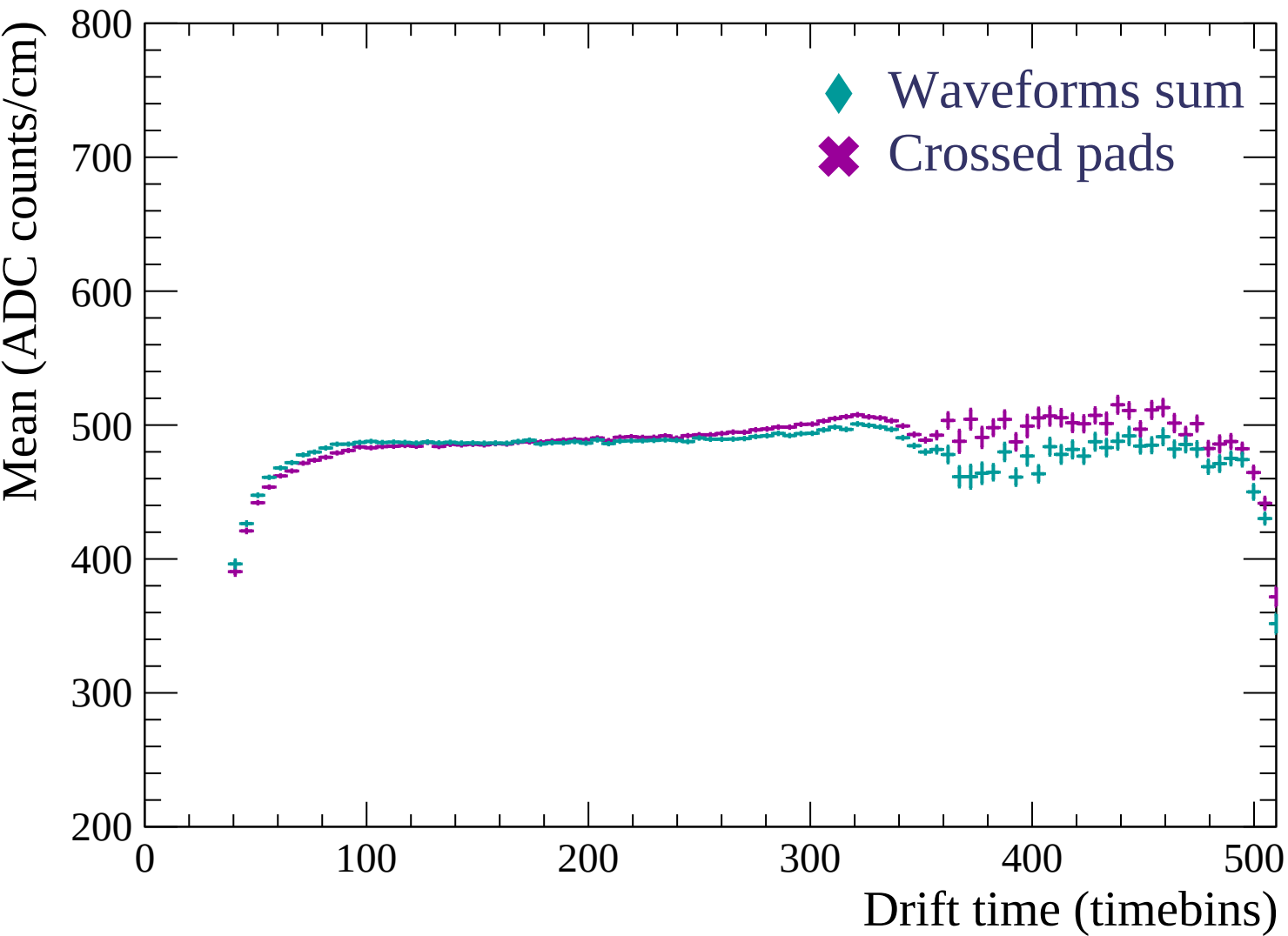


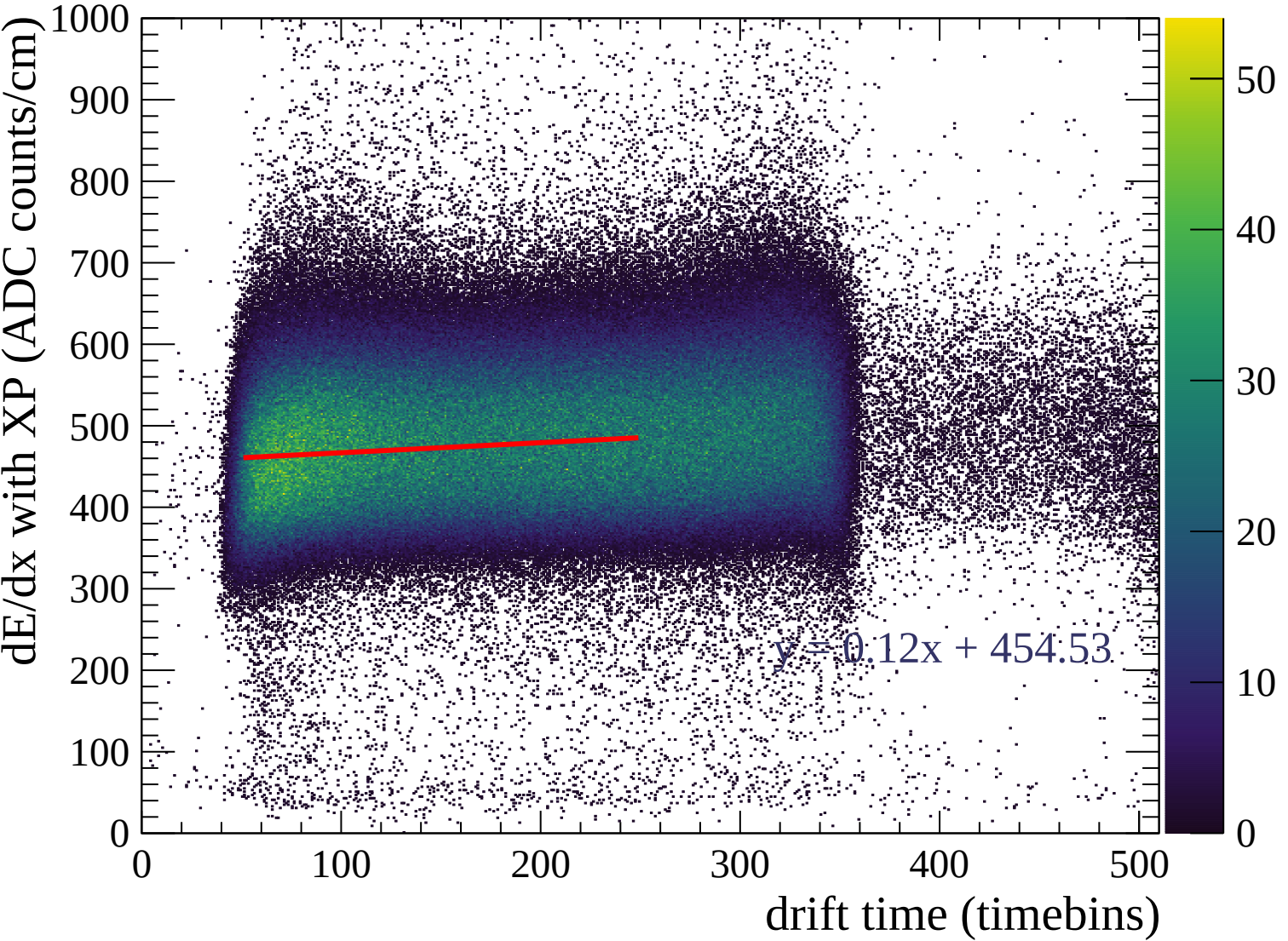


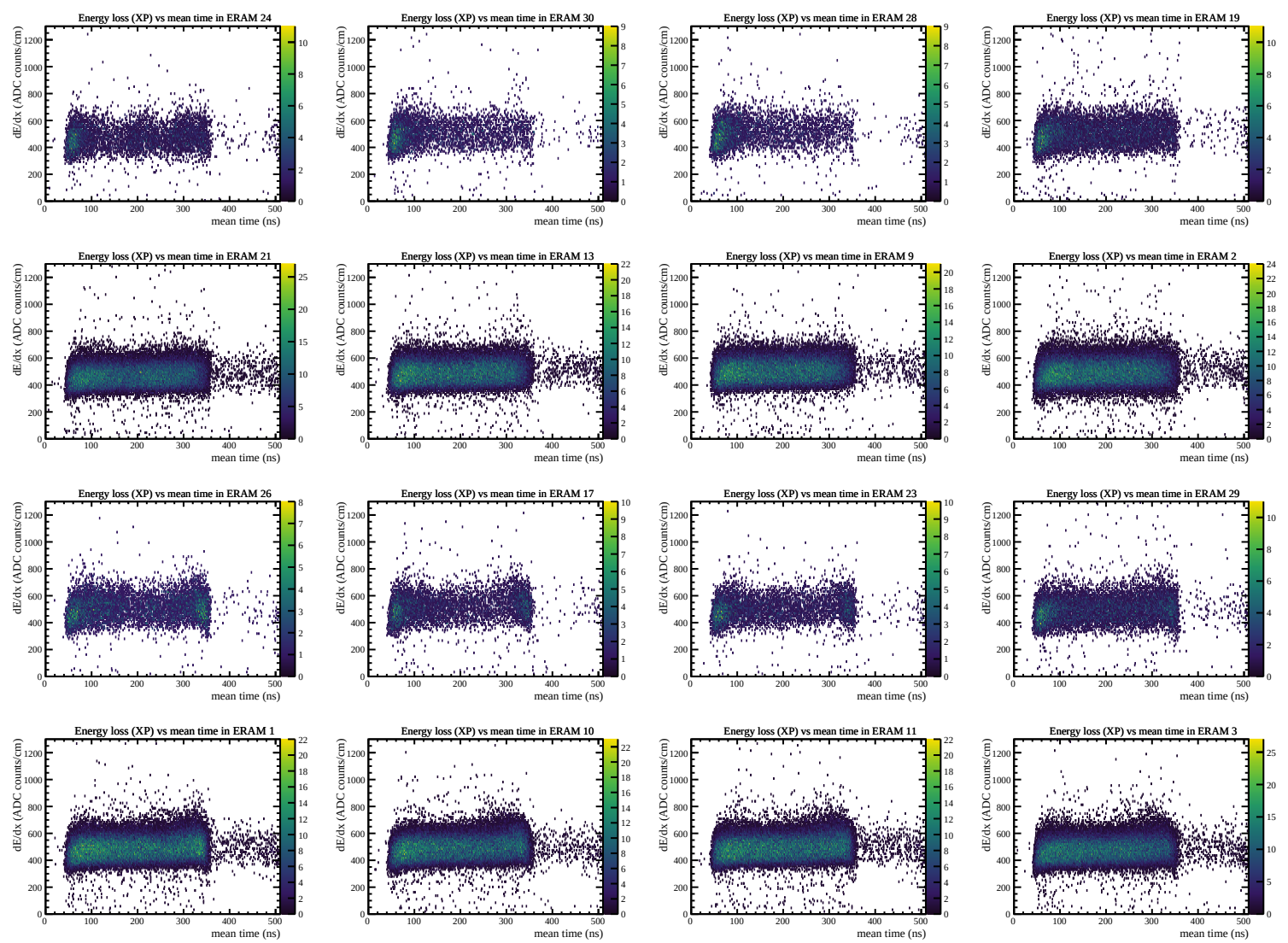


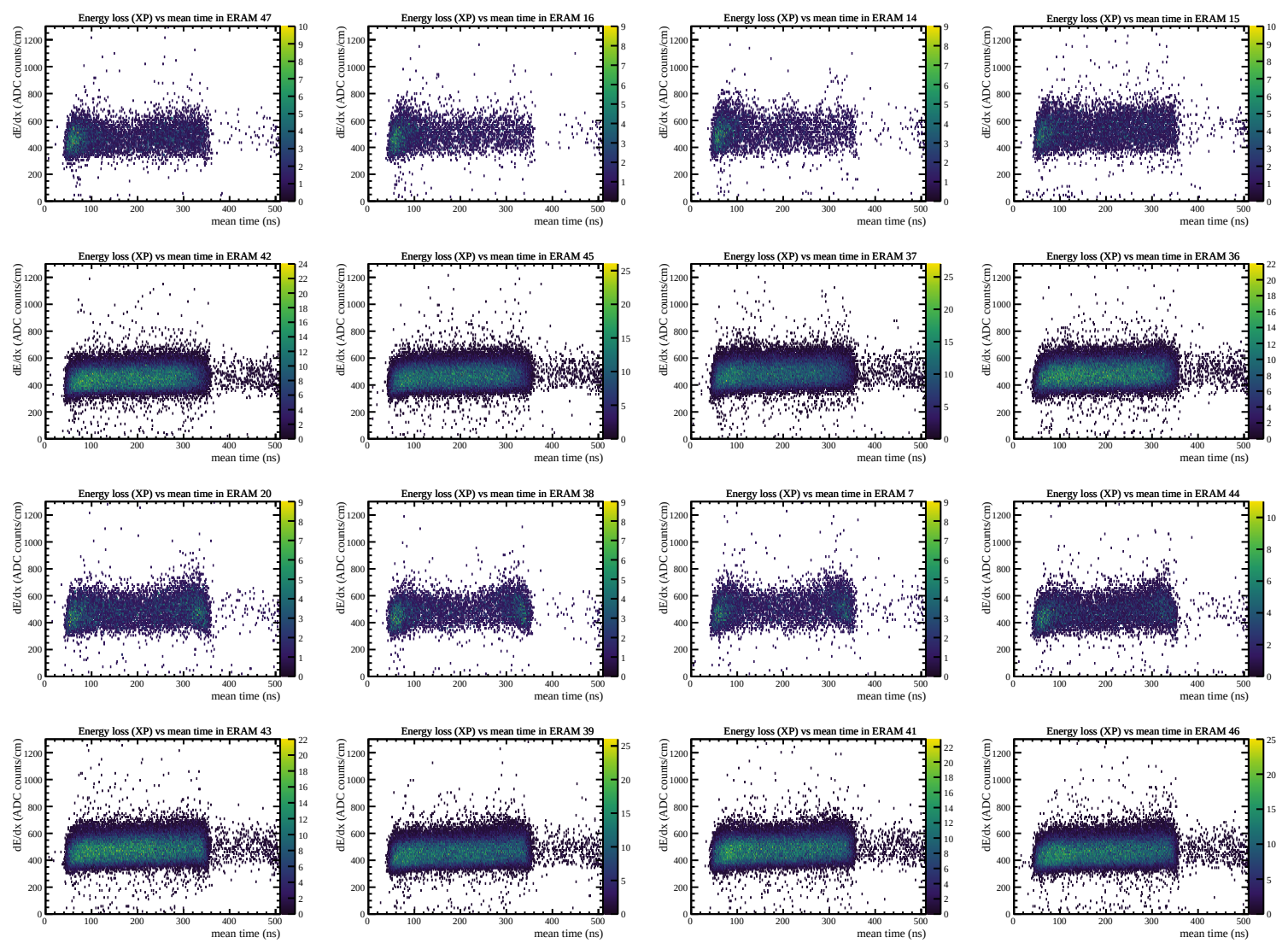
Resolution (%)

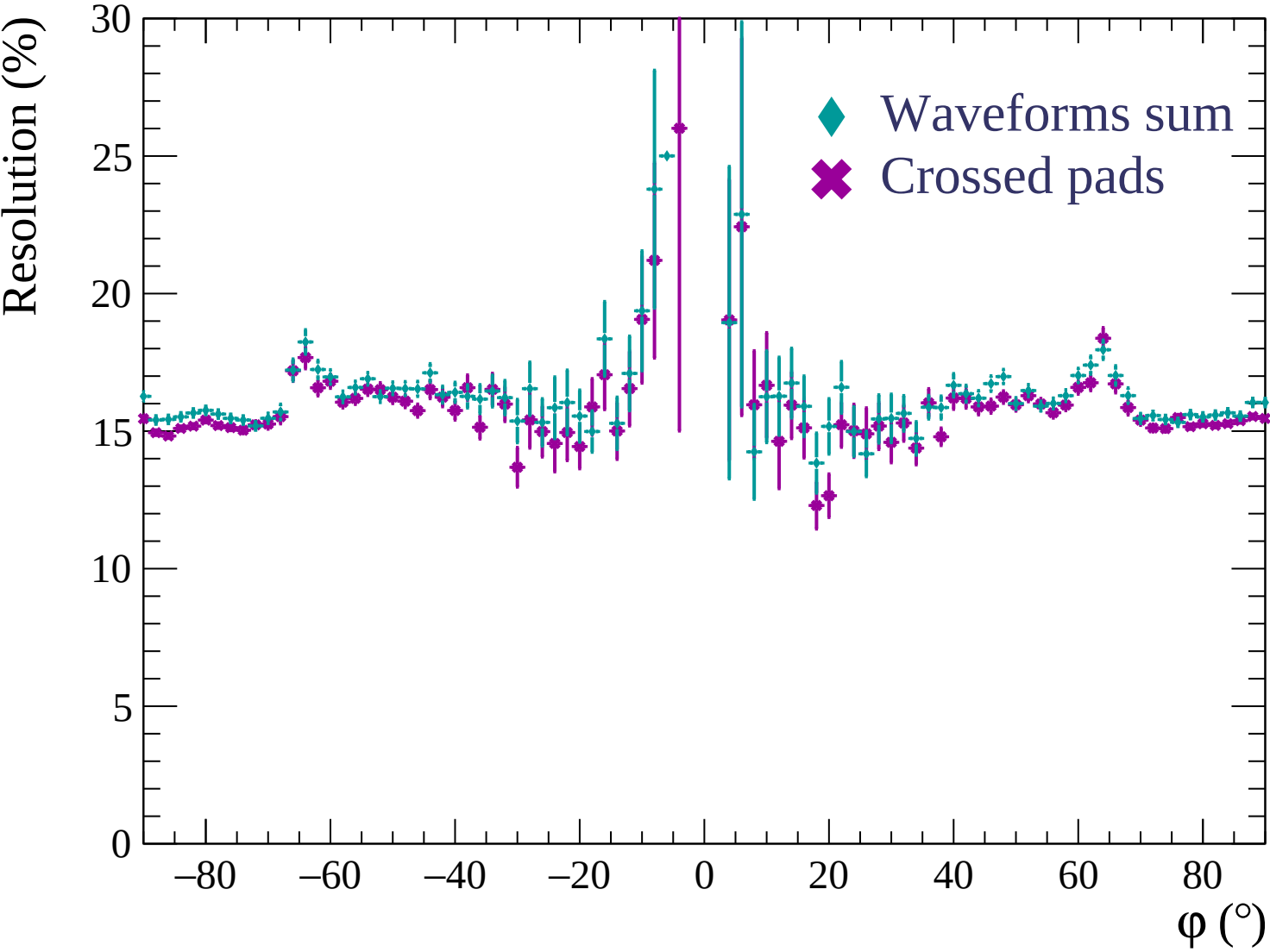


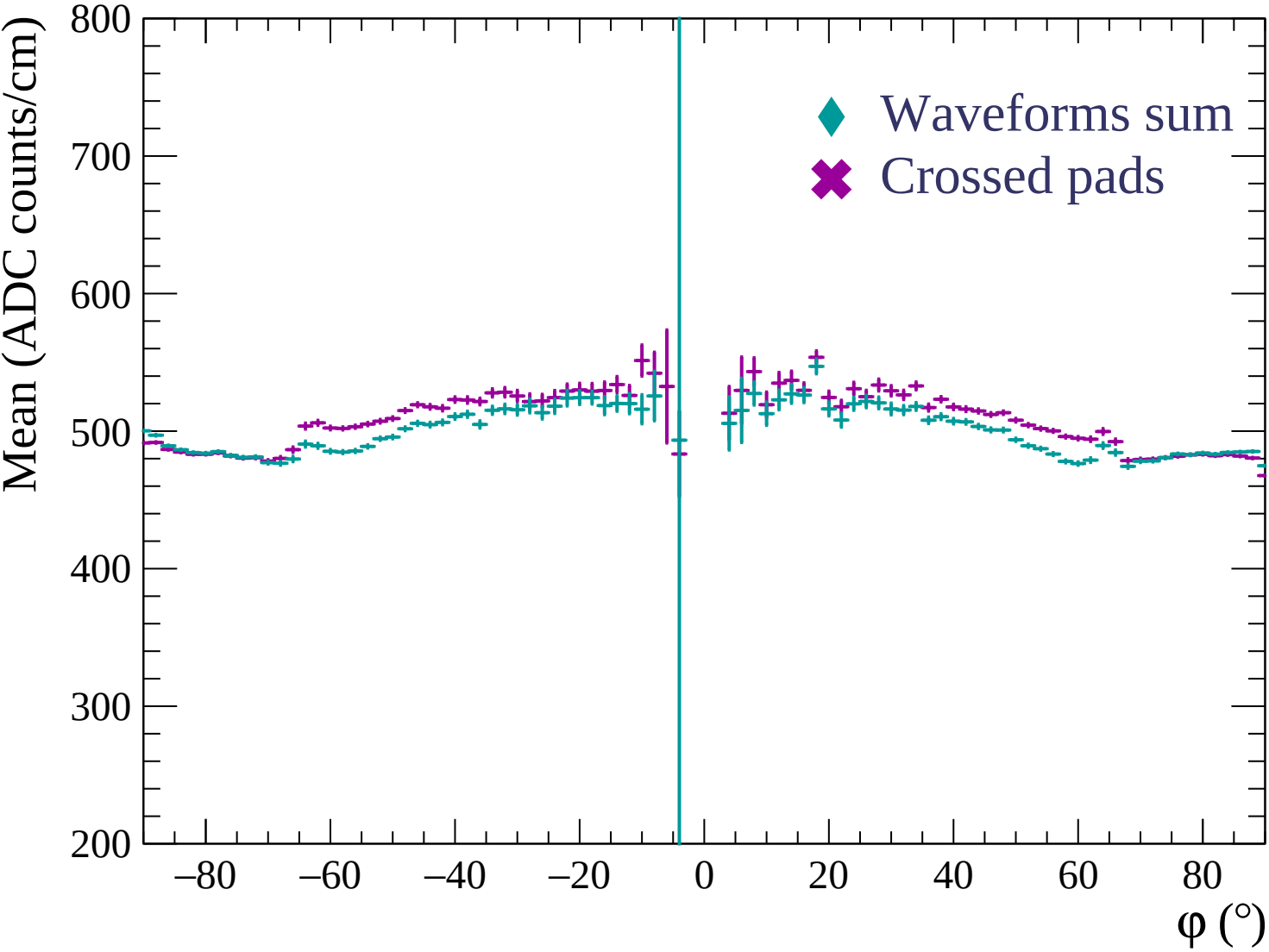


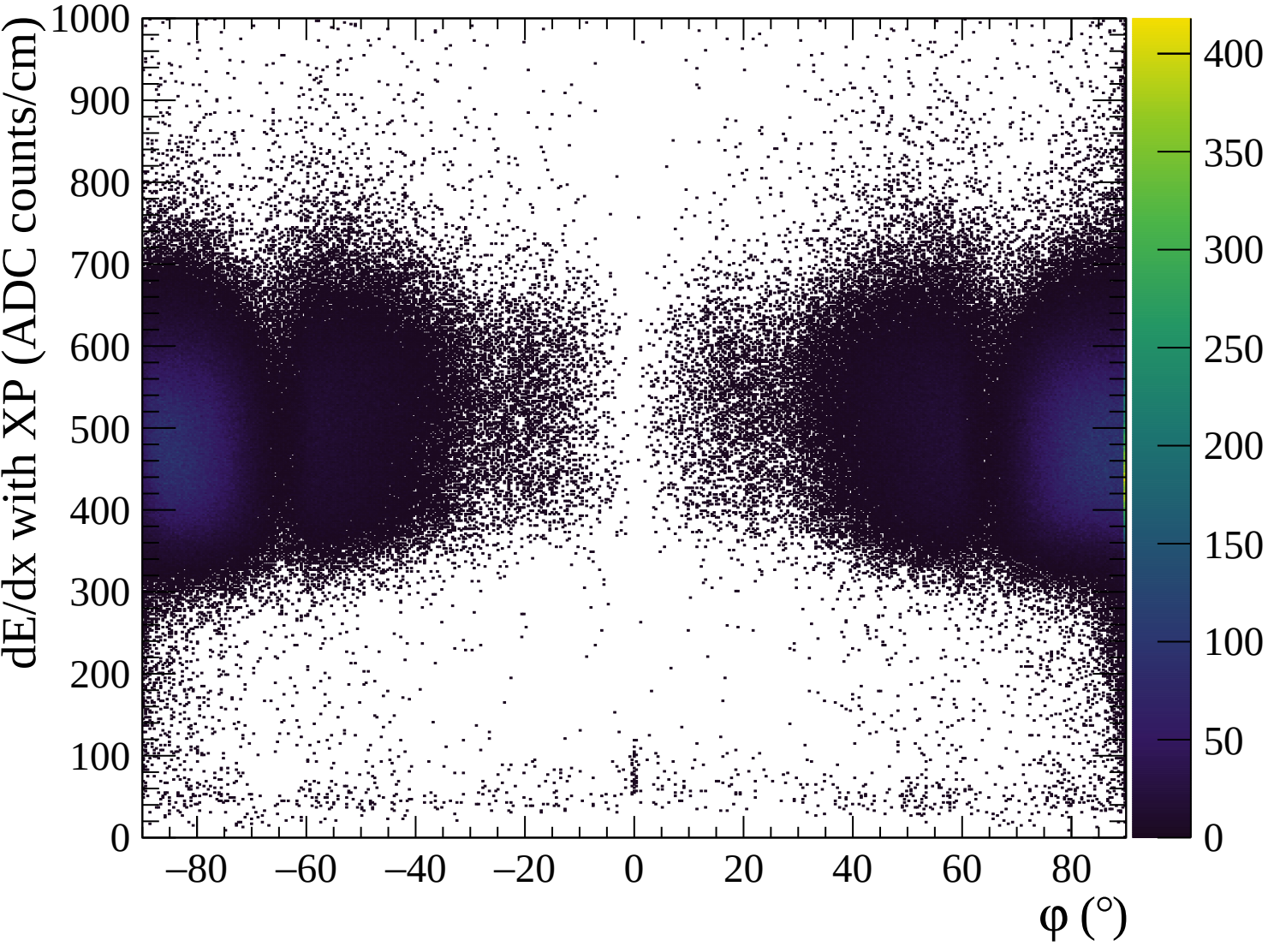


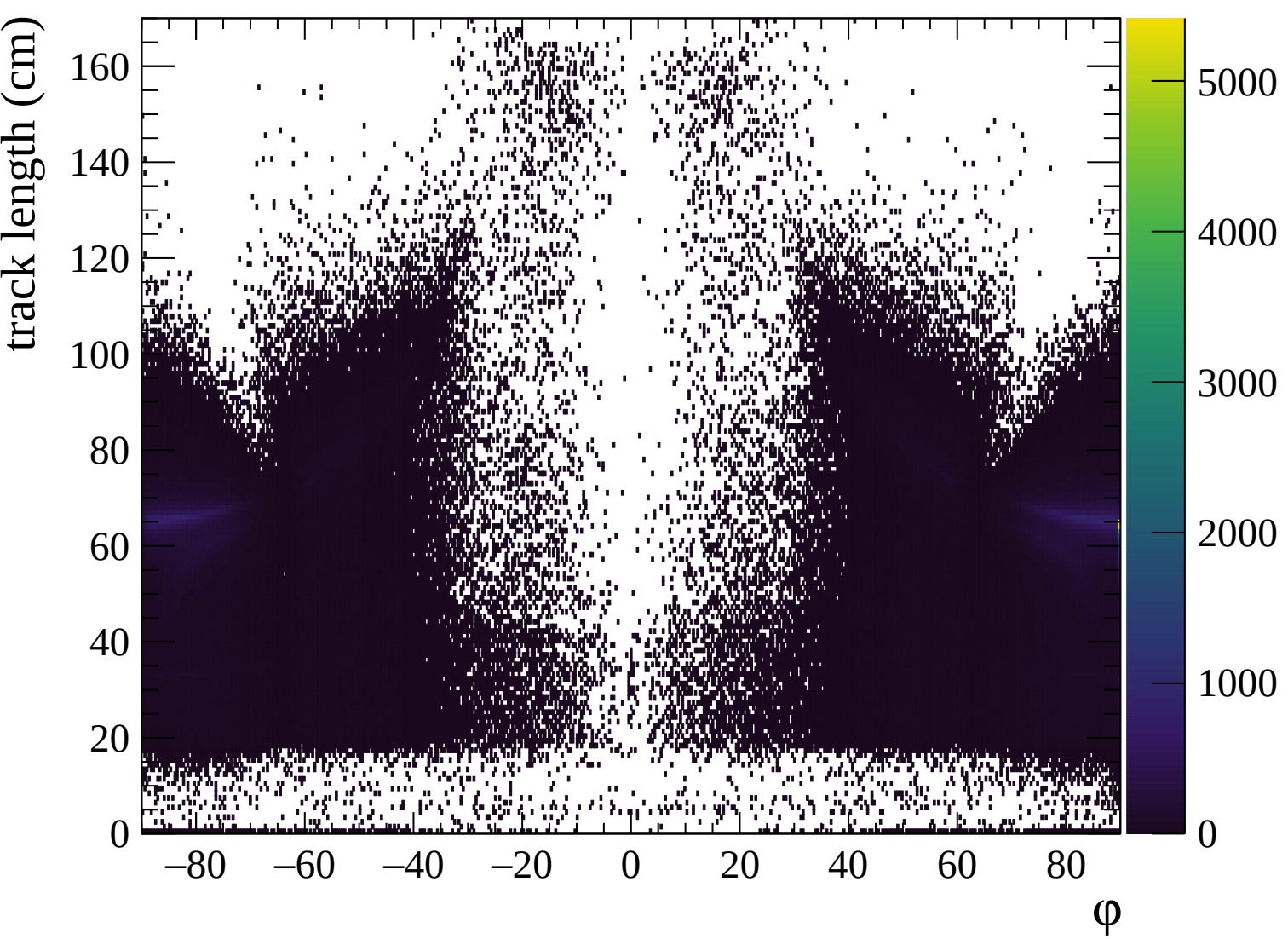


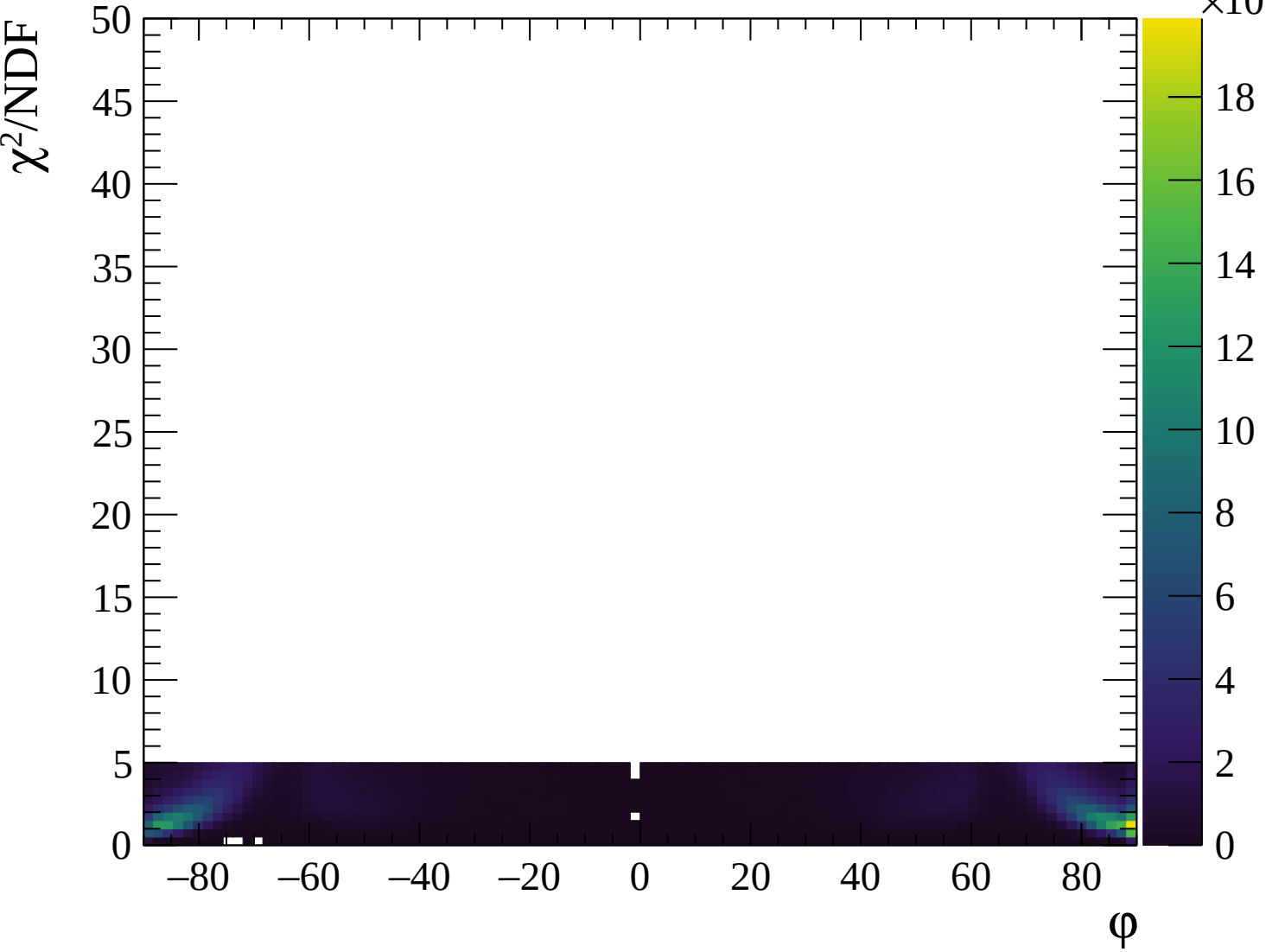


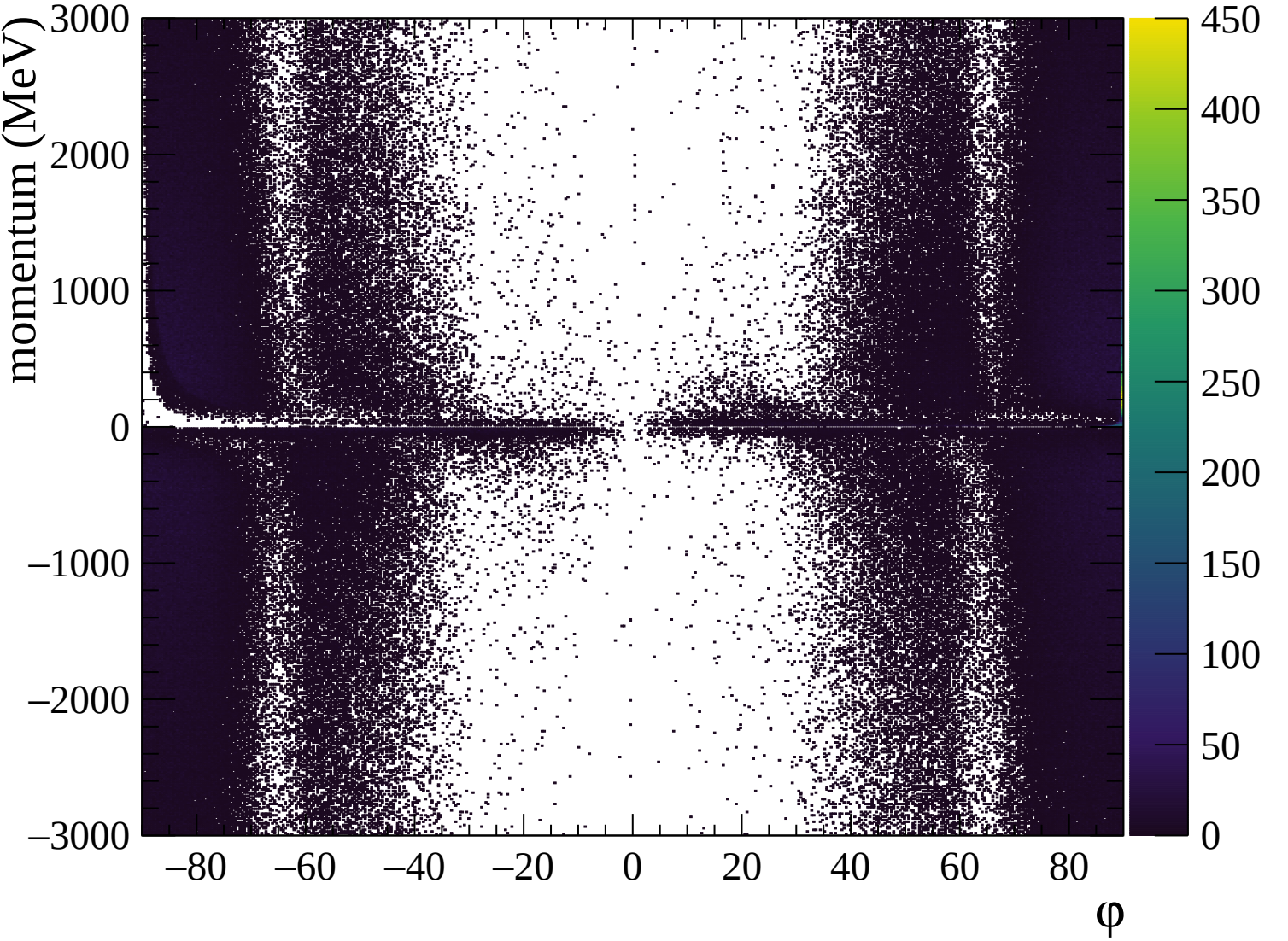


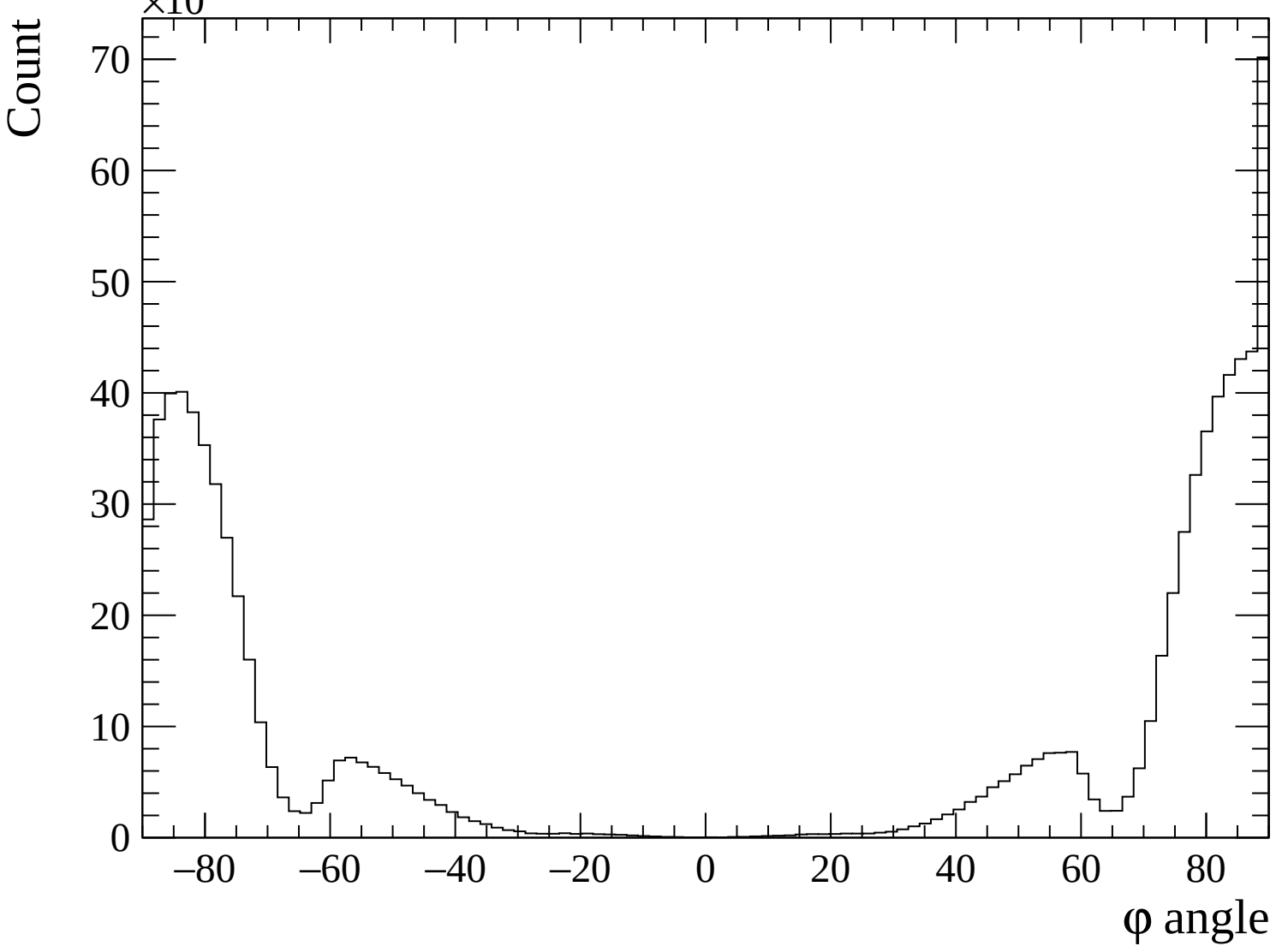


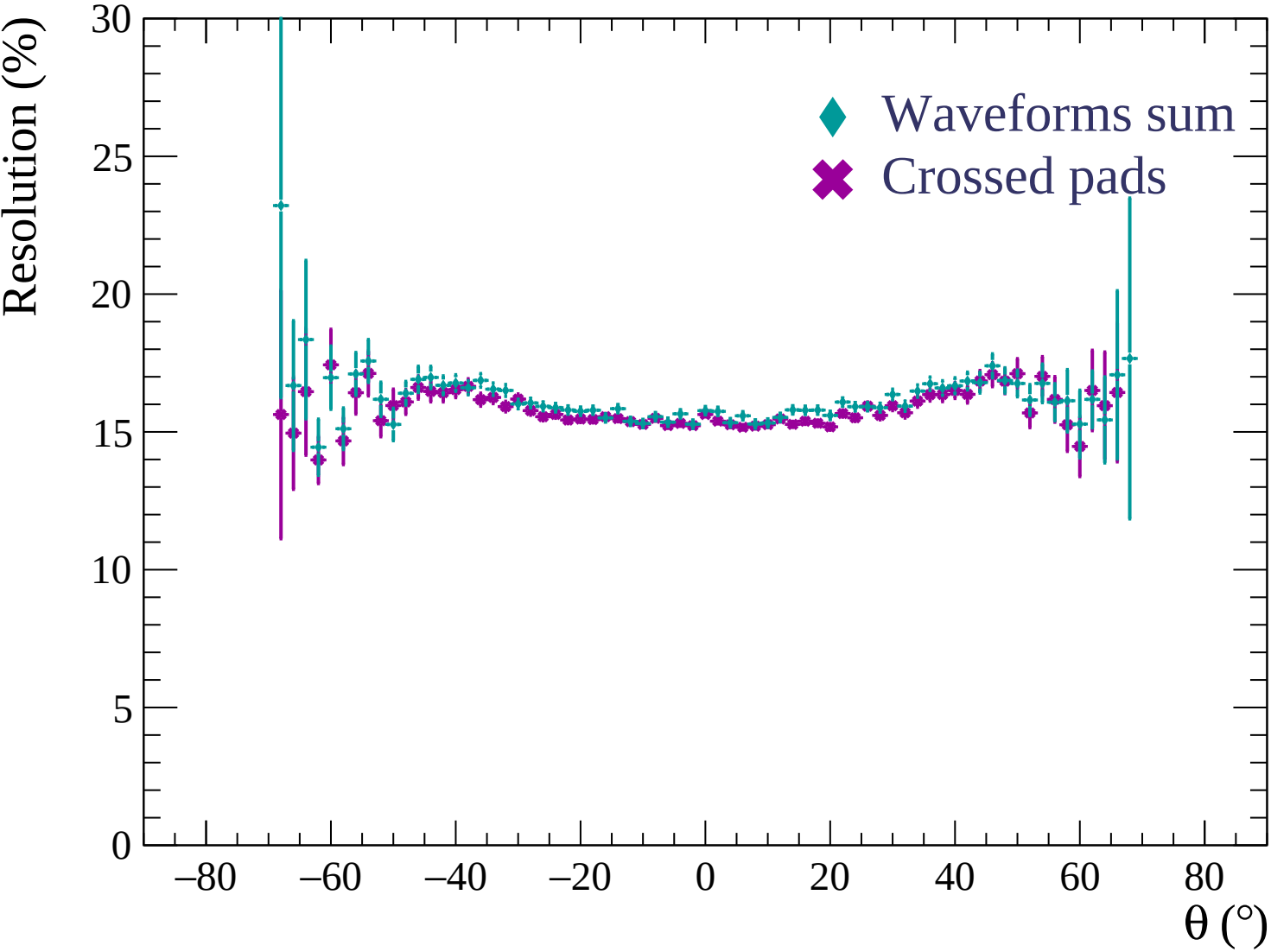


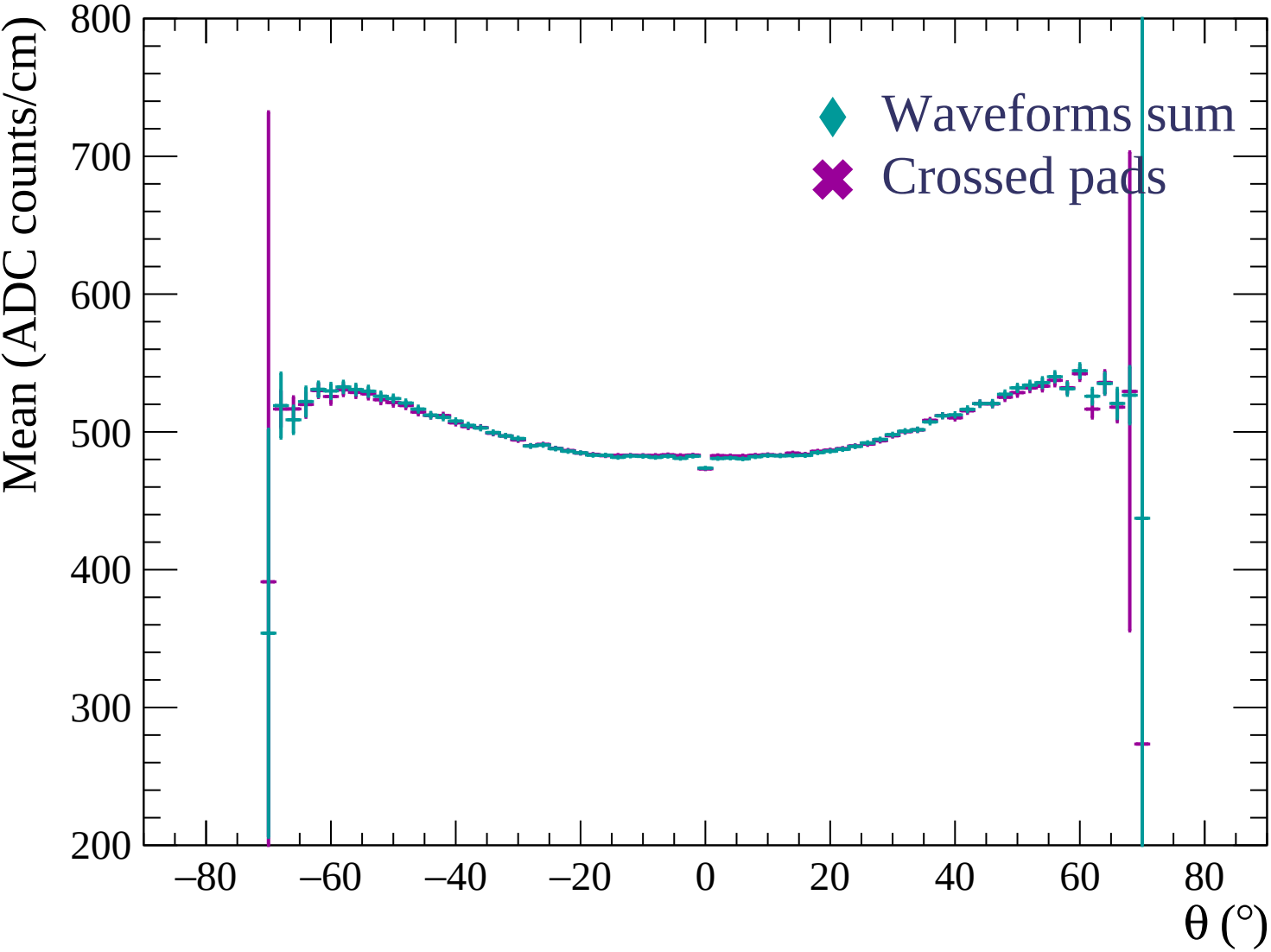


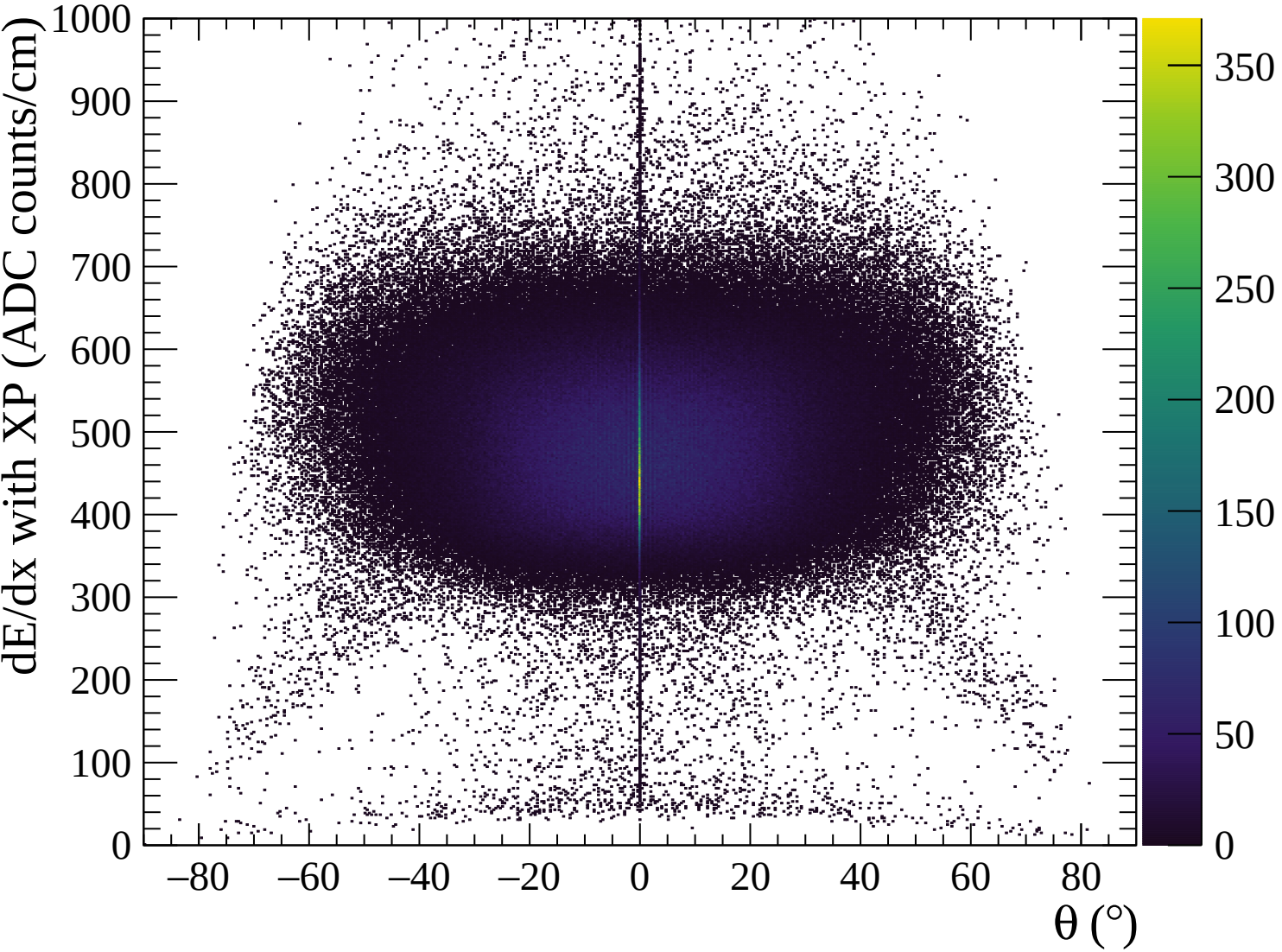


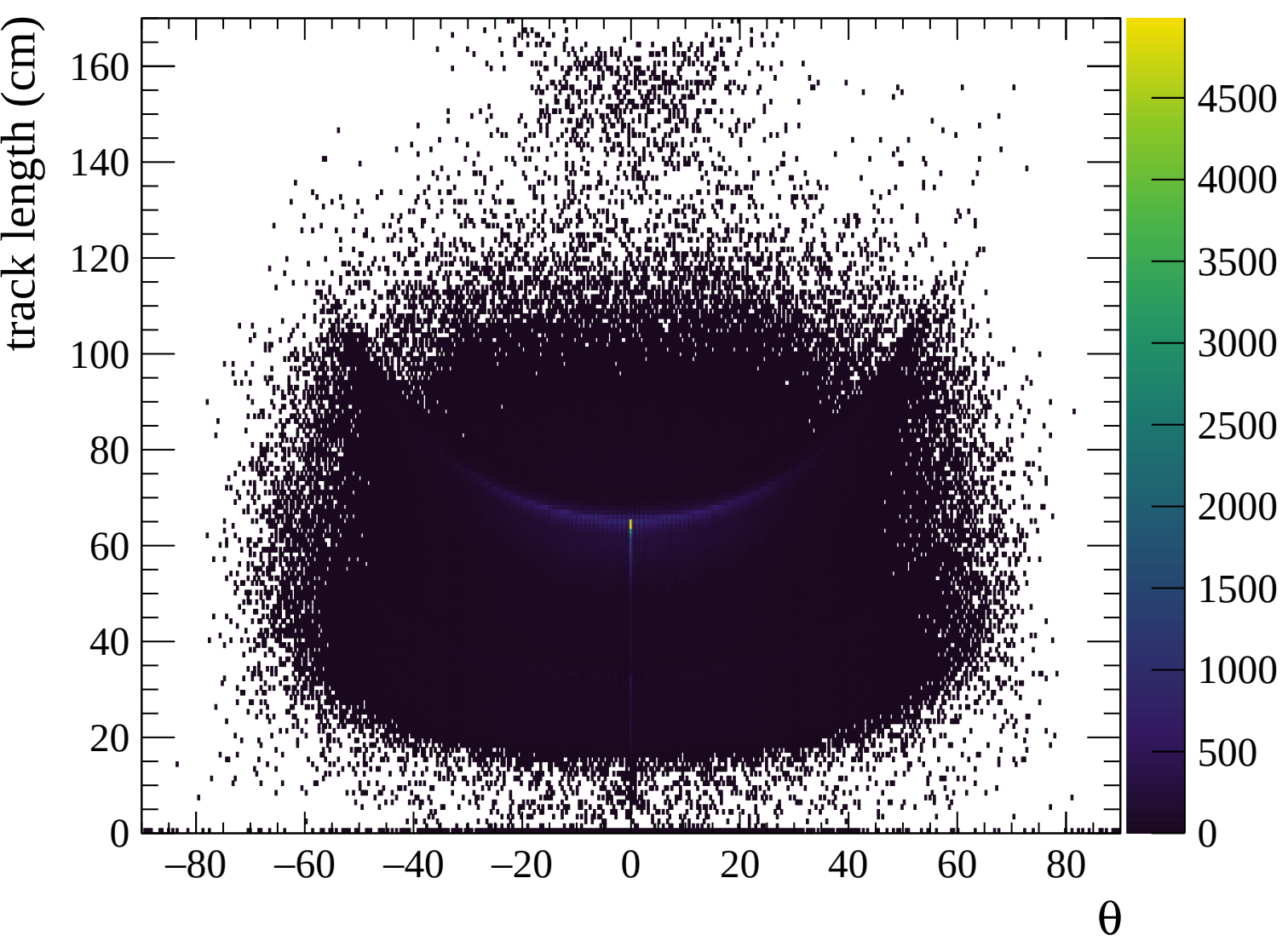


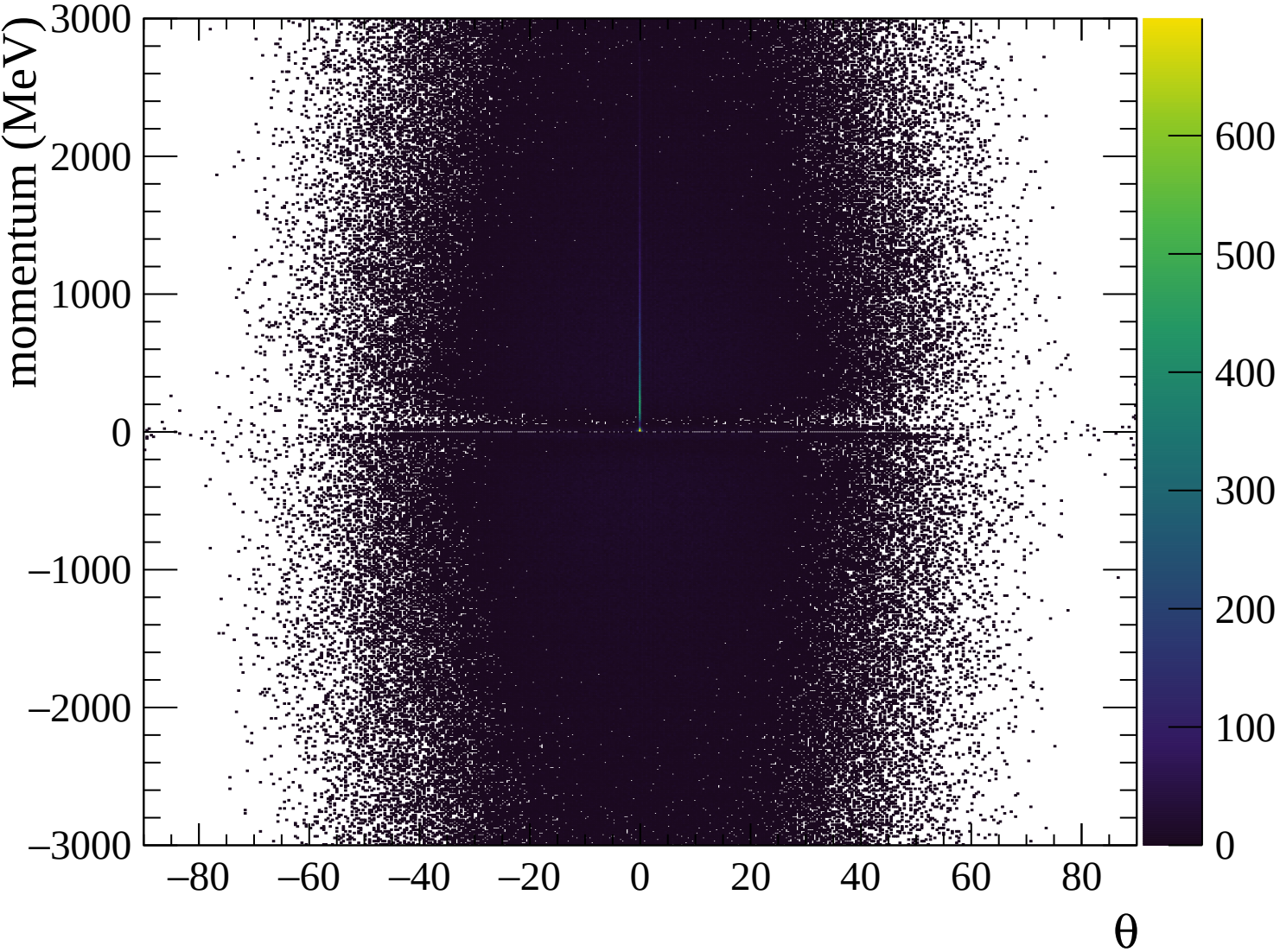


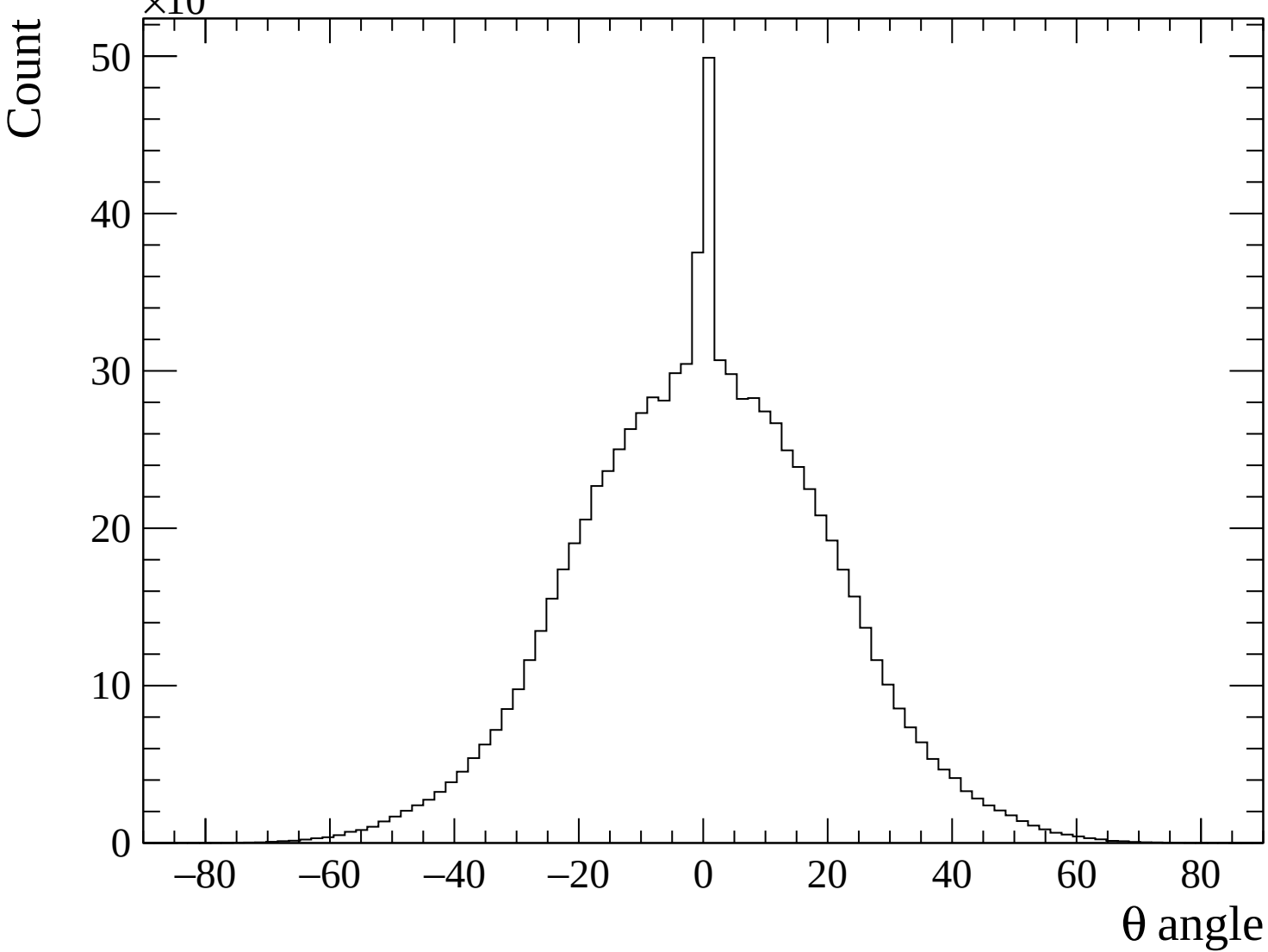


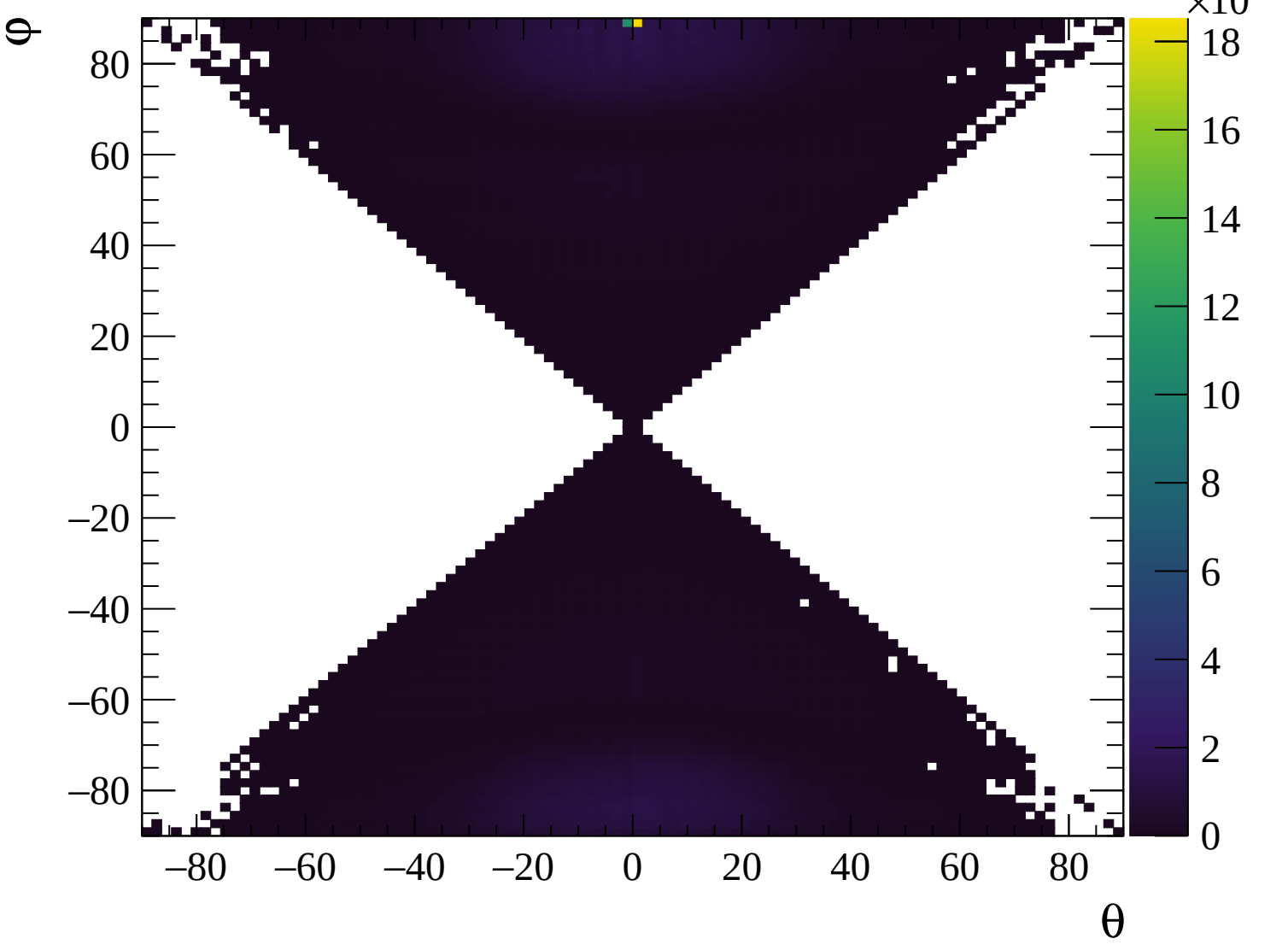


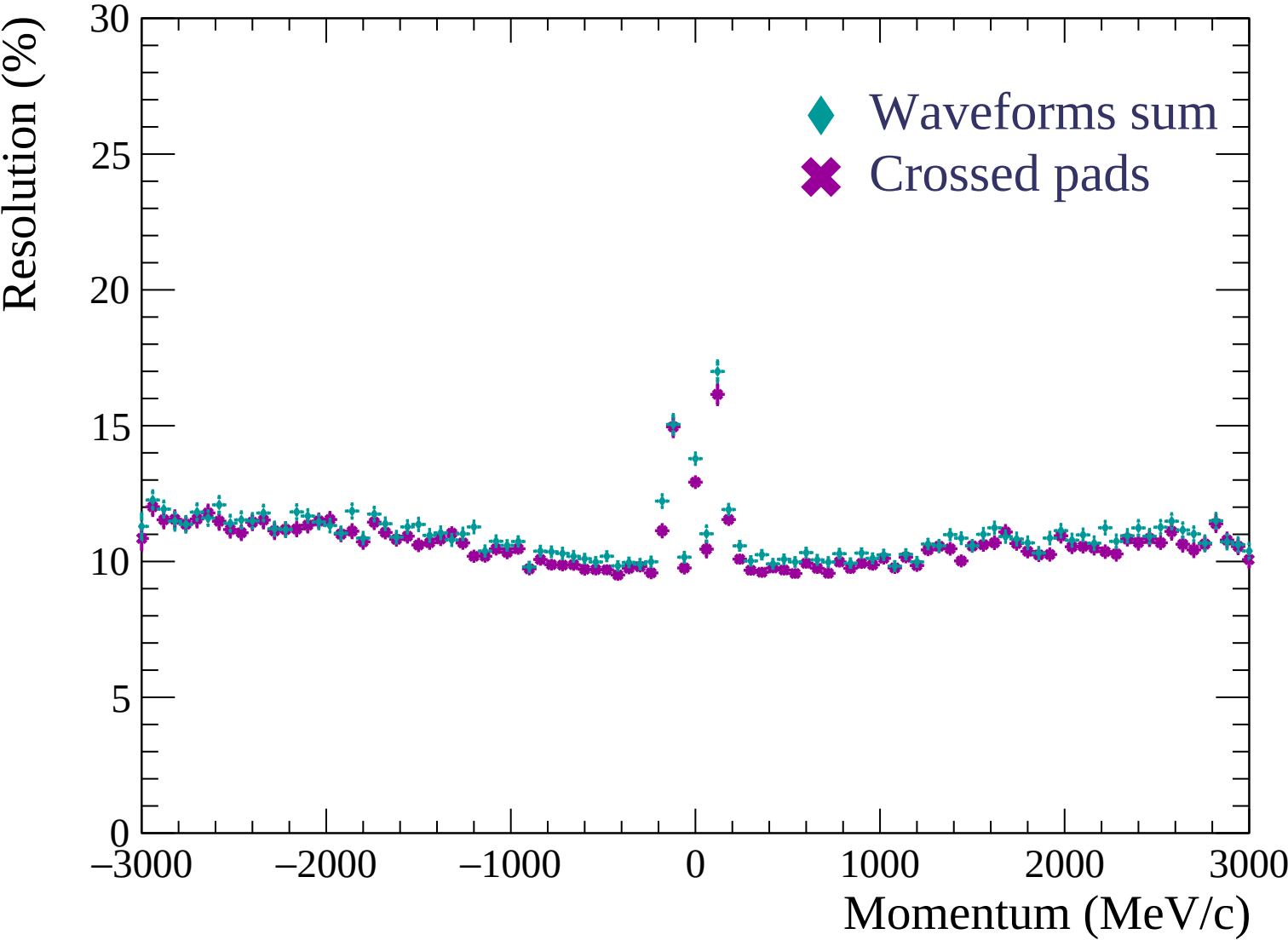


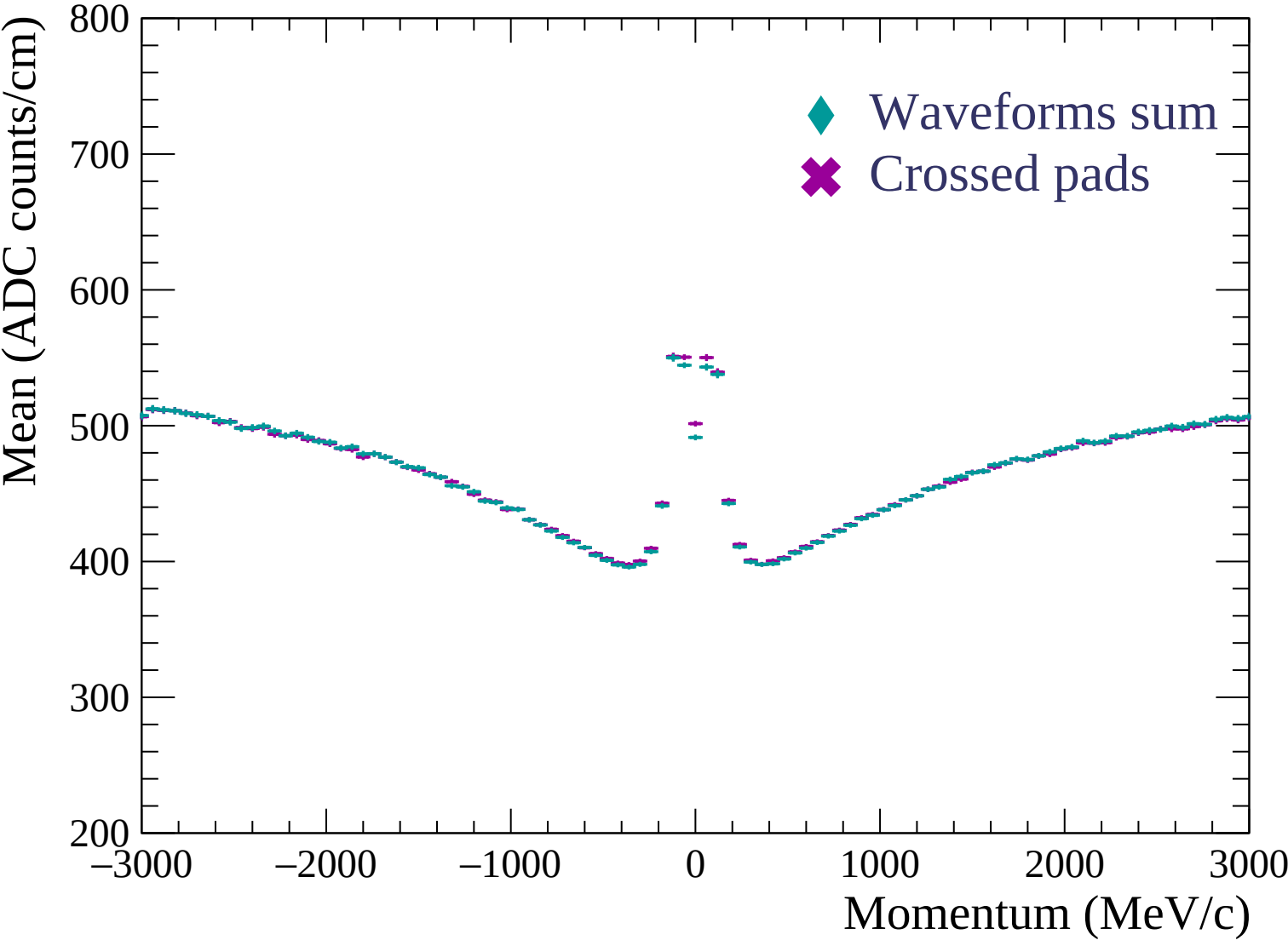


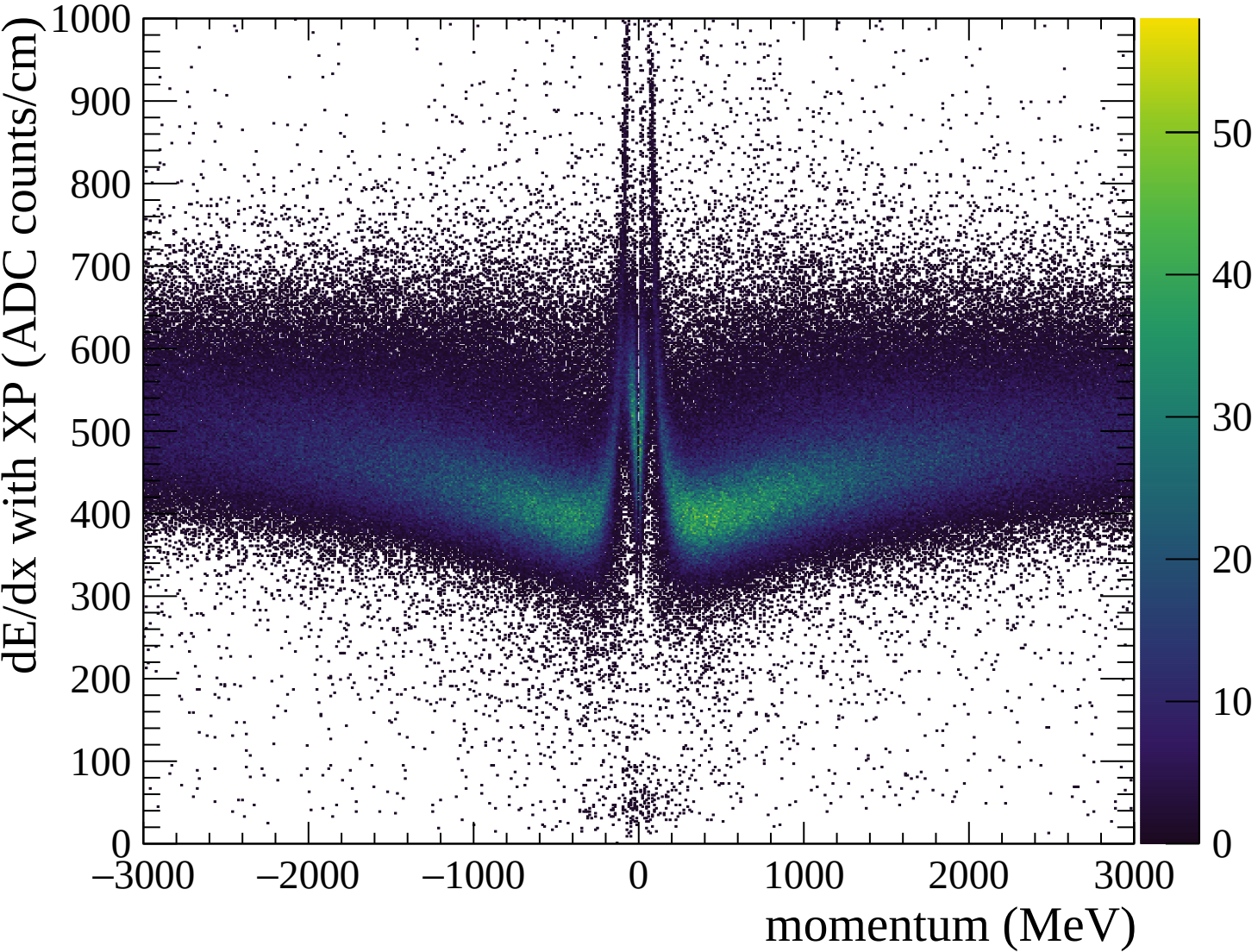


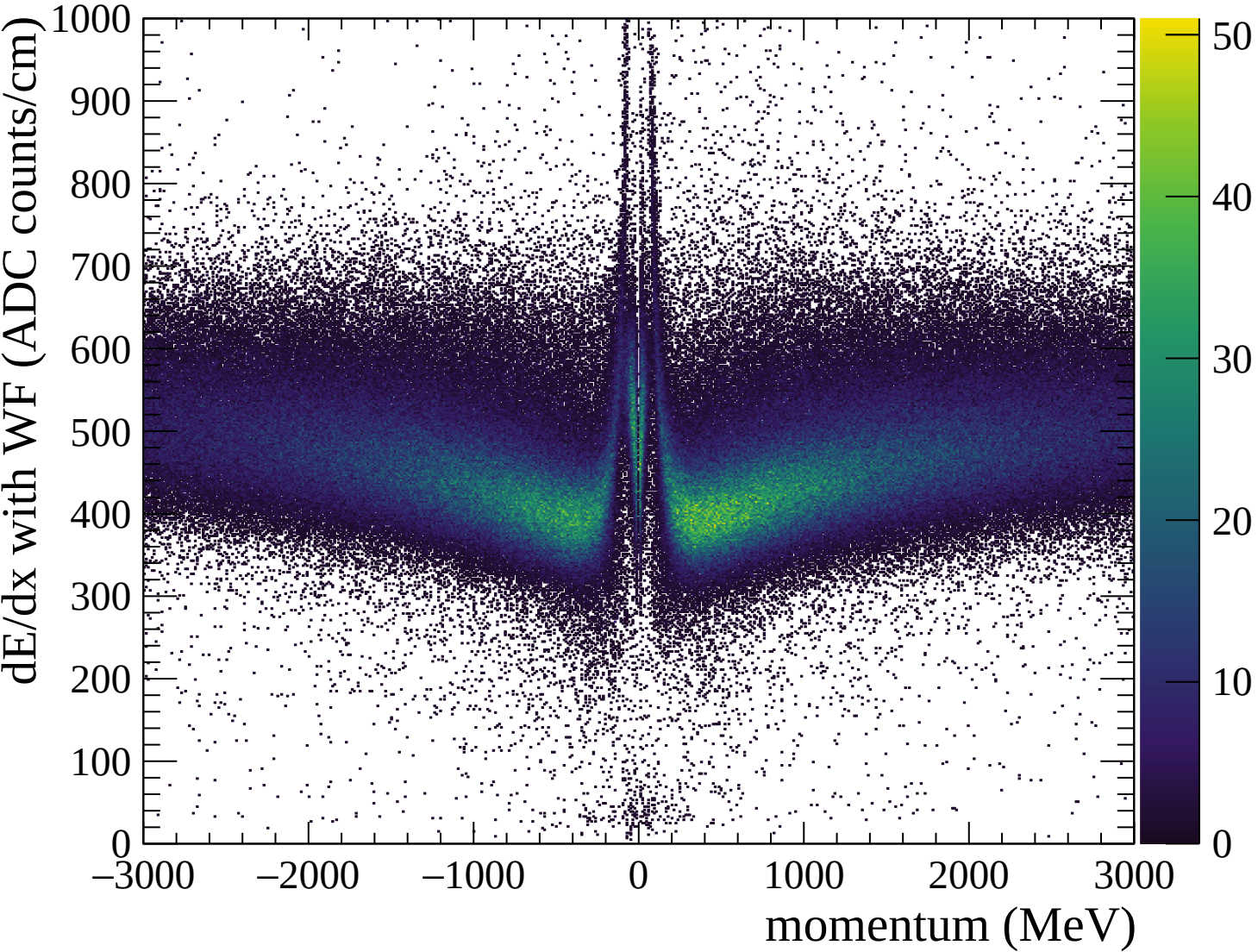


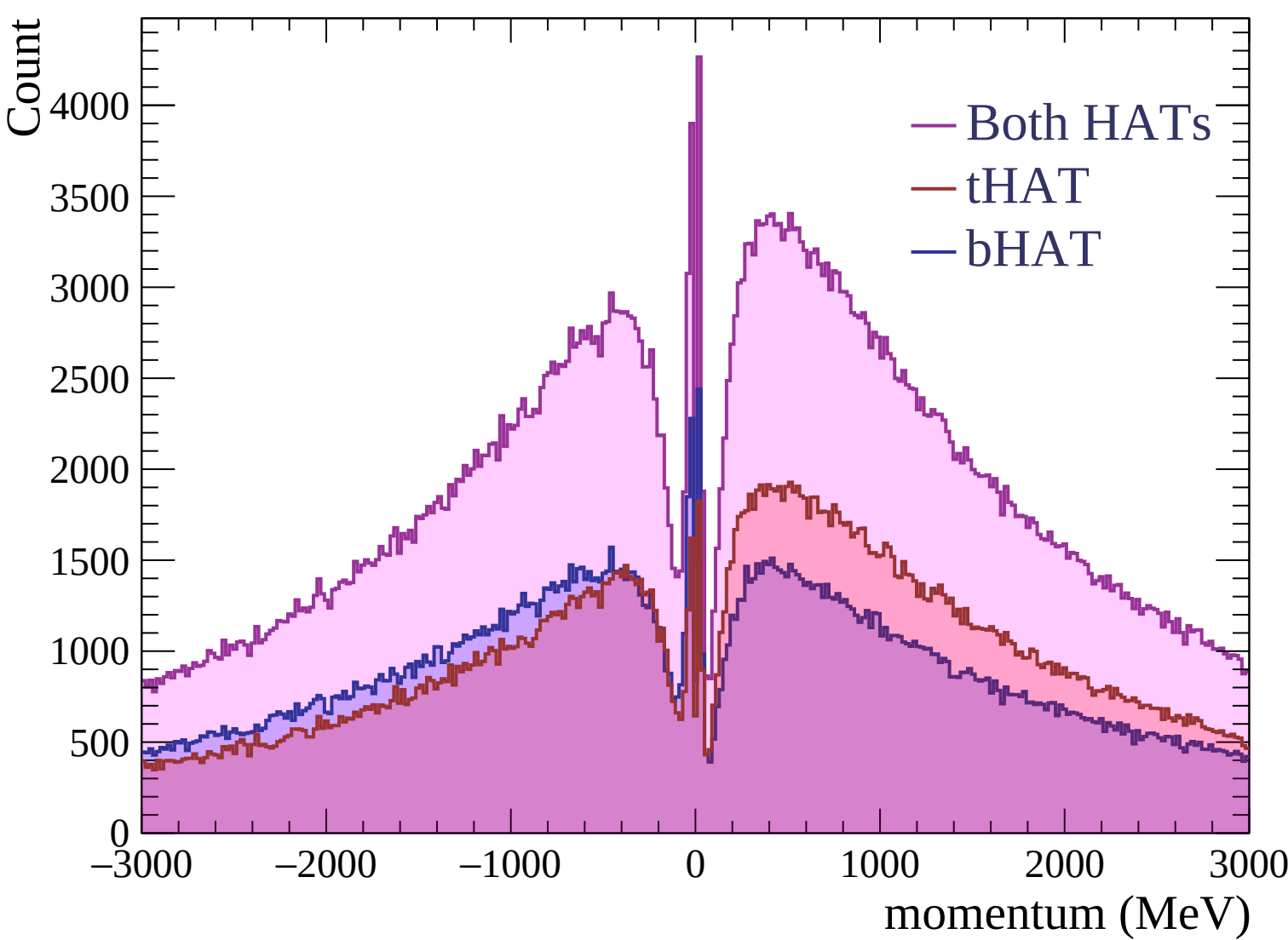


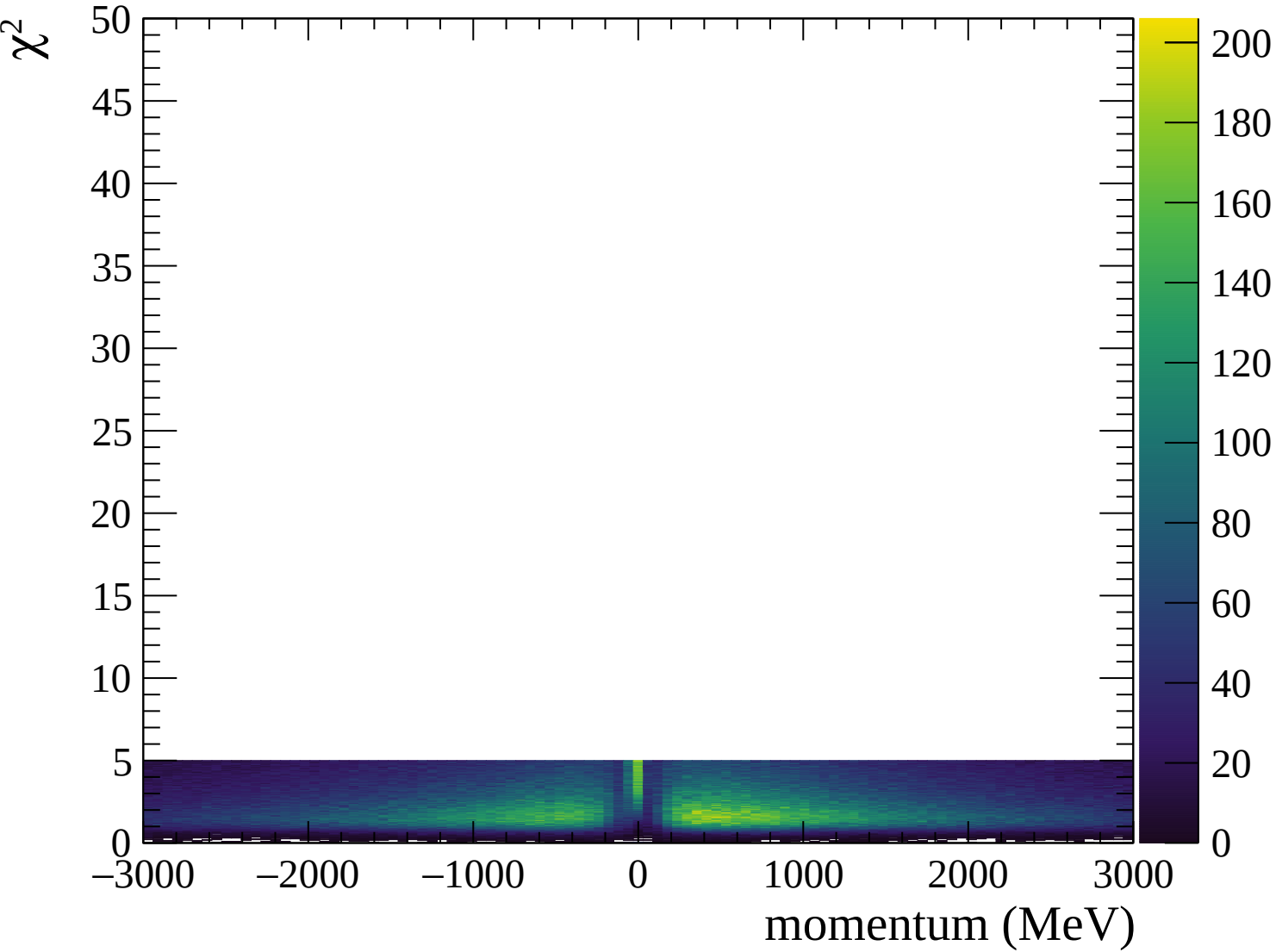


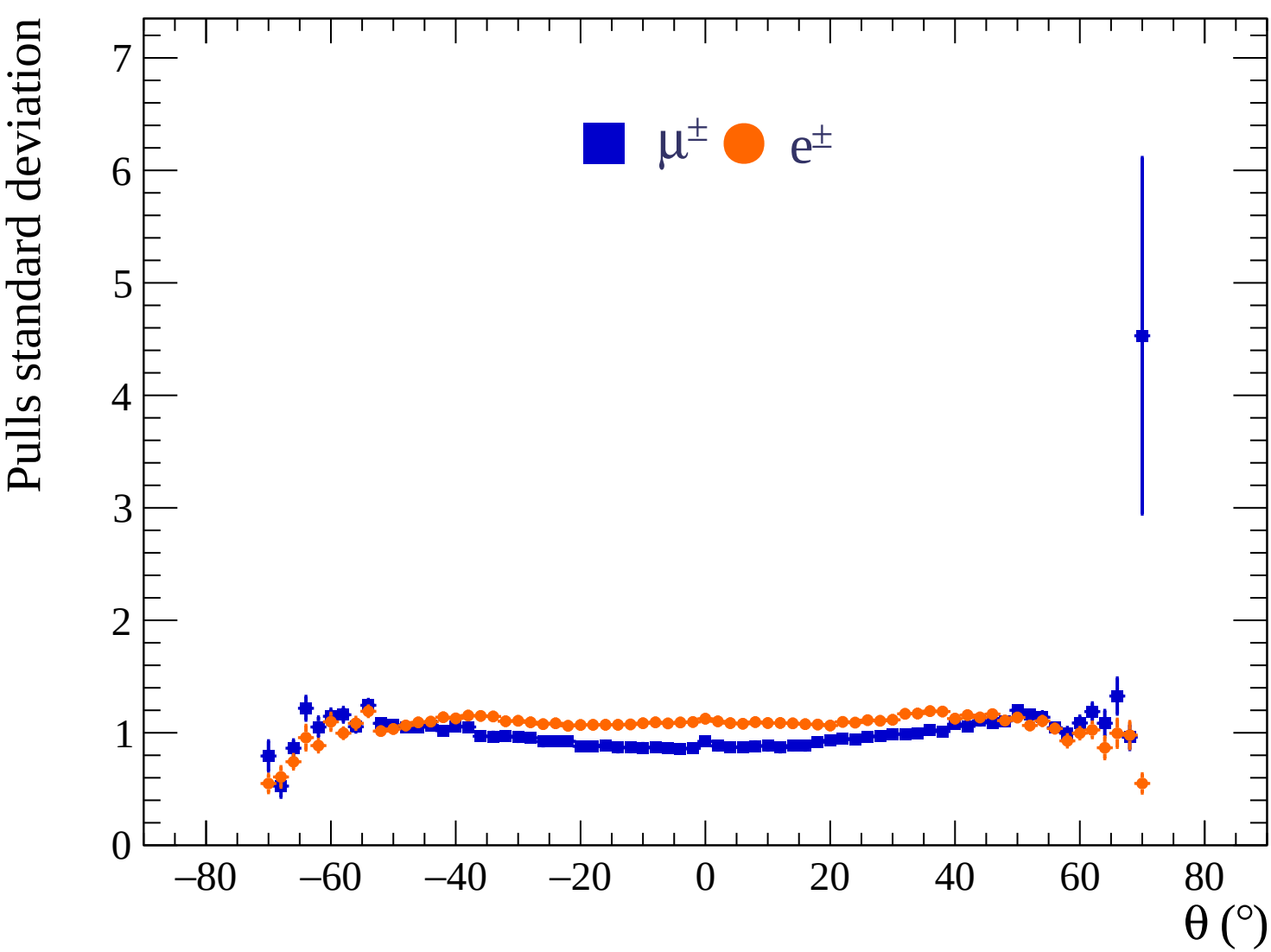


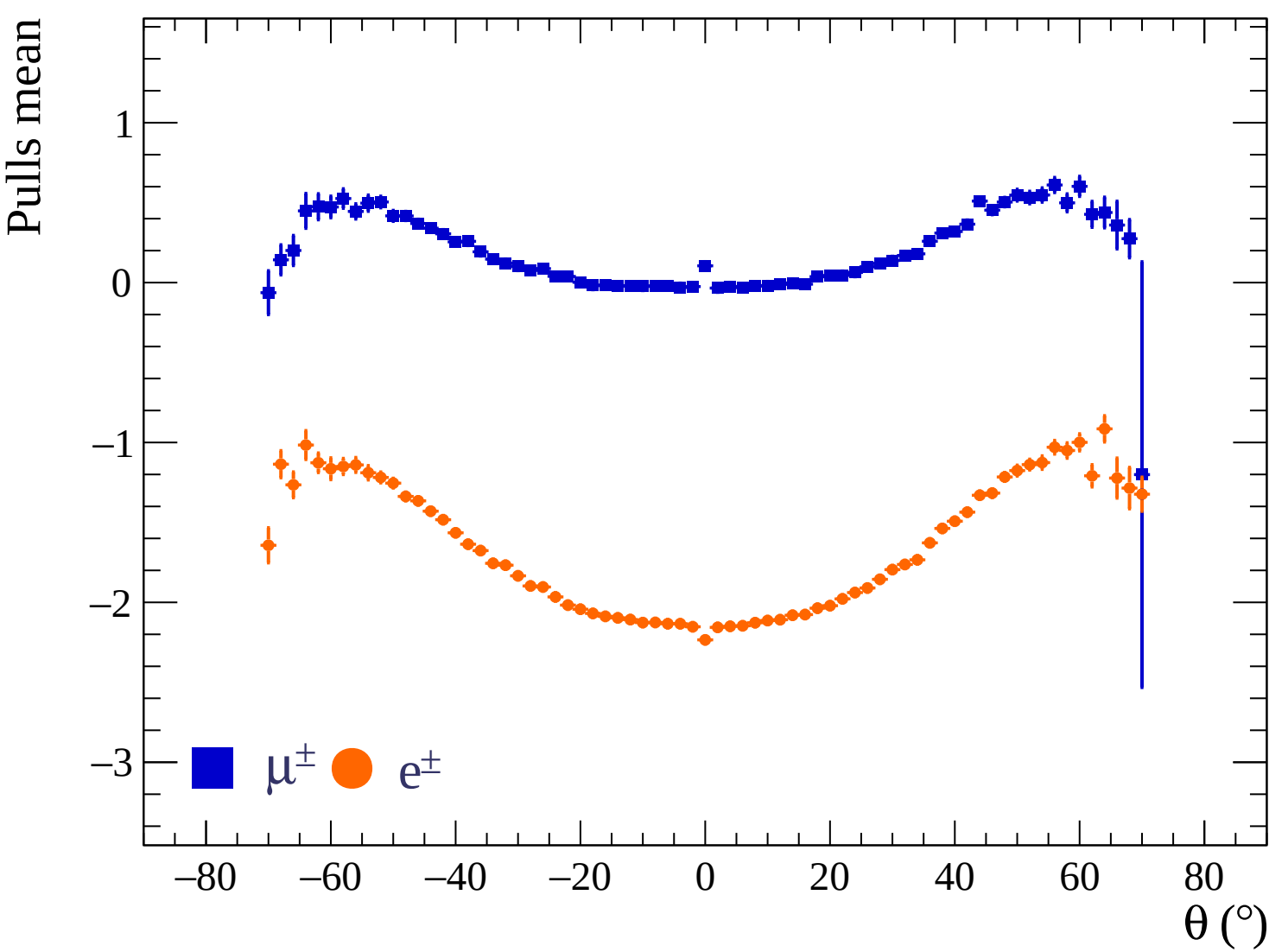


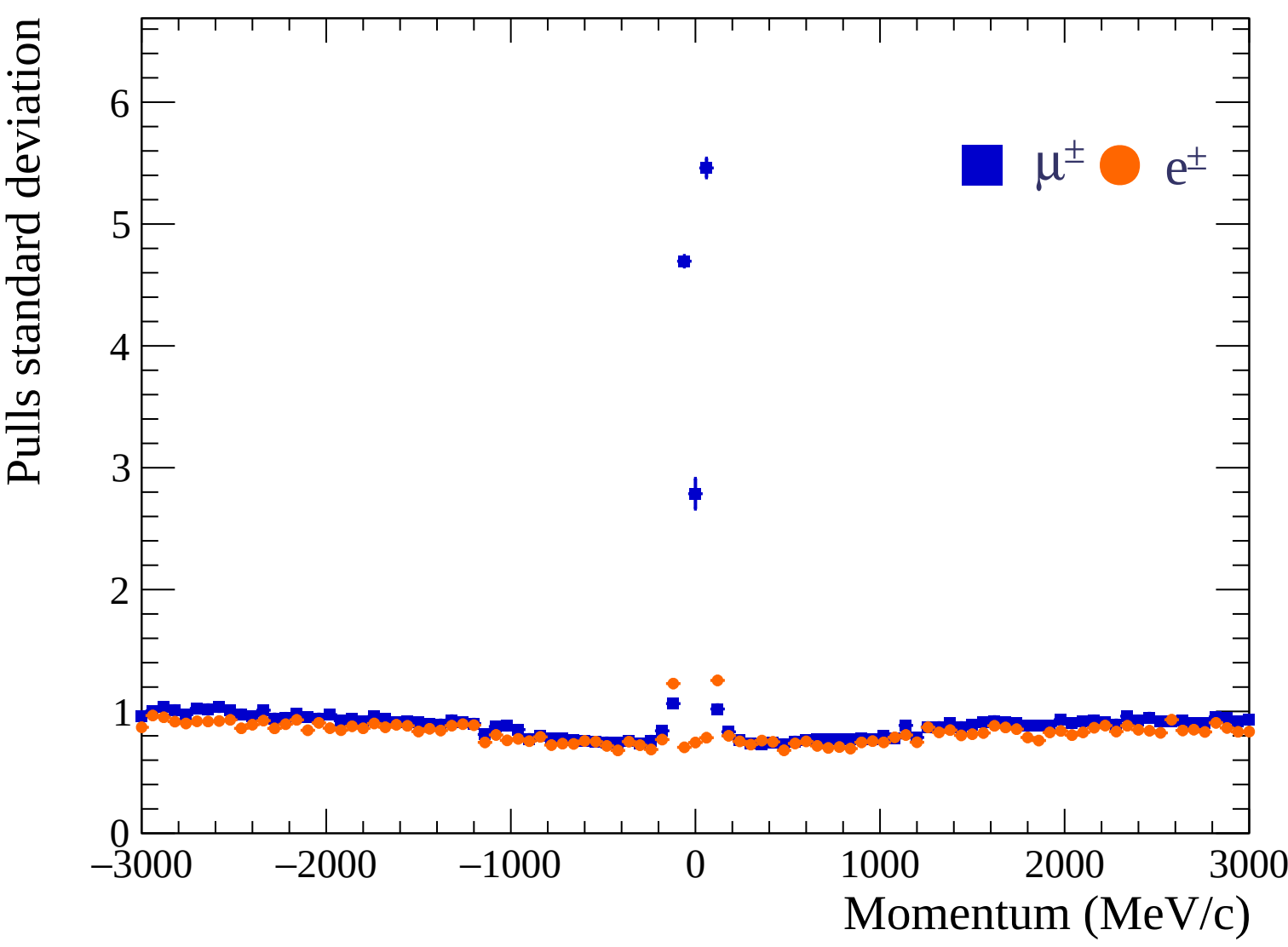


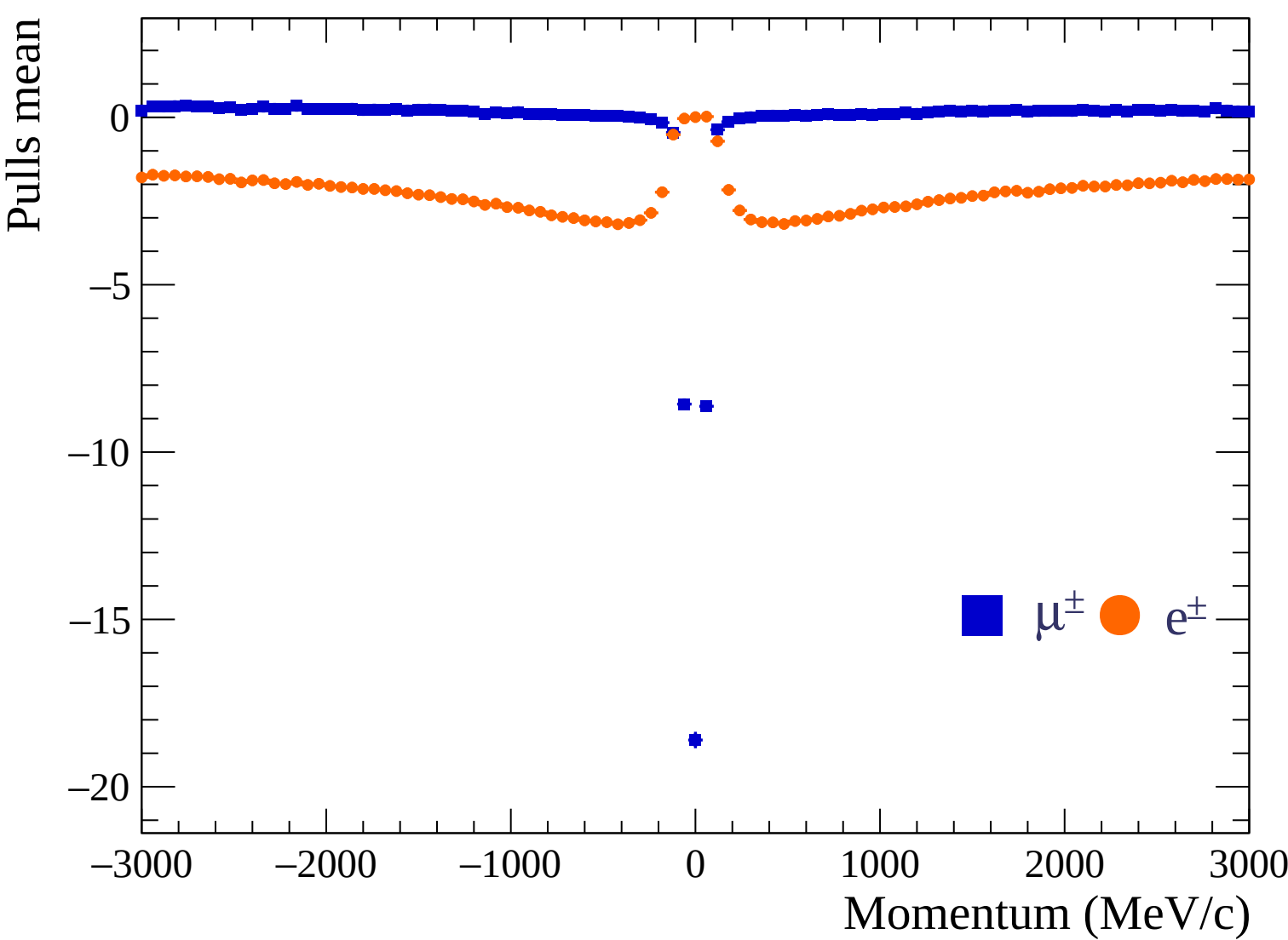


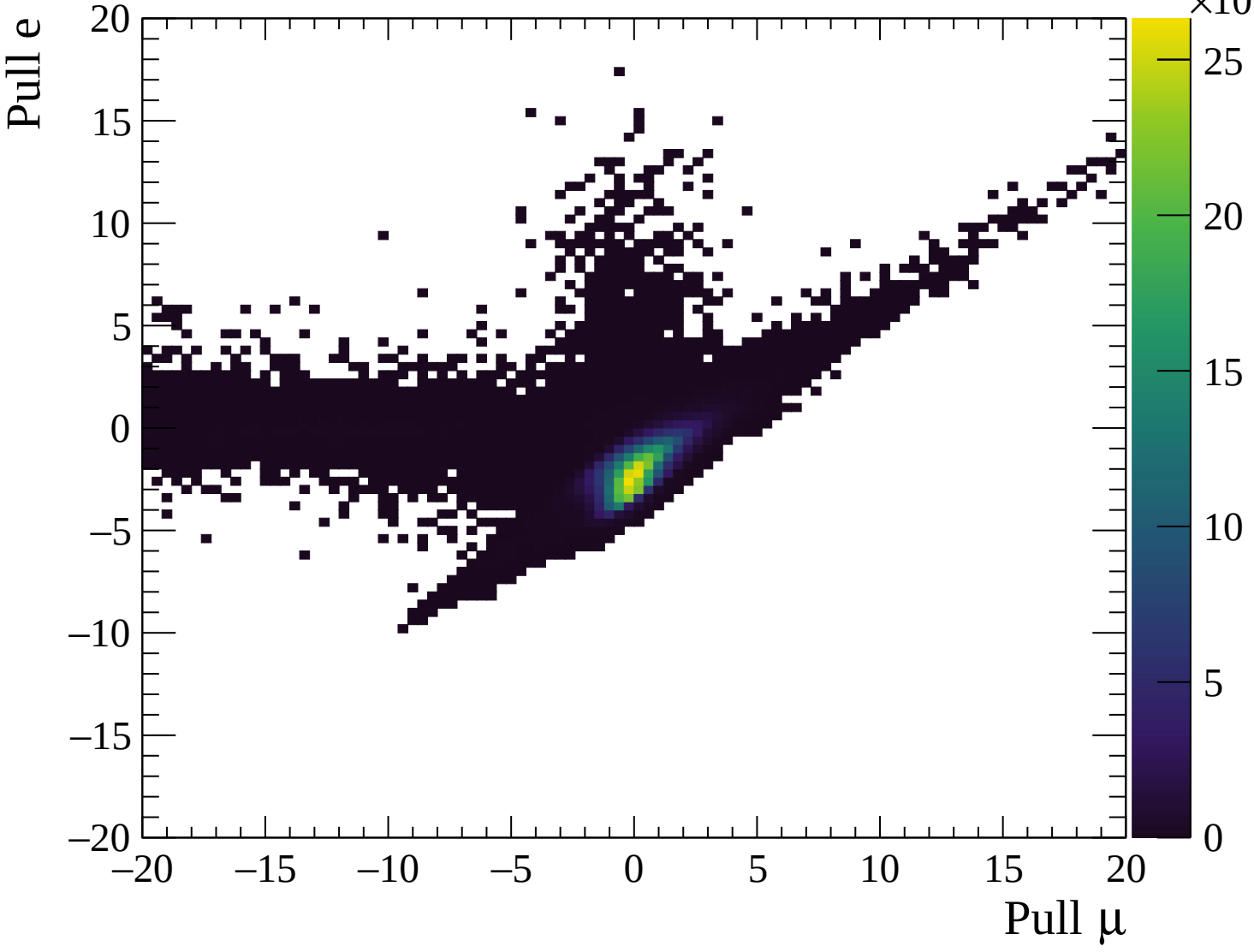


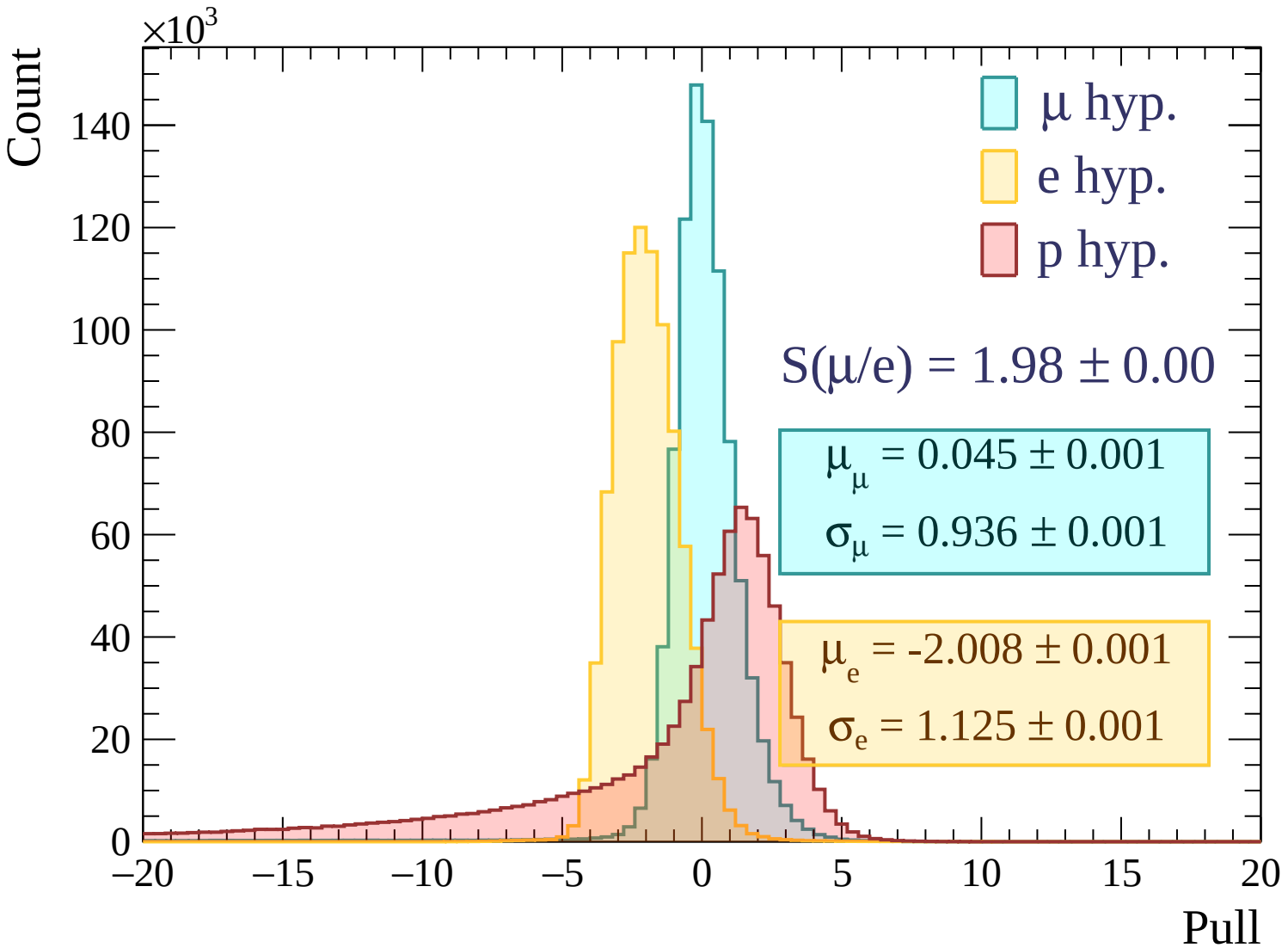


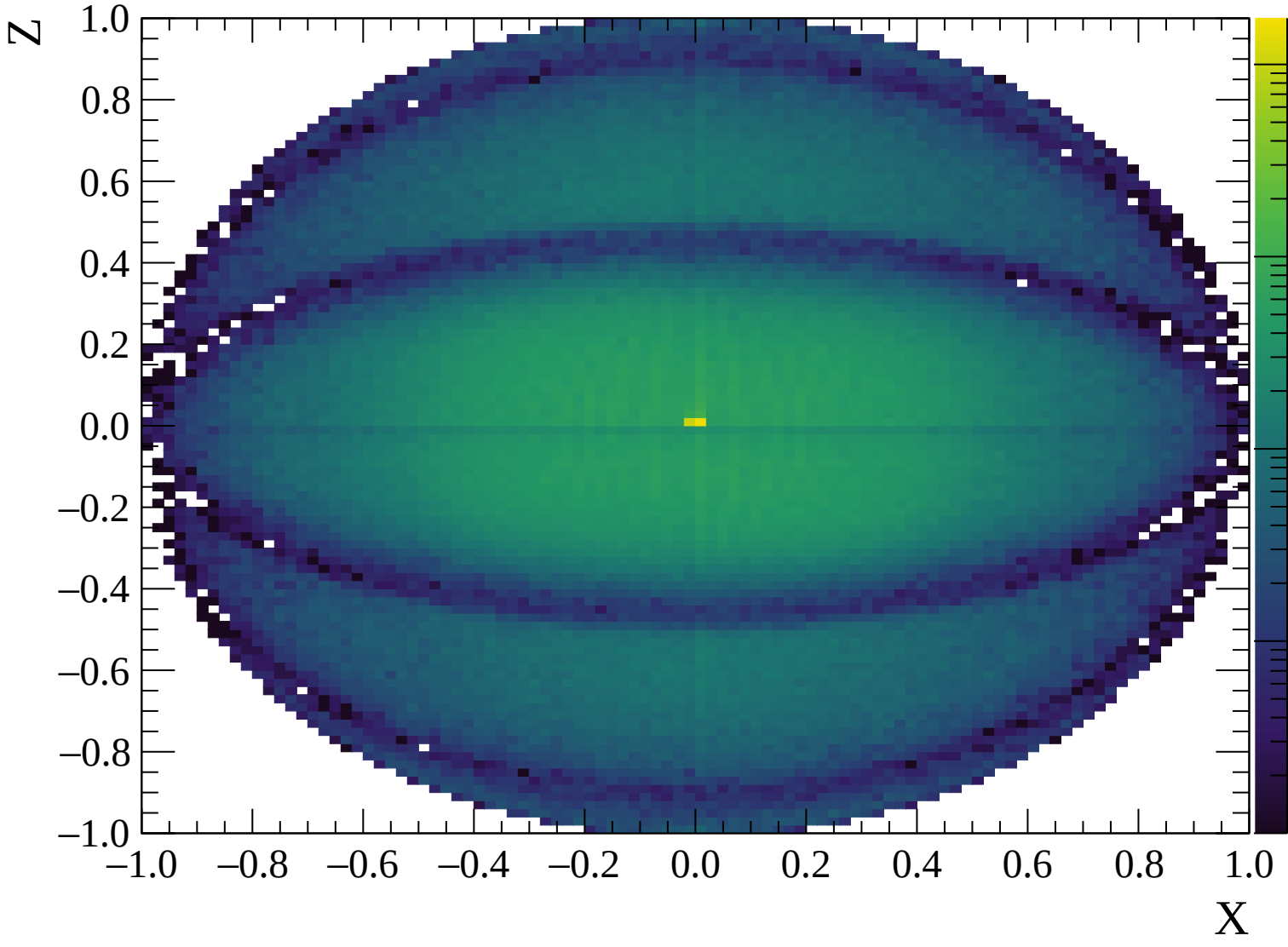


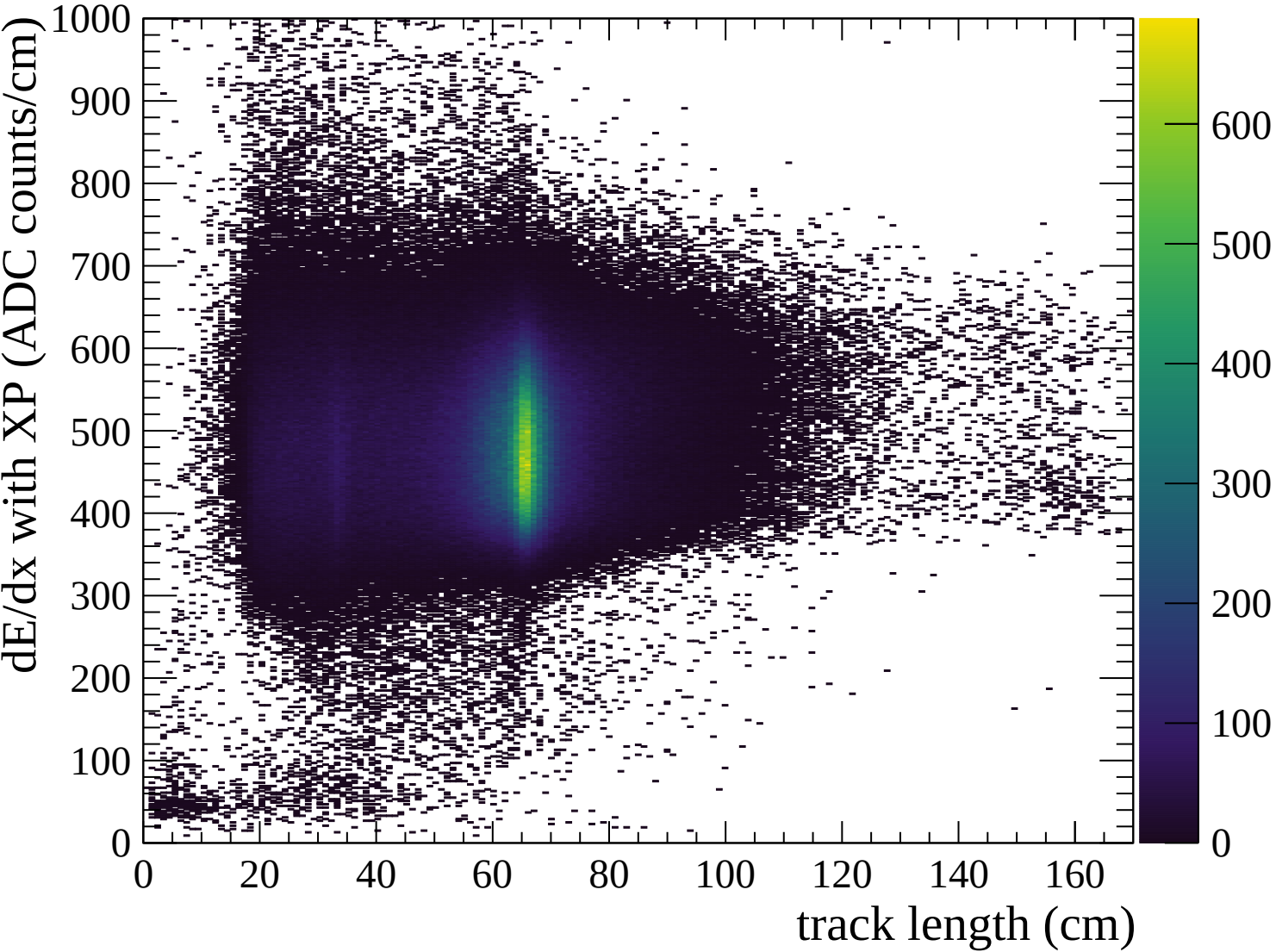


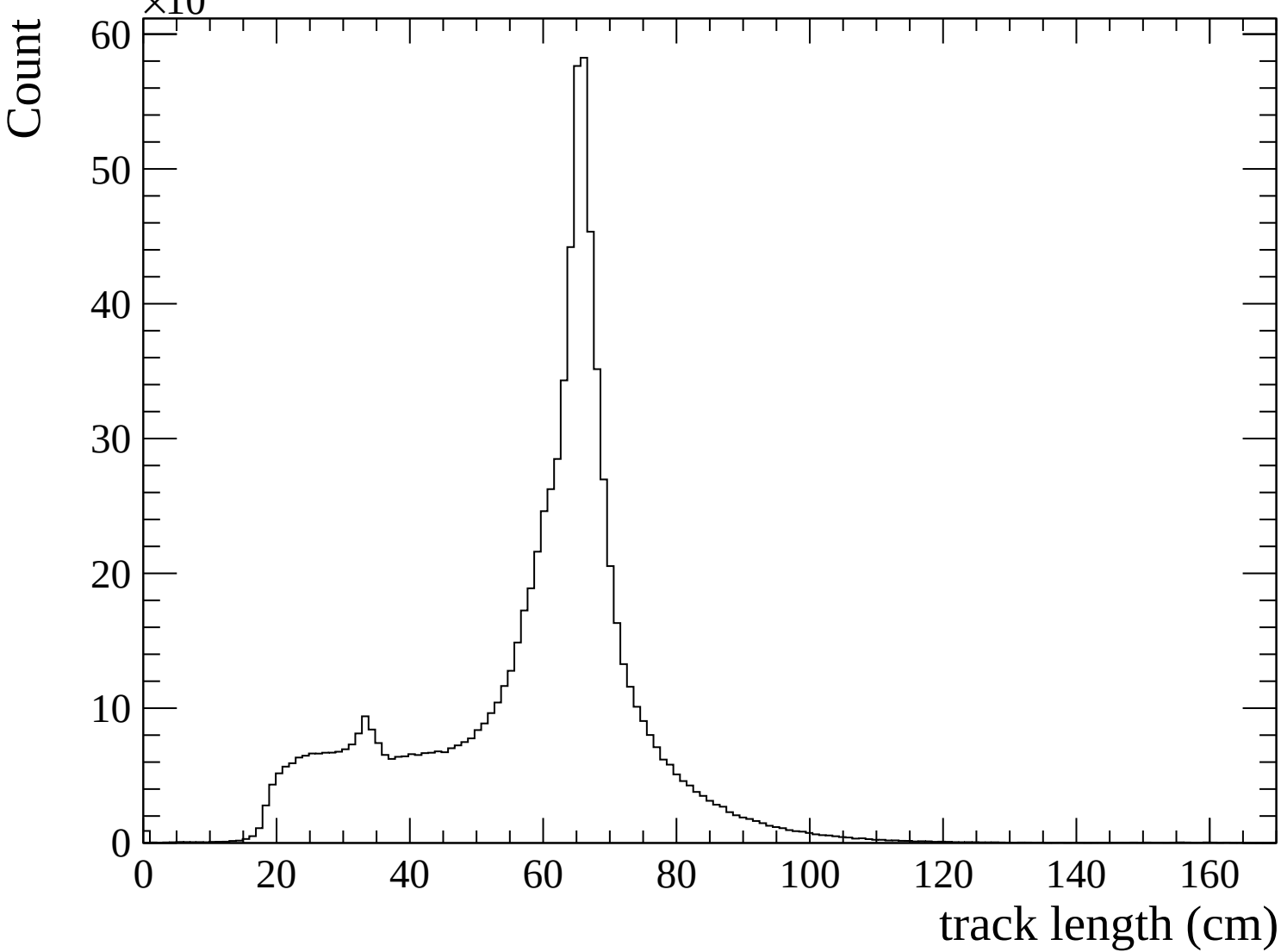


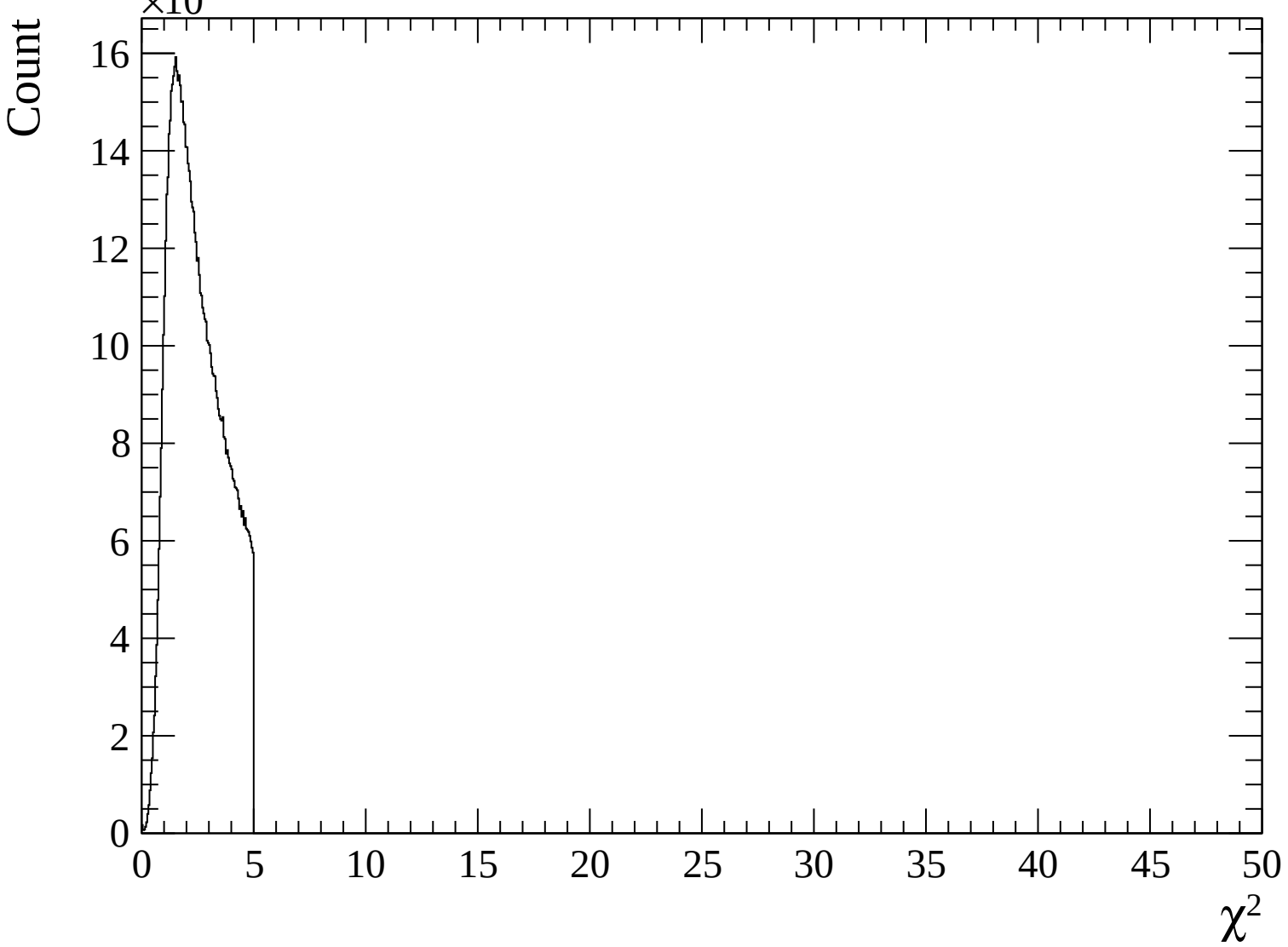


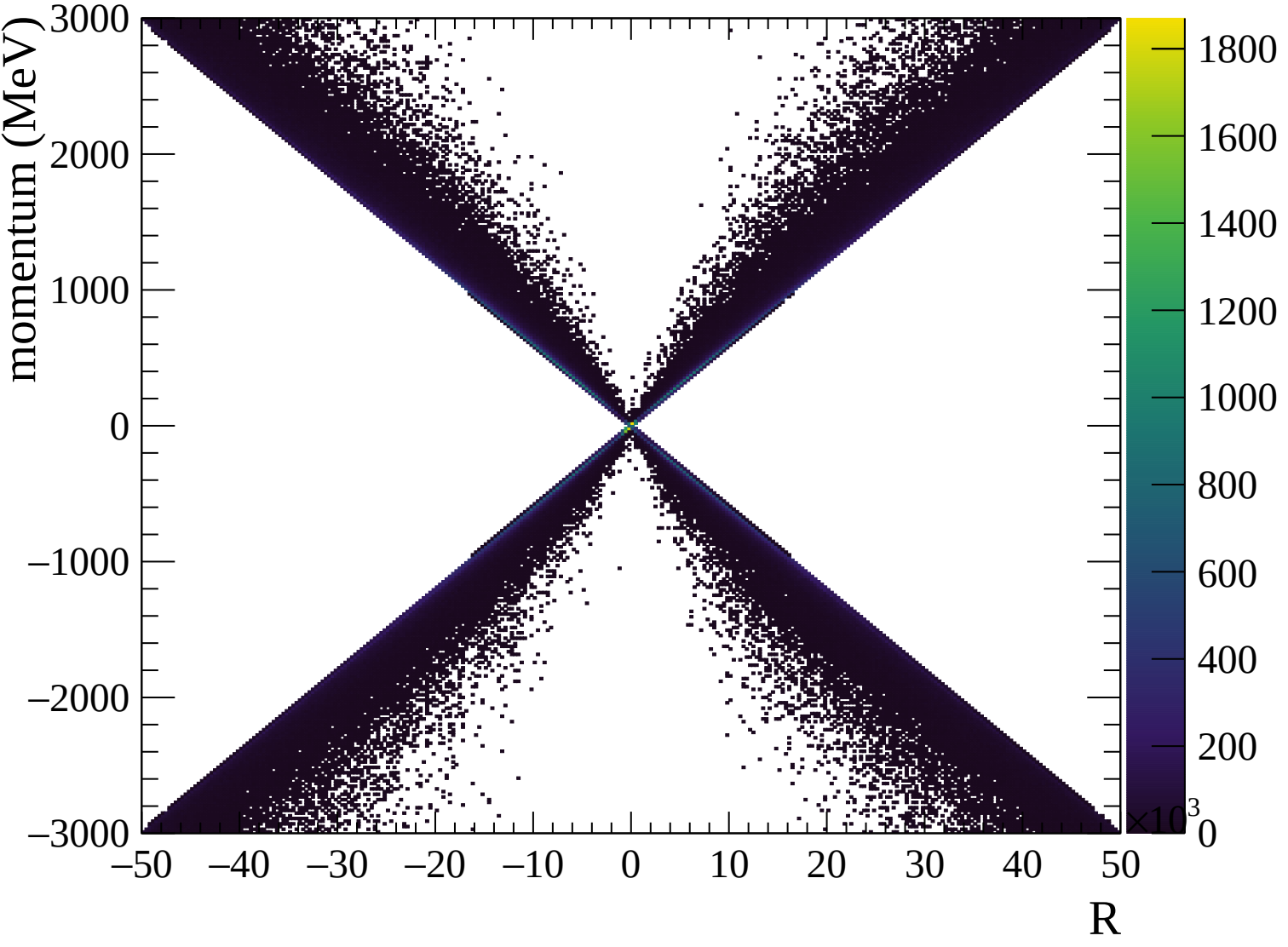


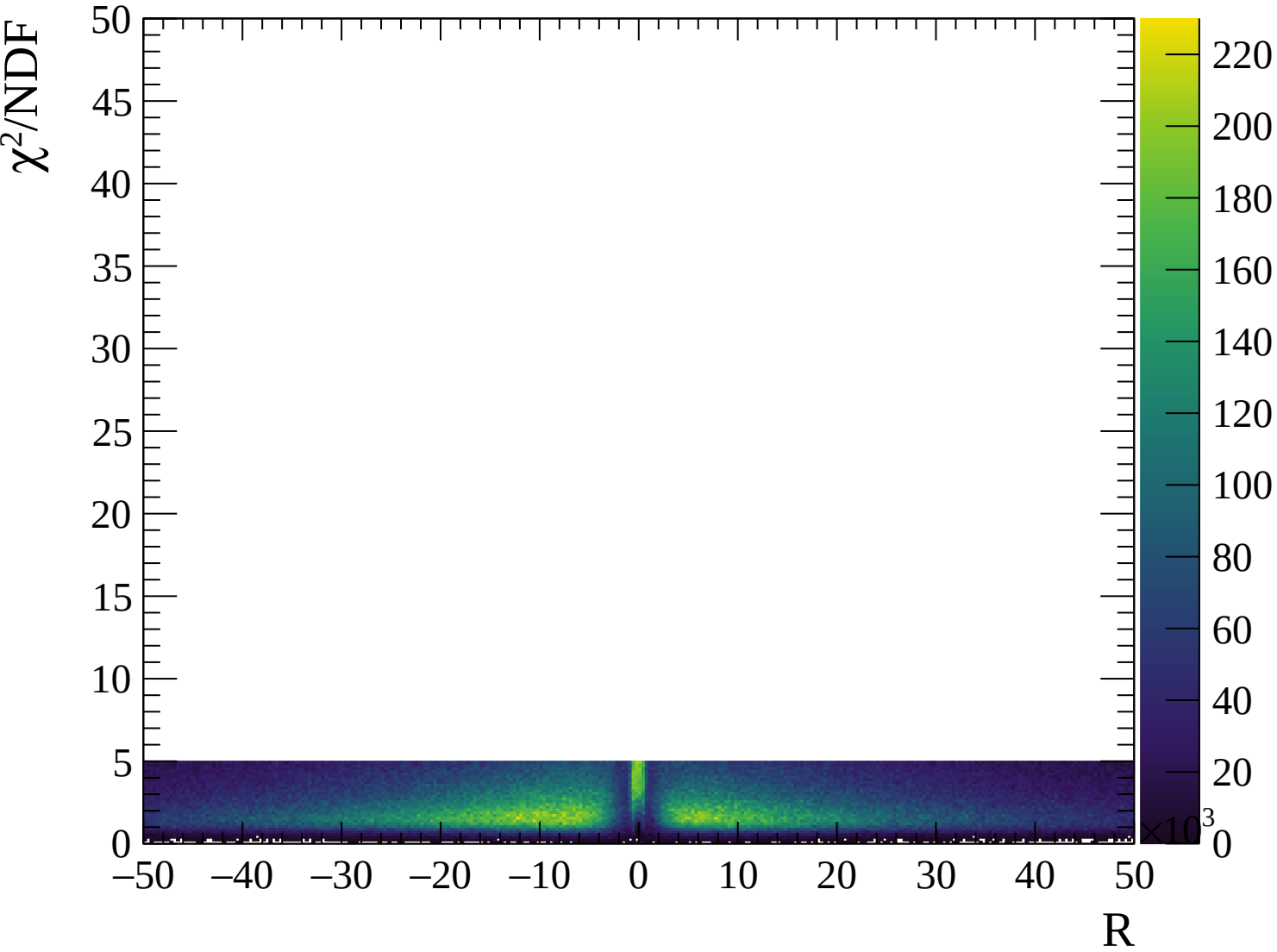






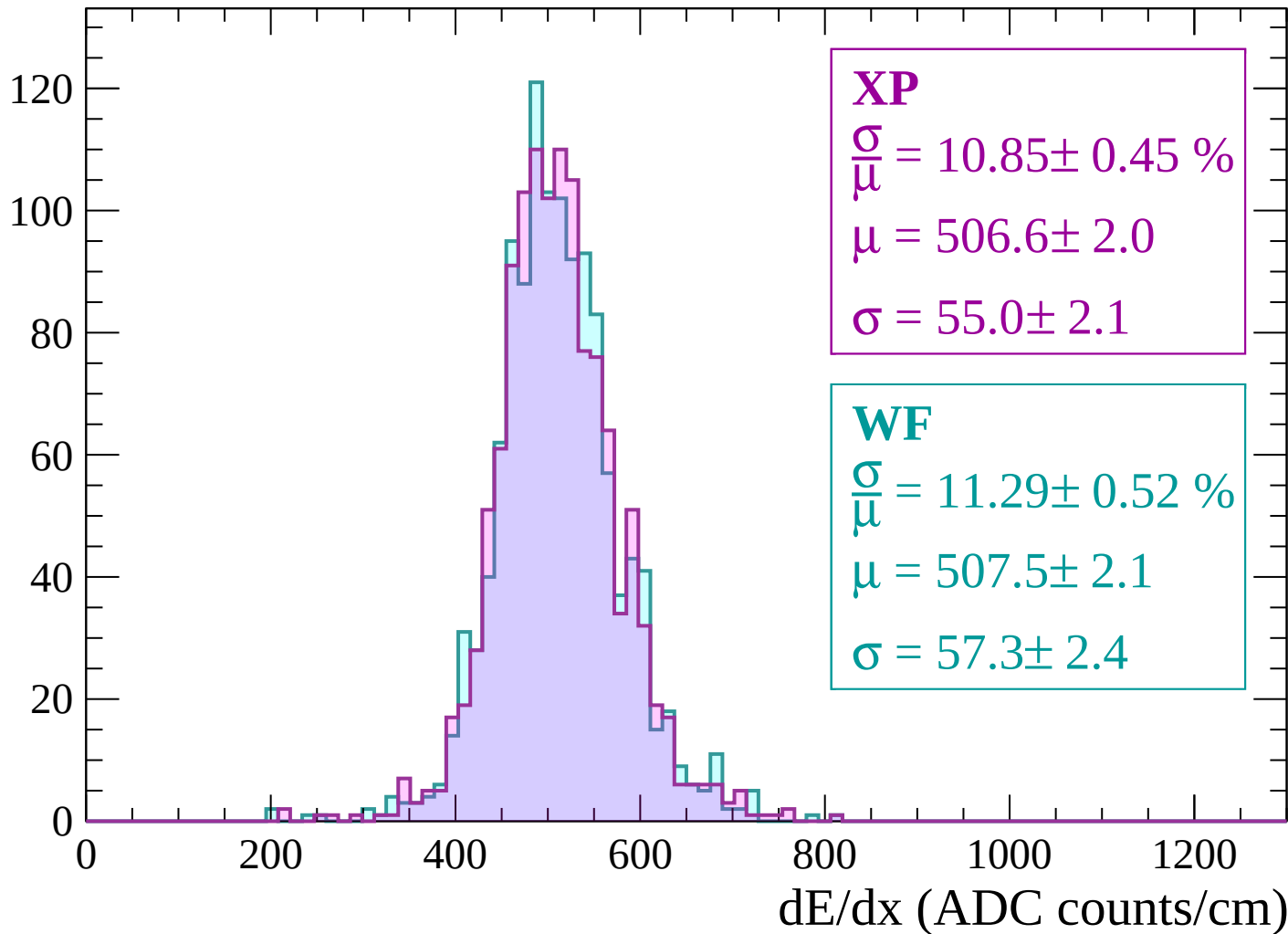




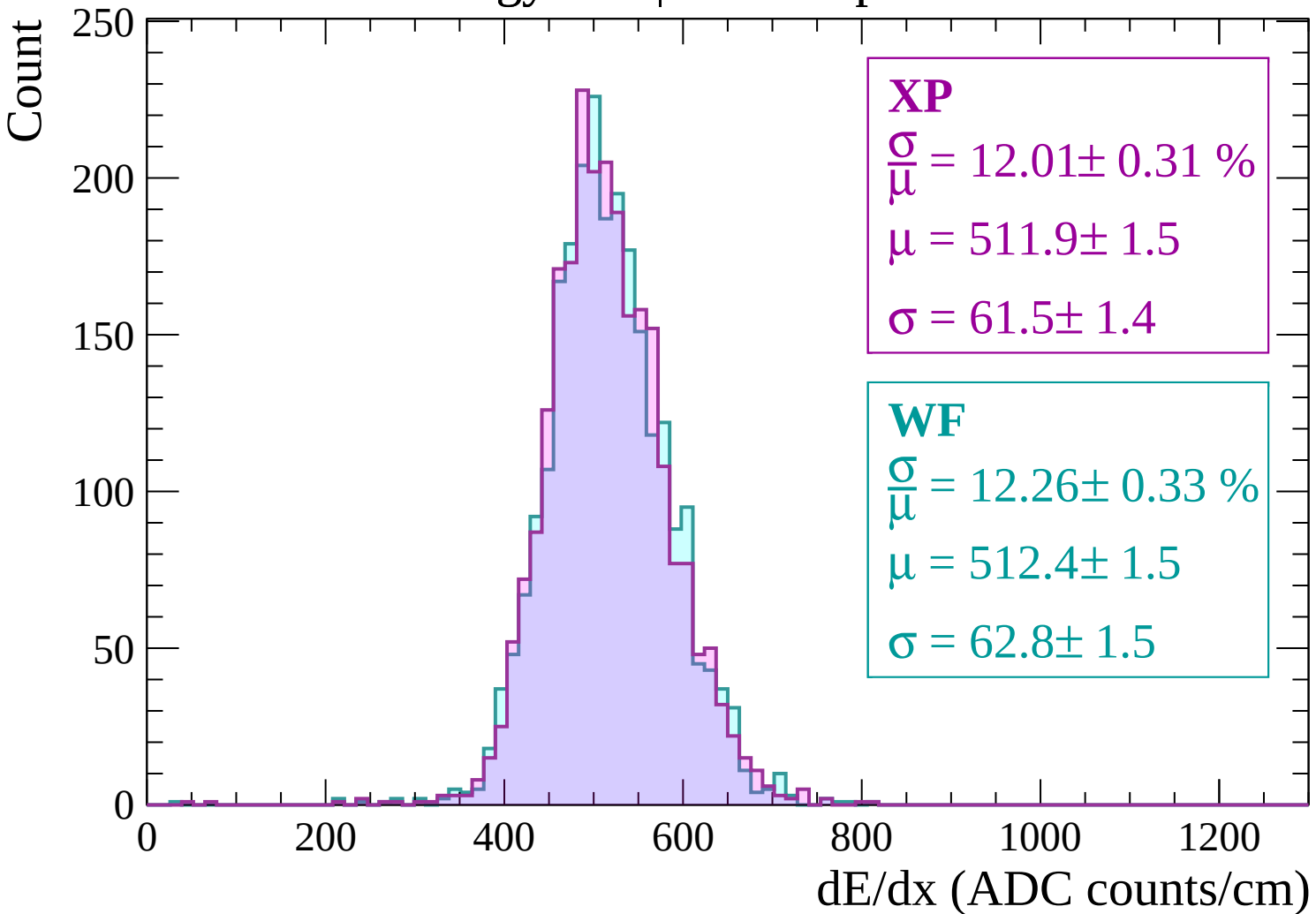


Energy loss | -3000 < p < -2940

Count

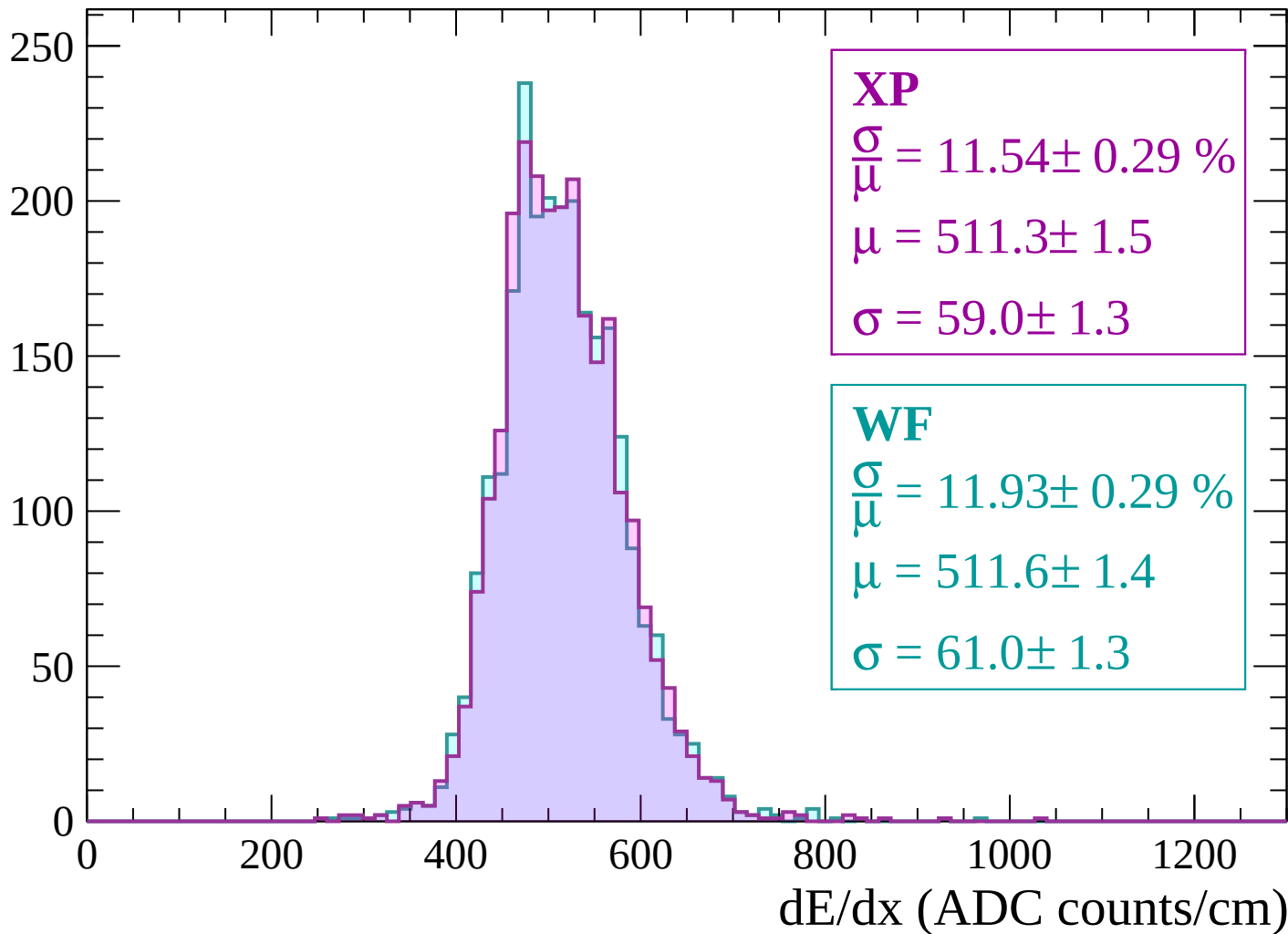


Energy loss | $-2940 < p < -2880$



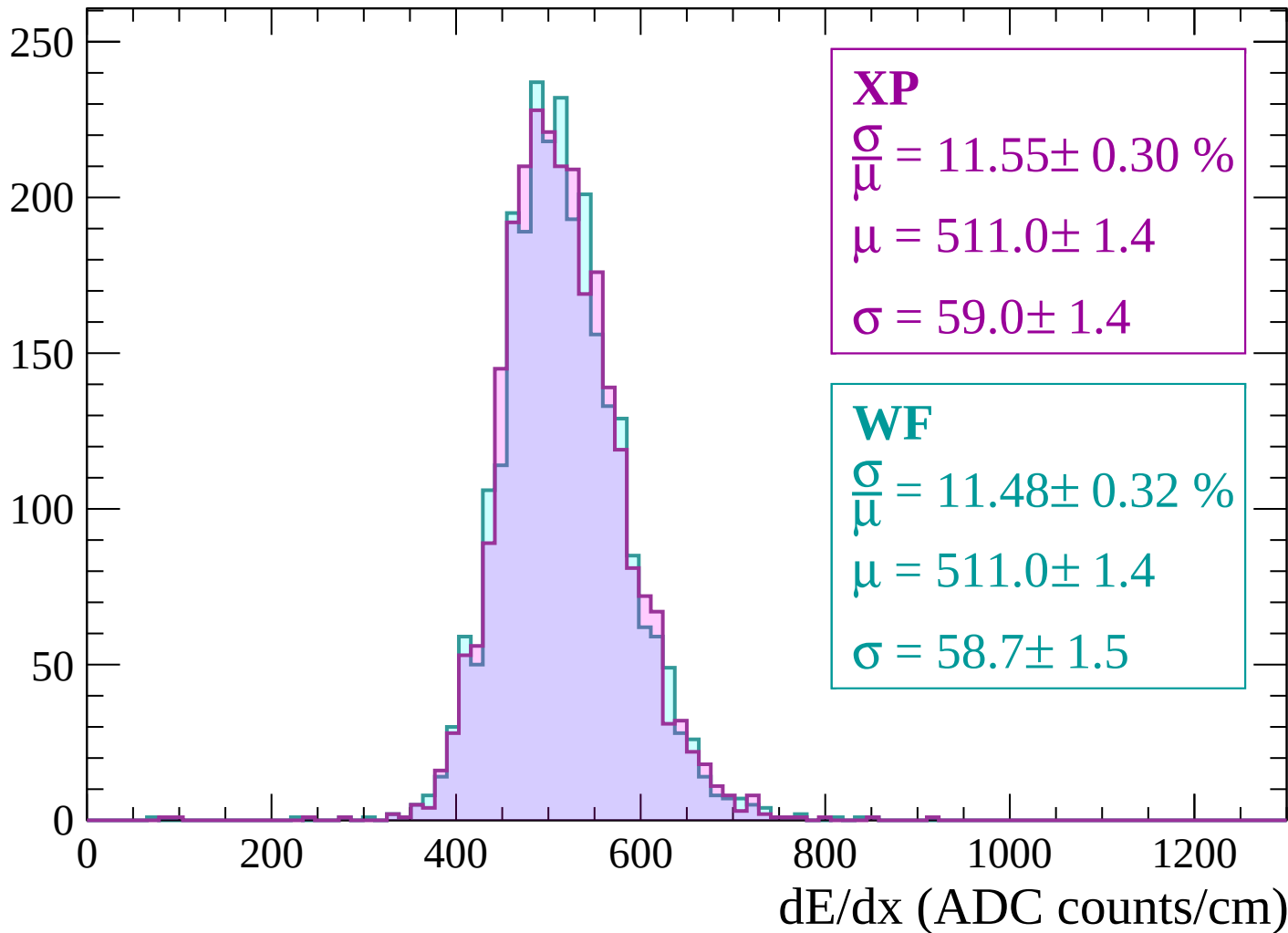
Energy loss | $-2880 < p < -2820$

Count



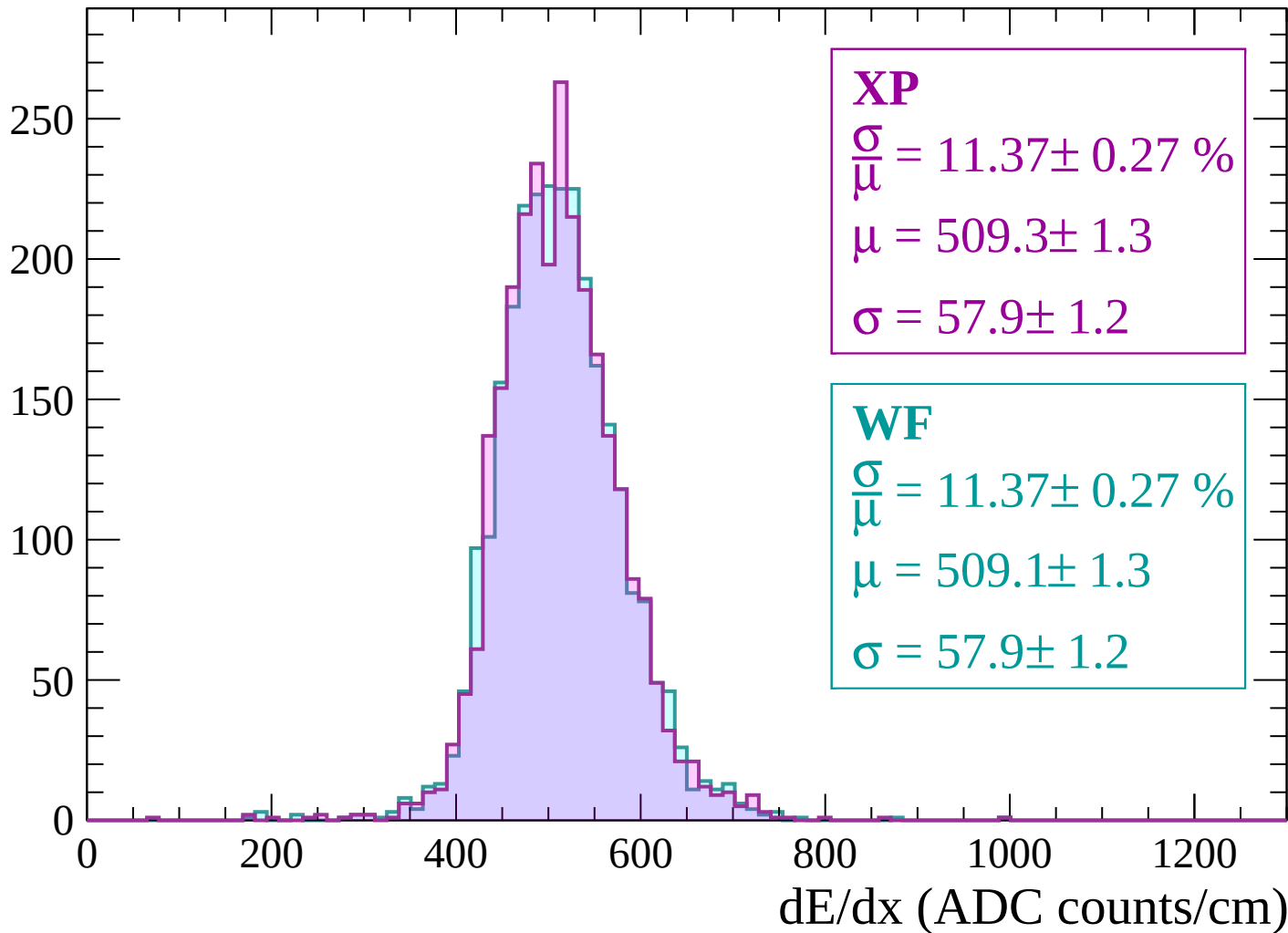
Energy loss | $-2820 < p < -2760$

Count



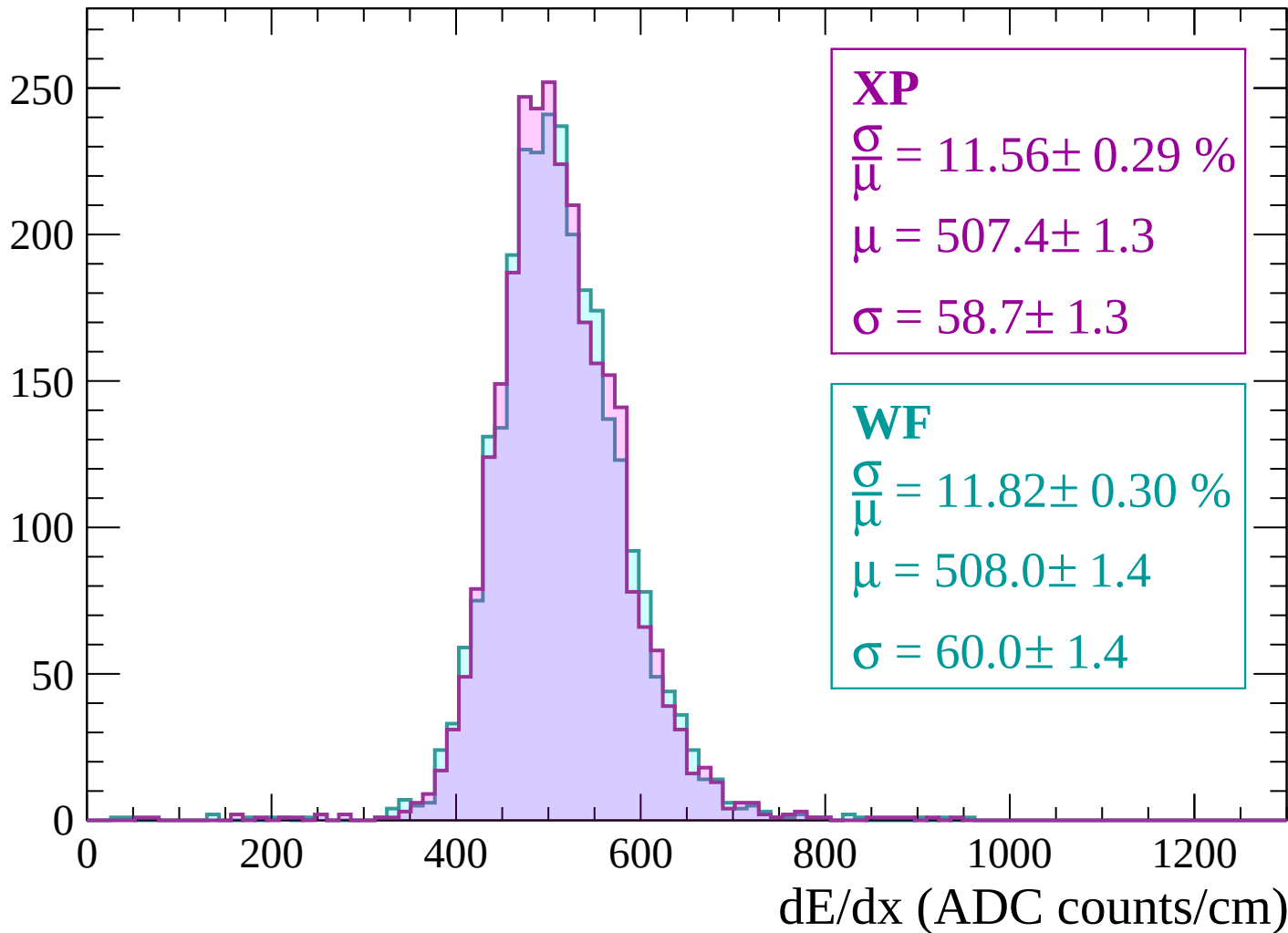
Energy loss | $-2760 < p < -2700$

Count



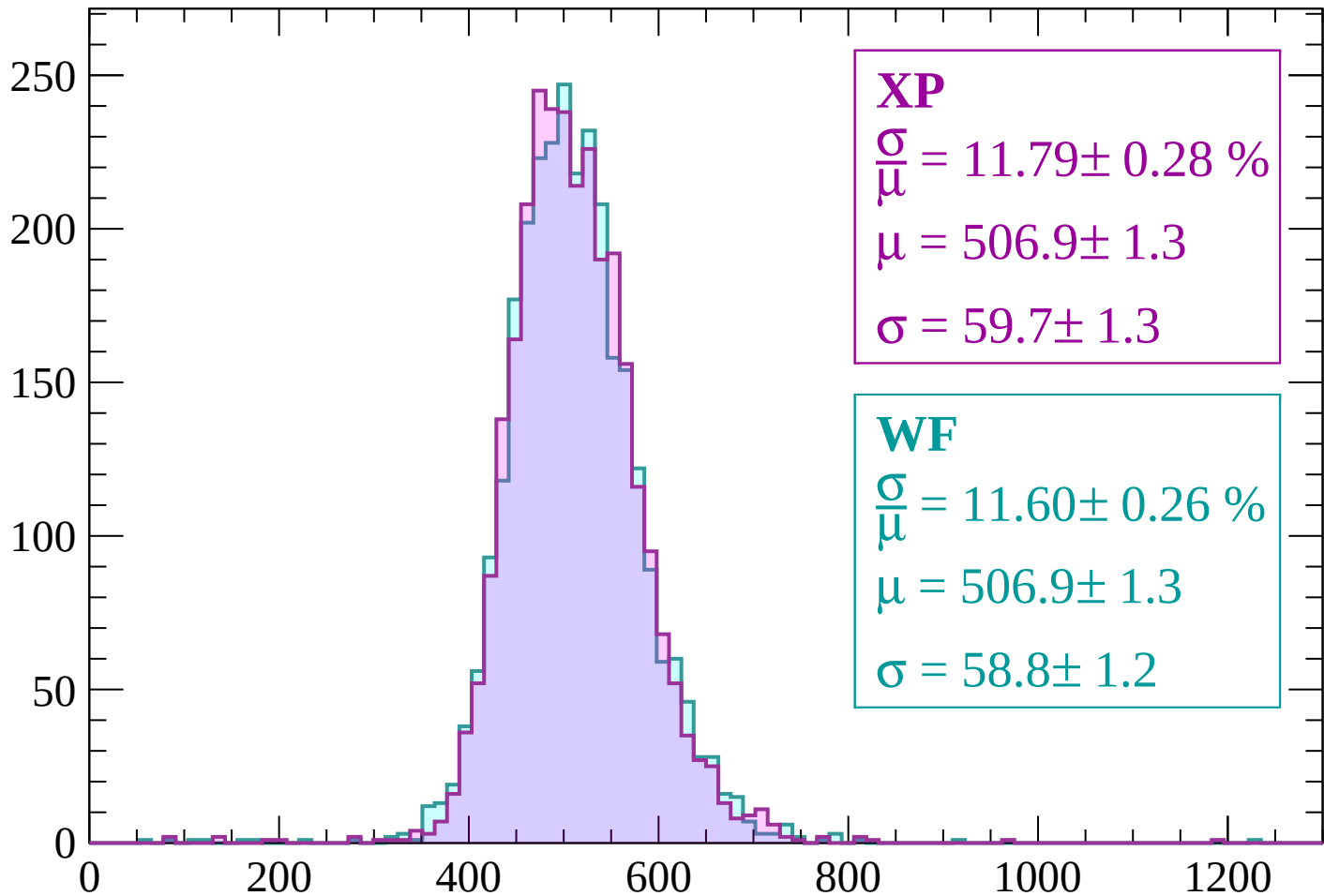
Energy loss | $-2700 < p < -2640$

Count



Energy loss | $-2640 < p < -2580$

Count



XP

$$\frac{\sigma}{\mu} = 11.79 \pm 0.28 \%$$

$$\mu = 506.9 \pm 1.3$$

$$\sigma = 59.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.60 \pm 0.26 \%$$

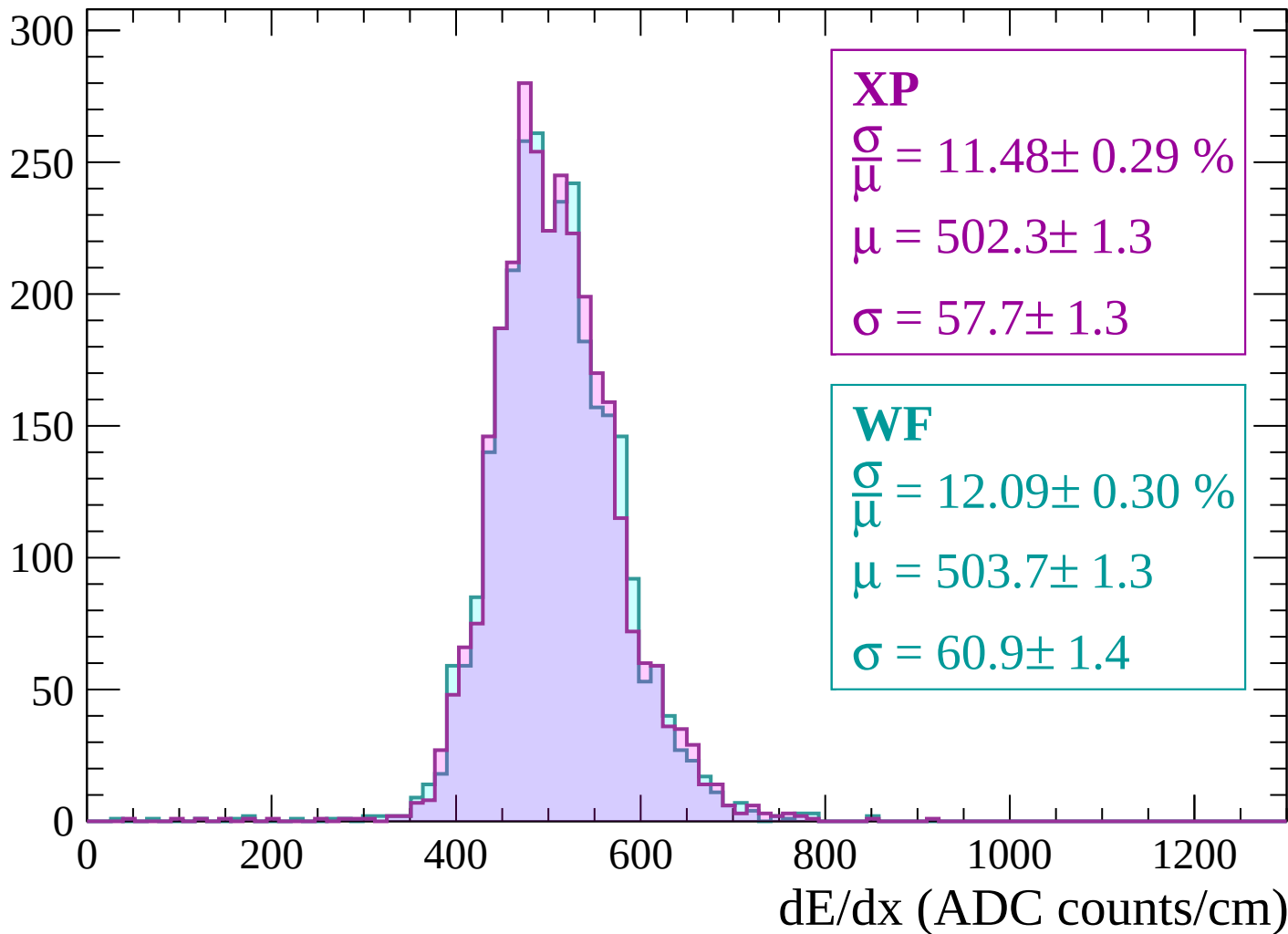
$$\mu = 506.9 \pm 1.3$$

$$\sigma = 58.8 \pm 1.2$$

dE/dx (ADC counts/cm)

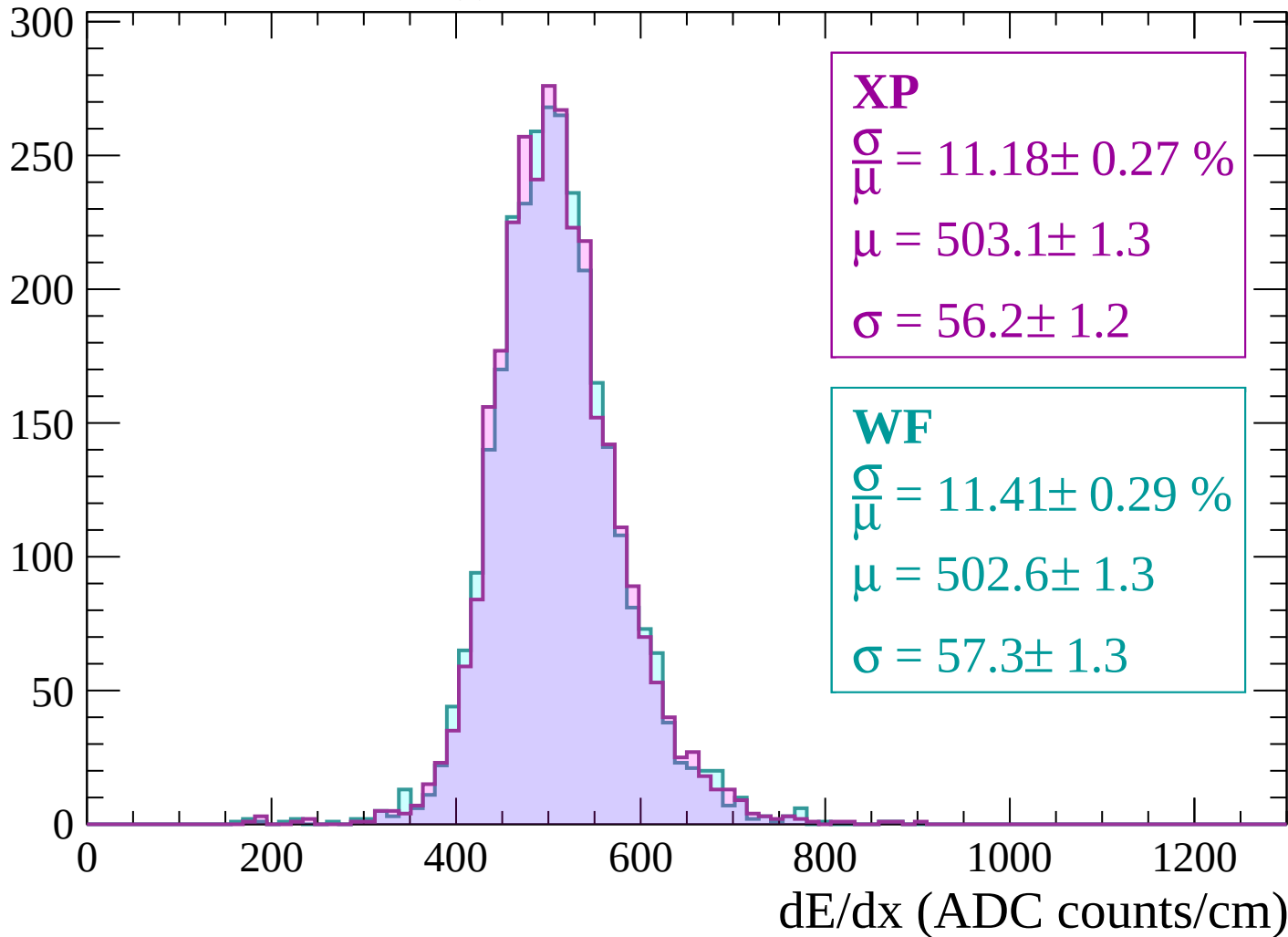
Energy loss | $-2580 < p < -2520$

Count



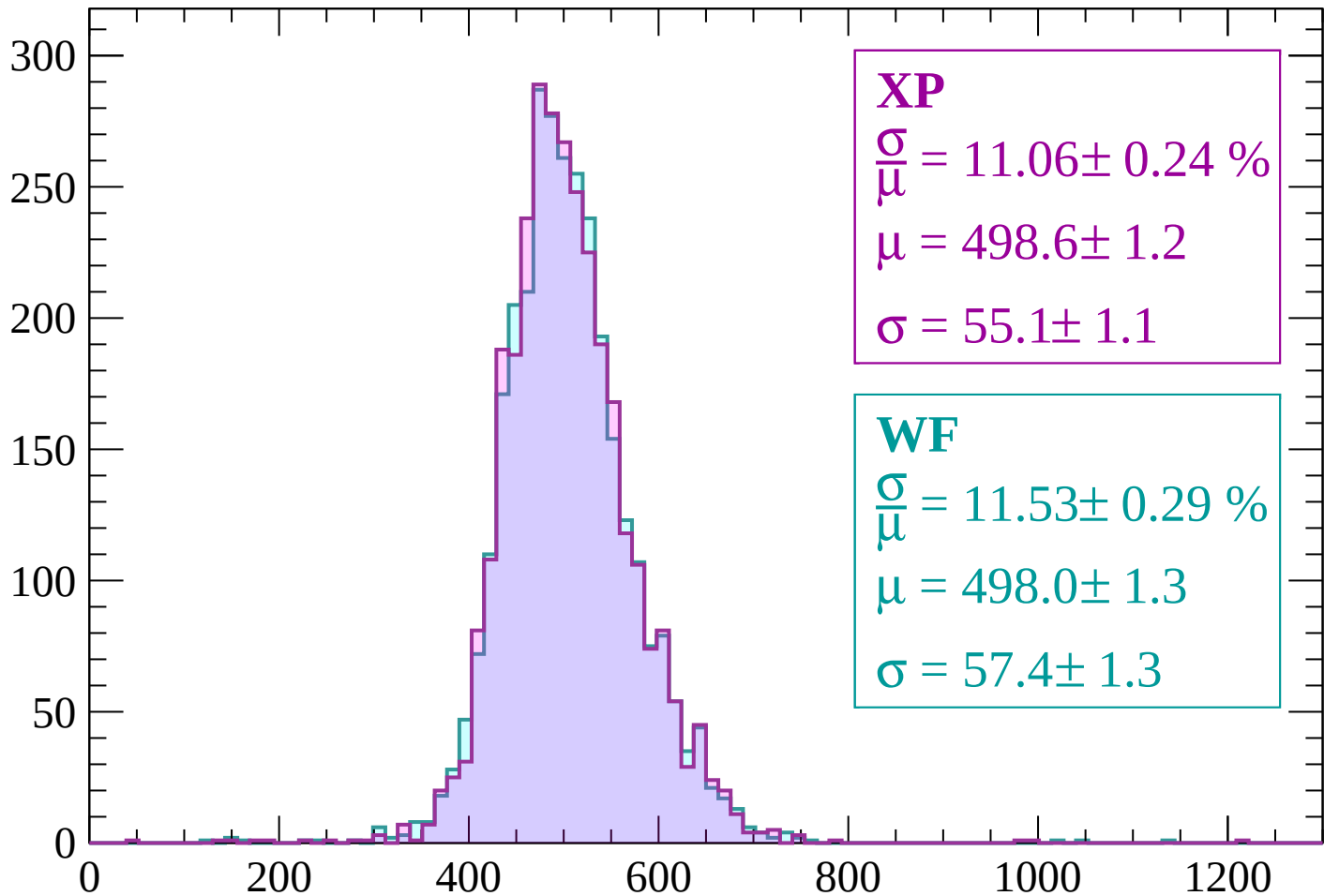
Energy loss | $-2520 < p < -2460$

Count



Energy loss | $-2460 < p < -2400$

Count



XP

$$\frac{\sigma}{\mu} = 11.06 \pm 0.24 \%$$

$$\mu = 498.6 \pm 1.2$$

$$\sigma = 55.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.53 \pm 0.29 \%$$

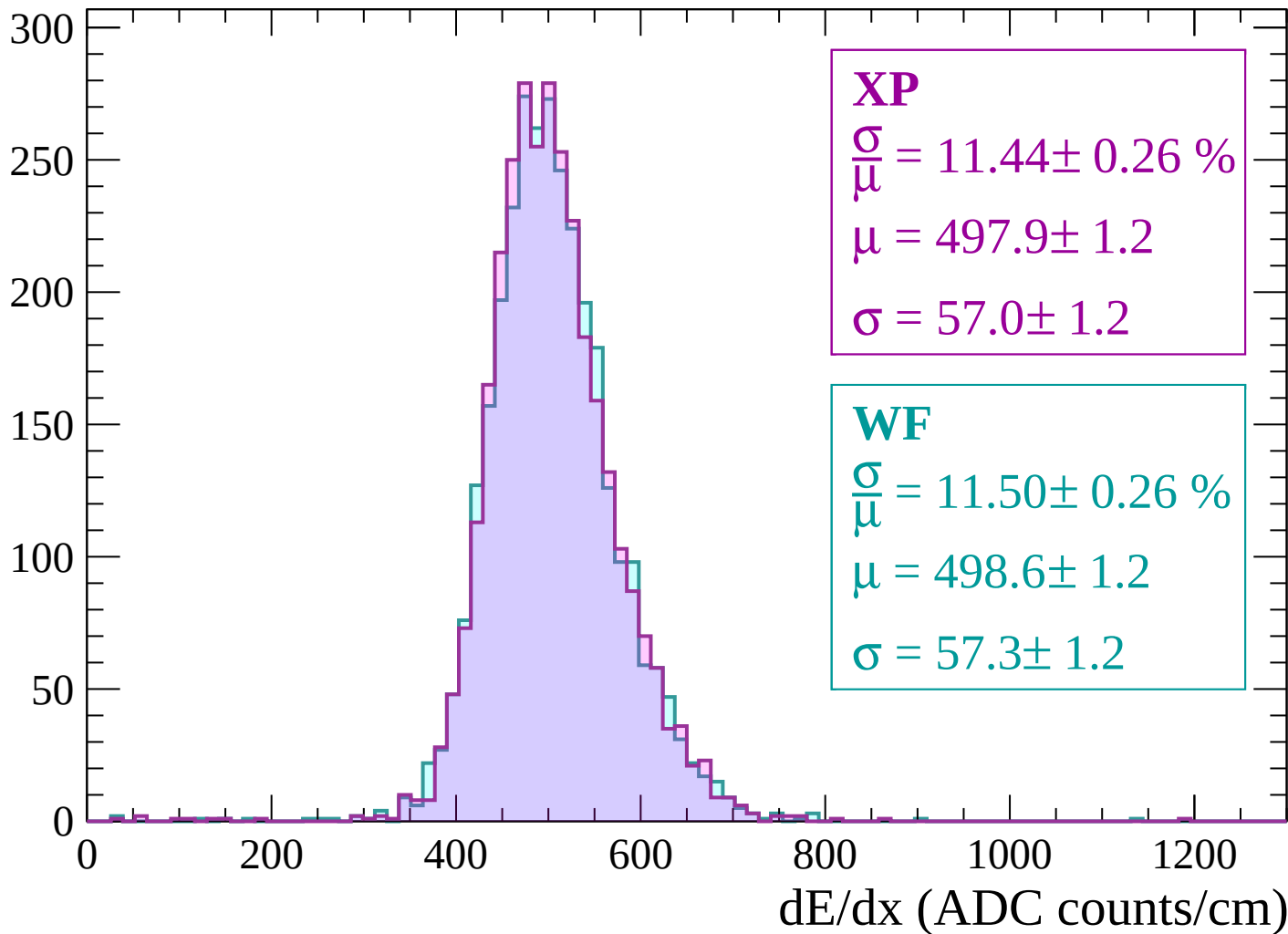
$$\mu = 498.0 \pm 1.3$$

$$\sigma = 57.4 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-2400 < p < -2340$

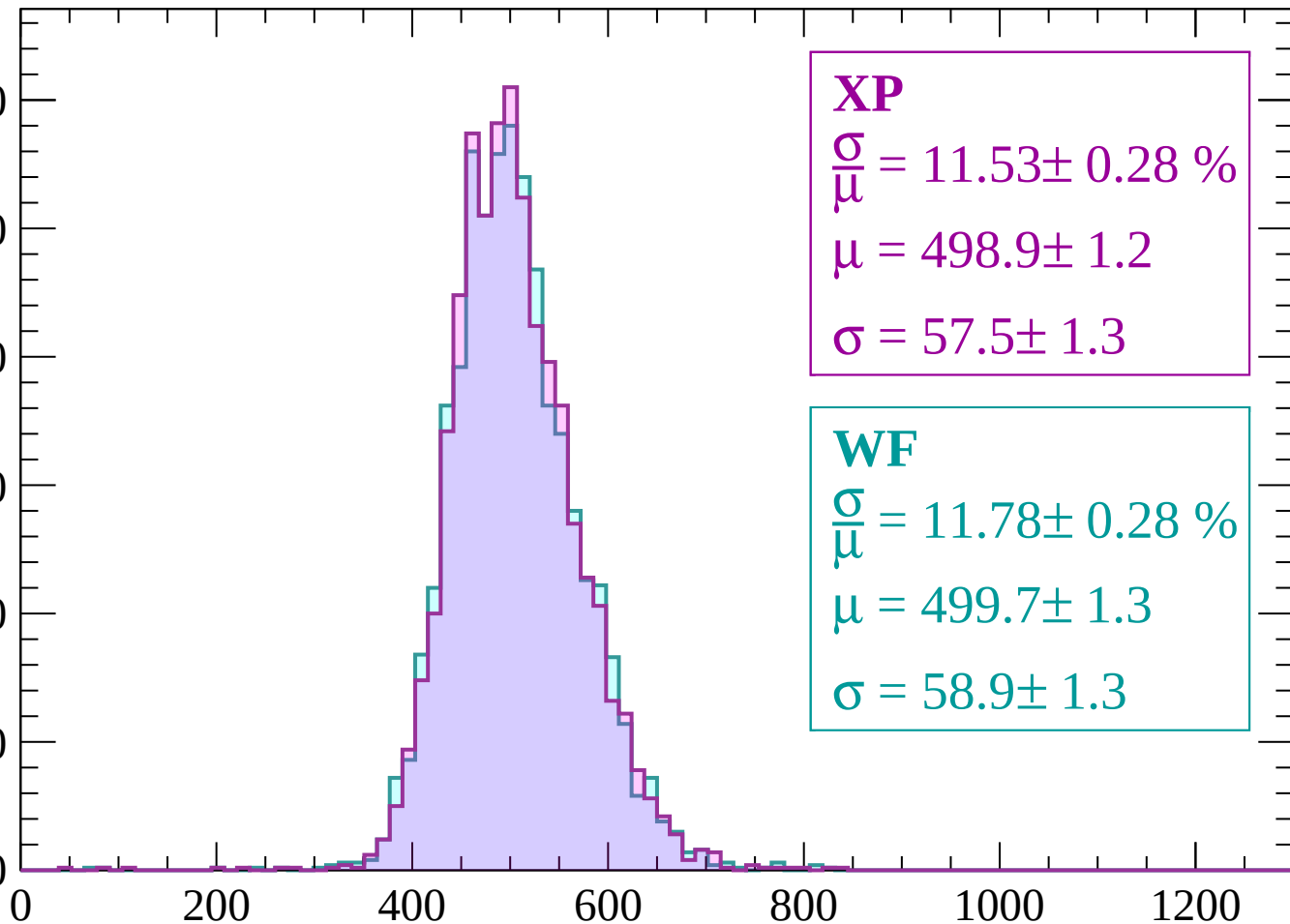
Count



Energy loss | $-2340 < p < -2280$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.53 \pm 0.28 \%$$

$$\mu = 498.9 \pm 1.2$$

$$\sigma = 57.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.78 \pm 0.28 \%$$

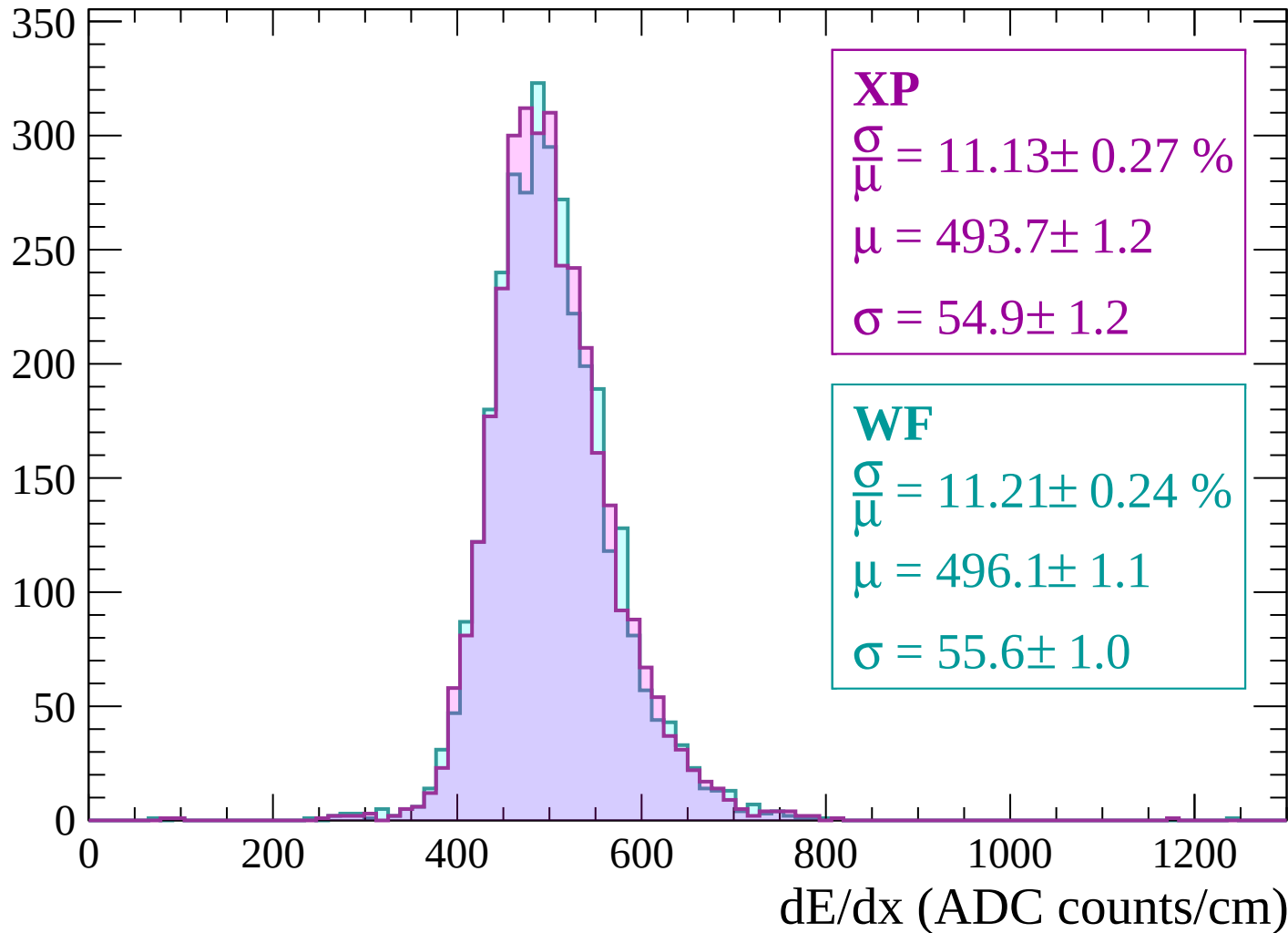
$$\mu = 499.7 \pm 1.3$$

$$\sigma = 58.9 \pm 1.3$$

dE/dx (ADC counts/cm)

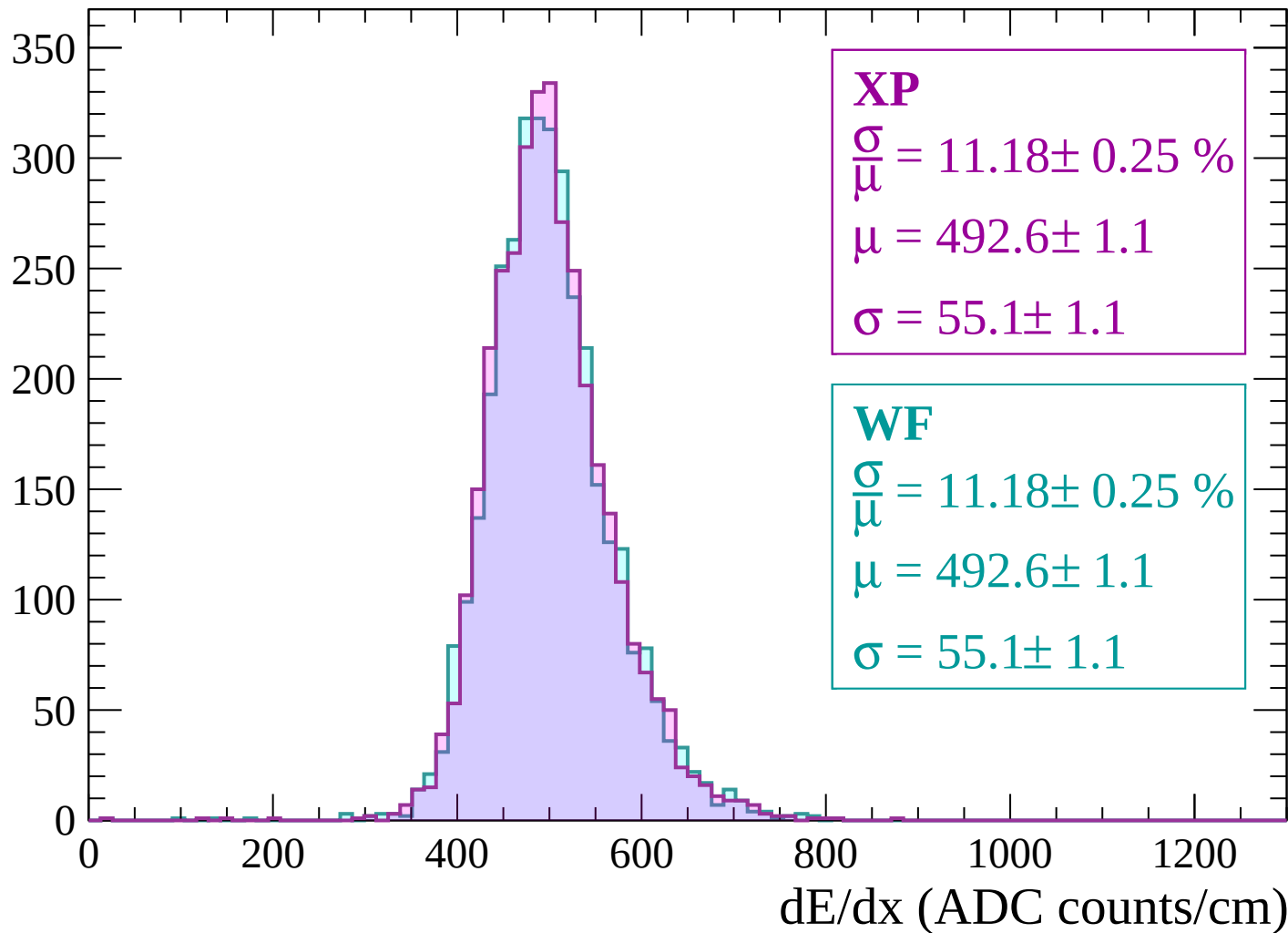
Energy loss | $-2280 < p < -2220$

Count



Energy loss | -2220 < p < -2160

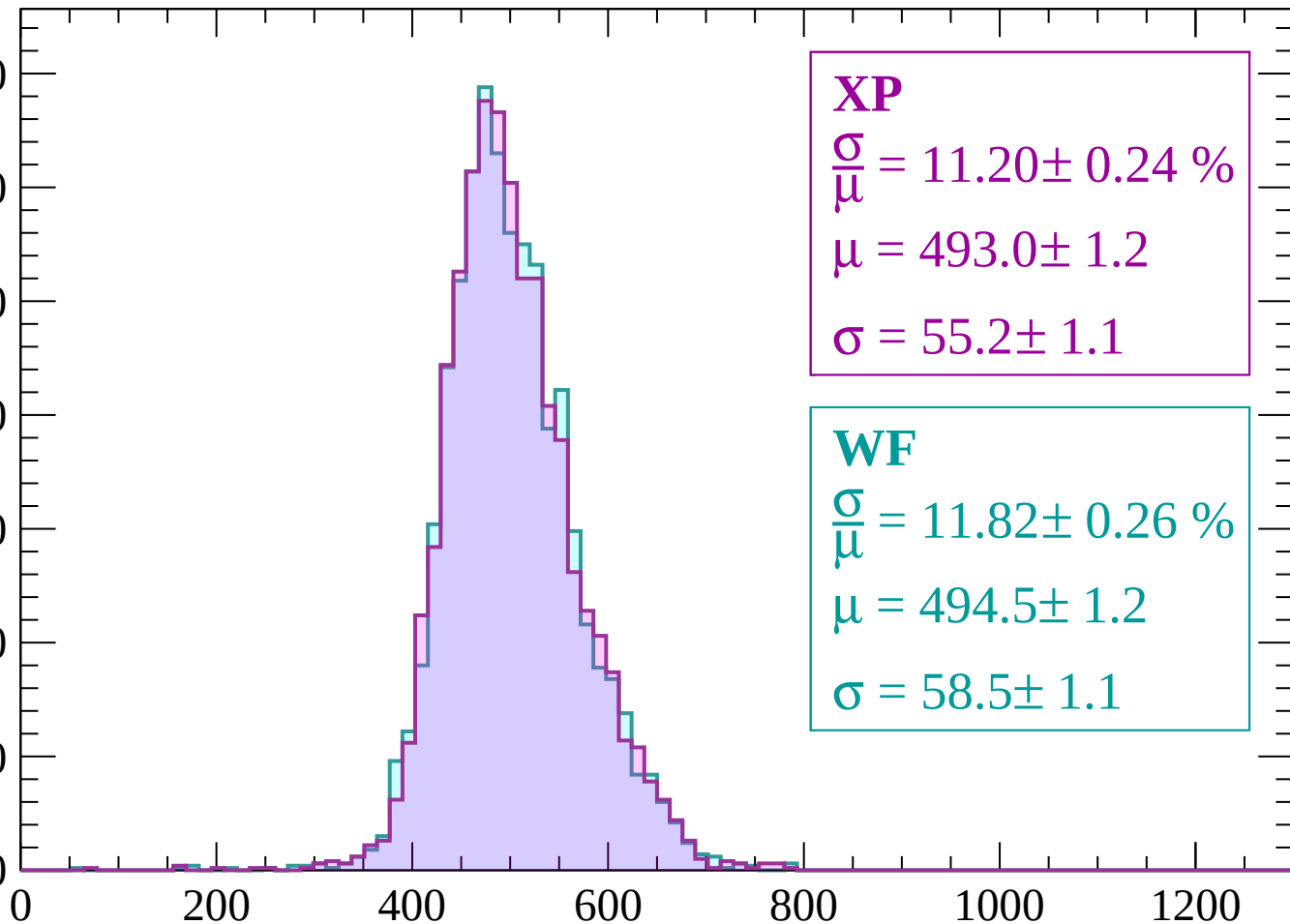
Count



Energy loss | $-2160 < p < -2100$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.20 \pm 0.24 \%$$

$$\mu = 493.0 \pm 1.2$$

$$\sigma = 55.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.82 \pm 0.26 \%$$

$$\mu = 494.5 \pm 1.2$$

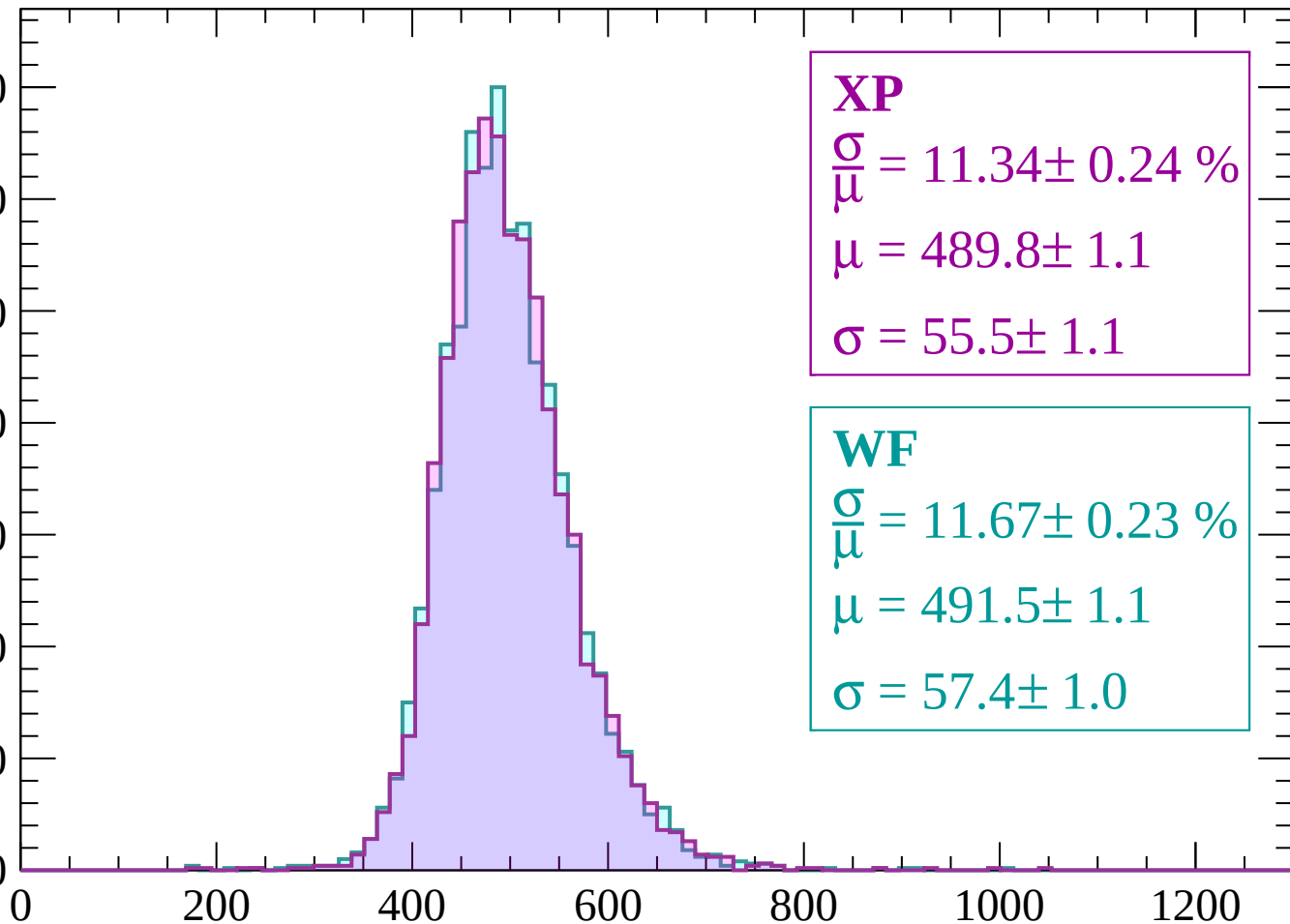
$$\sigma = 58.5 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -2100 < p < -2040

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 11.34 \pm 0.24 \%$$

$$\mu = 489.8 \pm 1.1$$

$$\sigma = 55.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 11.67 \pm 0.23 \%$$

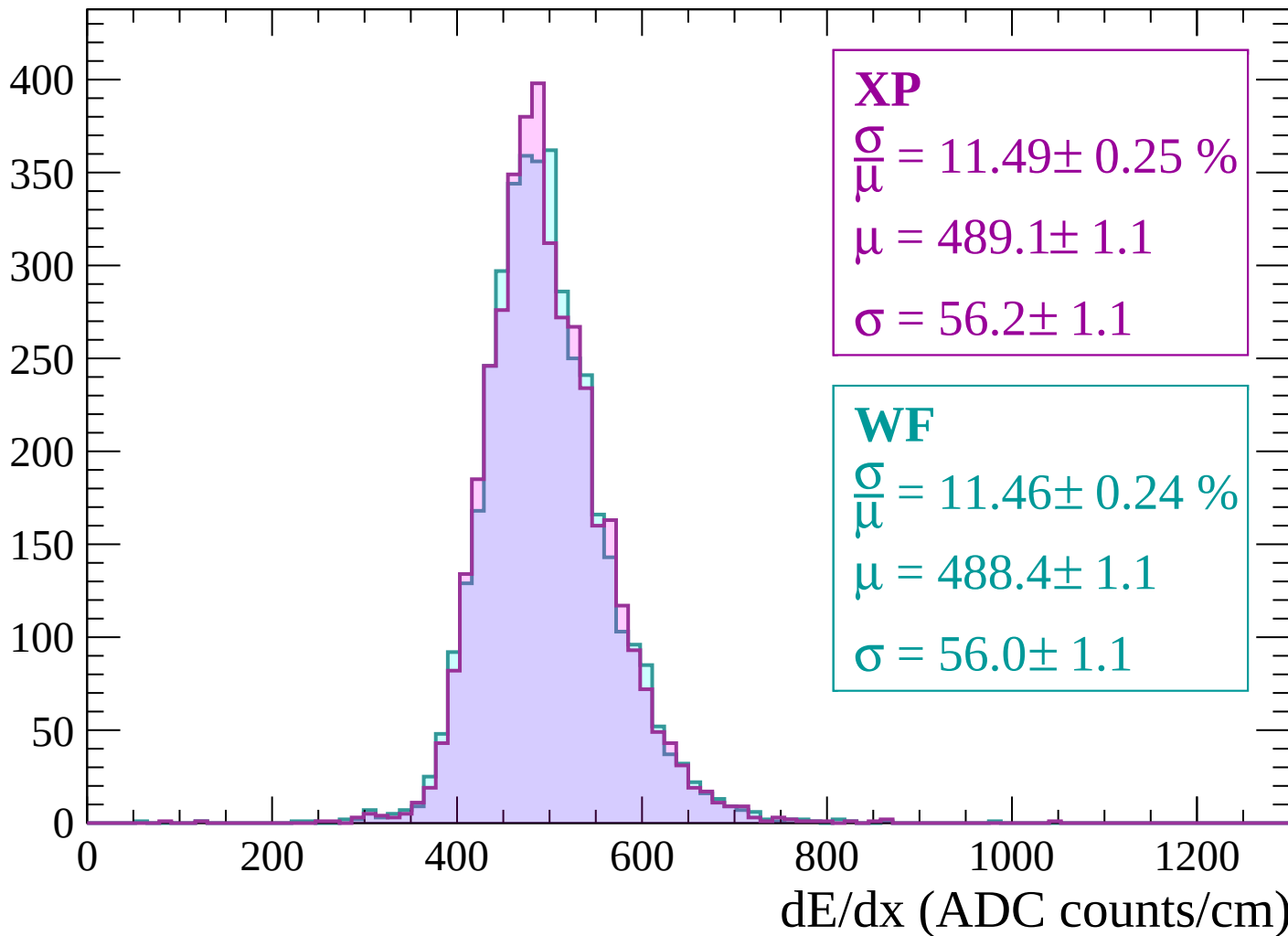
$$\mu = 491.5 \pm 1.1$$

$$\sigma = 57.4 \pm 1.0$$

dE/dx (ADC counts/cm)

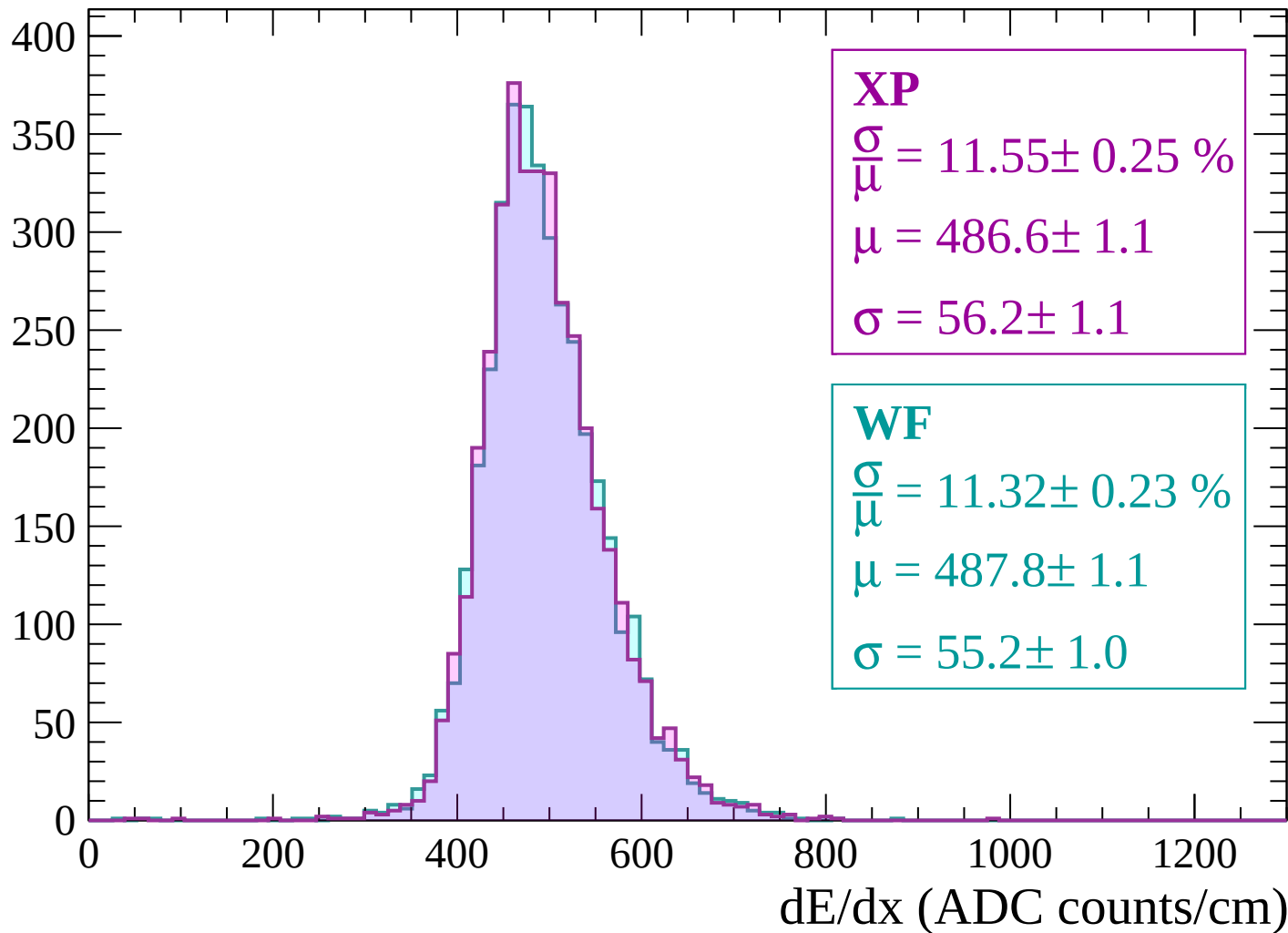
Energy loss | $-2040 < p < -1980$

Count

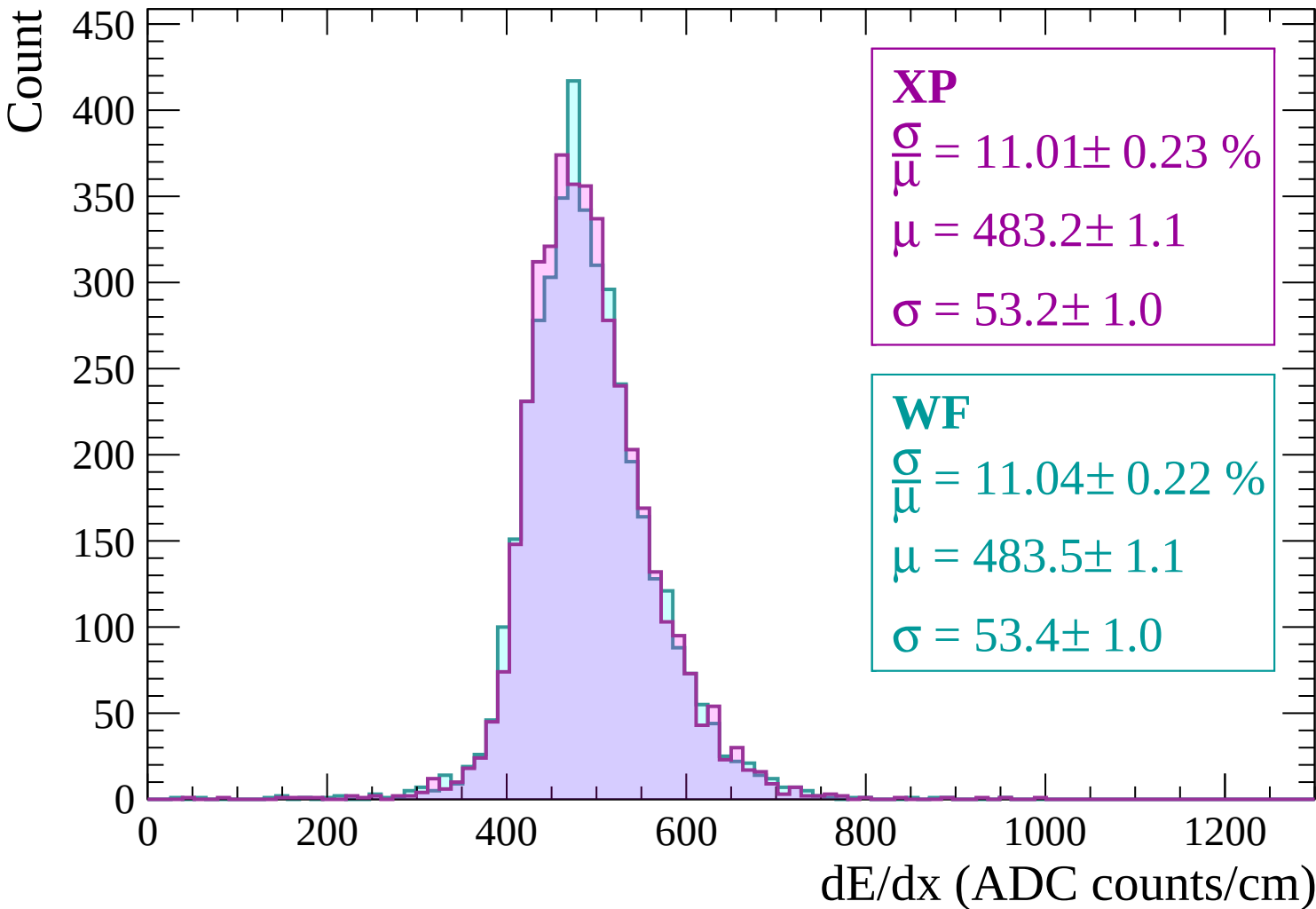


Energy loss | -1980 < p < -1920

Count

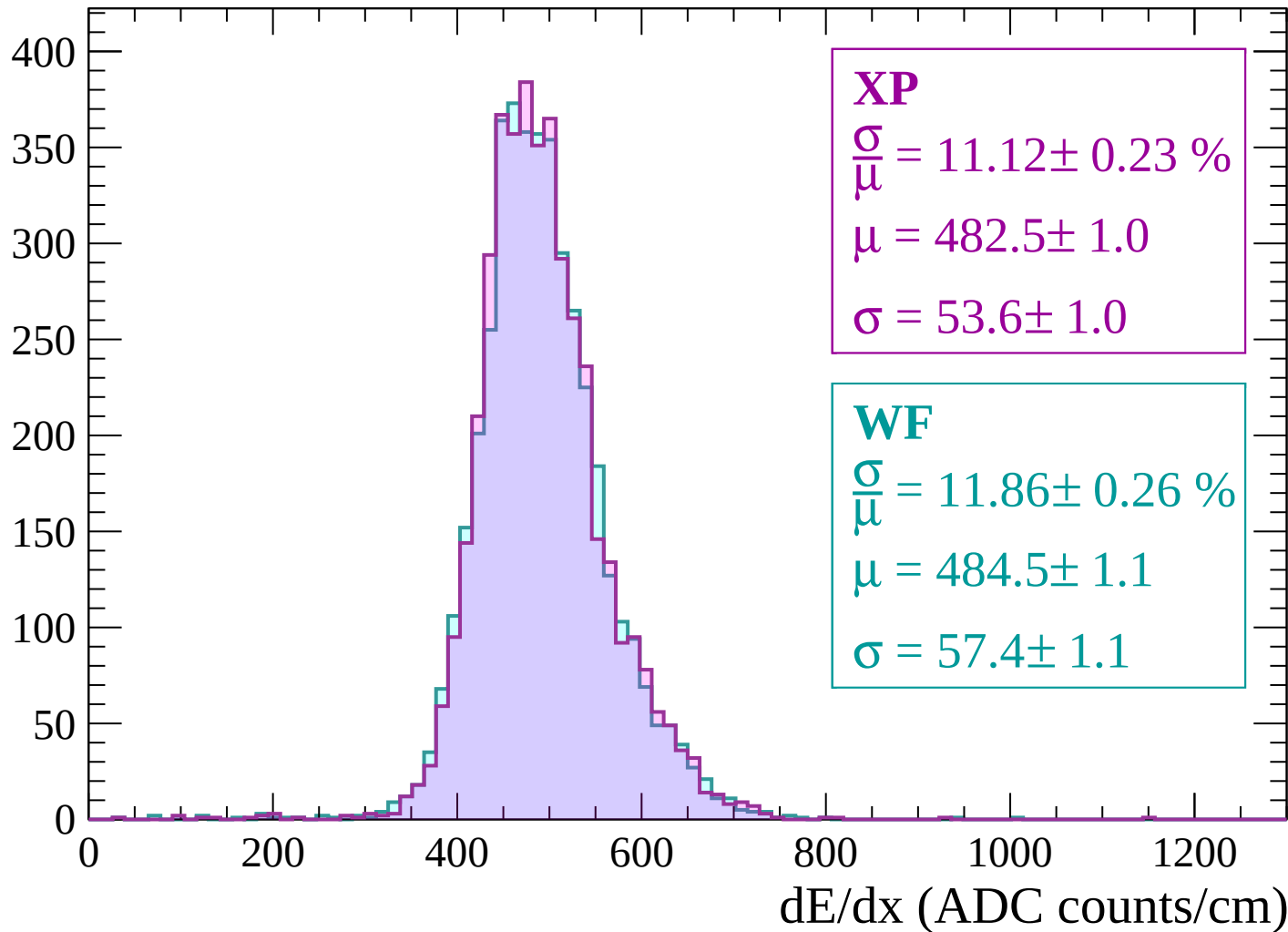


Energy loss | -1920 < p < -1860



Energy loss | $-1860 < p < -1800$

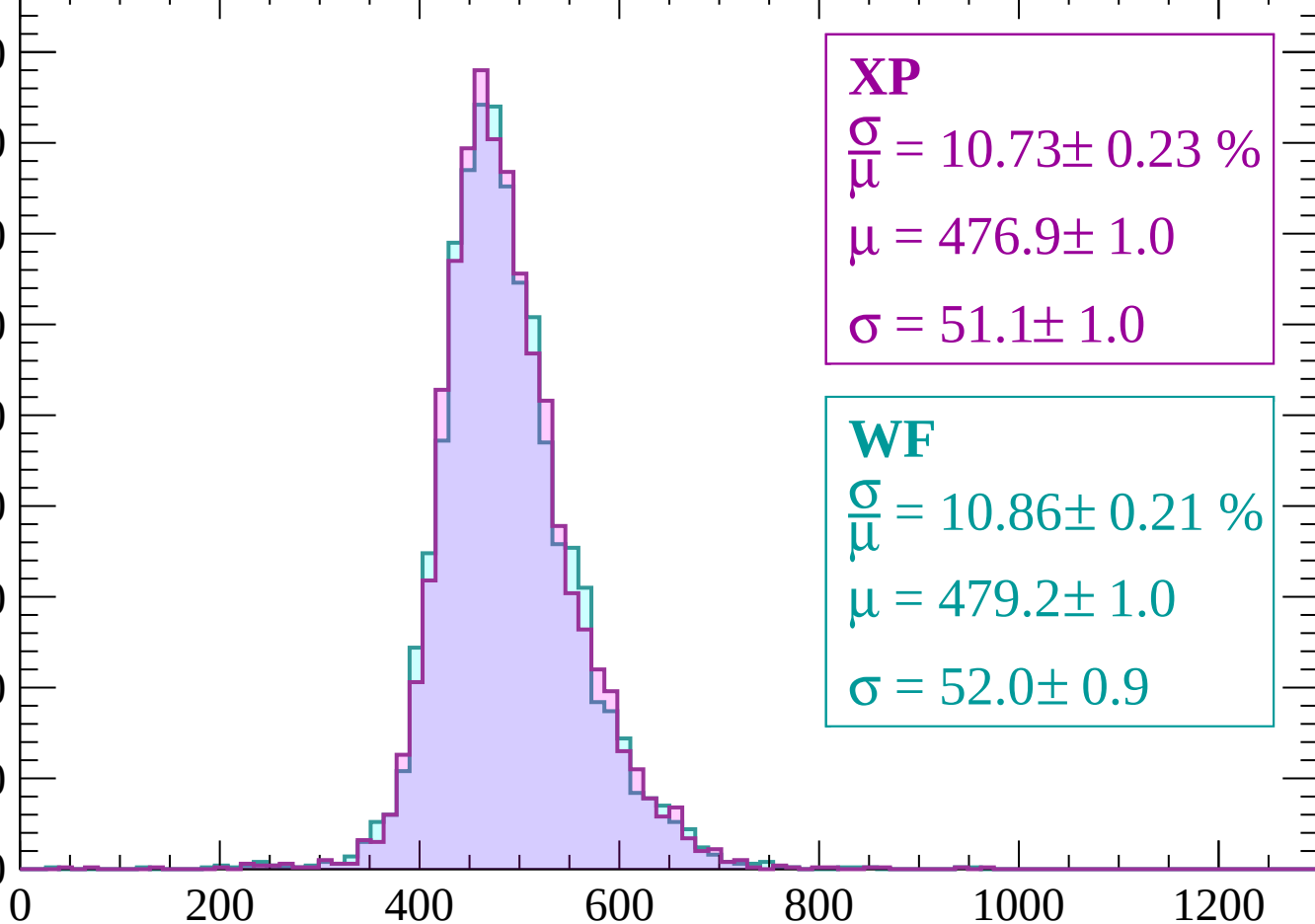
Count



Energy loss | -1800 < p < -1740

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 10.73 \pm 0.23 \%$$

$$\mu = 476.9 \pm 1.0$$

$$\sigma = 51.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.86 \pm 0.21 \%$$

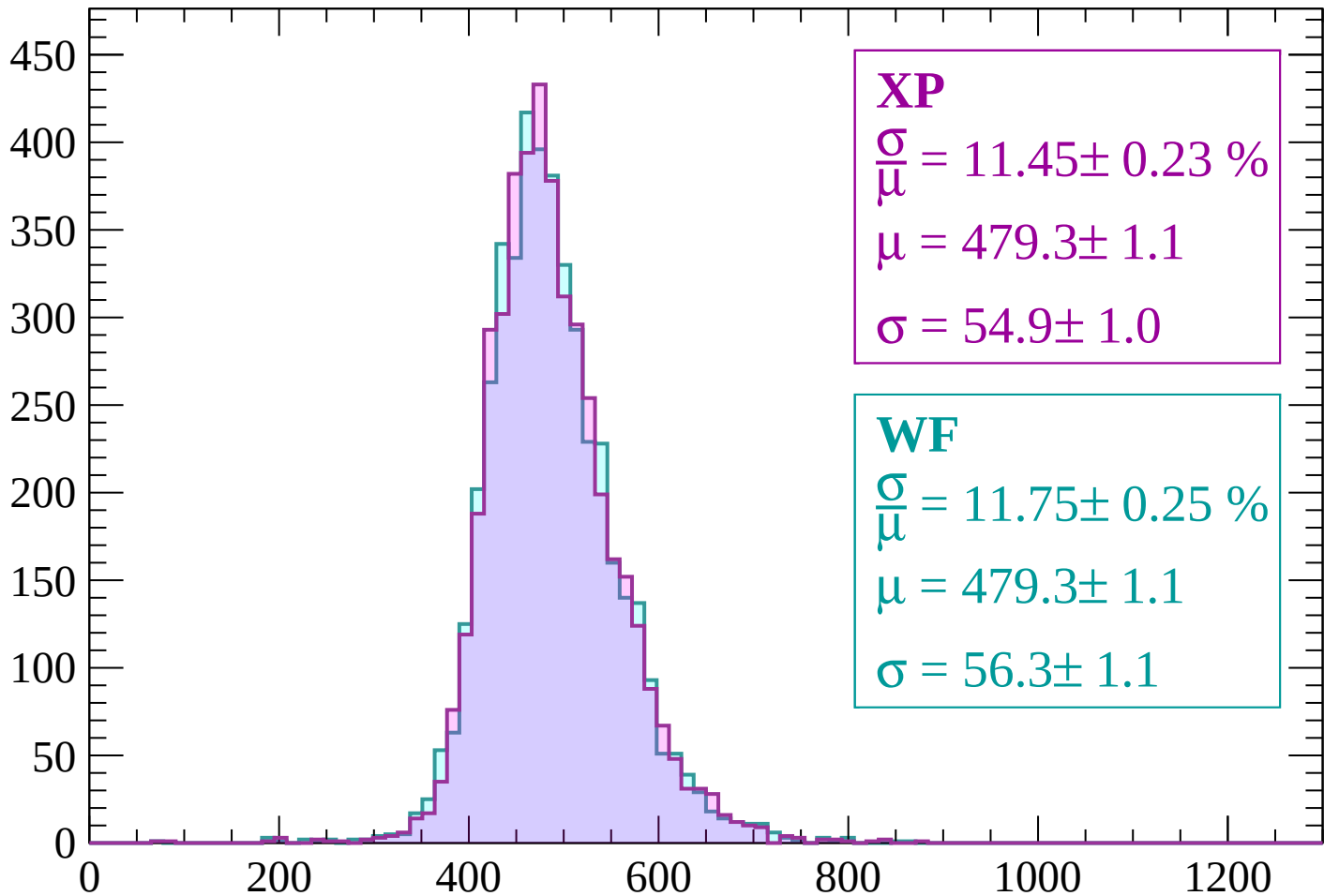
$$\mu = 479.2 \pm 1.0$$

$$\sigma = 52.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1740 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 11.45 \pm 0.23 \%$$

$$\mu = 479.3 \pm 1.1$$

$$\sigma = 54.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 11.75 \pm 0.25 \%$$

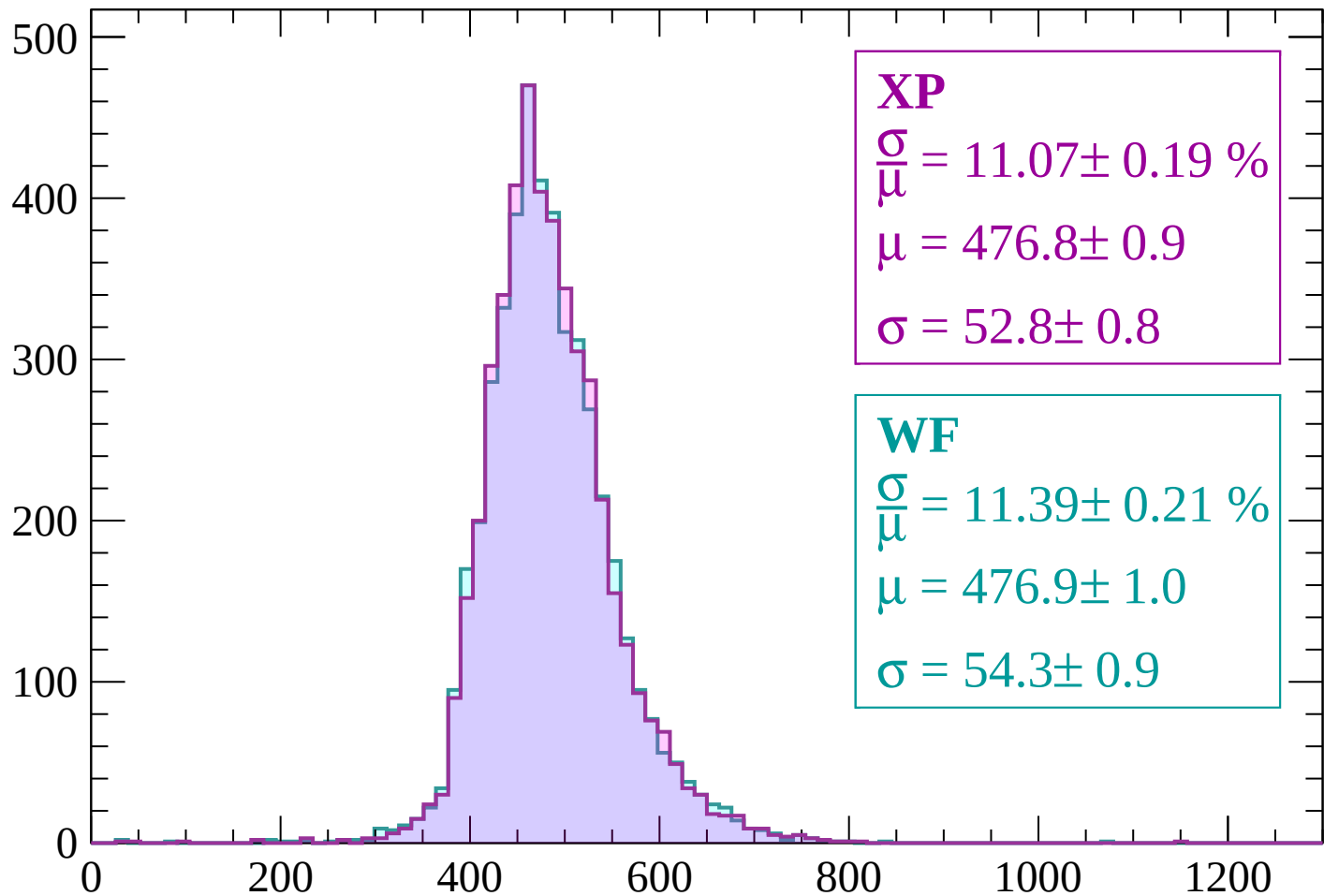
$$\mu = 479.3 \pm 1.1$$

$$\sigma = 56.3 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1620

Count



XP

$$\frac{\sigma}{\mu} = 11.07 \pm 0.19 \%$$

$$\mu = 476.8 \pm 0.9$$

$$\sigma = 52.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.39 \pm 0.21 \%$$

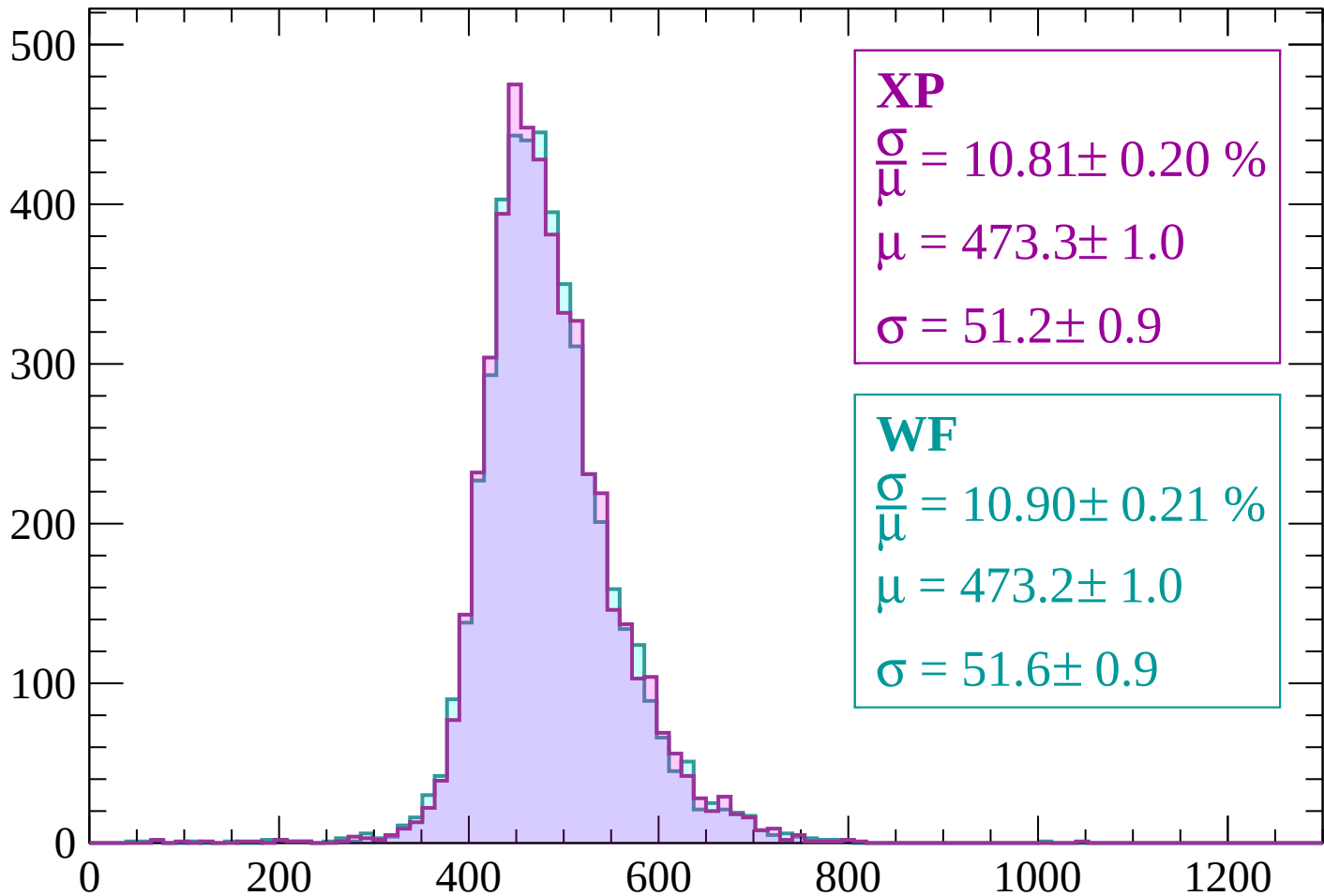
$$\mu = 476.9 \pm 1.0$$

$$\sigma = 54.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1620 < p < -1560$

Count



XP

$$\frac{\sigma}{\mu} = 10.81 \pm 0.20 \%$$

$$\mu = 473.3 \pm 1.0$$

$$\sigma = 51.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.90 \pm 0.21 \%$$

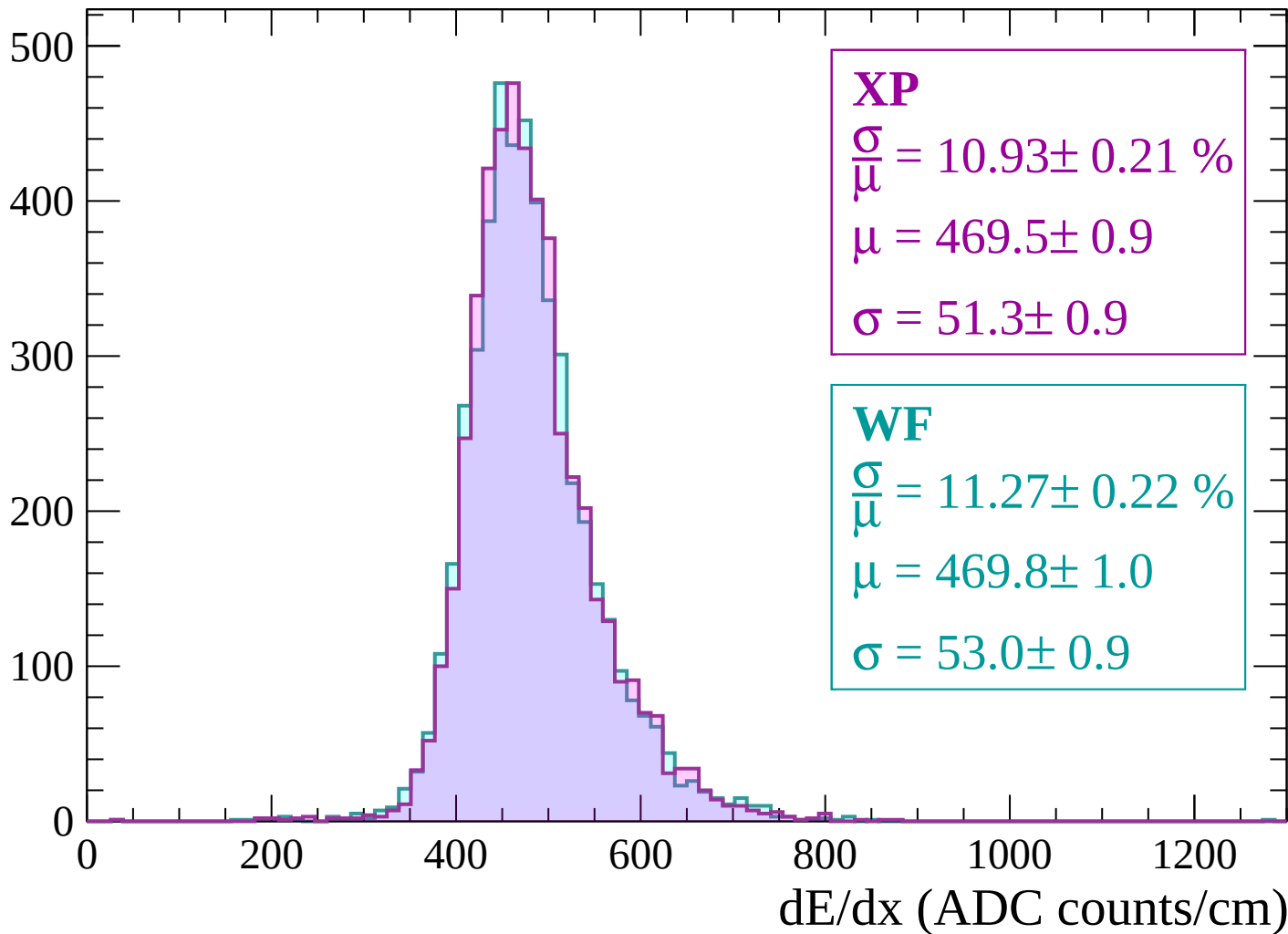
$$\mu = 473.2 \pm 1.0$$

$$\sigma = 51.6 \pm 0.9$$

dE/dx (ADC counts/cm)

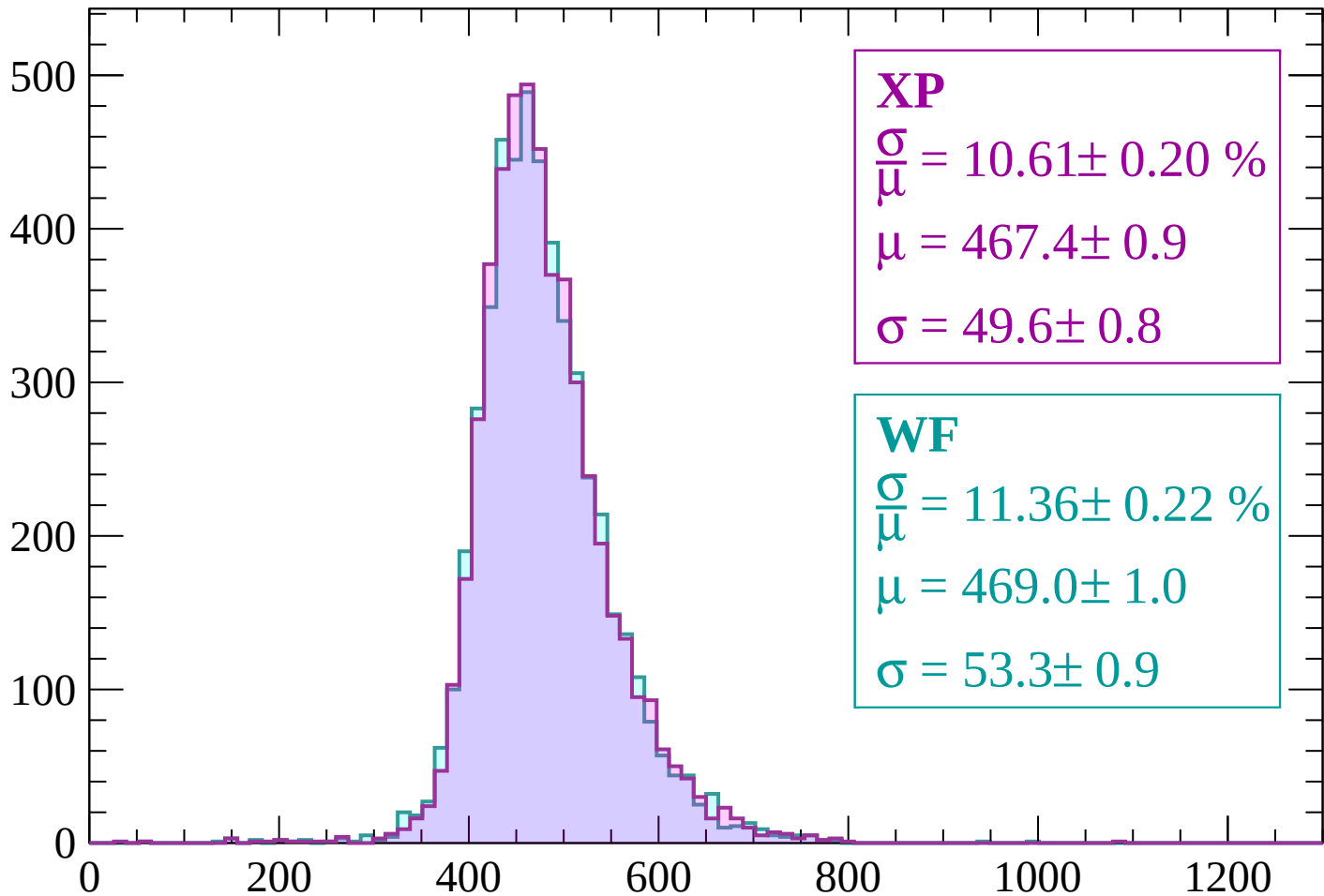
Energy loss | -1560 < p < -1500

Count



Energy loss | -1500 < p < -1440

Count



XP

$$\frac{\sigma}{\mu} = 10.61 \pm 0.20 \%$$

$$\mu = 467.4 \pm 0.9$$

$$\sigma = 49.6 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 11.36 \pm 0.22 \%$$

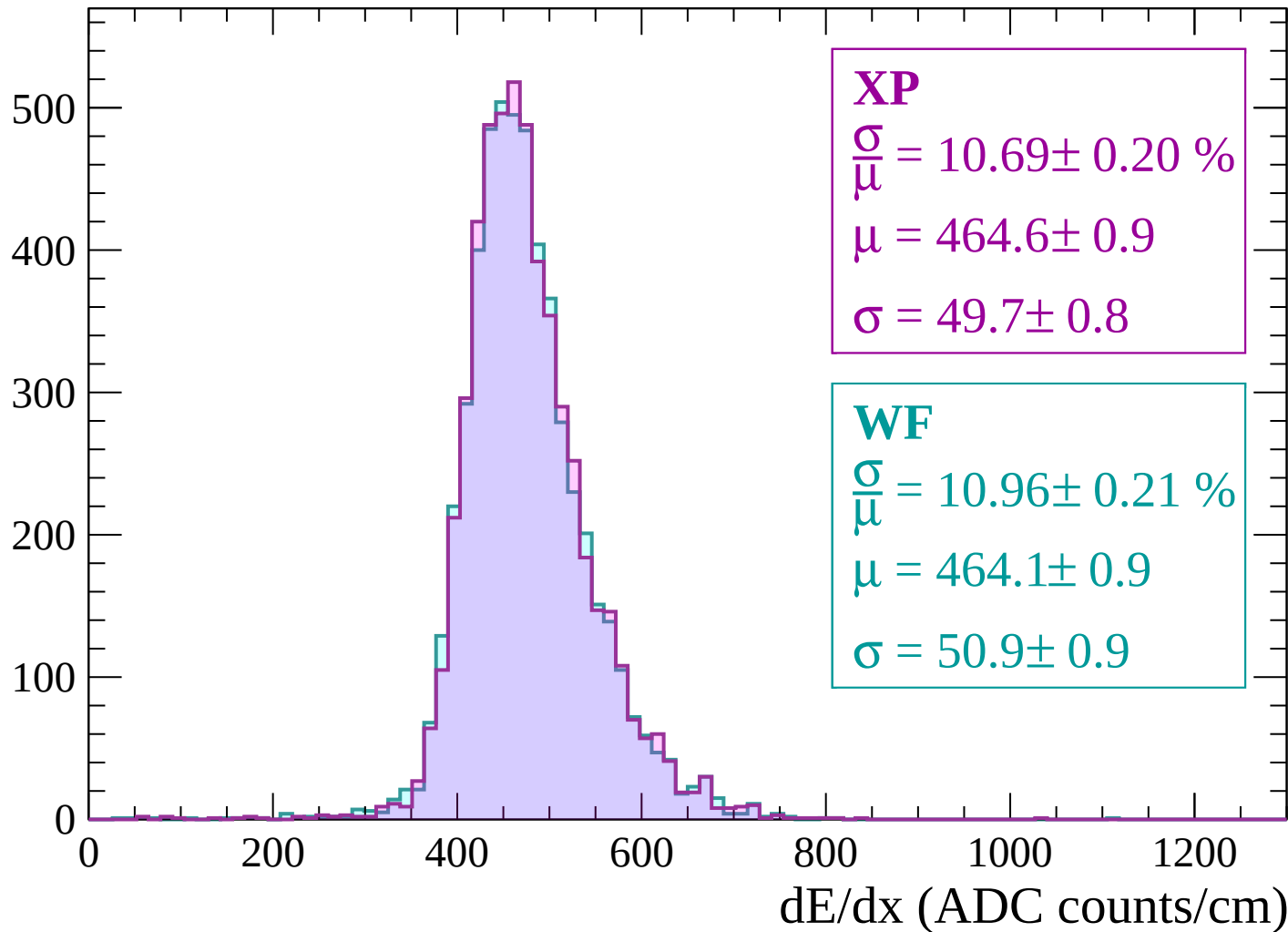
$$\mu = 469.0 \pm 1.0$$

$$\sigma = 53.3 \pm 0.9$$

dE/dx (ADC counts/cm)

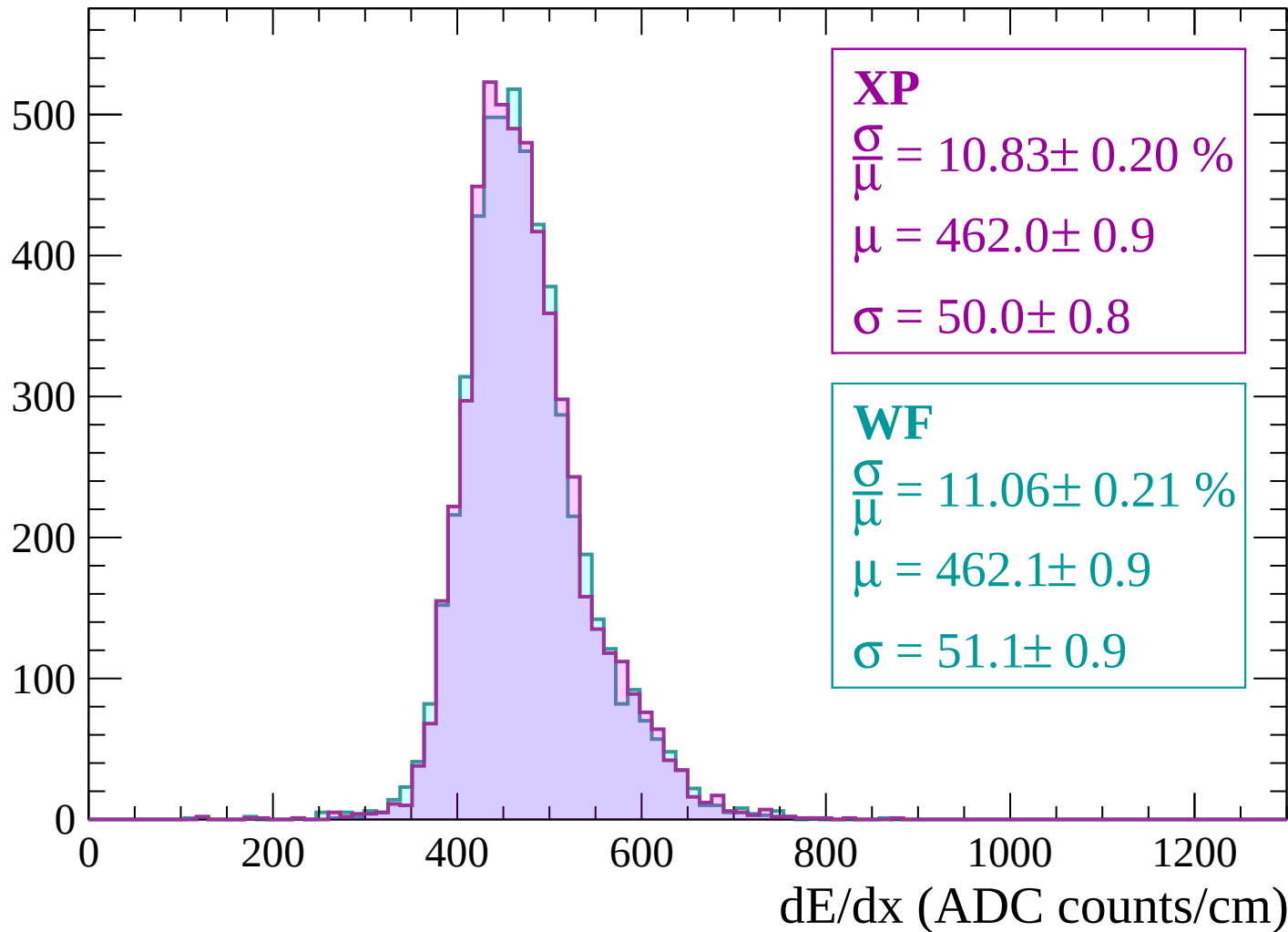
Energy loss | $-1440 < p < -1380$

Count



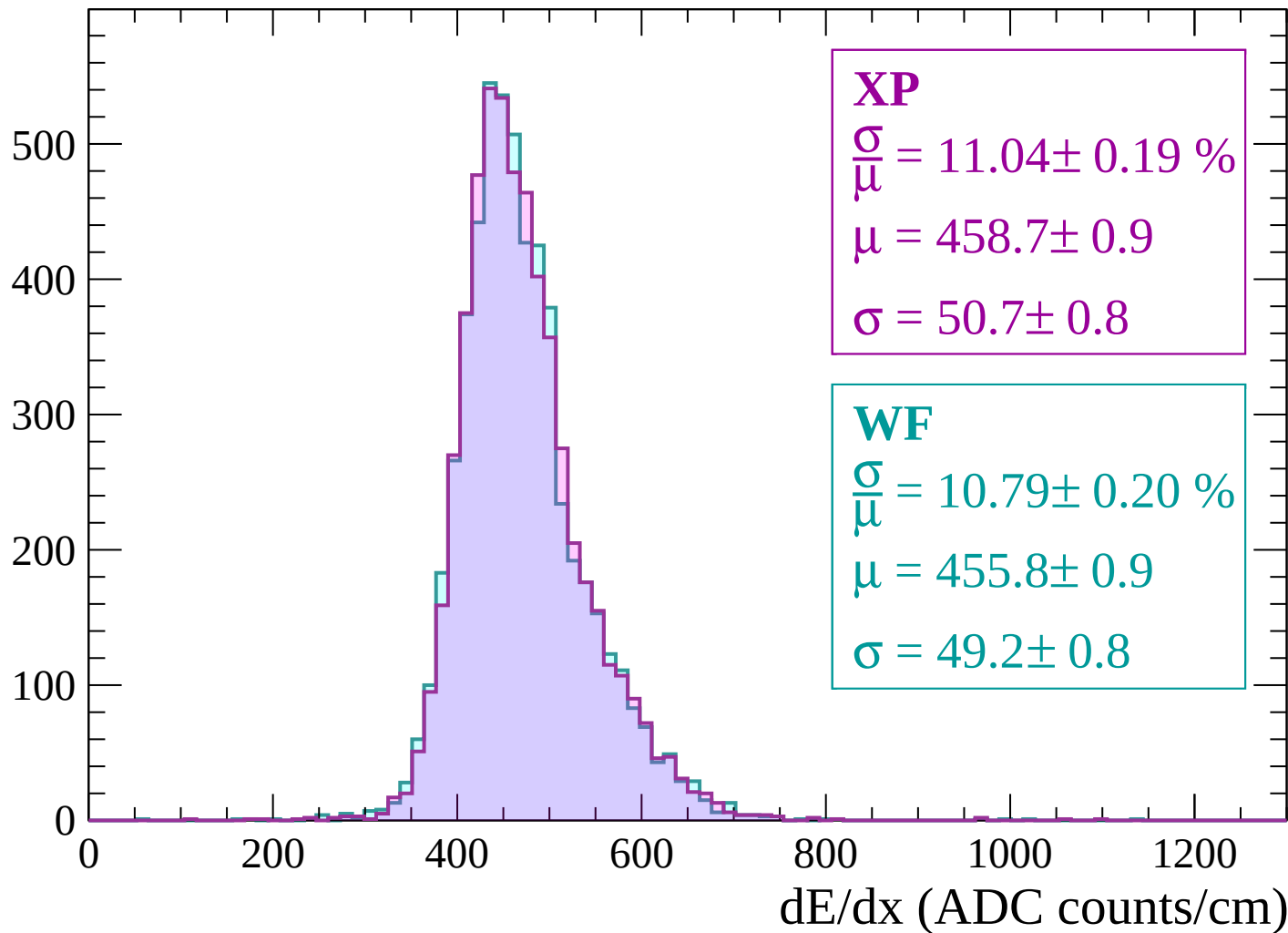
Energy loss | $-1380 < p < -1320$

Count

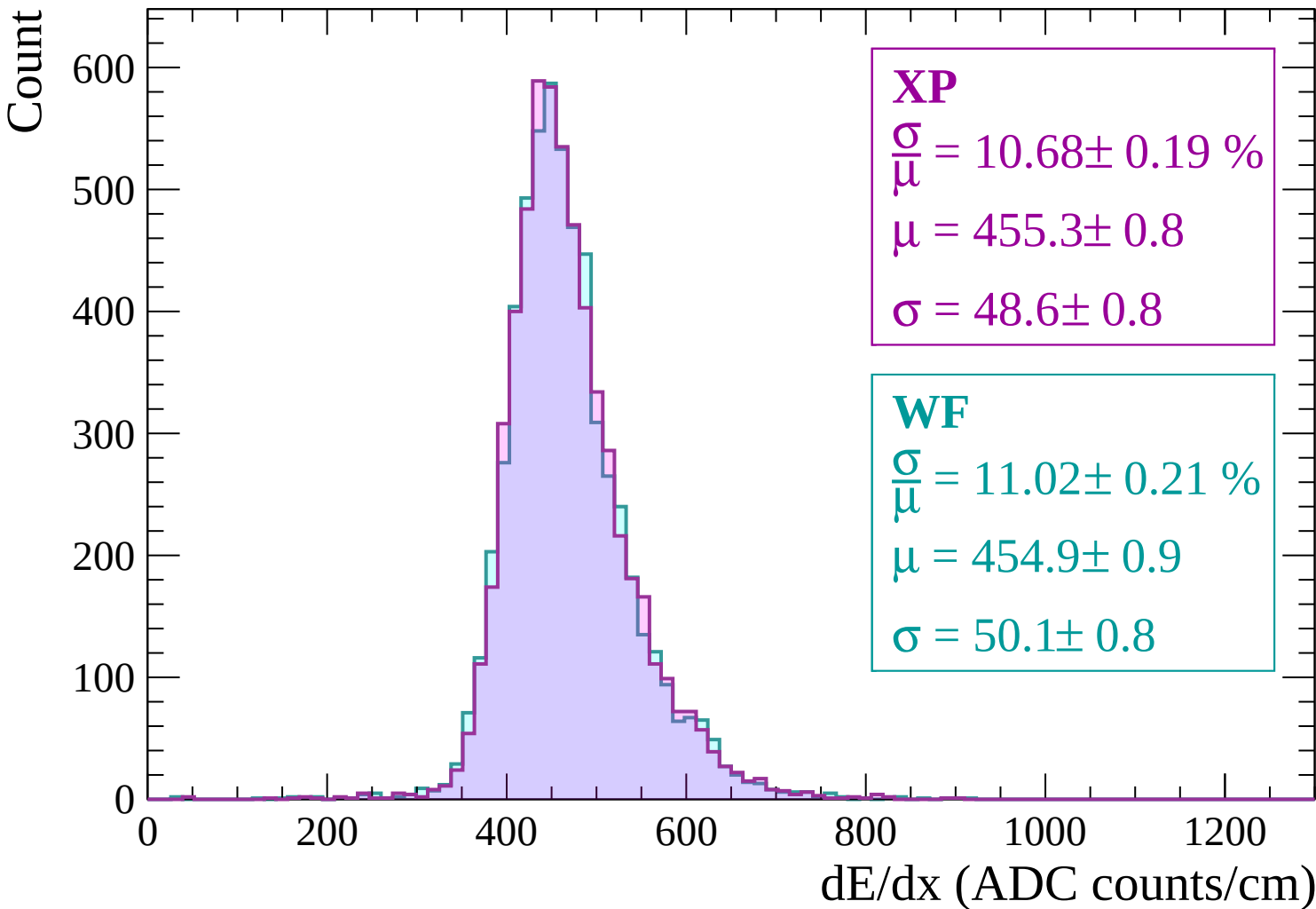


Energy loss | $-1320 < p < -1260$

Count

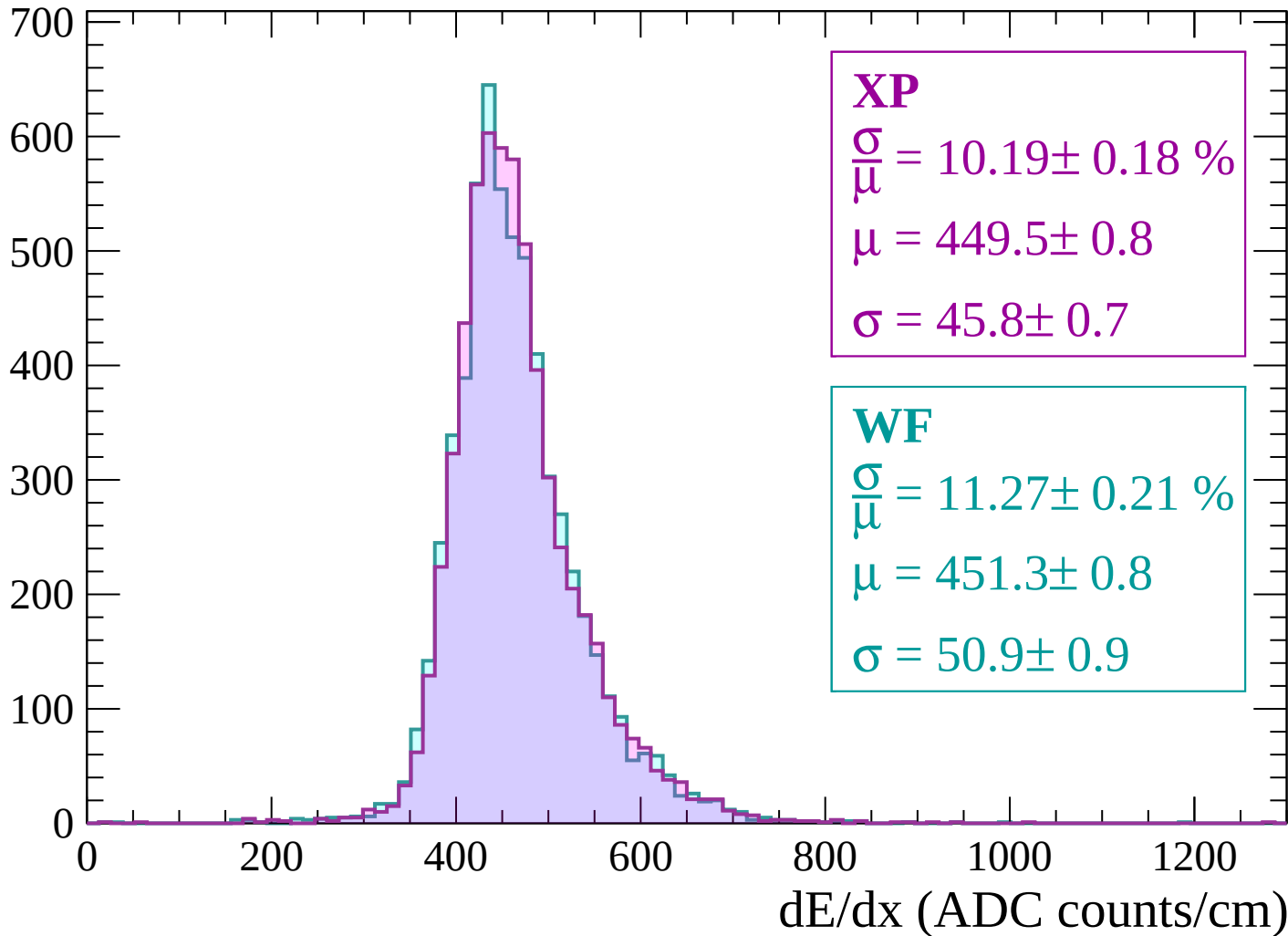


Energy loss | -1260 < p < -1200



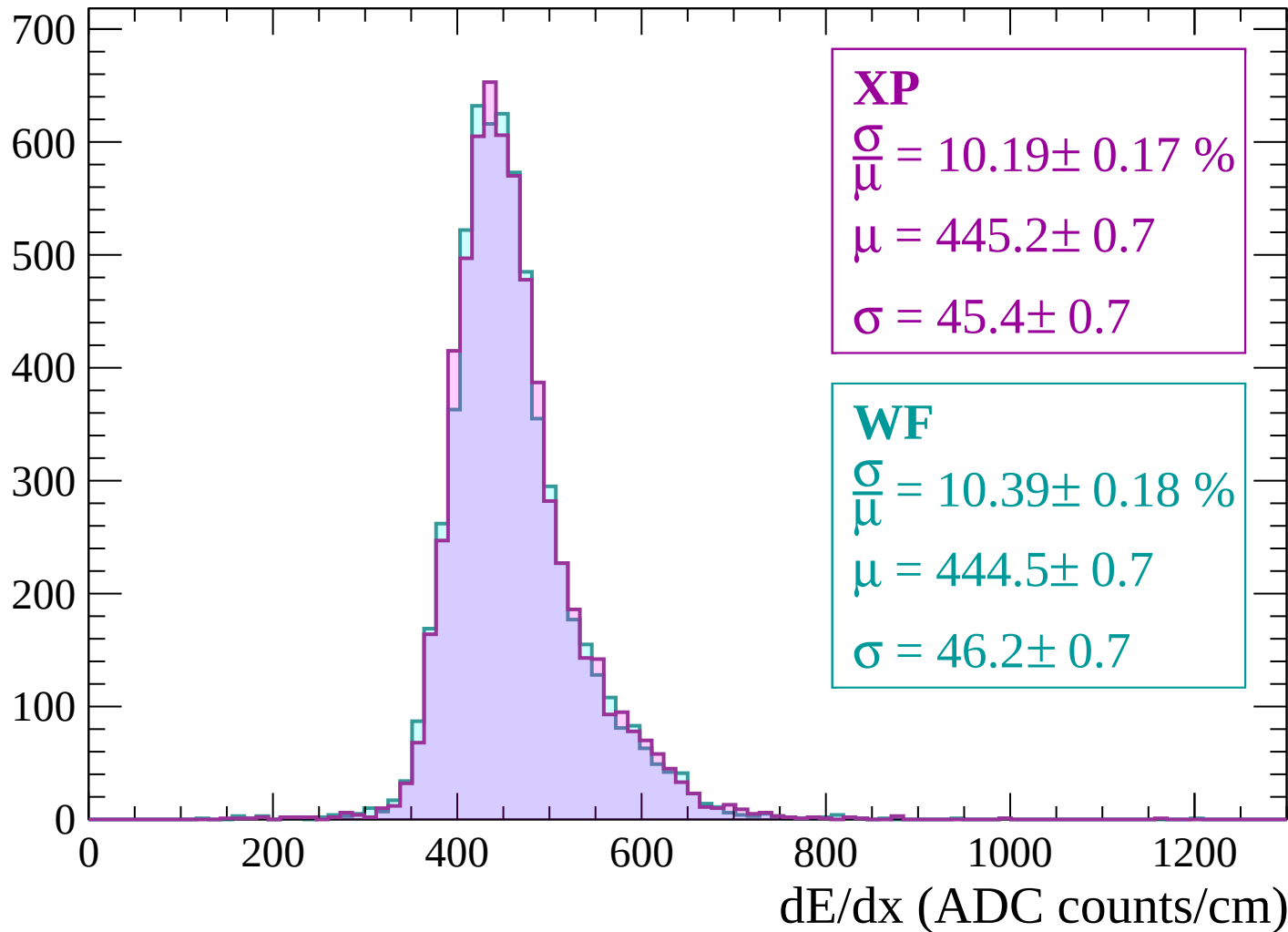
Energy loss | -1200 < p < -1140

Count



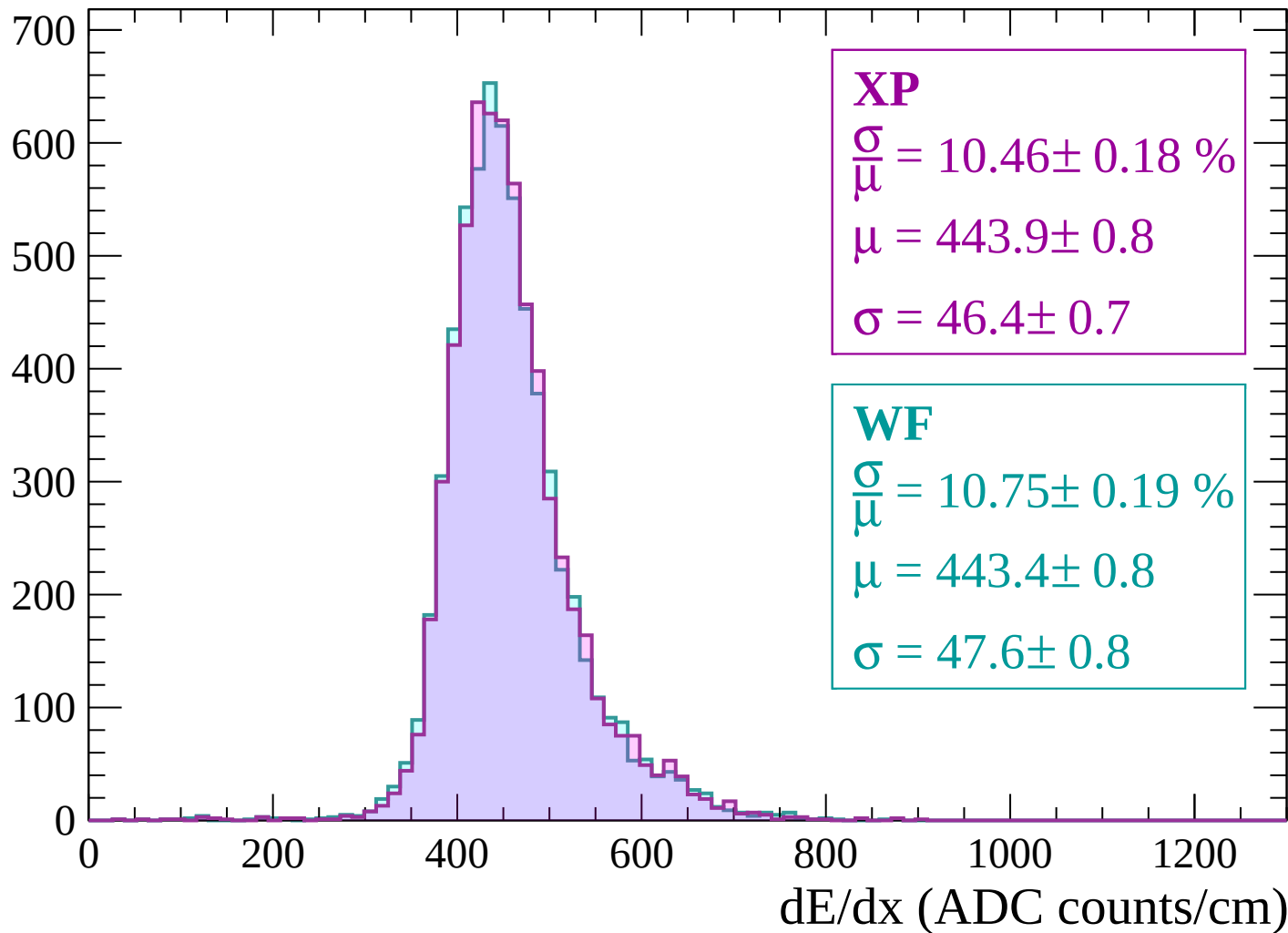
Energy loss | $-1140 < p < -1080$

Count



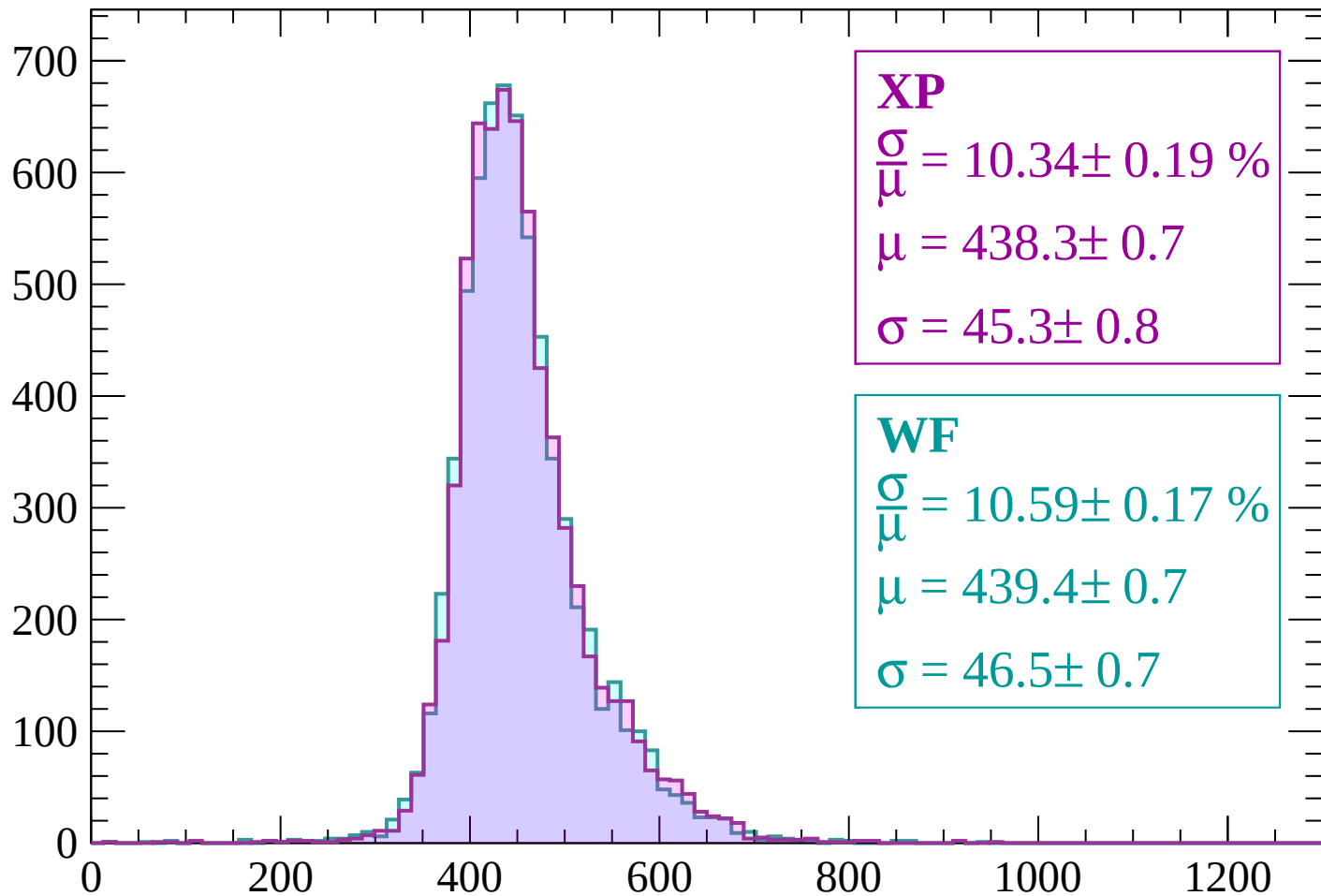
Energy loss | -1080 < p < -1020

Count



Energy loss | $-1020 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 10.34 \pm 0.19 \%$$

$$\mu = 438.3 \pm 0.7$$

$$\sigma = 45.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.59 \pm 0.17 \%$$

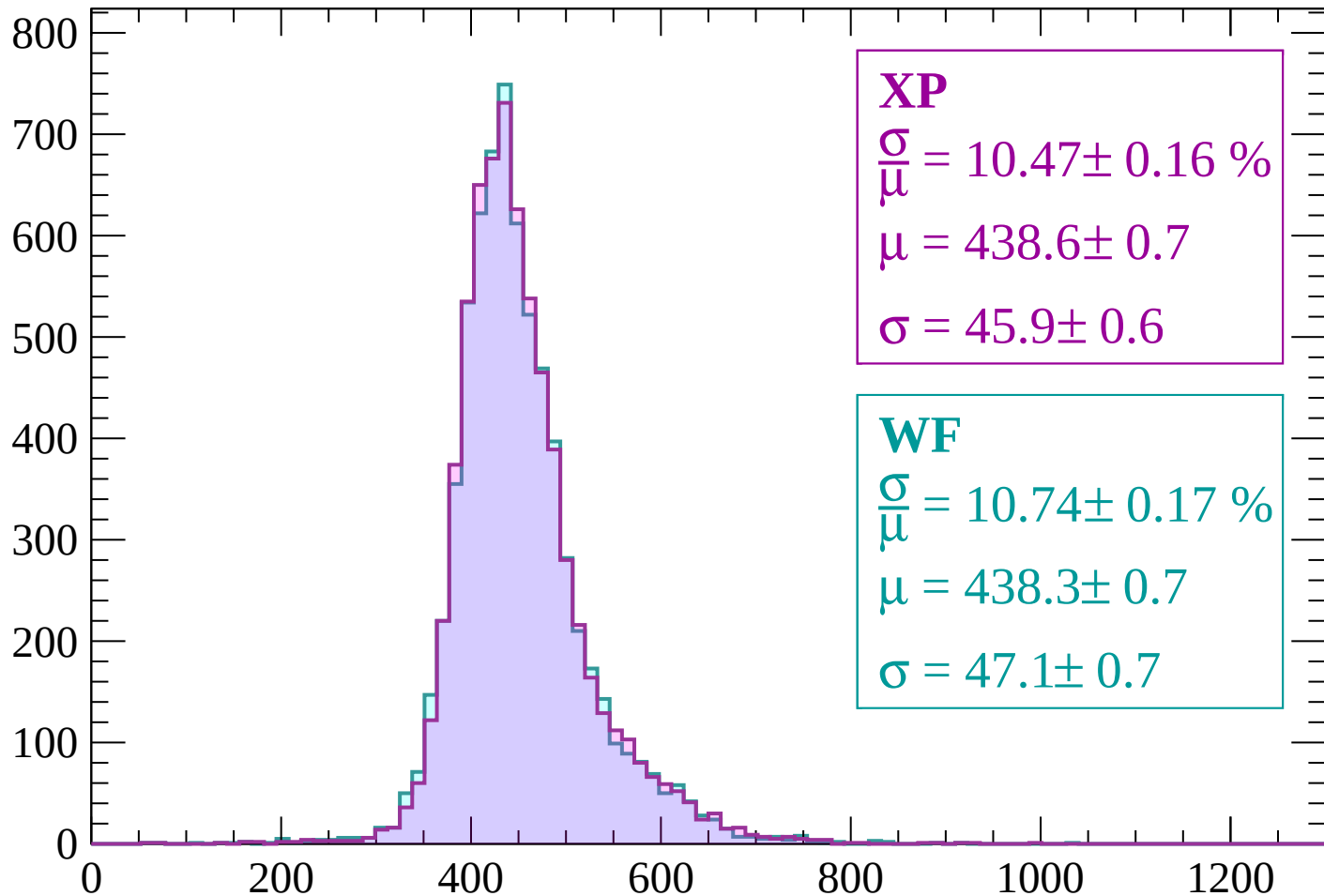
$$\mu = 439.4 \pm 0.7$$

$$\sigma = 46.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

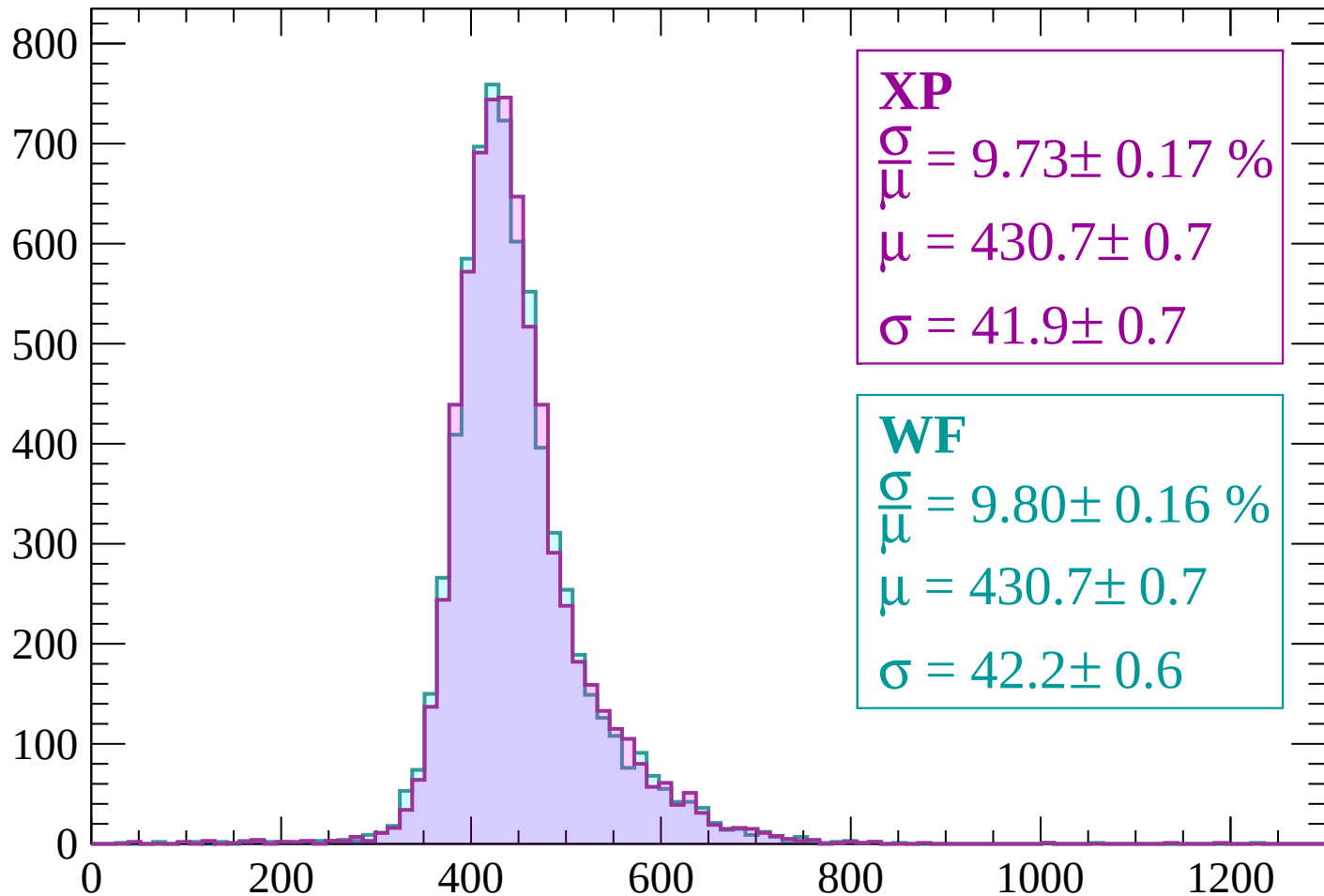
Count



dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 9.73 \pm 0.17 \%$$

$$\mu = 430.7 \pm 0.7$$

$$\sigma = 41.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.80 \pm 0.16 \%$$

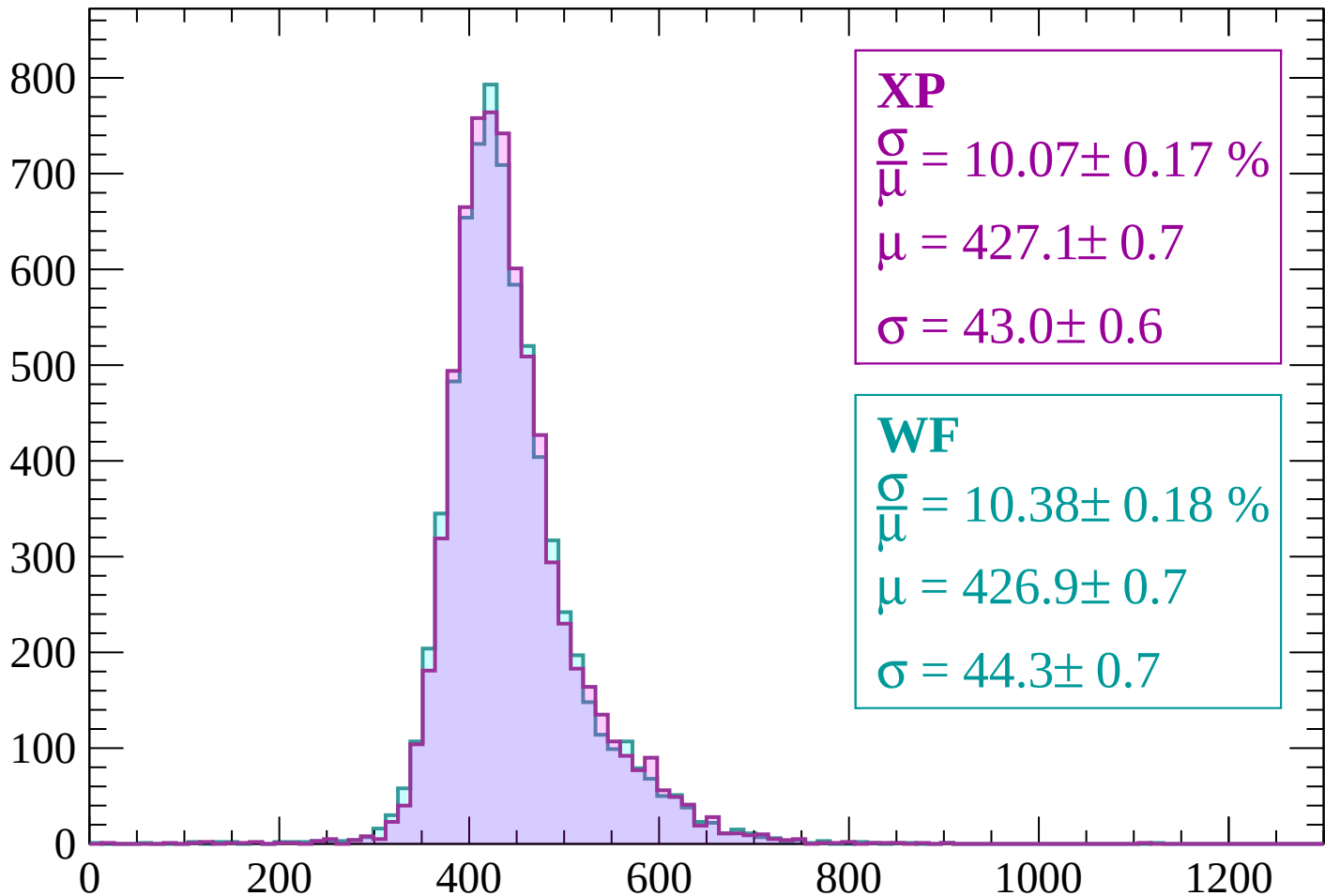
$$\mu = 430.7 \pm 0.7$$

$$\sigma = 42.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP

$$\frac{\sigma}{\mu} = 10.07 \pm 0.17 \%$$

$$\mu = 427.1 \pm 0.7$$

$$\sigma = 43.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.38 \pm 0.18 \%$$

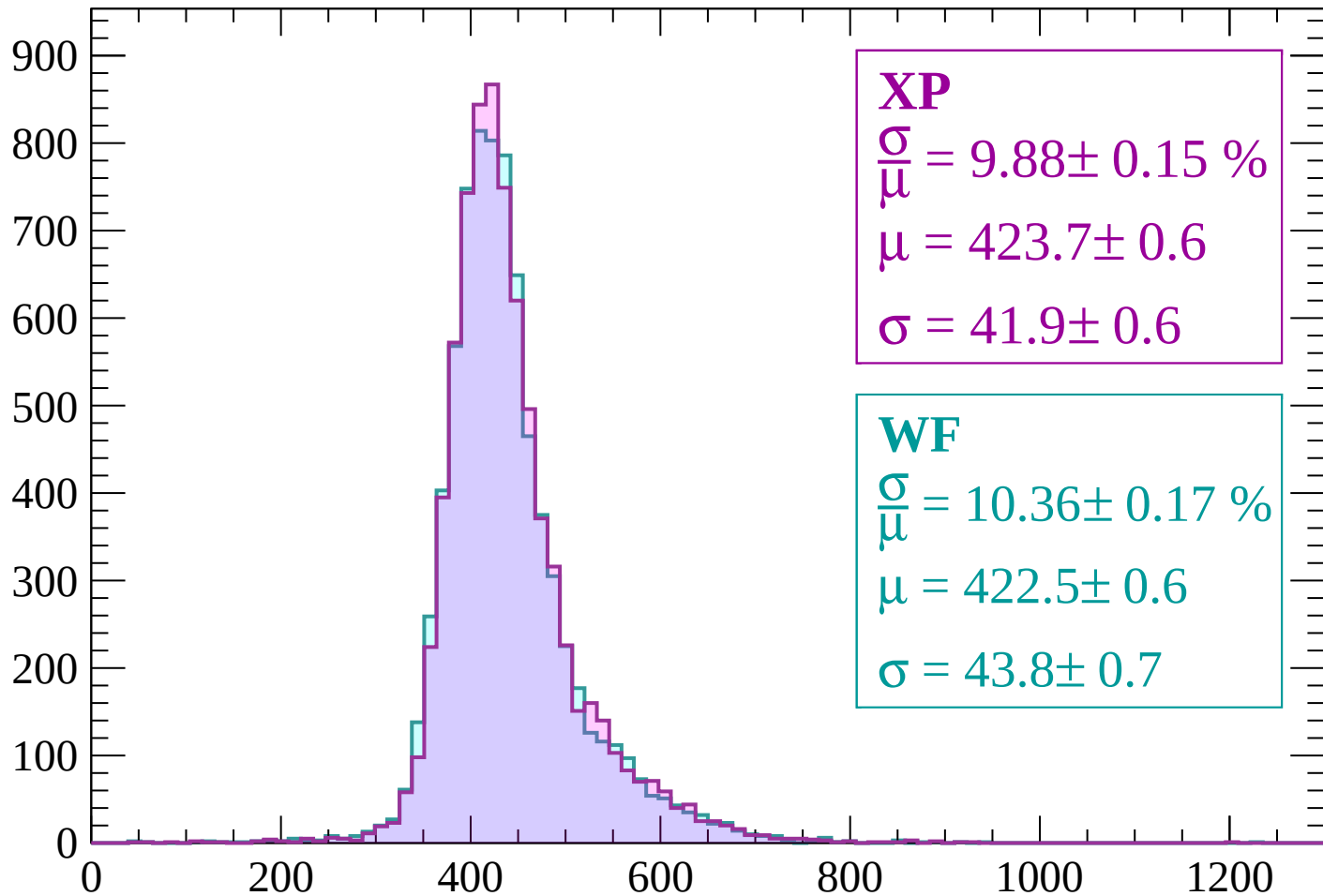
$$\mu = 426.9 \pm 0.7$$

$$\sigma = 44.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 9.88 \pm 0.15 \%$$

$$\mu = 423.7 \pm 0.6$$

$$\sigma = 41.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.36 \pm 0.17 \%$$

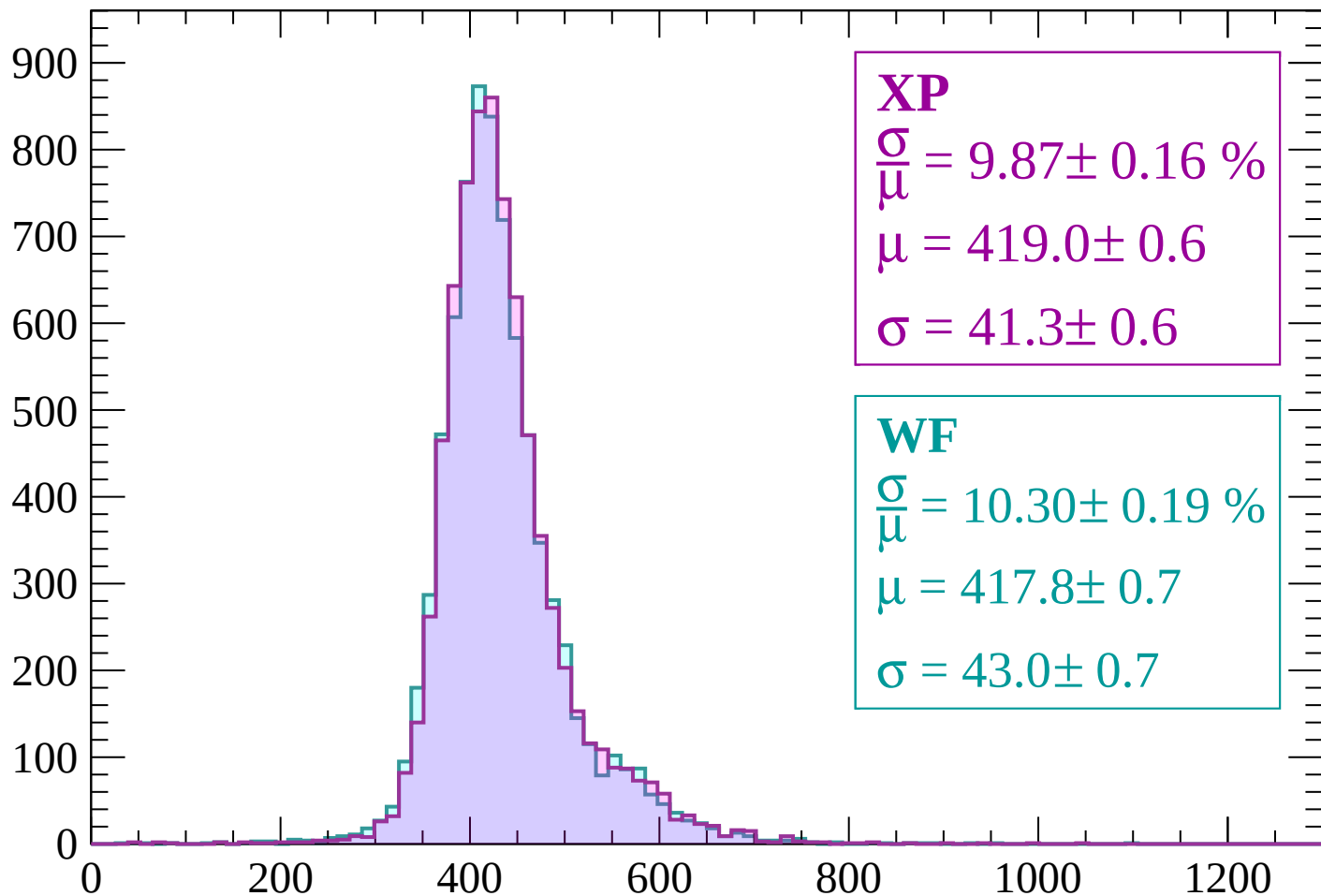
$$\mu = 422.5 \pm 0.6$$

$$\sigma = 43.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP

$$\frac{\sigma}{\mu} = 9.87 \pm 0.16 \%$$

$$\mu = 419.0 \pm 0.6$$

$$\sigma = 41.3 \pm 0.6$$

WF

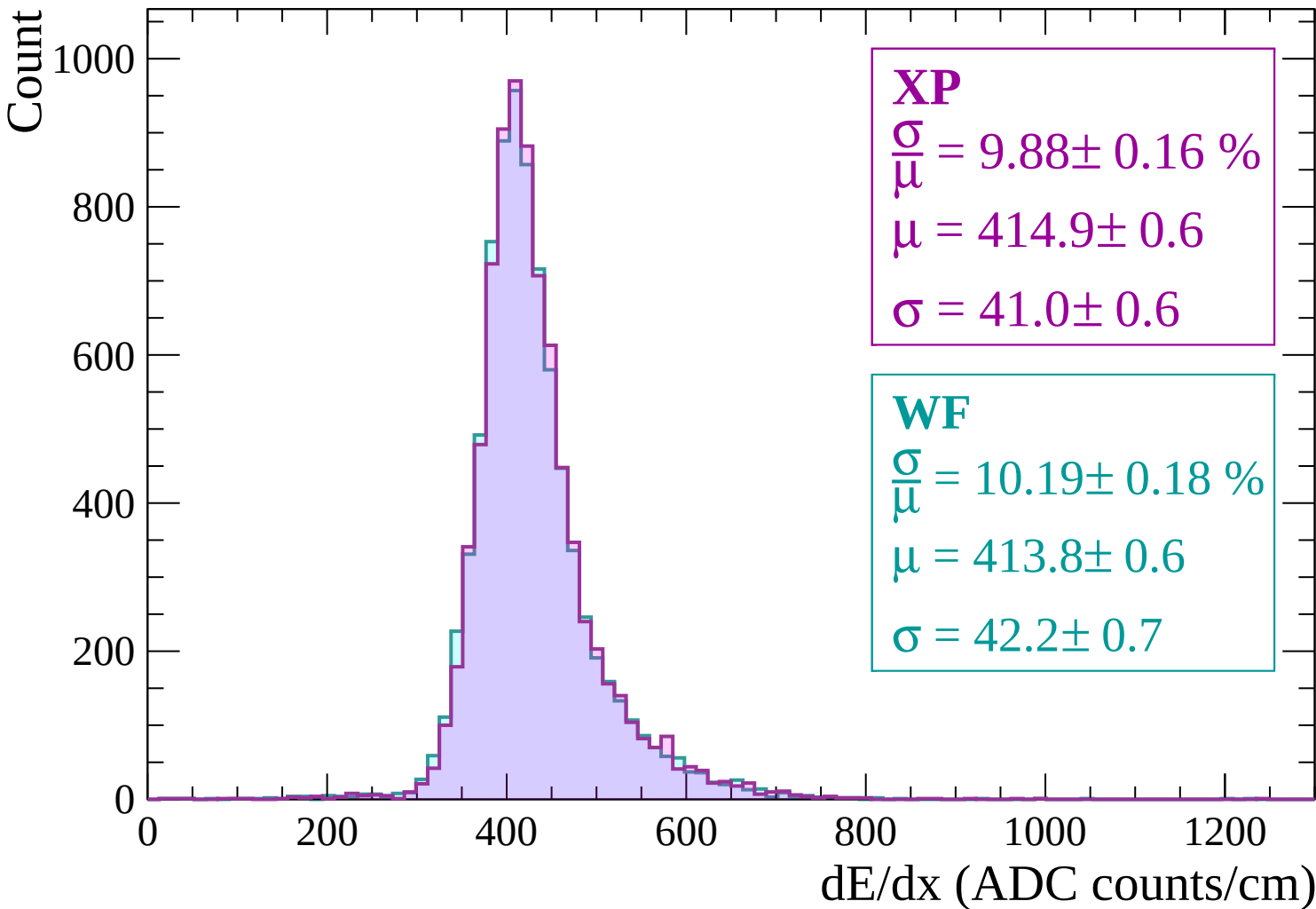
$$\frac{\sigma}{\mu} = 10.30 \pm 0.19 \%$$

$$\mu = 417.8 \pm 0.7$$

$$\sigma = 43.0 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$



Energy loss | $-600 < p < -540$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.71 \pm 0.17 \%$$

$$\mu = 410.1 \pm 0.6$$

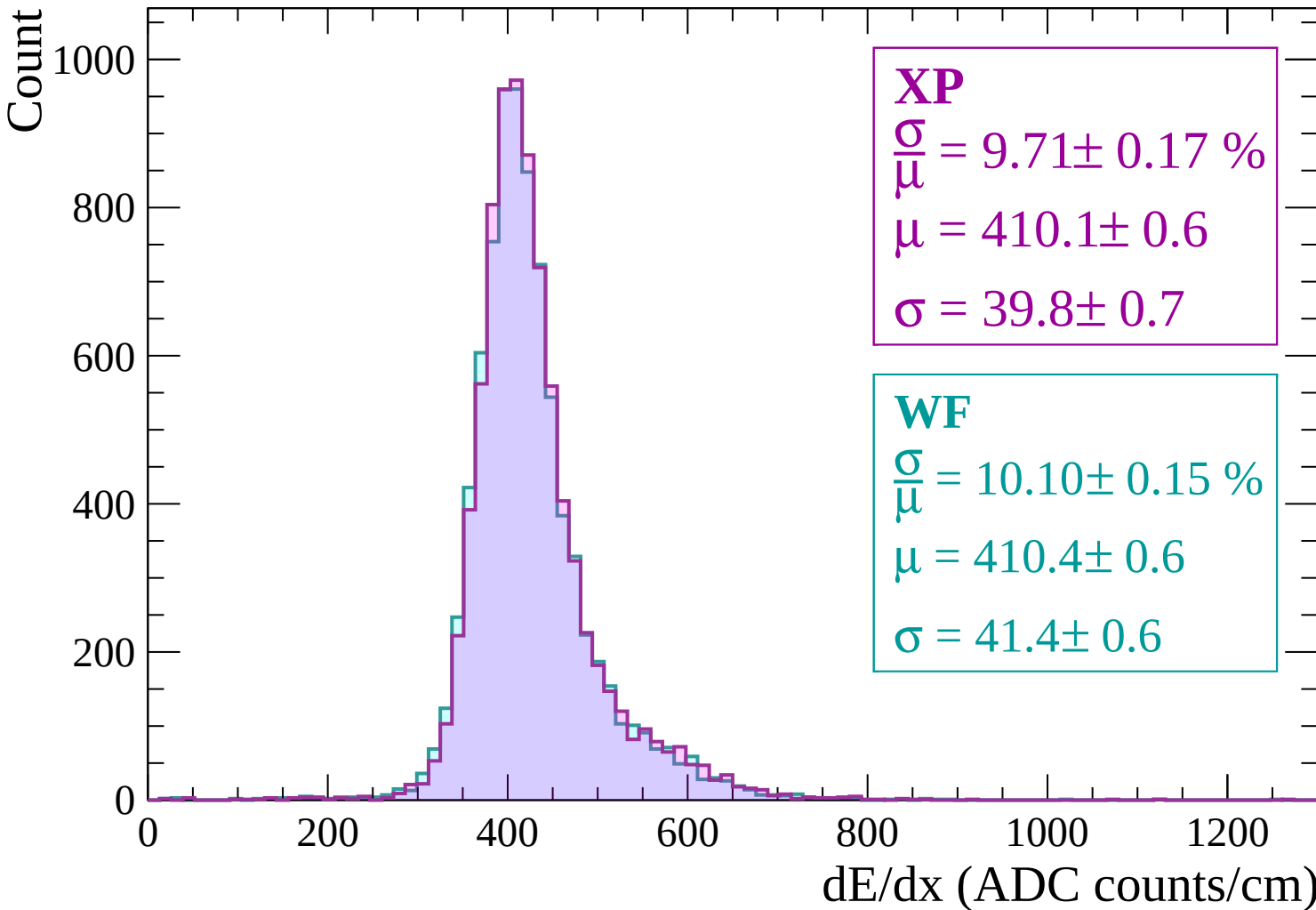
$$\sigma = 39.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.10 \pm 0.15 \%$$

$$\mu = 410.4 \pm 0.6$$

$$\sigma = 41.4 \pm 0.6$$



Energy loss | $-540 < p < -480$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.70 \pm 0.15 \%$$

$$\mu = 405.8 \pm 0.6$$

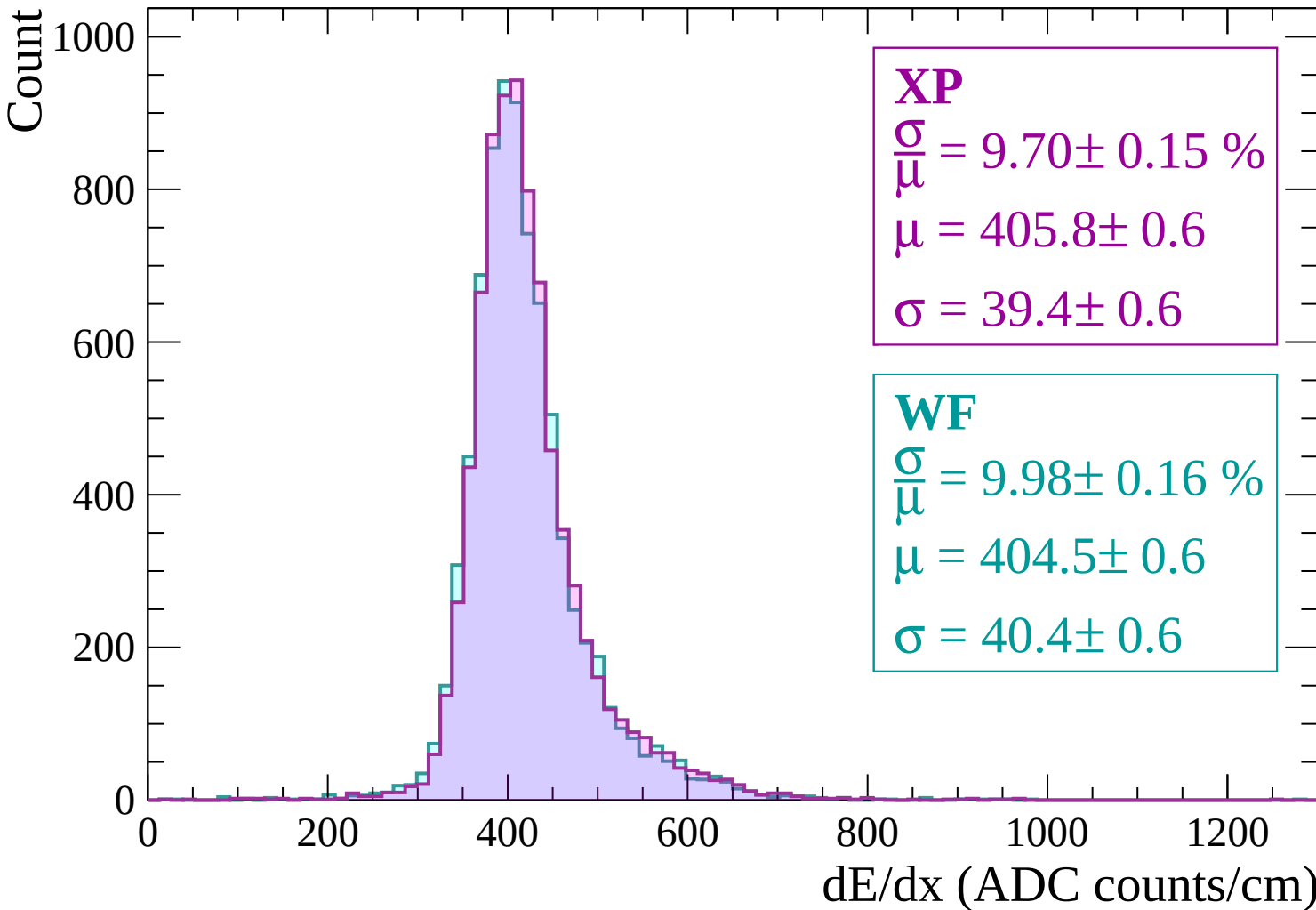
$$\sigma = 39.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.98 \pm 0.16 \%$$

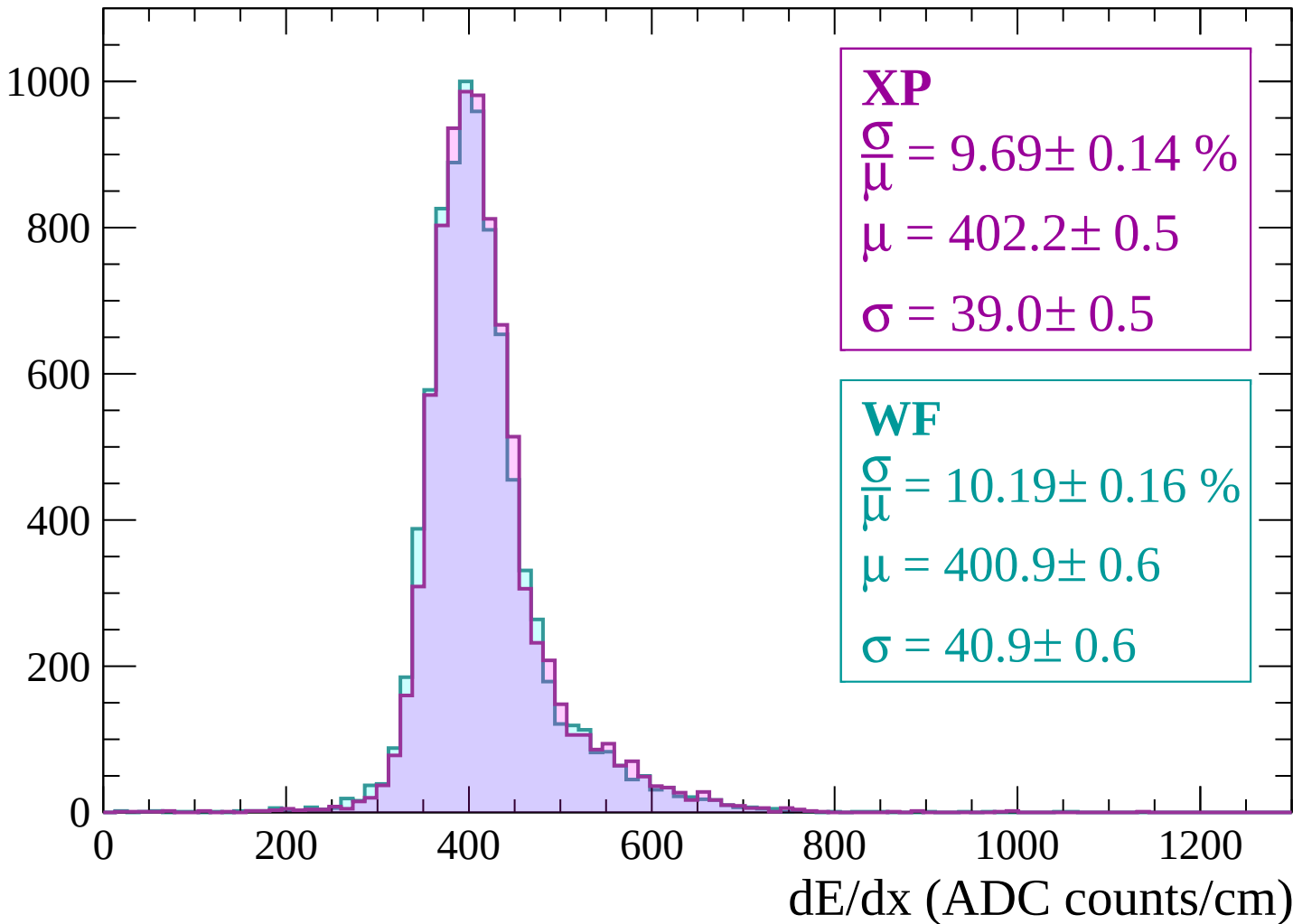
$$\mu = 404.5 \pm 0.6$$

$$\sigma = 40.4 \pm 0.6$$



Energy loss | $-480 < p < -420$

Count



Energy loss | $-420 < p < -360$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.50 \pm 0.15 \%$$

$$\mu = 398.9 \pm 0.5$$

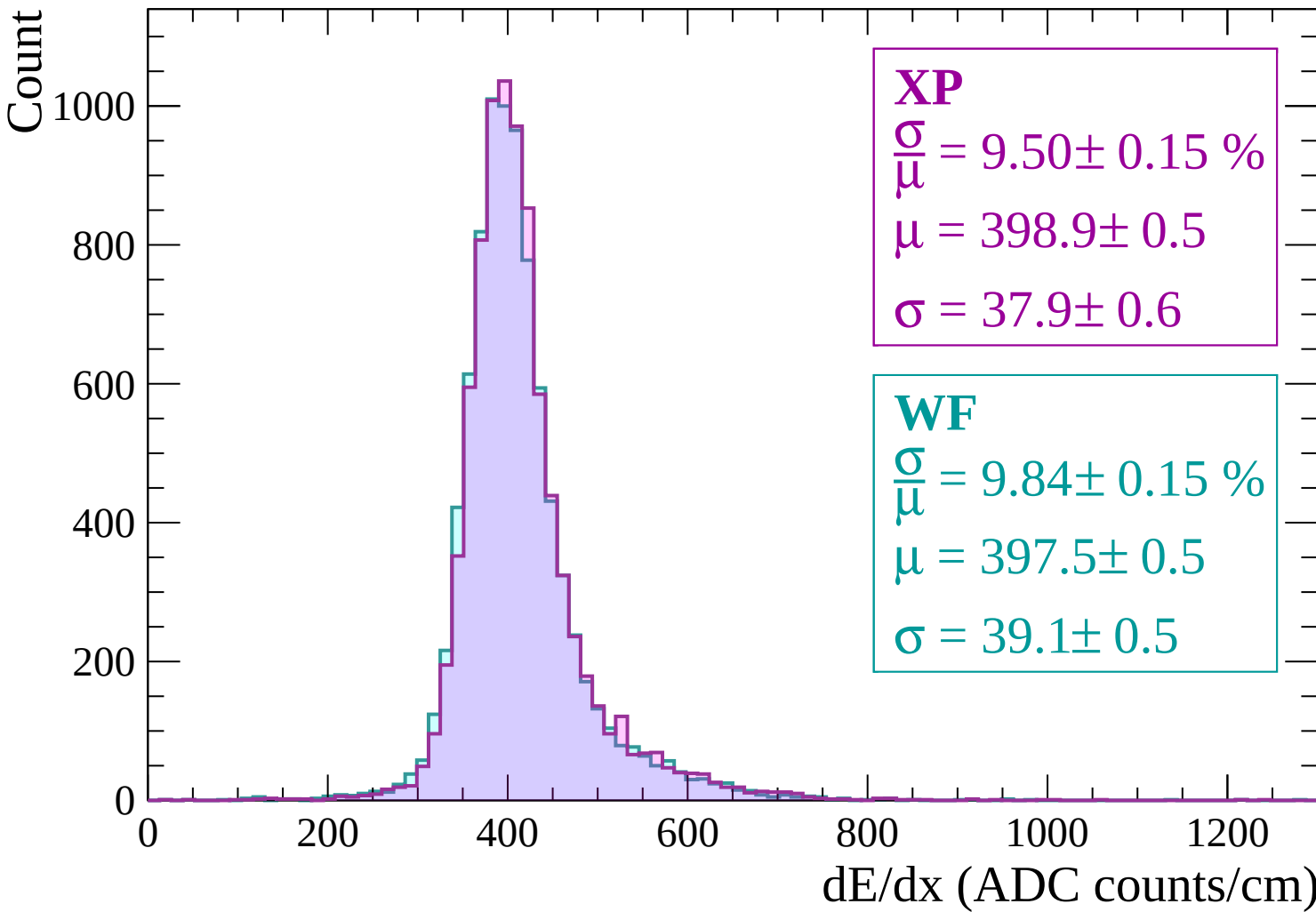
$$\sigma = 37.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.84 \pm 0.15 \%$$

$$\mu = 397.5 \pm 0.5$$

$$\sigma = 39.1 \pm 0.5$$



Energy loss | $-360 < p < -300$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.75 \pm 0.16 \%$$

$$\mu = 397.7 \pm 0.5$$

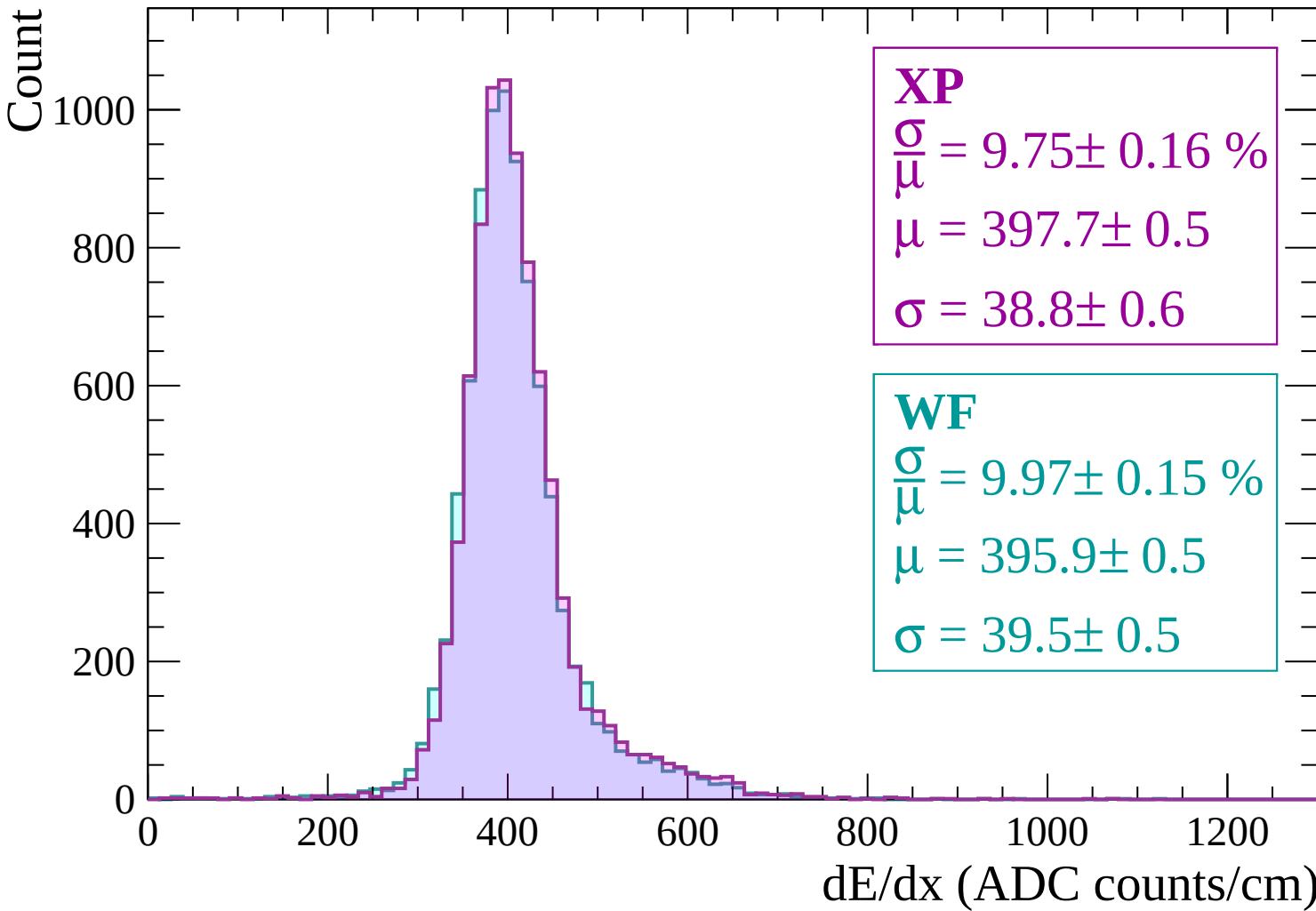
$$\sigma = 38.8 \pm 0.6$$

WF

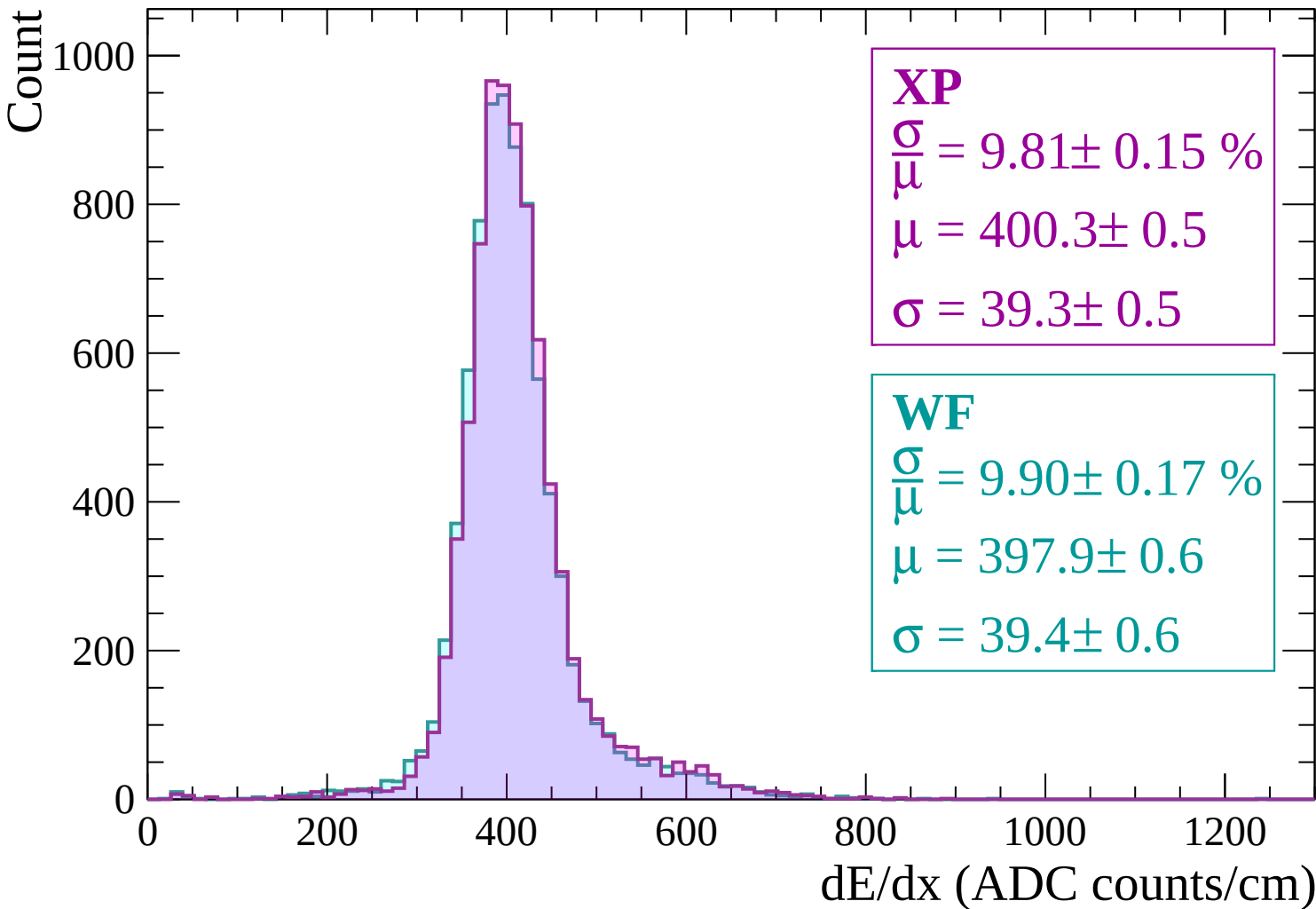
$$\frac{\sigma}{\mu} = 9.97 \pm 0.15 \%$$

$$\mu = 395.9 \pm 0.5$$

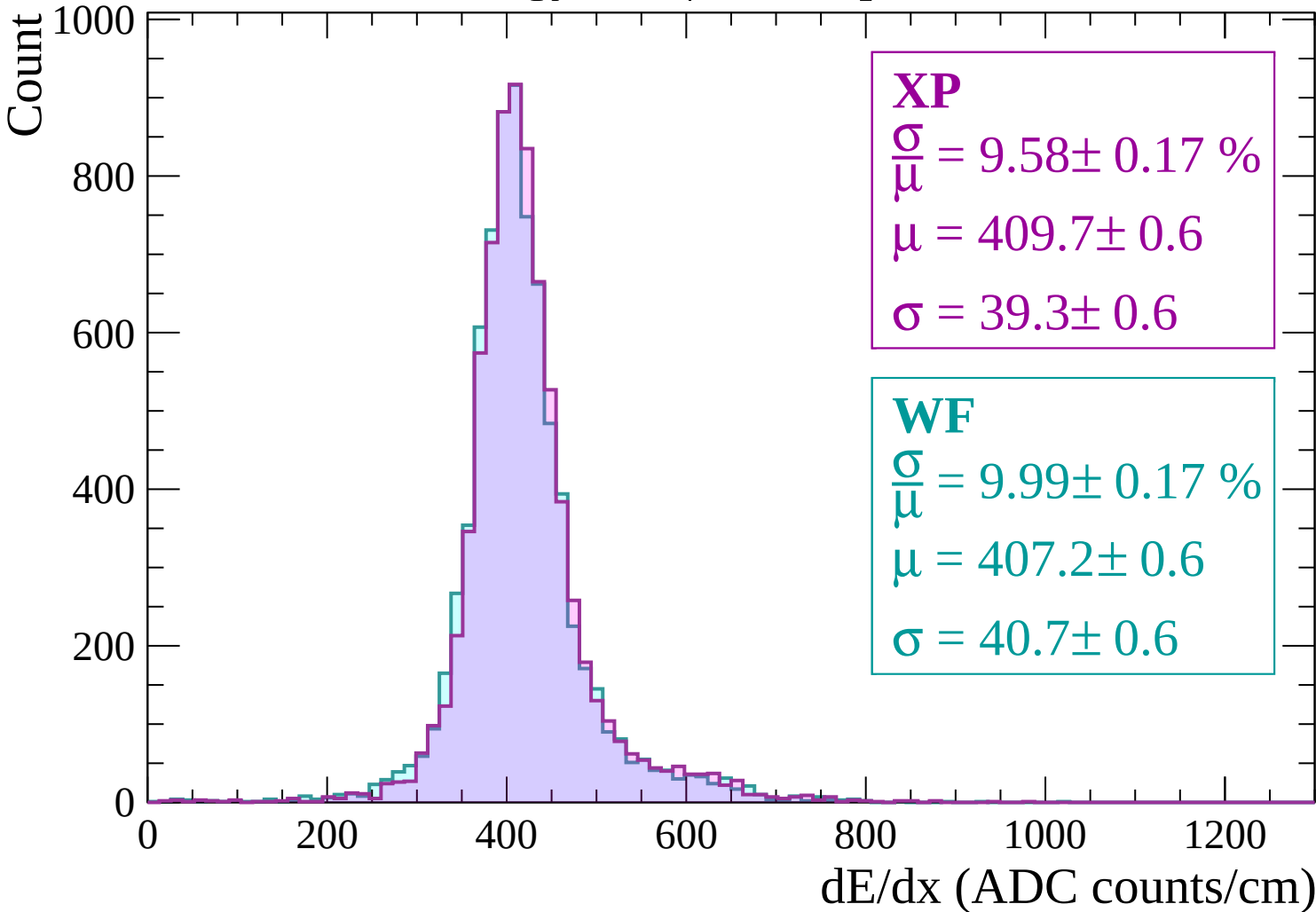
$$\sigma = 39.5 \pm 0.5$$



Energy loss | $-300 < p < -240$

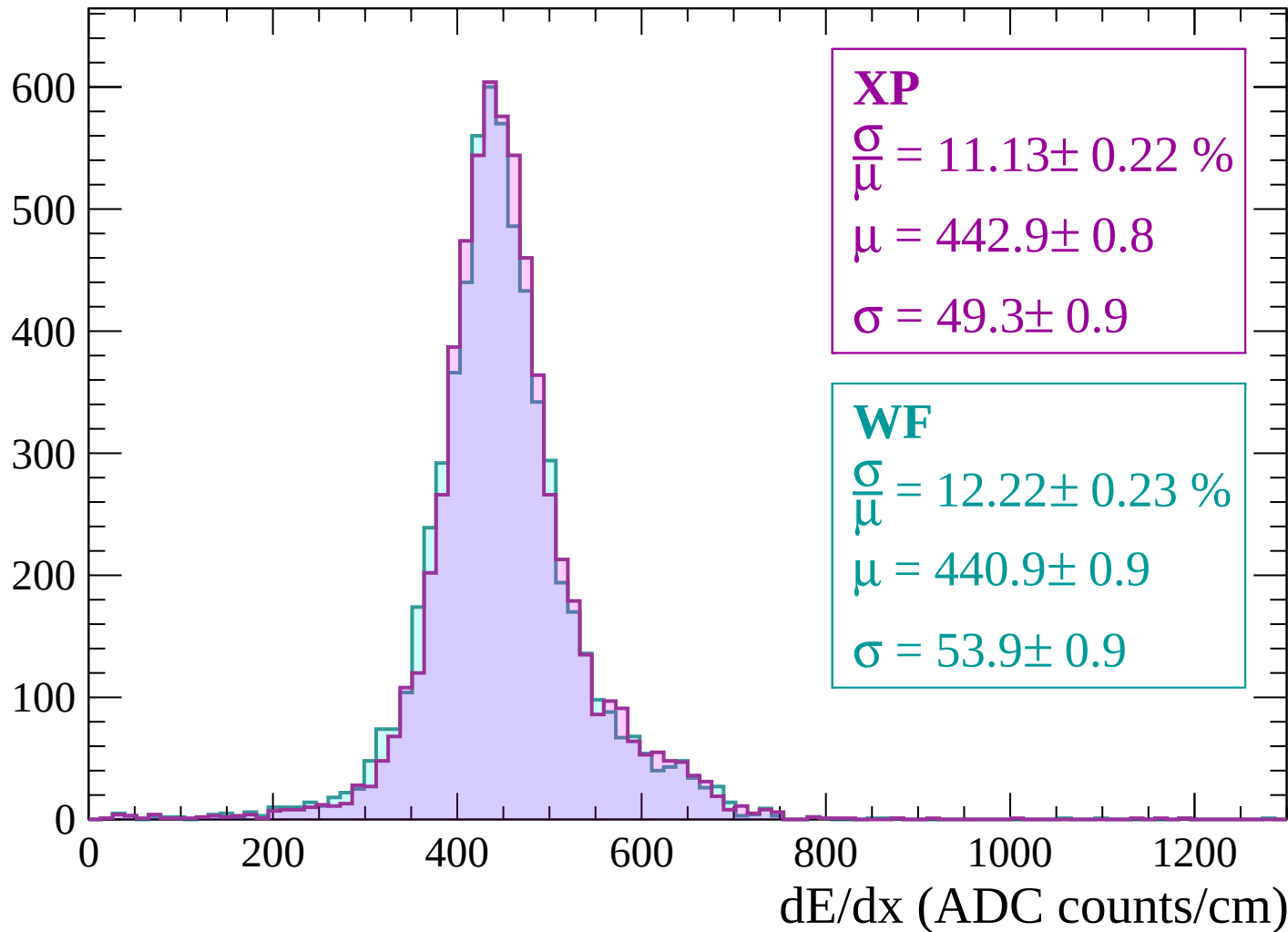


Energy loss | -240 < p < -180



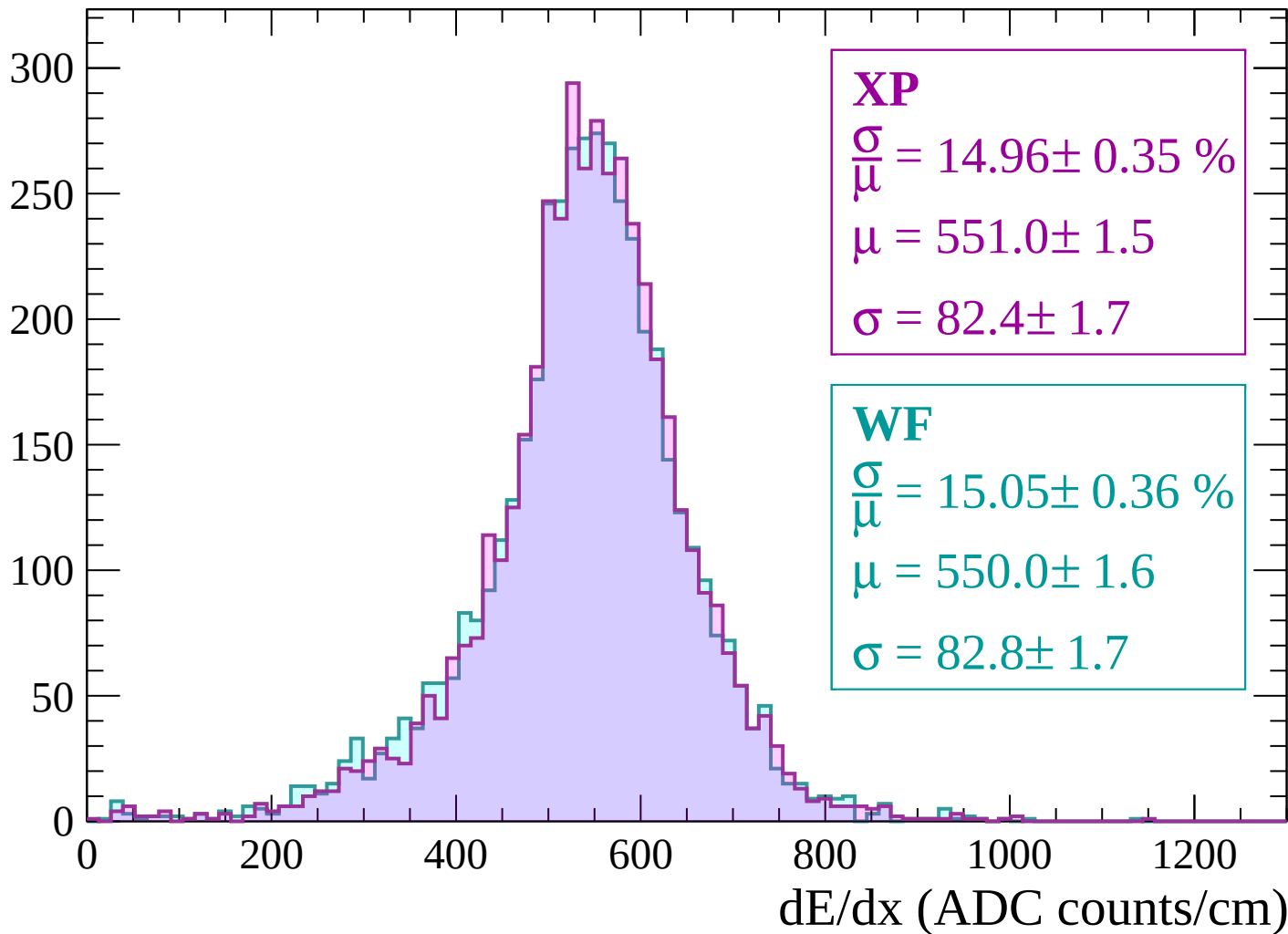
Energy loss | $-180 < p < -120$

Count



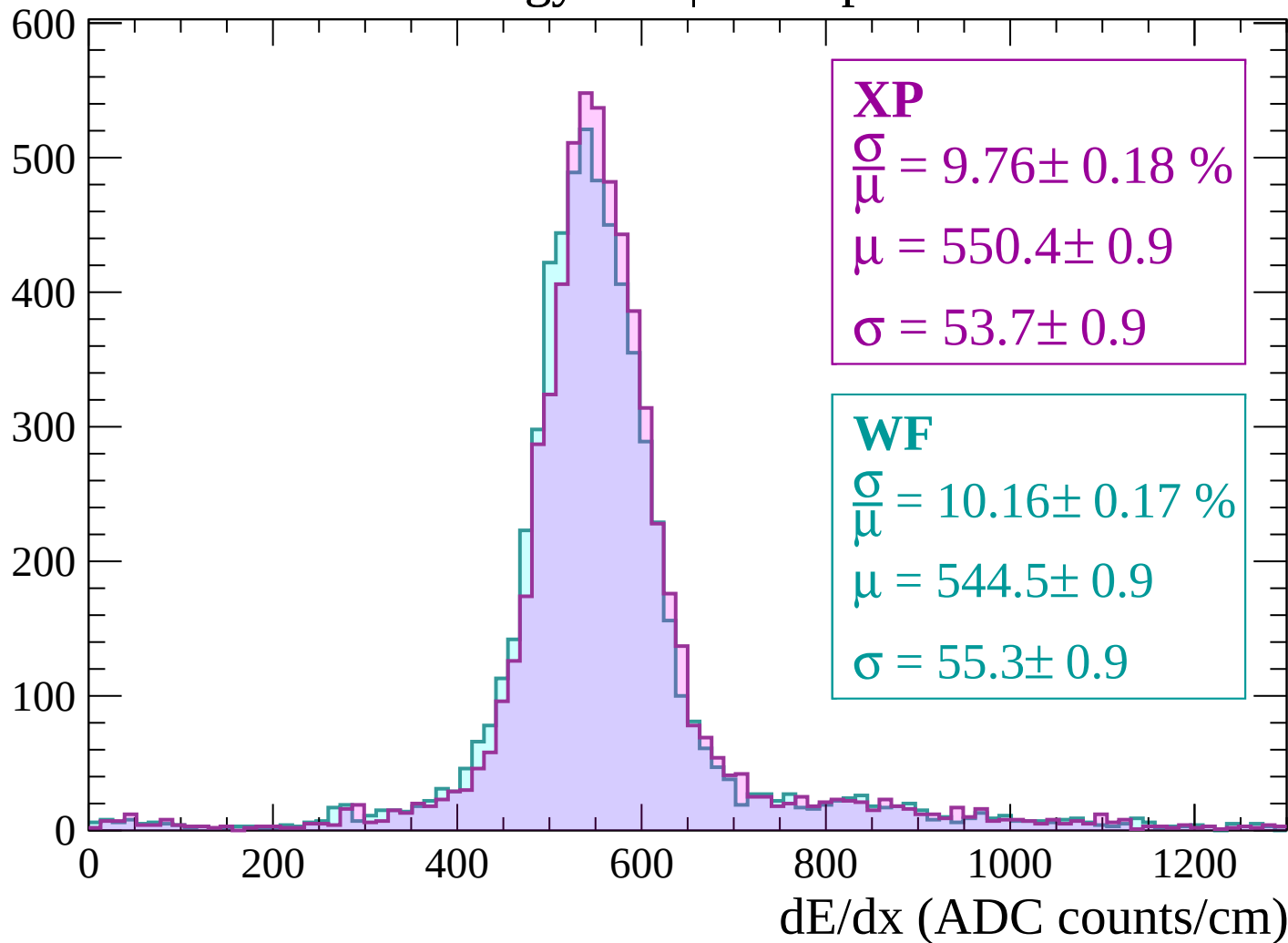
Energy loss | $-120 < p < -60$

Count



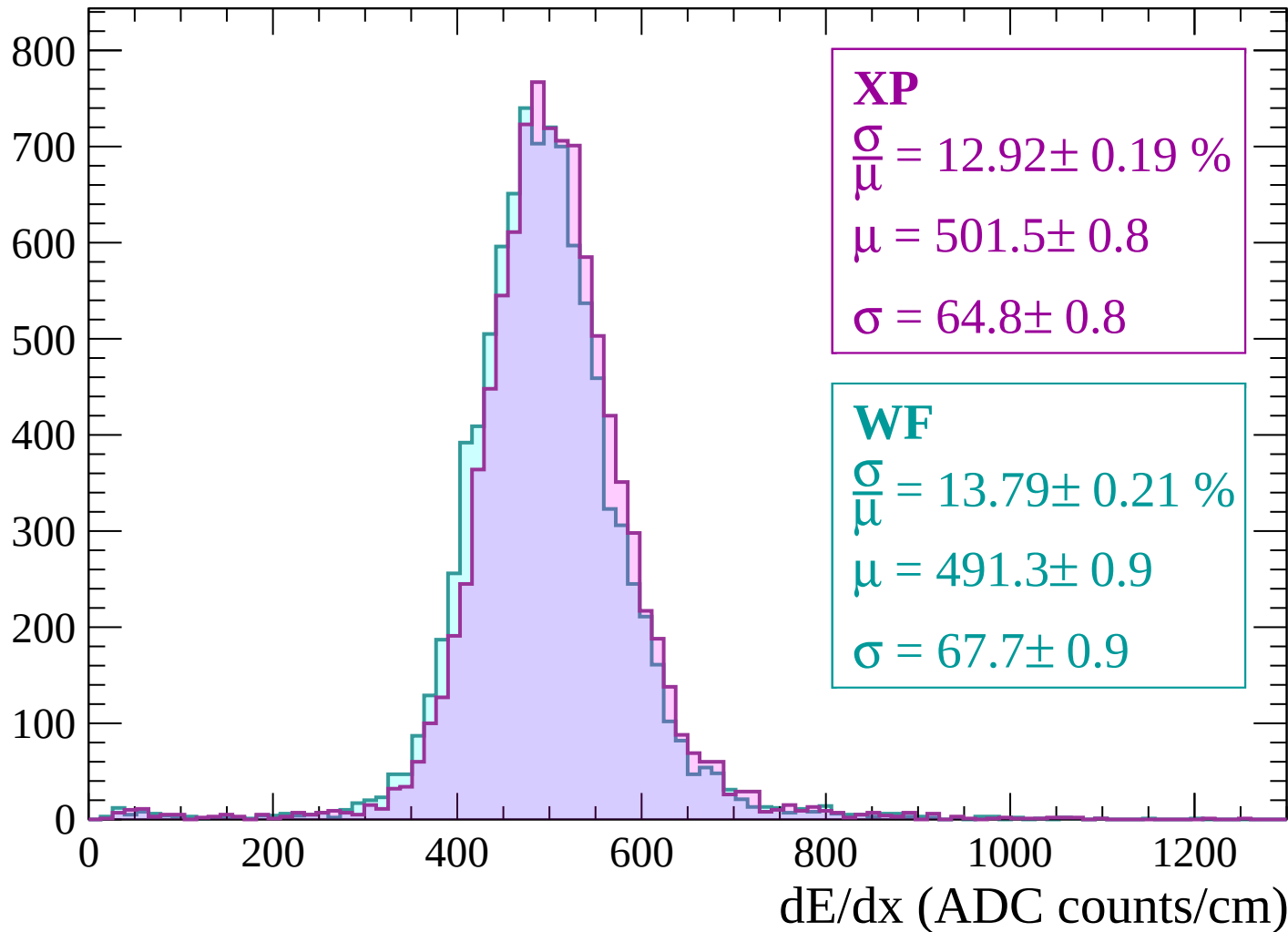
Energy loss | $-60 < p < 0$

Count



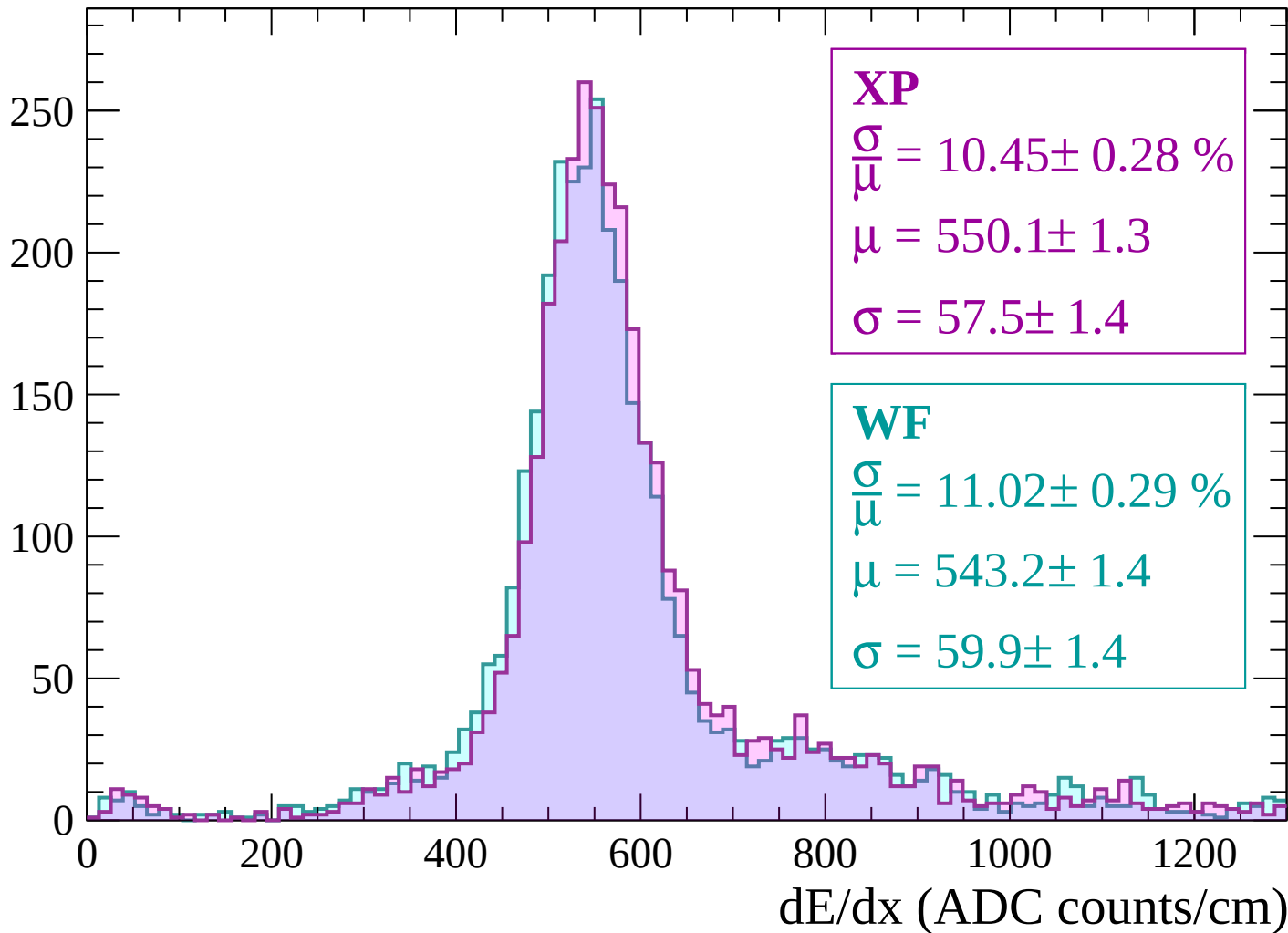
Energy loss | $0 < p < 60$

Count



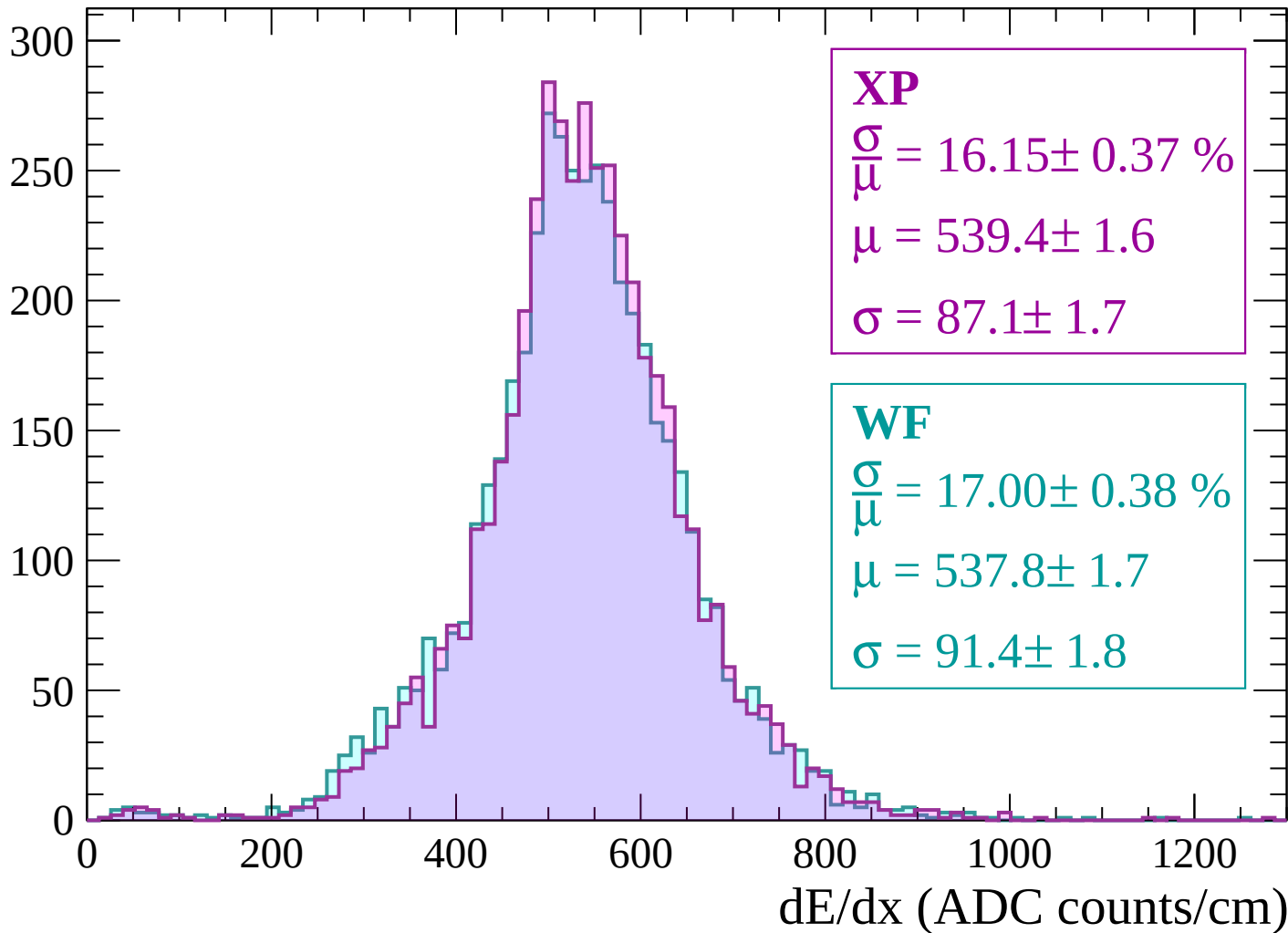
Energy loss | $60 < p < 120$

Count



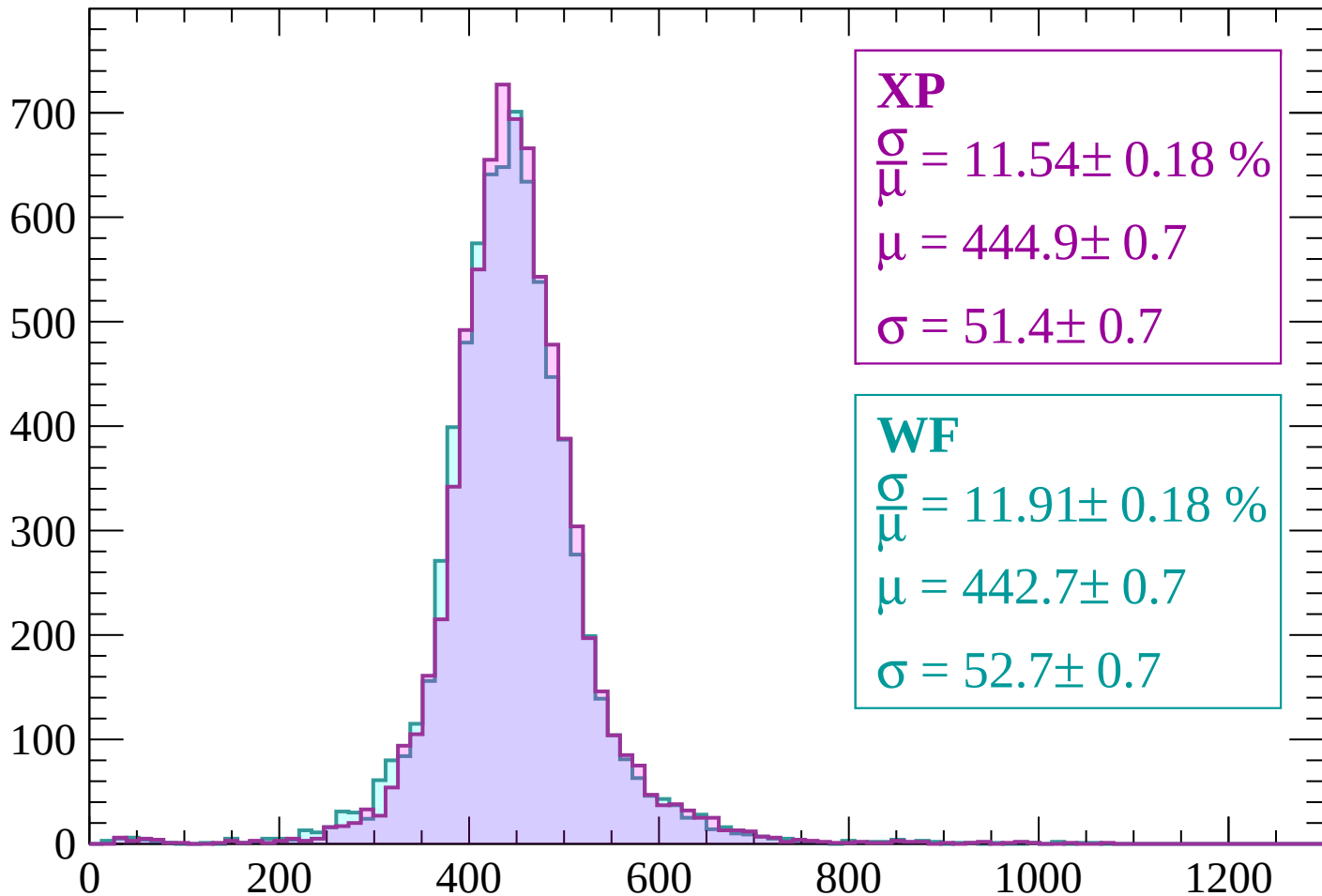
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 11.54 \pm 0.18 \%$$

$$\mu = 444.9 \pm 0.7$$

$$\sigma = 51.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.91 \pm 0.18 \%$$

$$\mu = 442.7 \pm 0.7$$

$$\sigma = 52.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 10.09 \pm 0.14 \%$$

$$\mu = 412.6 \pm 0.5$$

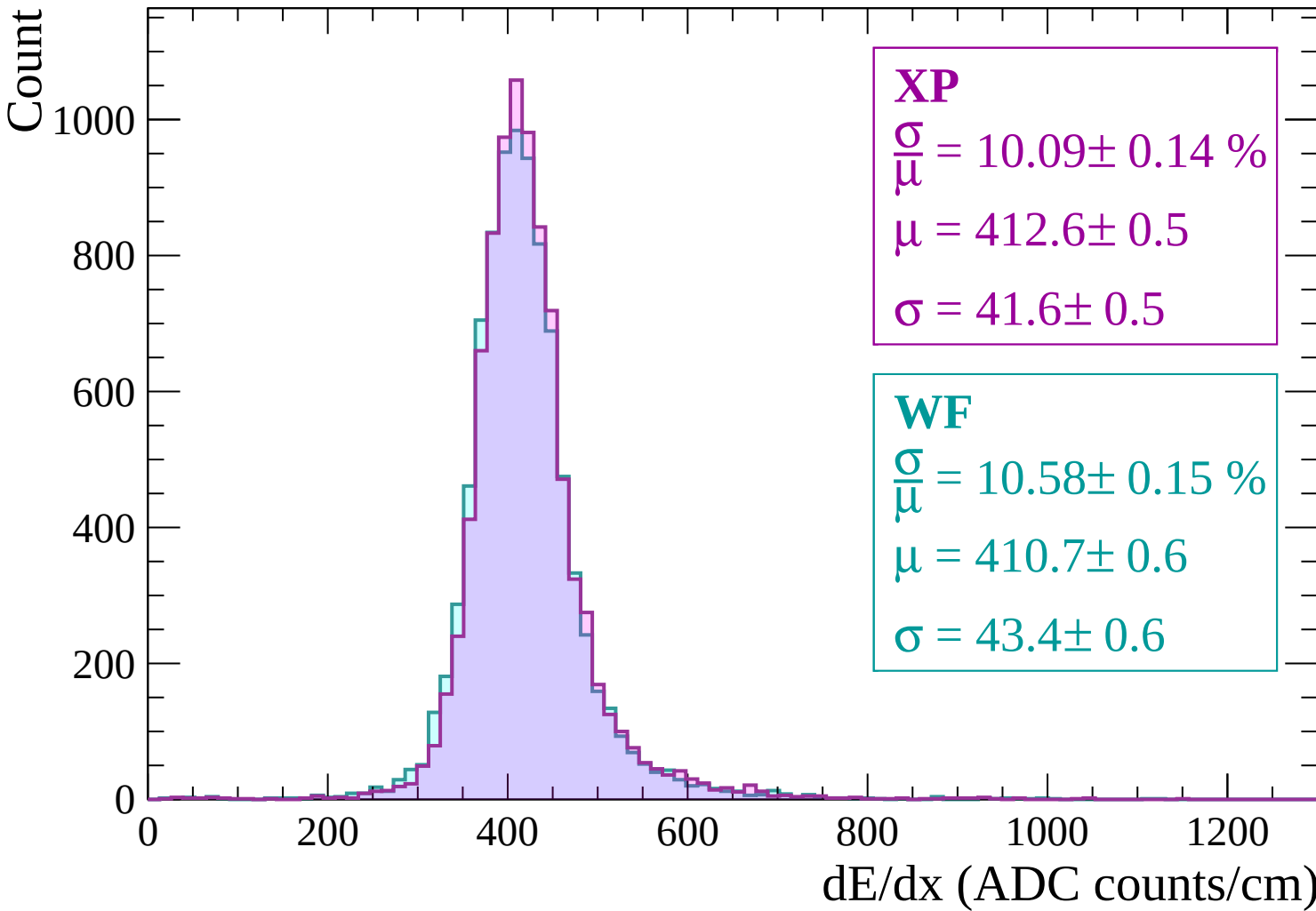
$$\sigma = 41.6 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 10.58 \pm 0.15 \%$$

$$\mu = 410.7 \pm 0.6$$

$$\sigma = 43.4 \pm 0.6$$



Energy loss | $300 < p < 360$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.68 \pm 0.13 \%$$

$$\mu = 401.0 \pm 0.5$$

$$\sigma = 38.8 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 10.02 \pm 0.14 \%$$

$$\mu = 399.6 \pm 0.5$$

$$\sigma = 40.0 \pm 0.5$$

Energy loss | $360 < p < 420$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.60 \pm 0.14 \%$$

$$\mu = 398.0 \pm 0.5$$

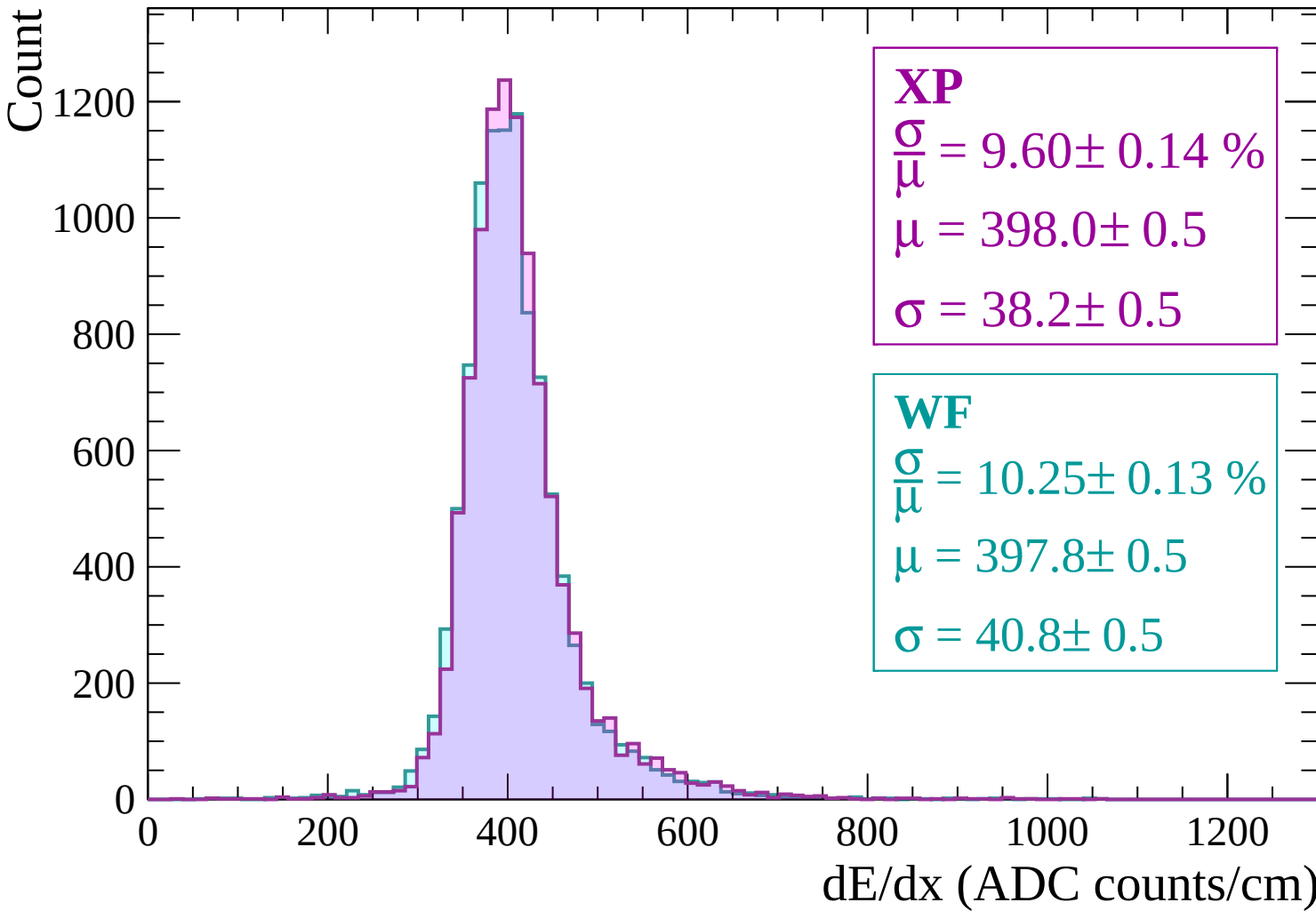
$$\sigma = 38.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 10.25 \pm 0.13 \%$$

$$\mu = 397.8 \pm 0.5$$

$$\sigma = 40.8 \pm 0.5$$



Energy loss | $420 < p < 480$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.76 \pm 0.13 \%$$

$$\mu = 400.5 \pm 0.5$$

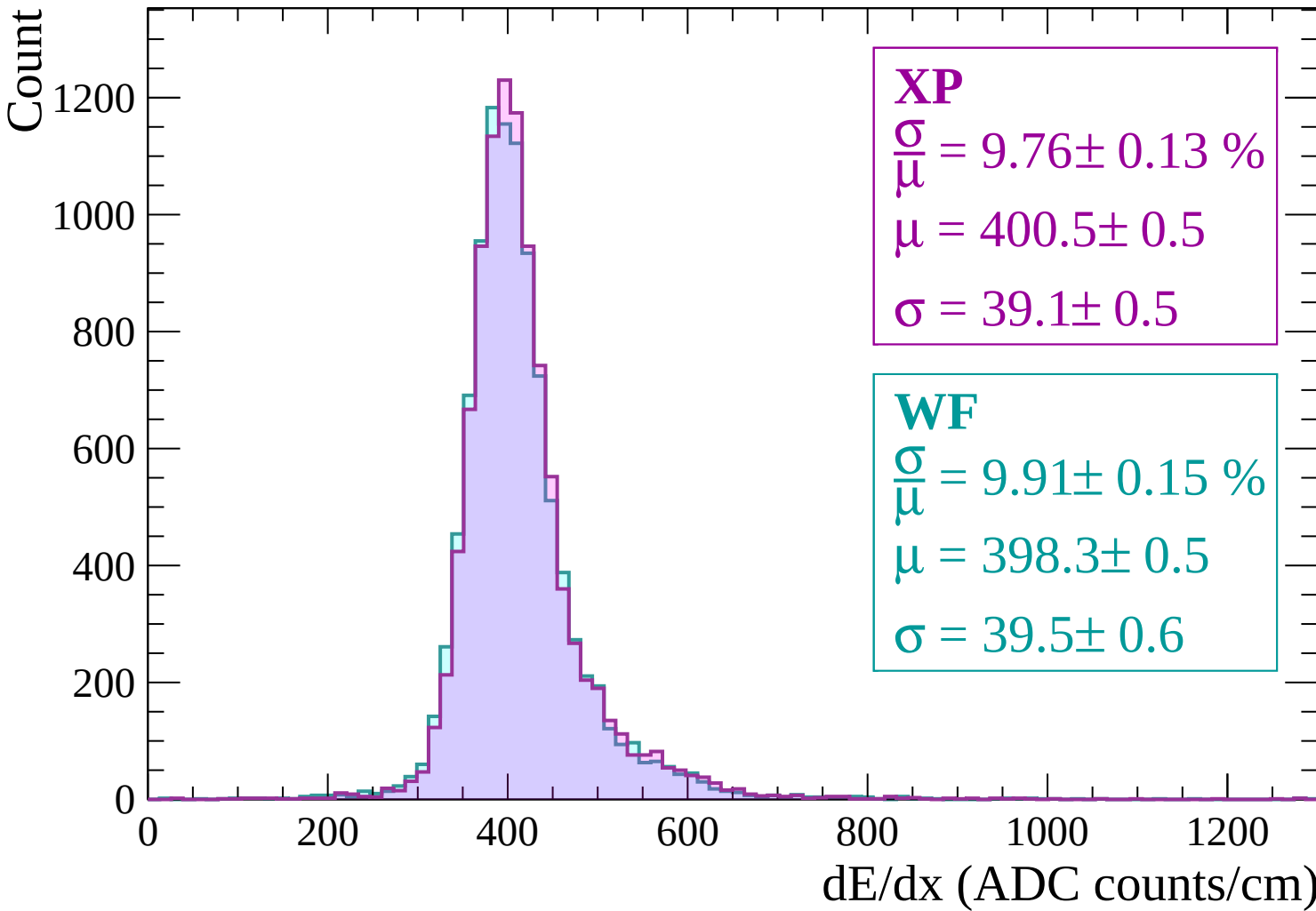
$$\sigma = 39.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.91 \pm 0.15 \%$$

$$\mu = 398.3 \pm 0.5$$

$$\sigma = 39.5 \pm 0.6$$



Energy loss | $480 < p < 540$

Count

1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 9.69 \pm 0.14 \%$$

$$\mu = 402.8 \pm 0.5$$

$$\sigma = 39.0 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 10.08 \pm 0.15 \%$$

$$\mu = 401.8 \pm 0.5$$

$$\sigma = 40.5 \pm 0.5$$

0

200

400

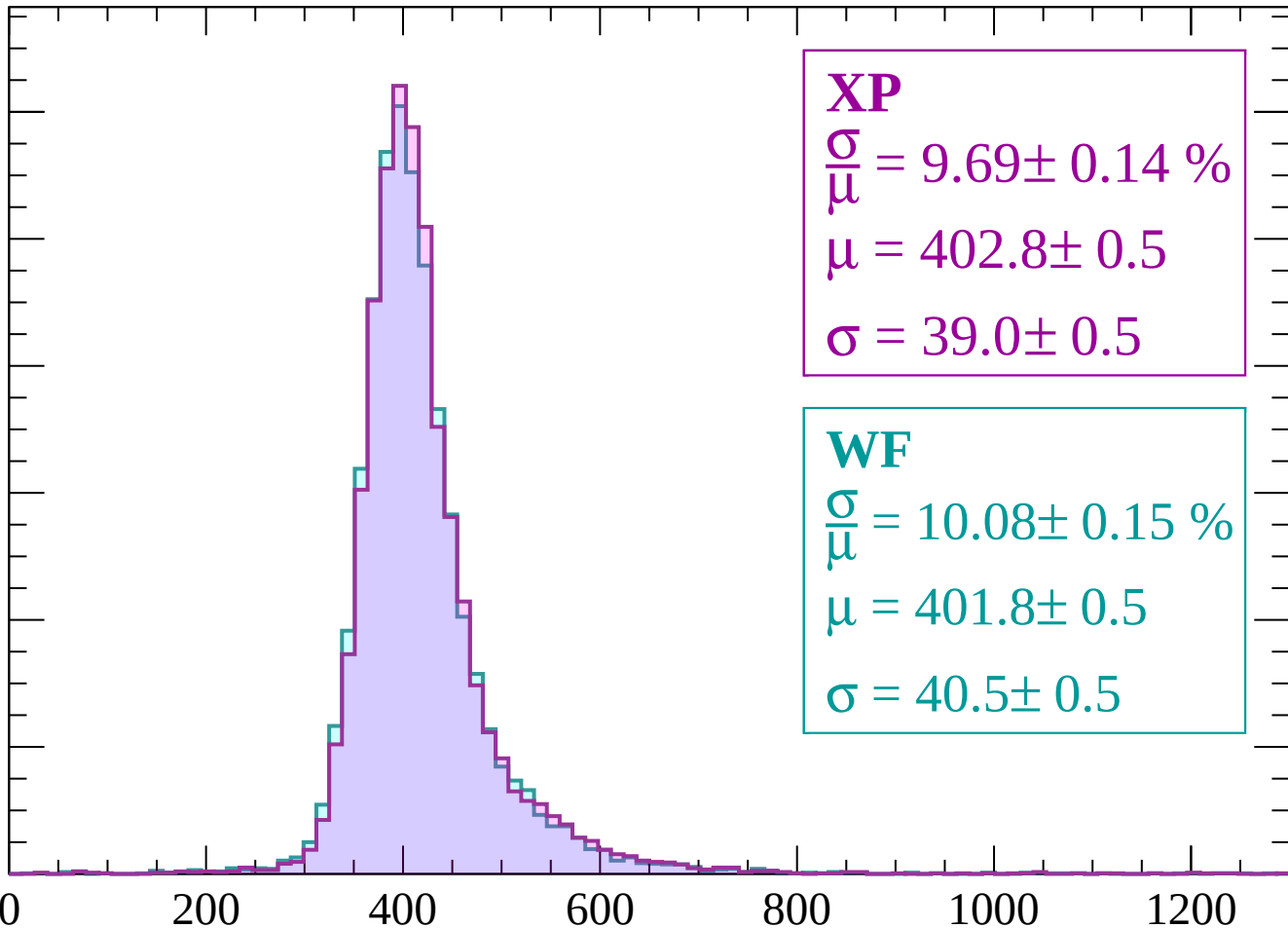
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $540 < p < 600$

Count

1200

1000

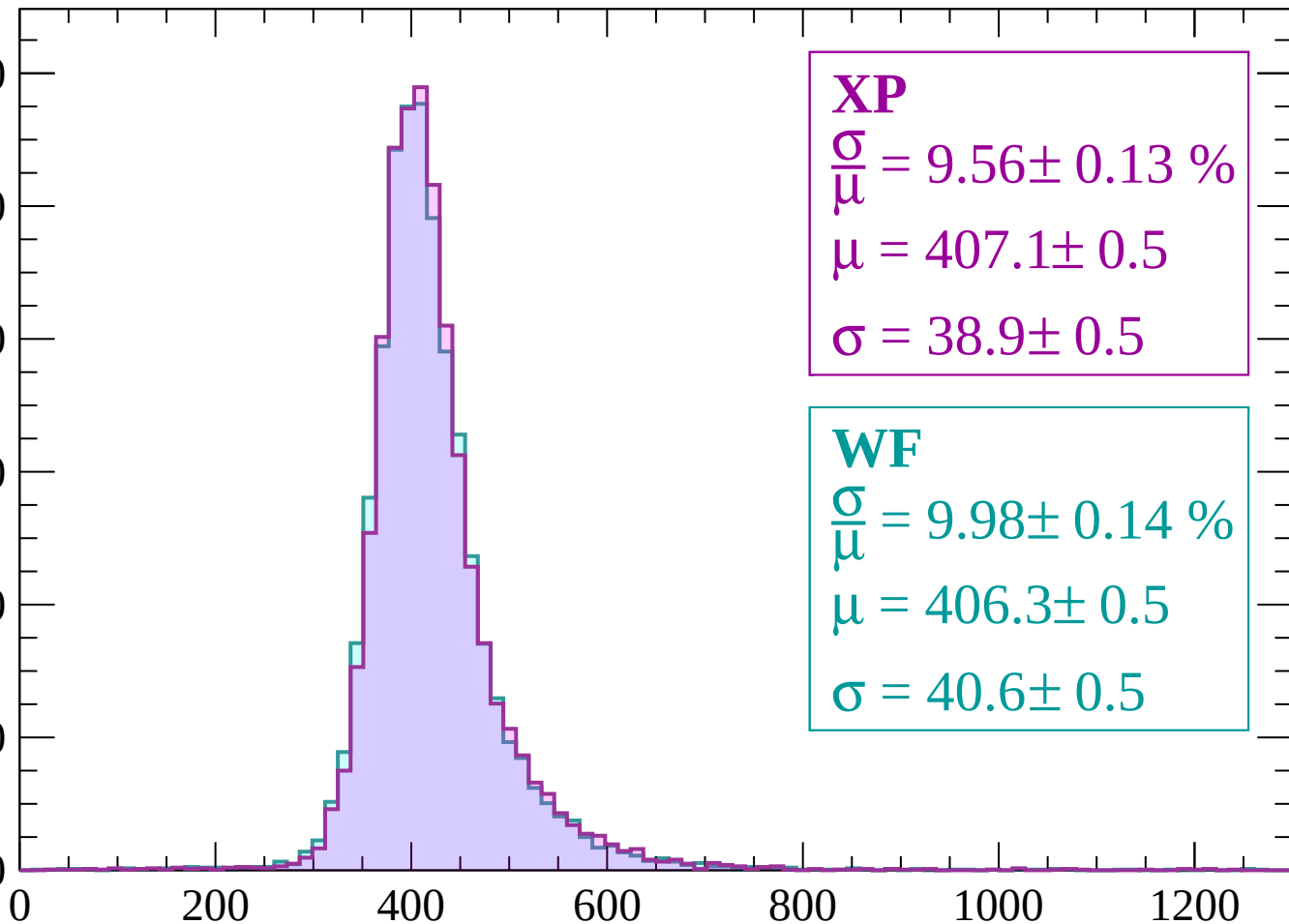
800

600

400

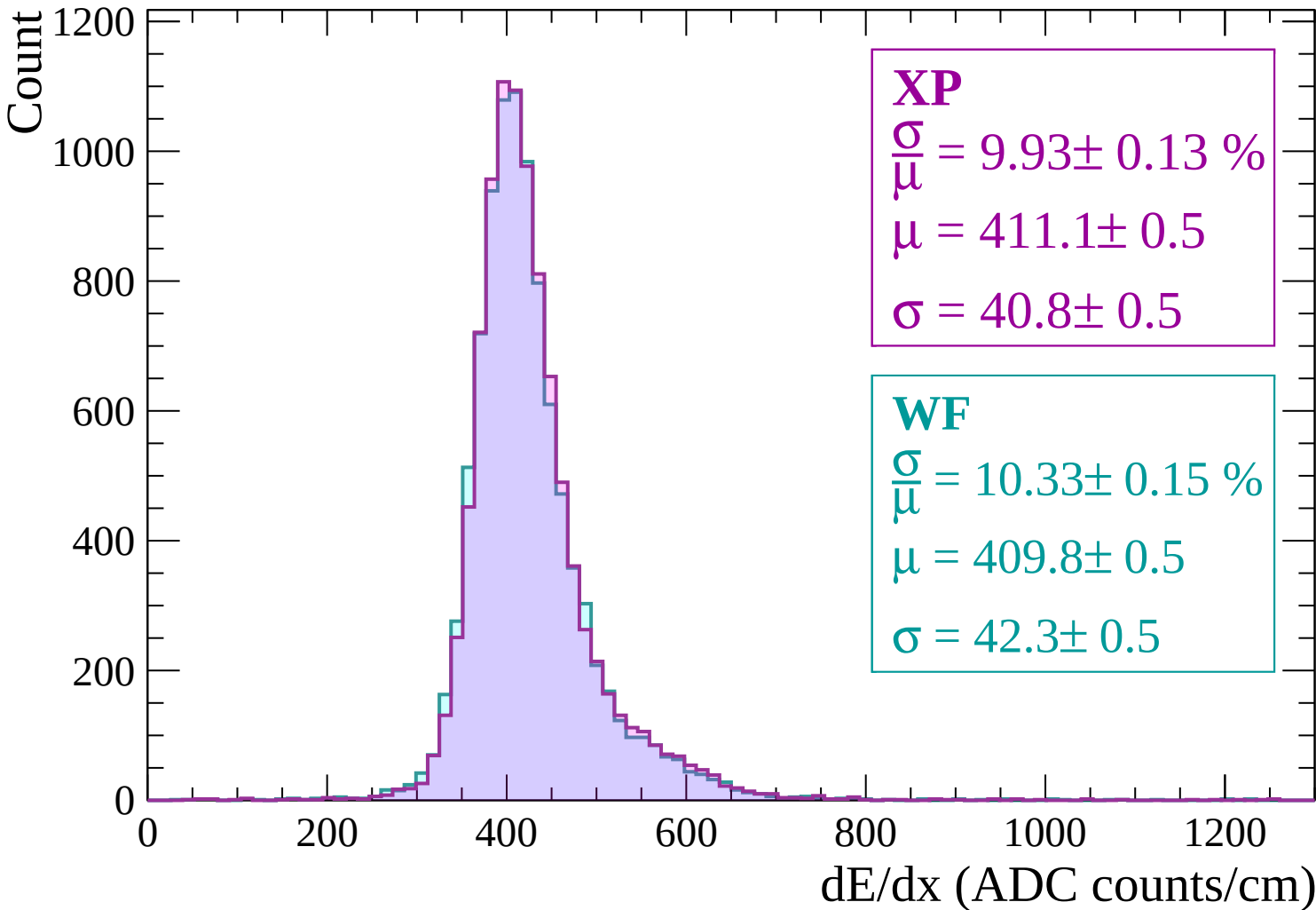
200

0

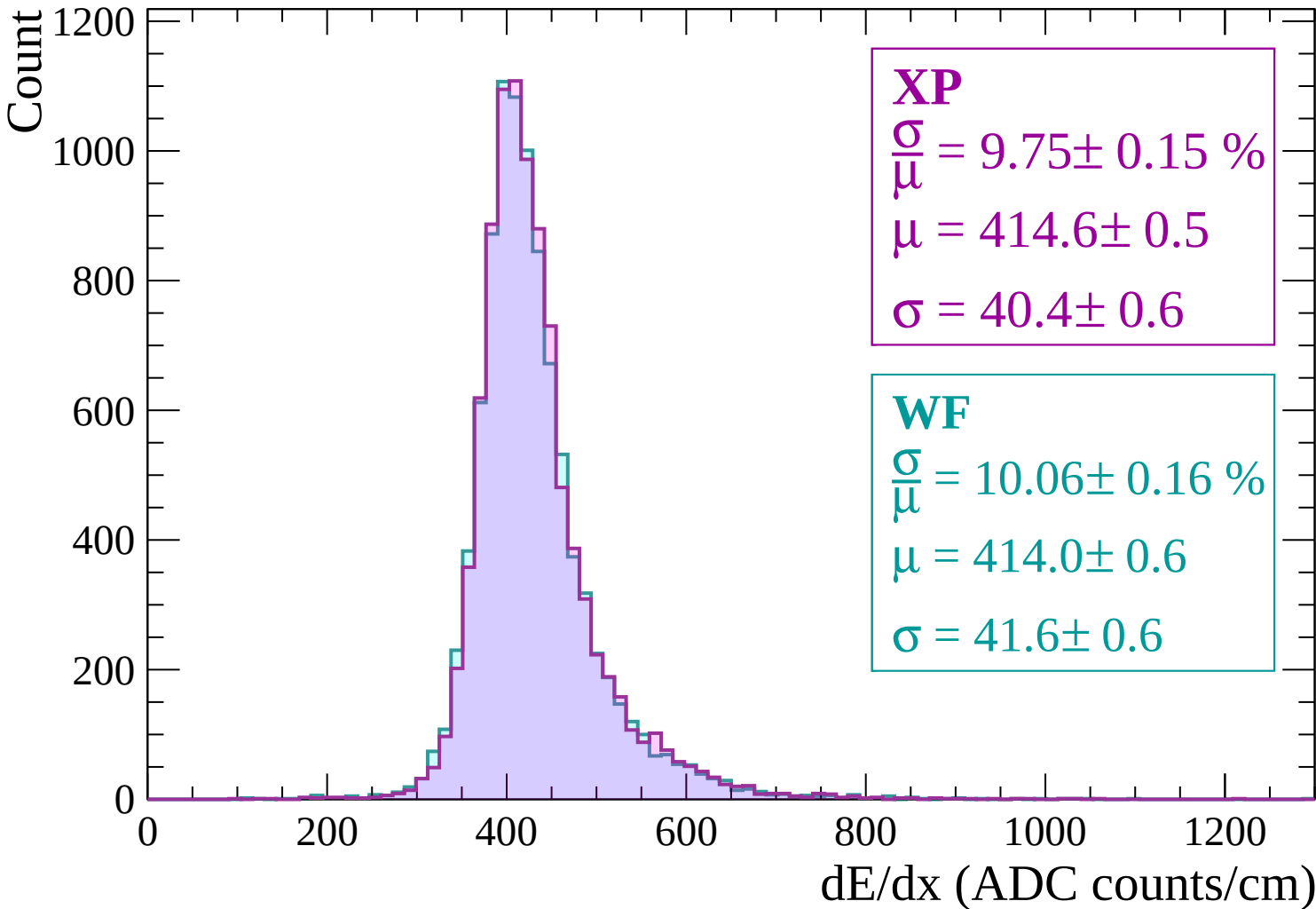


dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

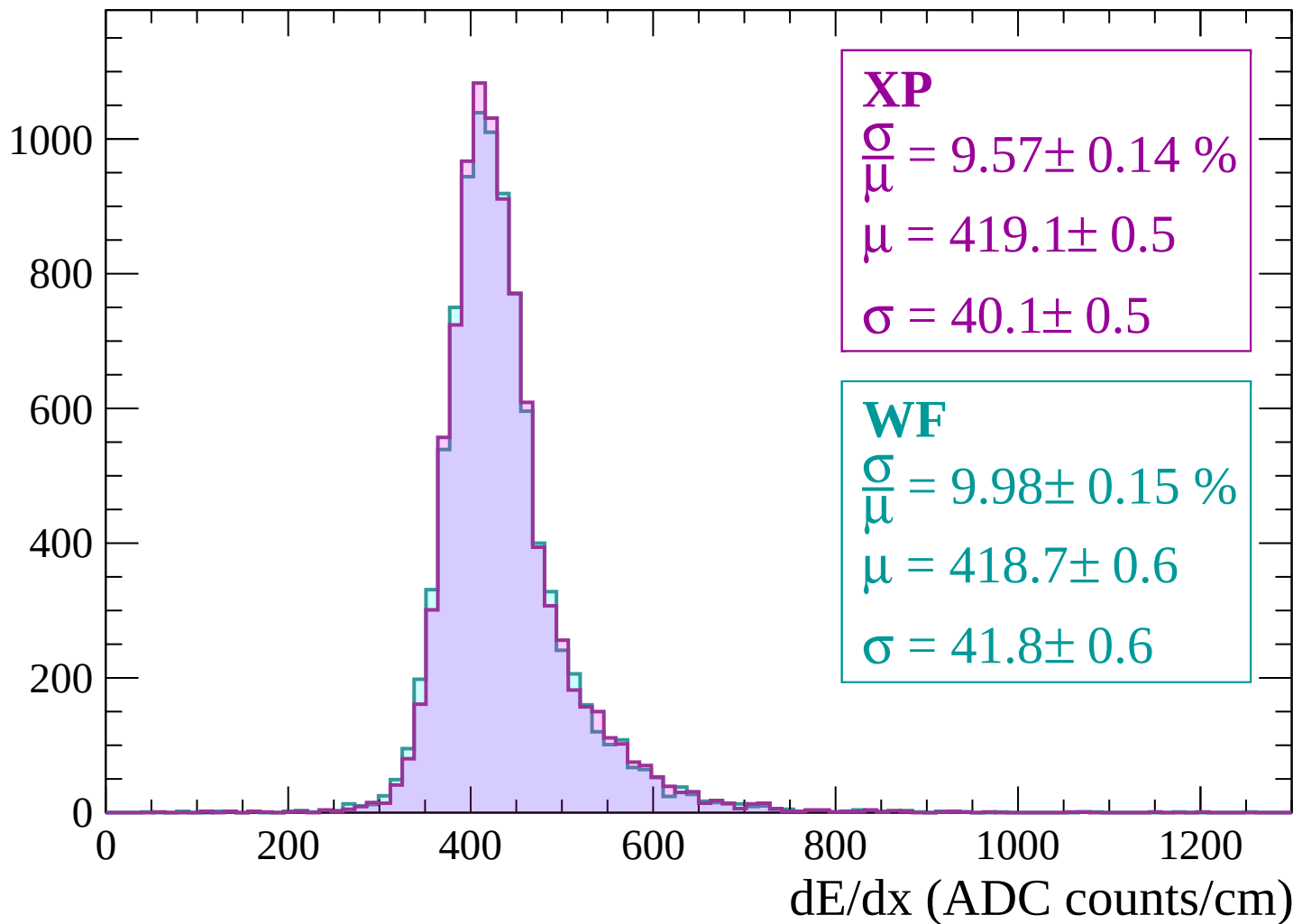


Energy loss | $660 < p < 720$



Energy loss | $720 < p < 780$

Count



Energy loss | $780 < p < 840$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.98 \pm 0.14 \%$$

$$\mu = 423.1 \pm 0.6$$

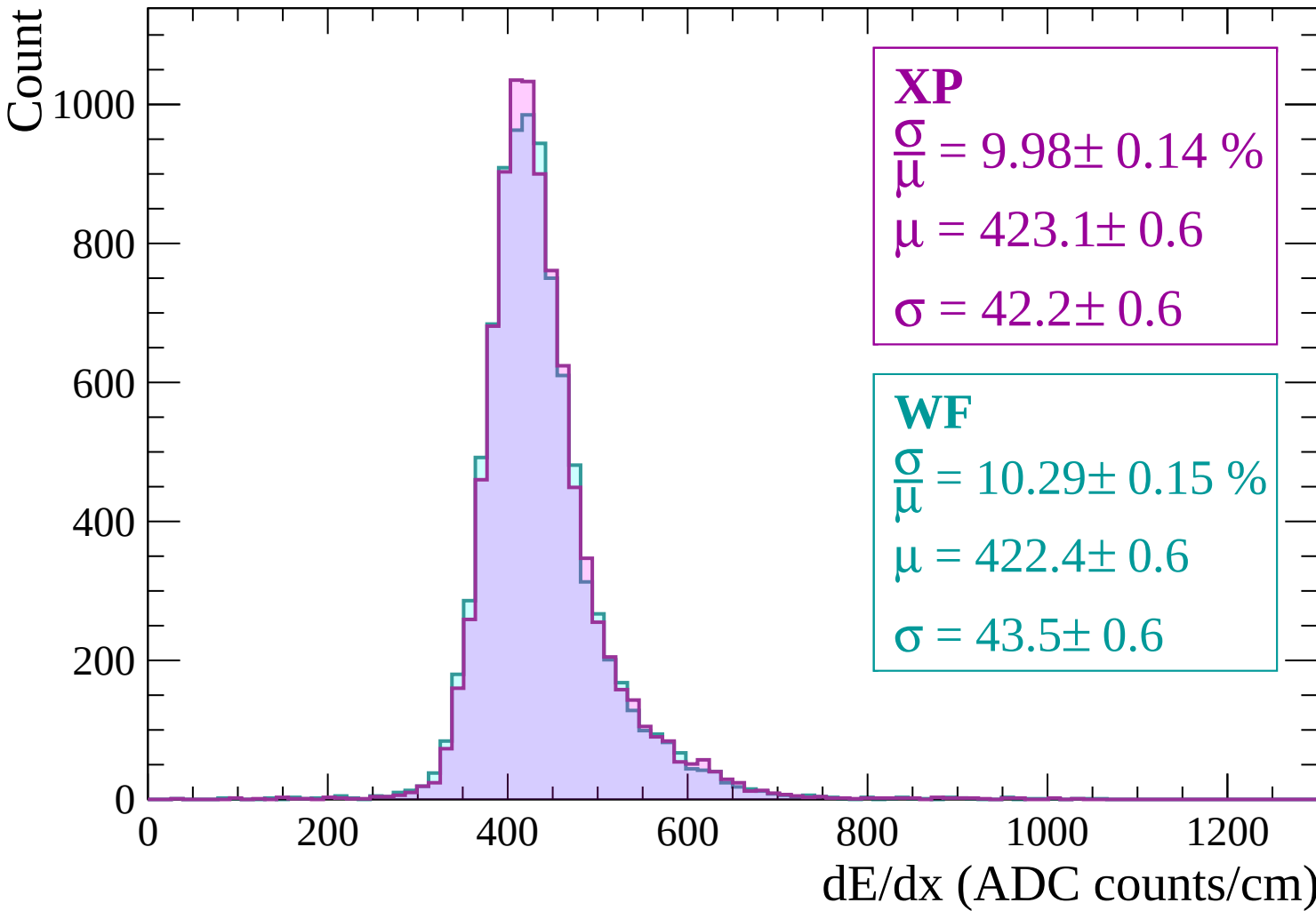
$$\sigma = 42.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 10.29 \pm 0.15 \%$$

$$\mu = 422.4 \pm 0.6$$

$$\sigma = 43.5 \pm 0.6$$



Energy loss | $840 < p < 900$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 9.75 \pm 0.14 \%$$

$$\mu = 427.3 \pm 0.6$$

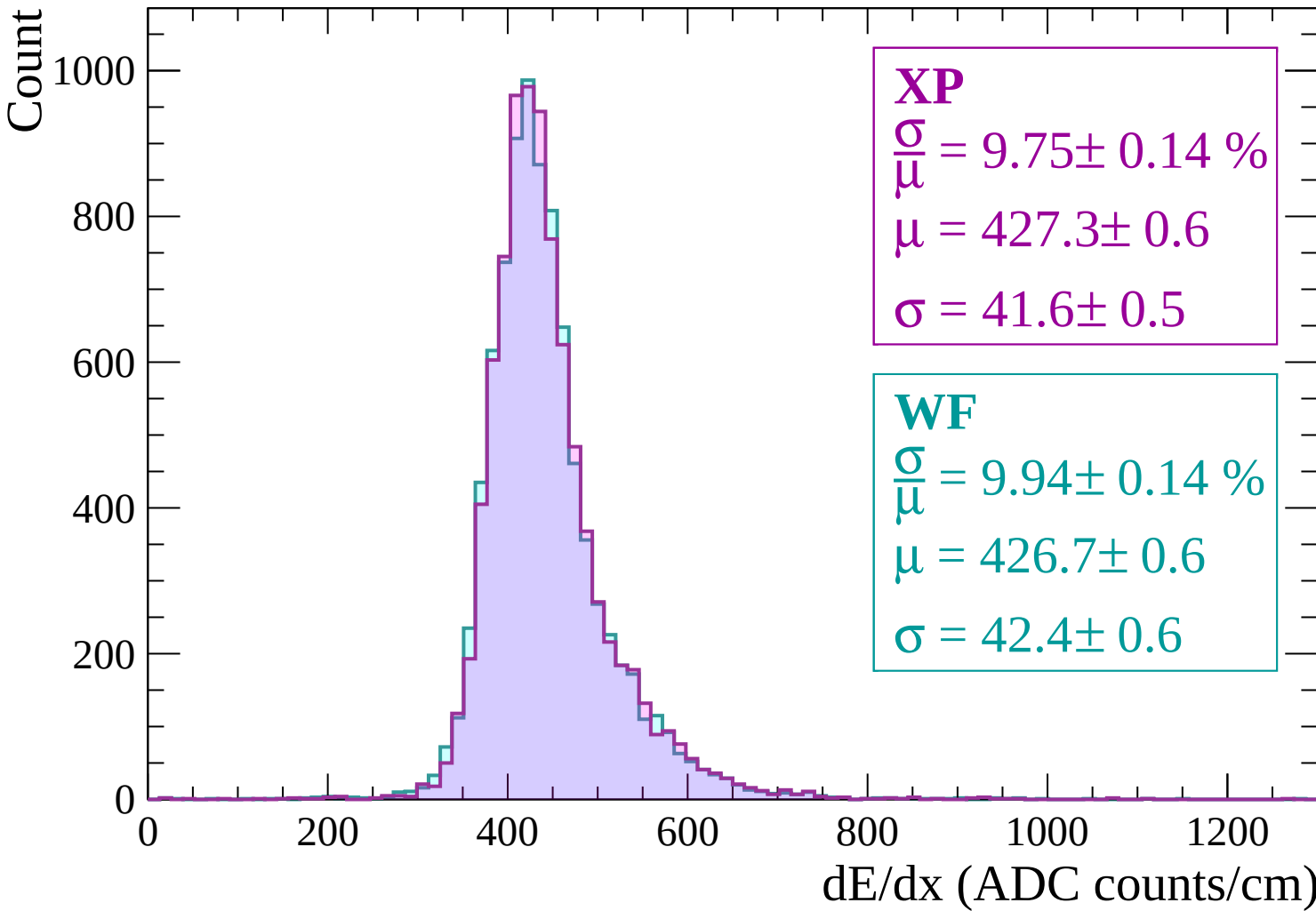
$$\sigma = 41.6 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.94 \pm 0.14 \%$$

$$\mu = 426.7 \pm 0.6$$

$$\sigma = 42.4 \pm 0.6$$



Energy loss | $900 < p < 960$

Count

1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 9.93 \pm 0.14 \%$$

$$\mu = 432.2 \pm 0.6$$

$$\sigma = 42.9 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 10.31 \pm 0.15 \%$$

$$\mu = 431.5 \pm 0.6$$

$$\sigma = 44.5 \pm 0.6$$

0

200

400

600

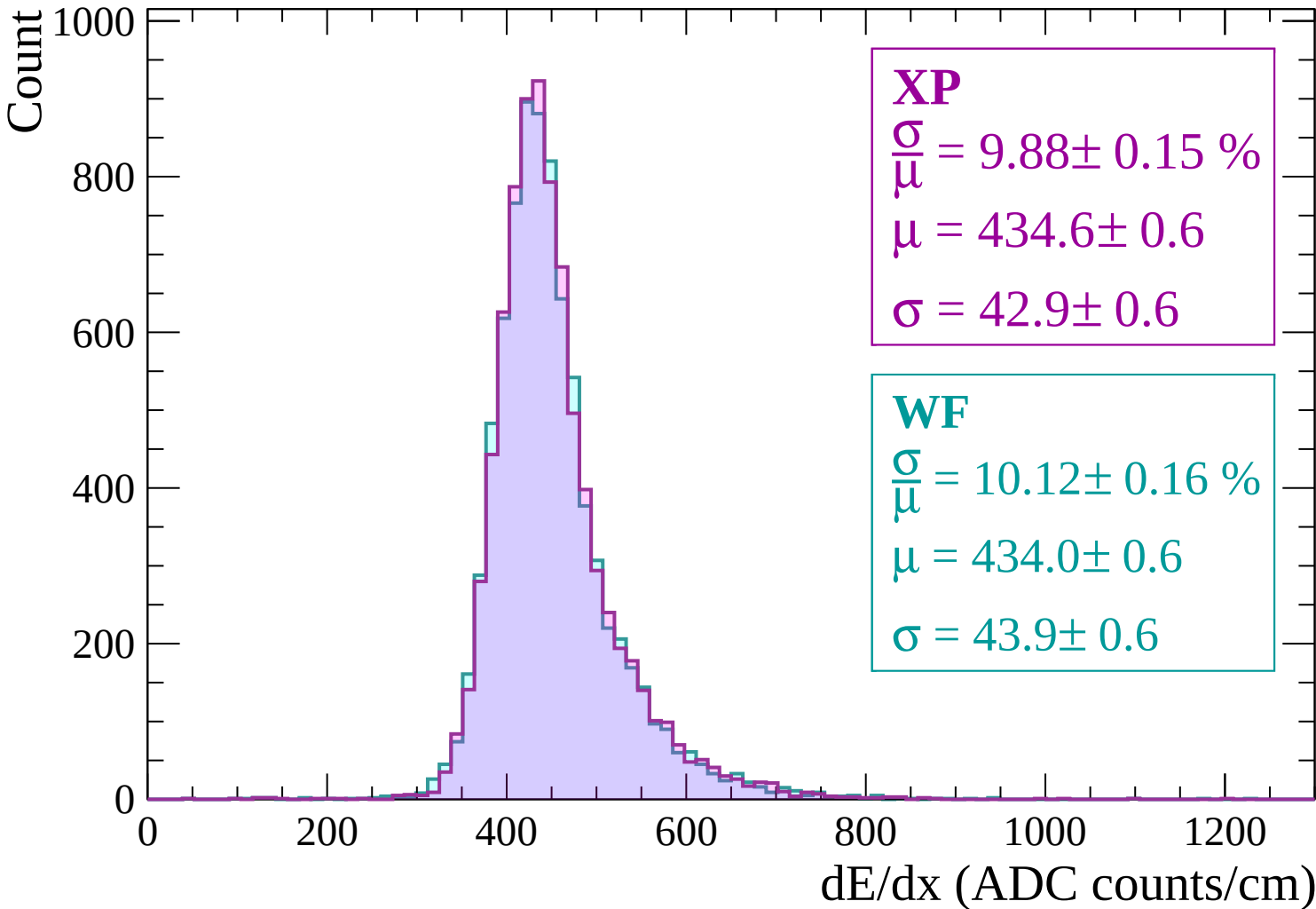
800

1000

1200

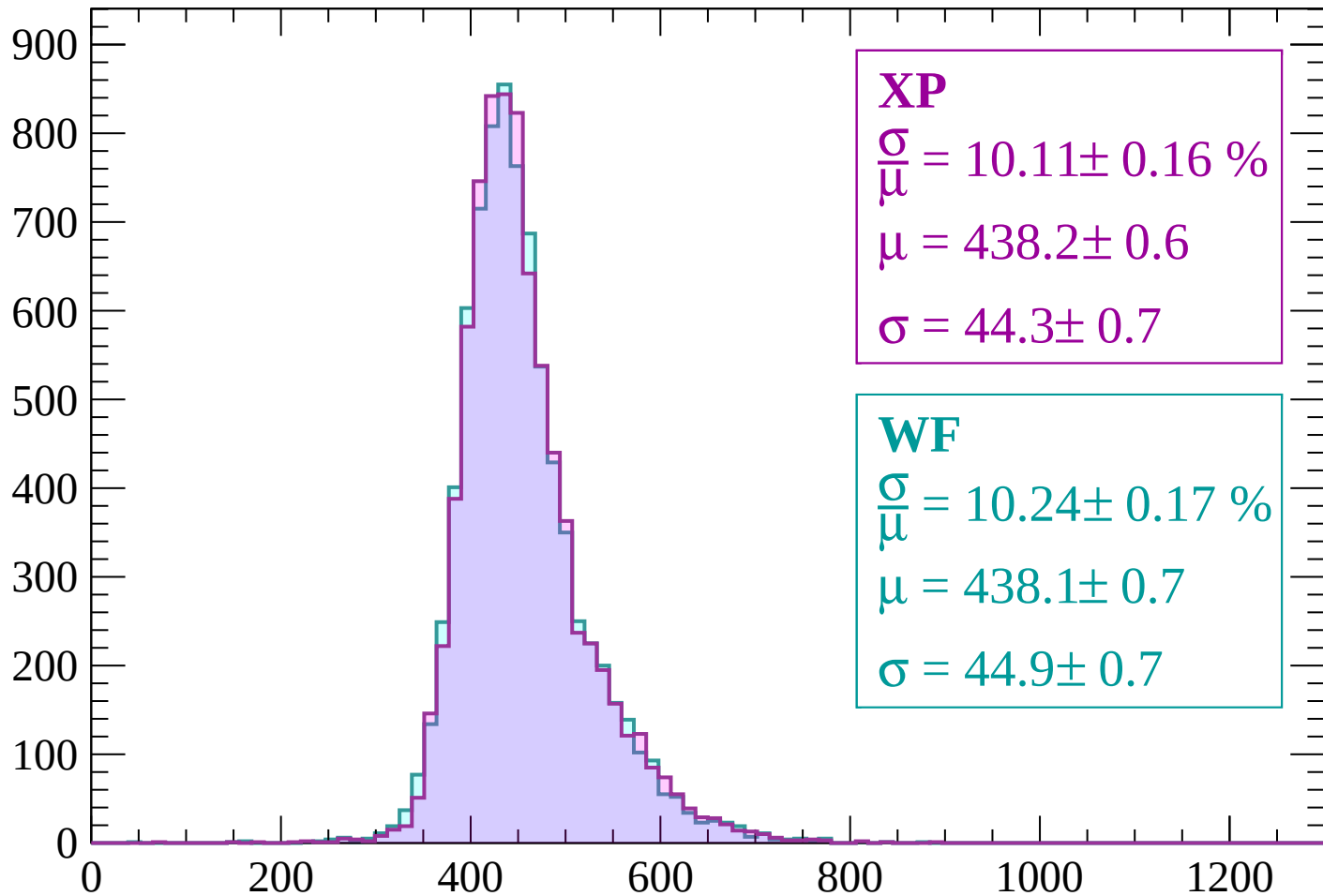
dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$



Energy loss | $1020 < p < 1080$

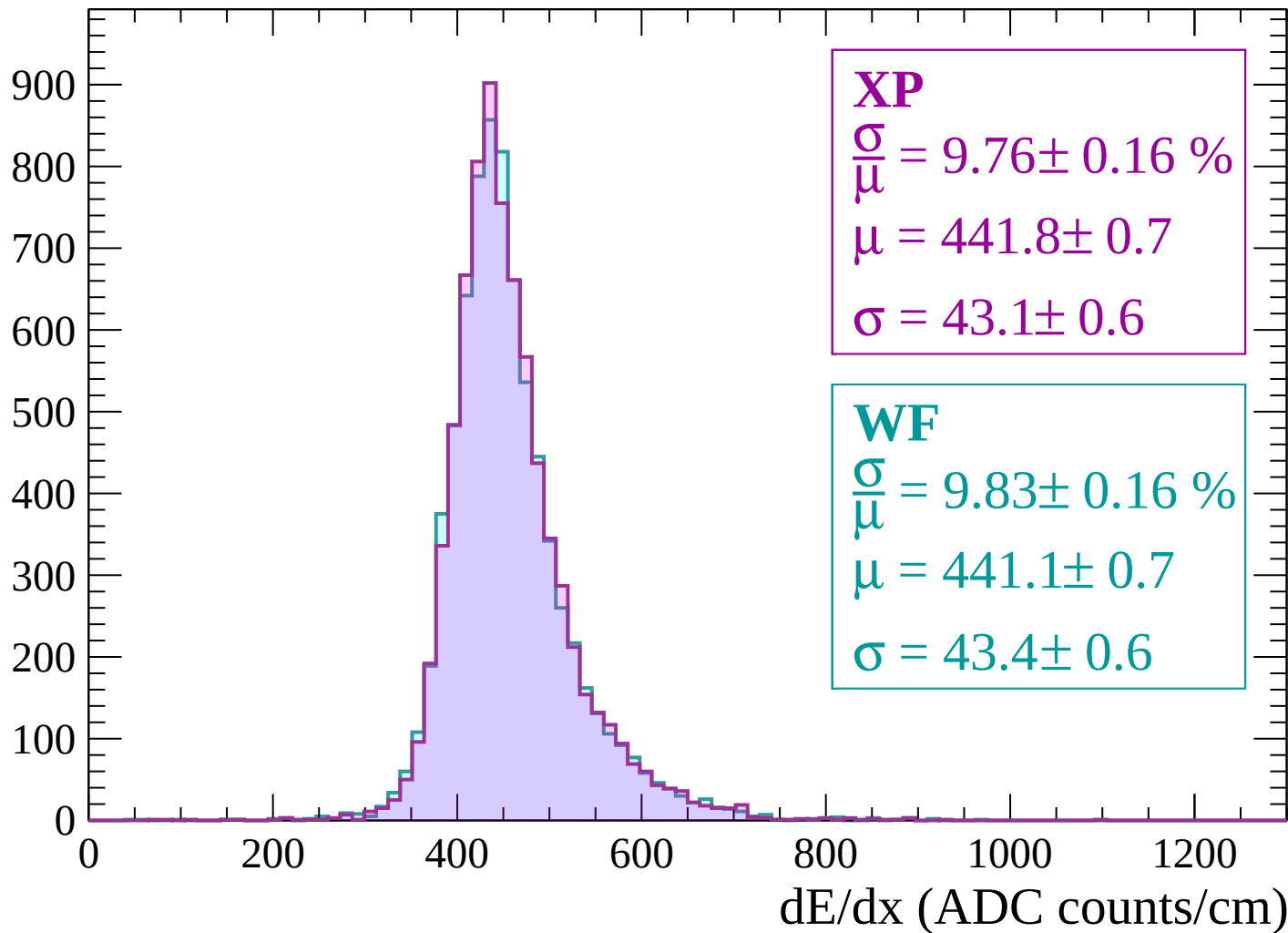
Count



dE/dx (ADC counts/cm)

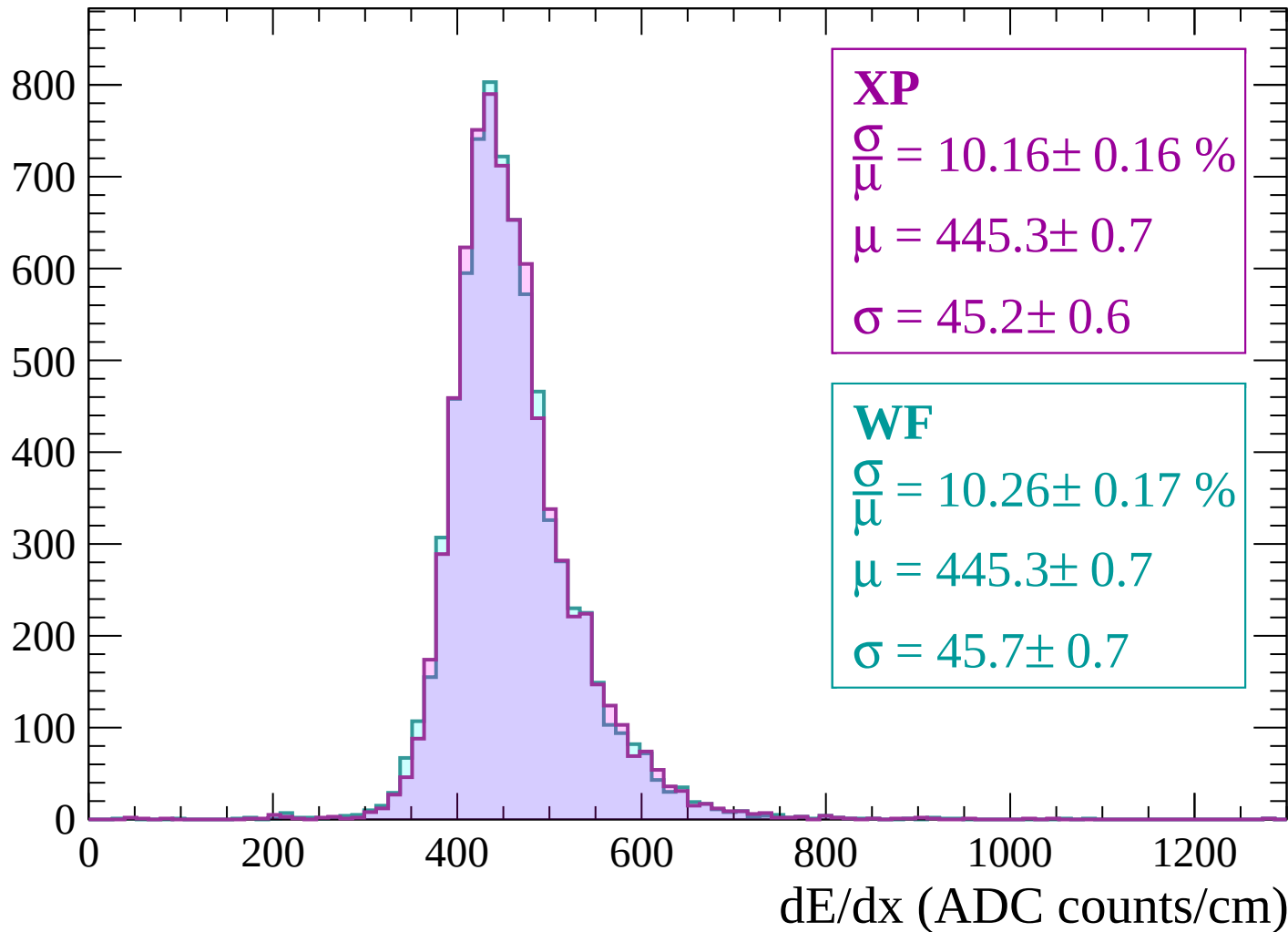
Energy loss | $1080 < p < 1140$

Count



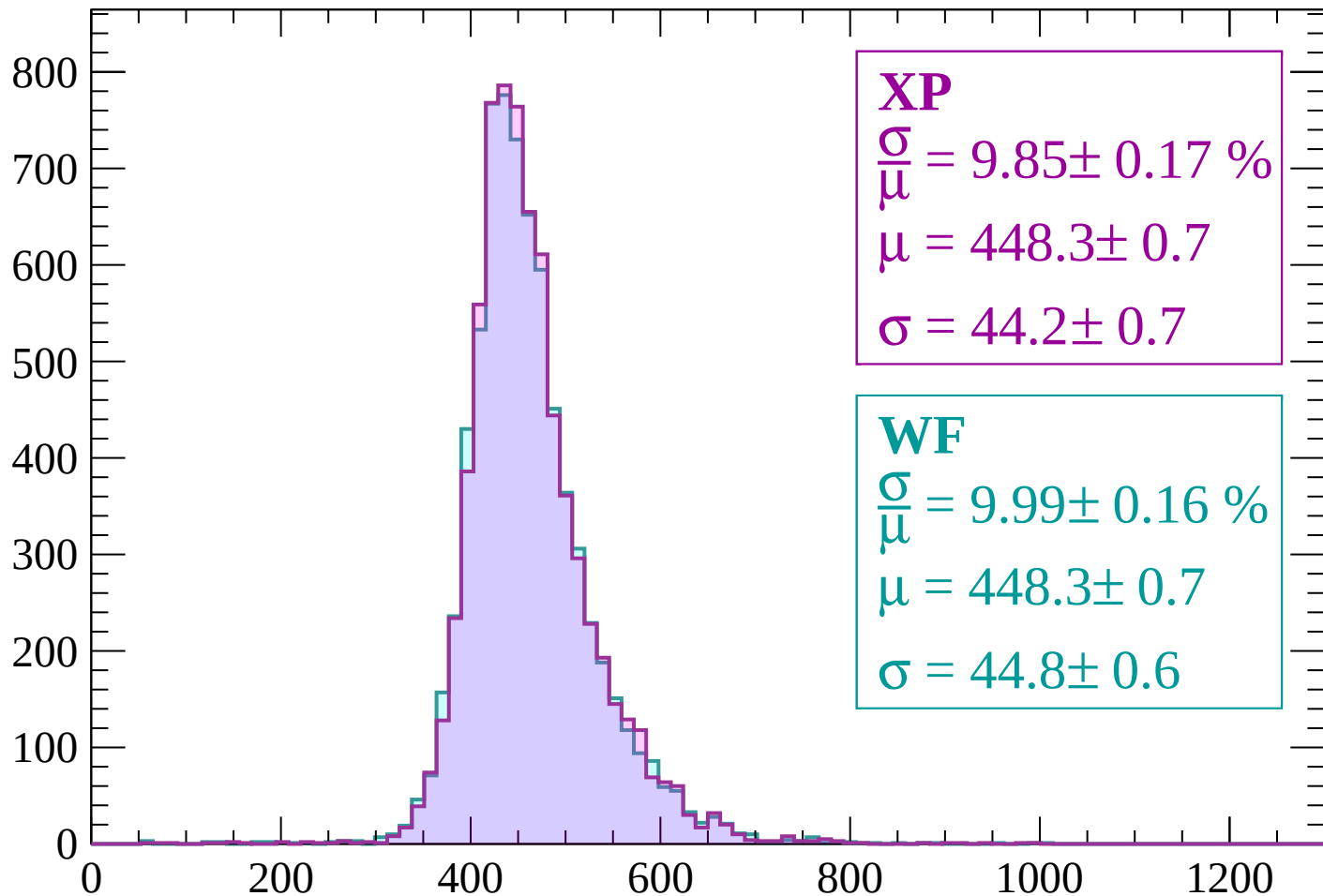
Energy loss | $1140 < p < 1200$

Count



Energy loss | $1200 < p < 1260$

Count



XP

$$\frac{\sigma}{\mu} = 9.85 \pm 0.17 \%$$

$$\mu = 448.3 \pm 0.7$$

$$\sigma = 44.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.99 \pm 0.16 \%$$

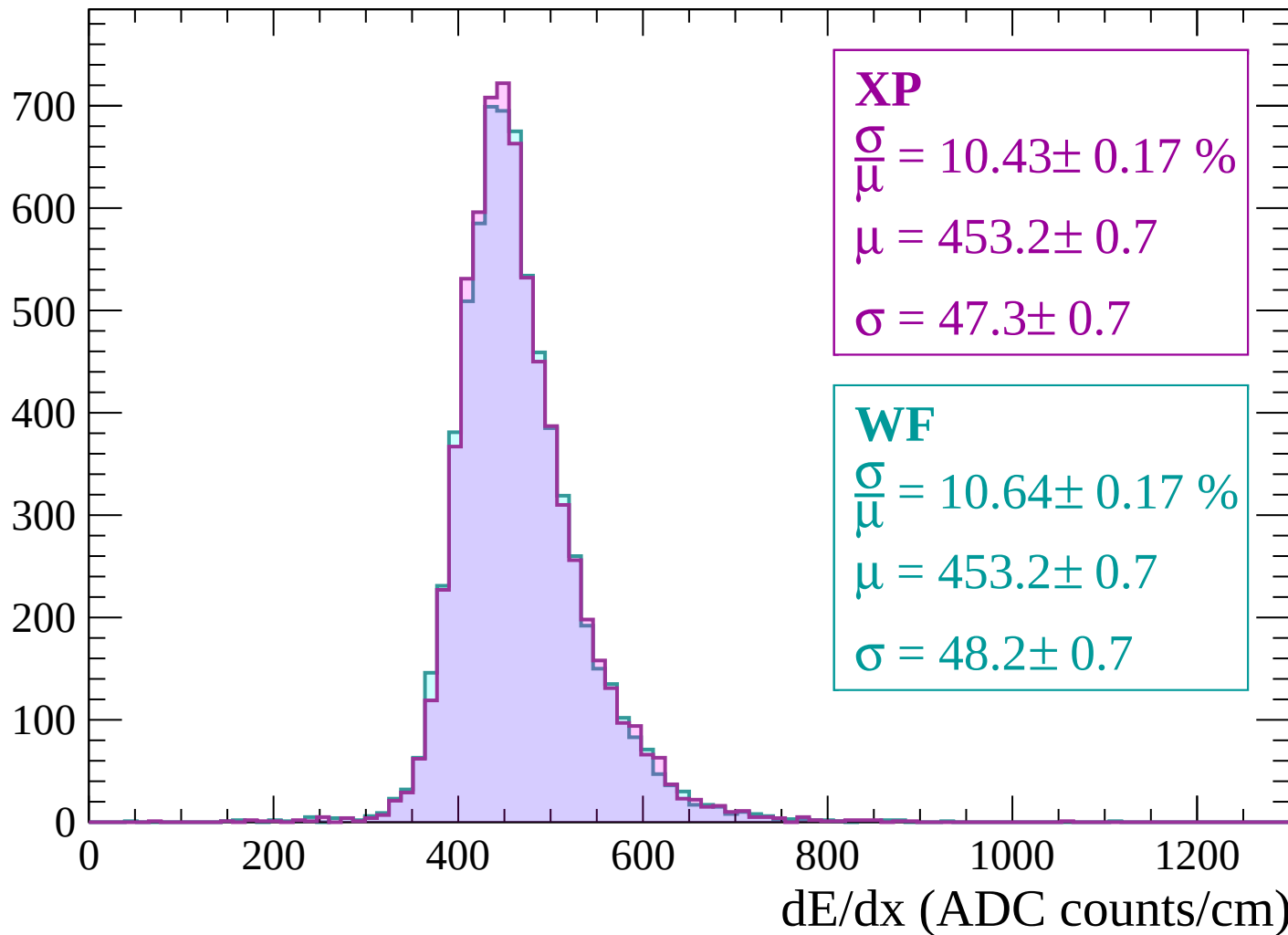
$$\mu = 448.3 \pm 0.7$$

$$\sigma = 44.8 \pm 0.6$$

dE/dx (ADC counts/cm)

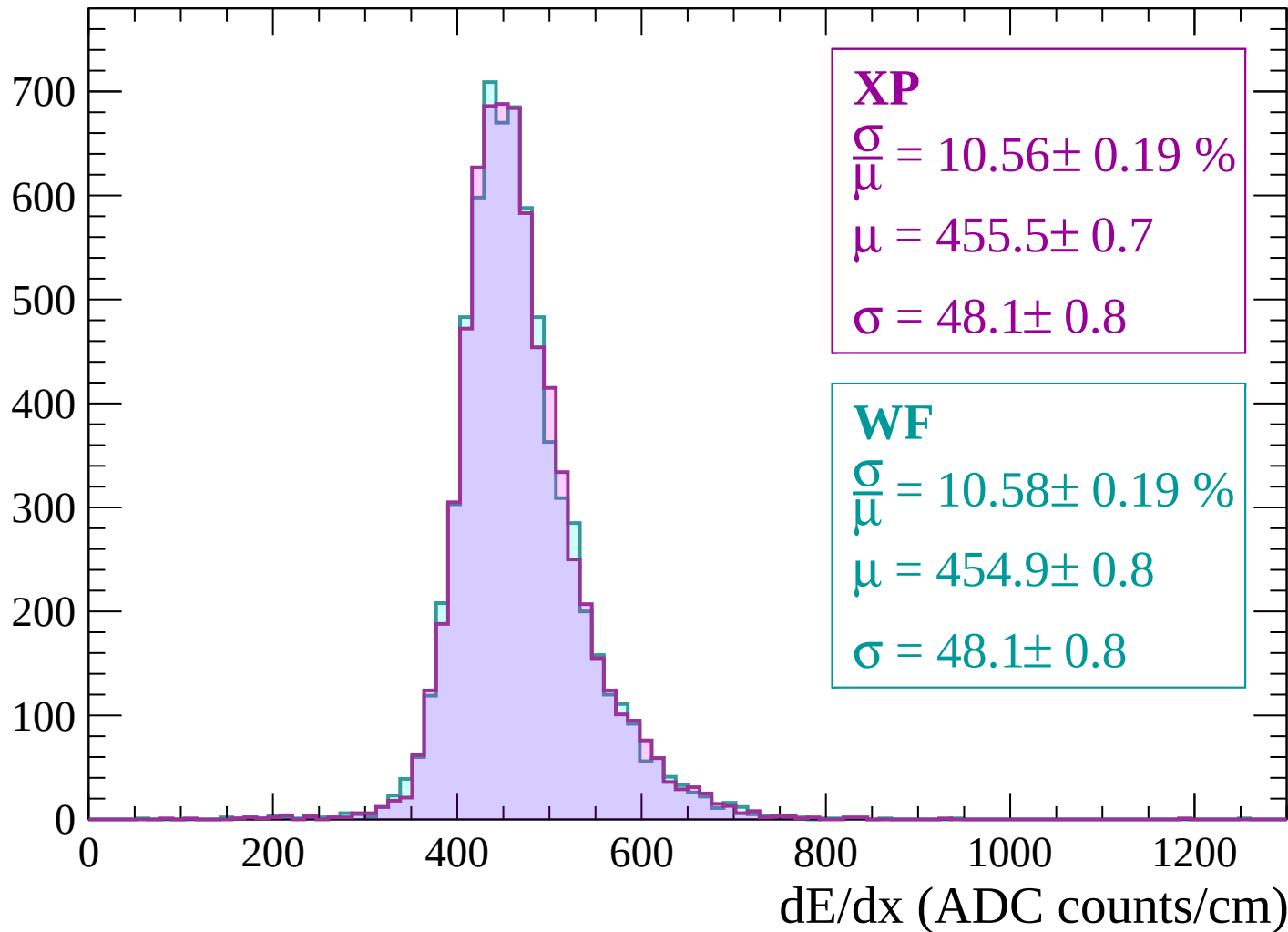
Energy loss | $1260 < p < 1320$

Count



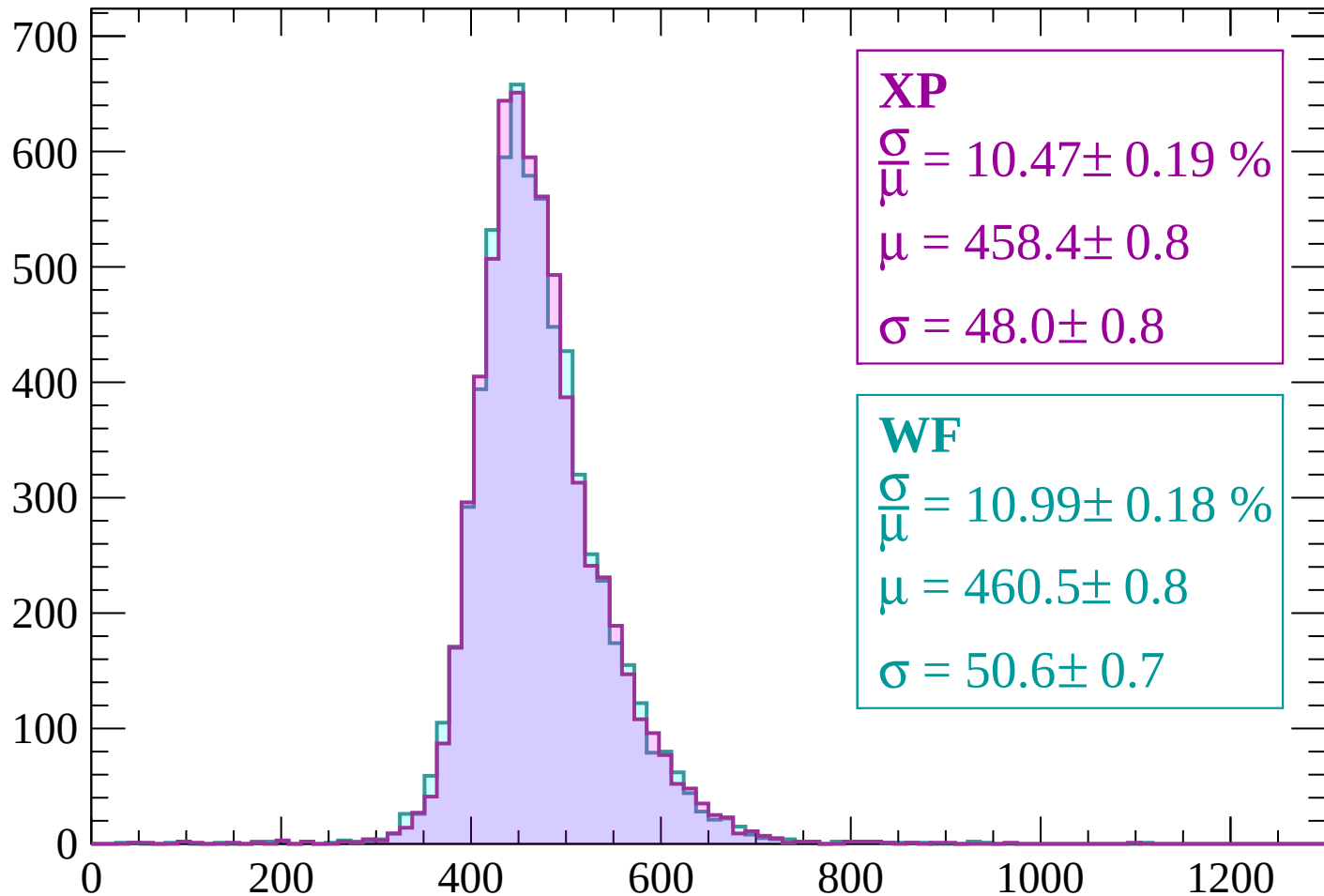
Energy loss | $1320 < p < 1380$

Count



Energy loss | $1380 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 10.47 \pm 0.19 \%$$

$$\mu = 458.4 \pm 0.8$$

$$\sigma = 48.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 10.99 \pm 0.18 \%$$

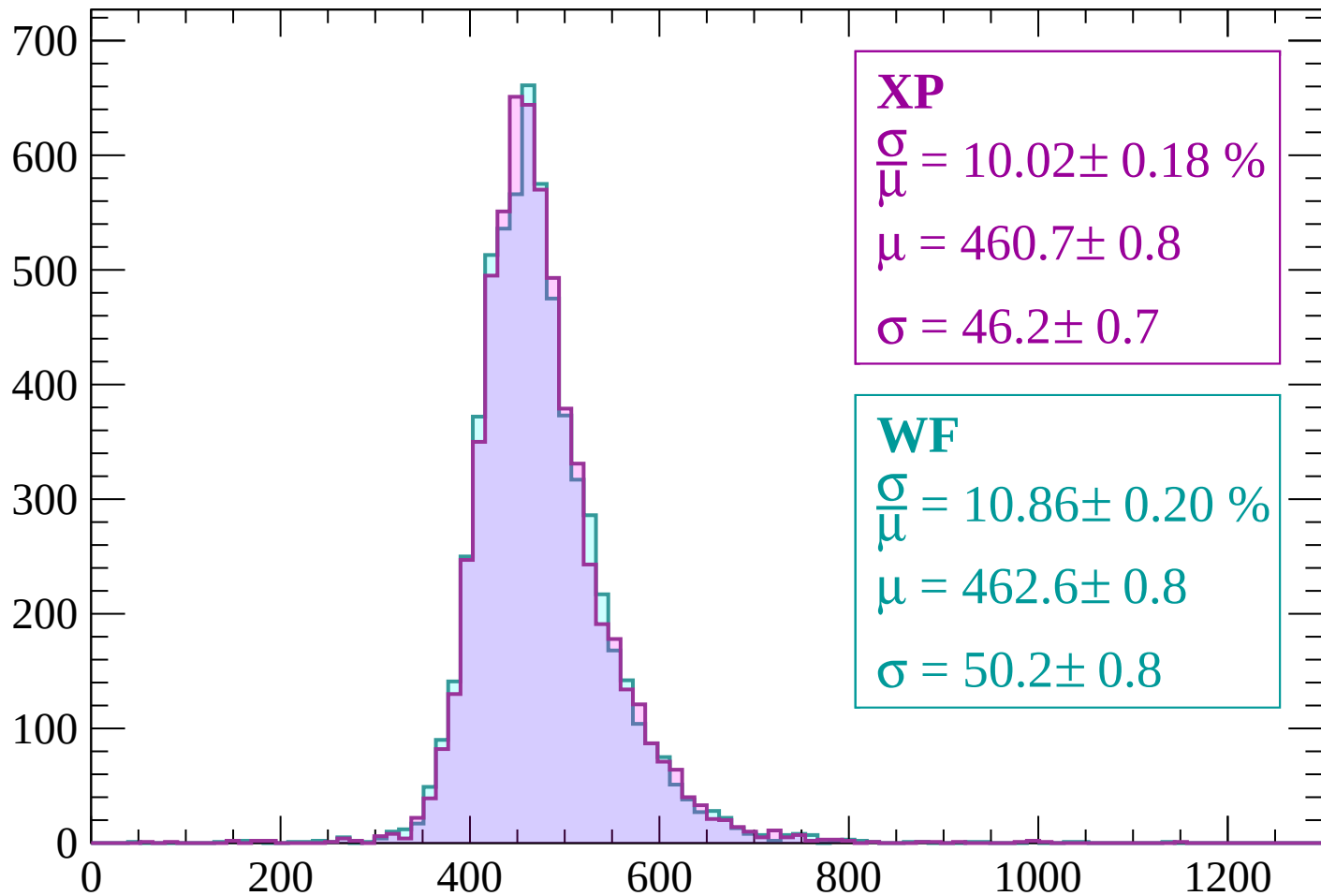
$$\mu = 460.5 \pm 0.8$$

$$\sigma = 50.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP

$$\frac{\sigma}{\mu} = 10.02 \pm 0.18 \%$$

$$\mu = 460.7 \pm 0.8$$

$$\sigma = 46.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 10.86 \pm 0.20 \%$$

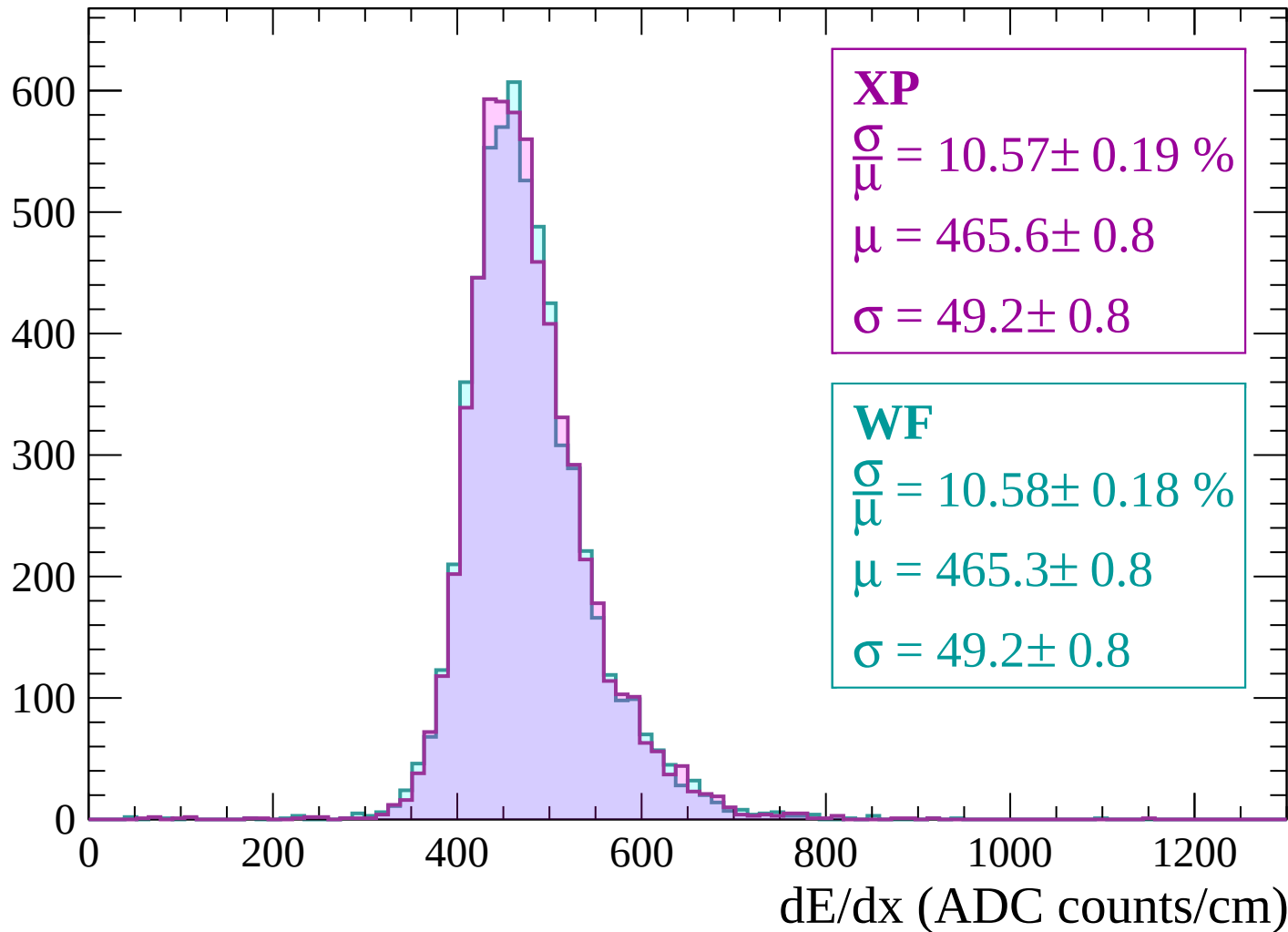
$$\mu = 462.6 \pm 0.8$$

$$\sigma = 50.2 \pm 0.8$$

dE/dx (ADC counts/cm)

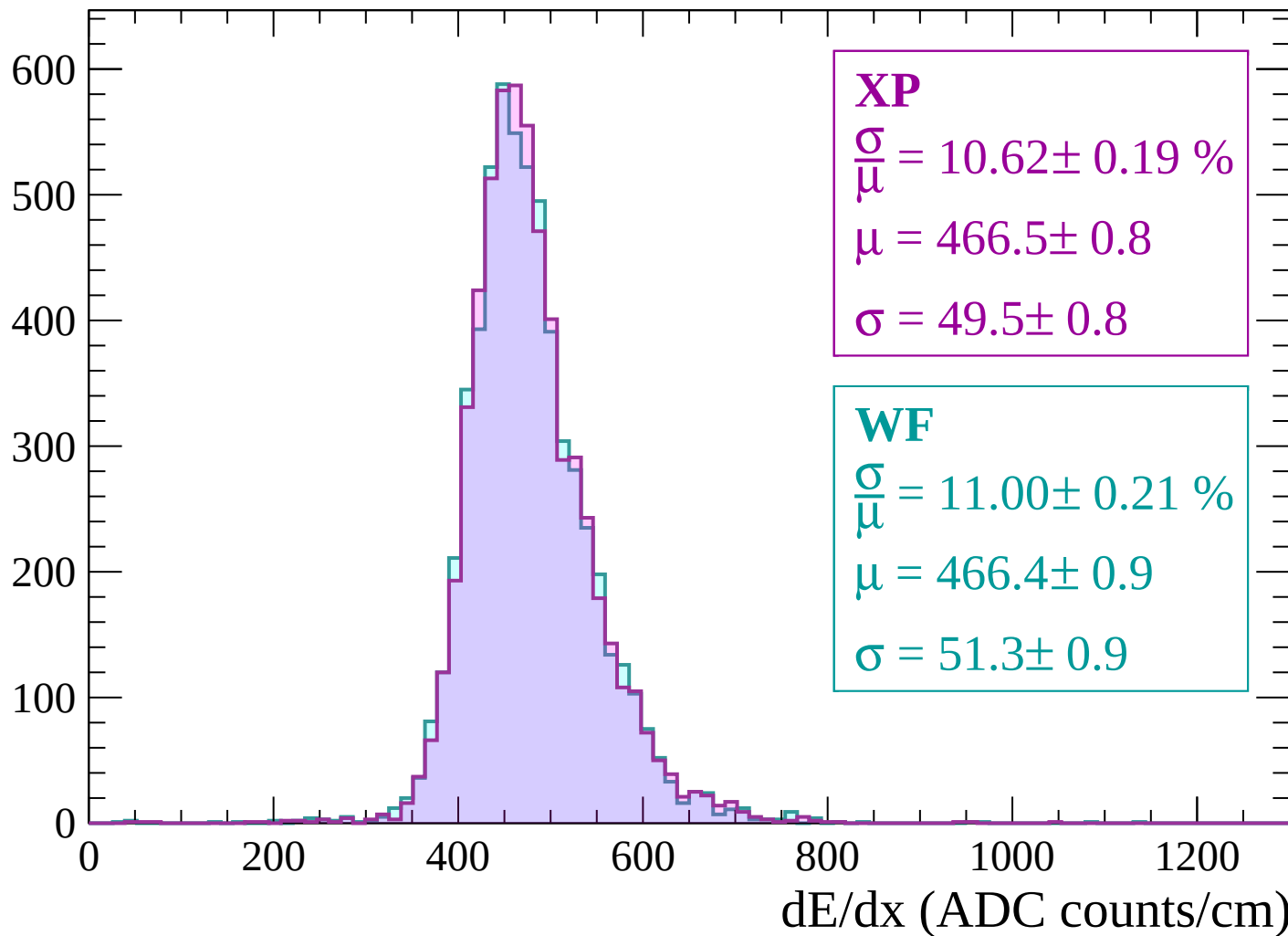
Energy loss | $1500 < p < 1560$

Count



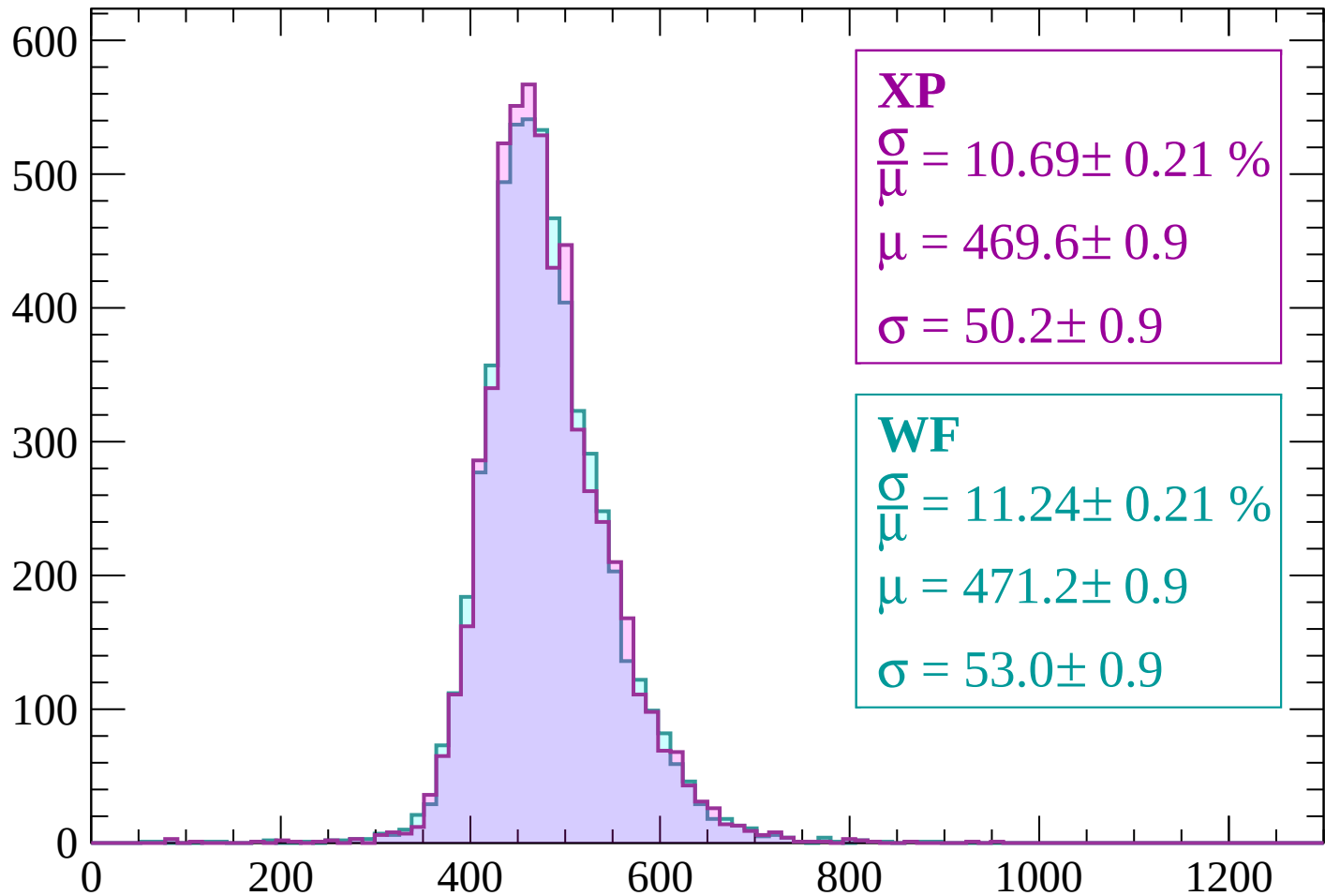
Energy loss | $1560 < p < 1620$

Count



Energy loss | $1620 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 10.69 \pm 0.21 \%$$

$$\mu = 469.6 \pm 0.9$$

$$\sigma = 50.2 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 11.24 \pm 0.21 \%$$

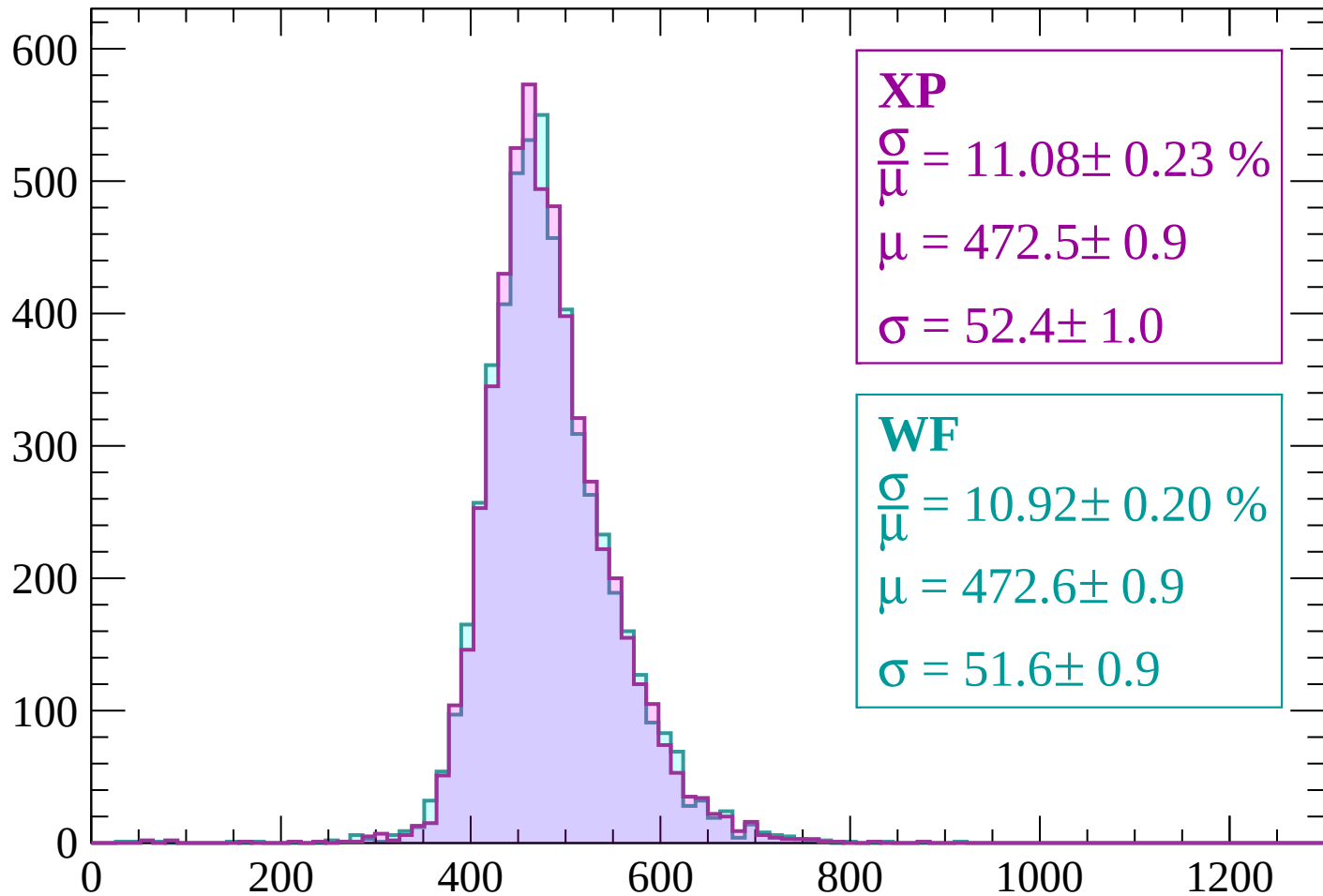
$$\mu = 471.2 \pm 0.9$$

$$\sigma = 53.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP

$$\frac{\sigma}{\mu} = 11.08 \pm 0.23 \%$$

$$\mu = 472.5 \pm 0.9$$

$$\sigma = 52.4 \pm 1.0$$

WF

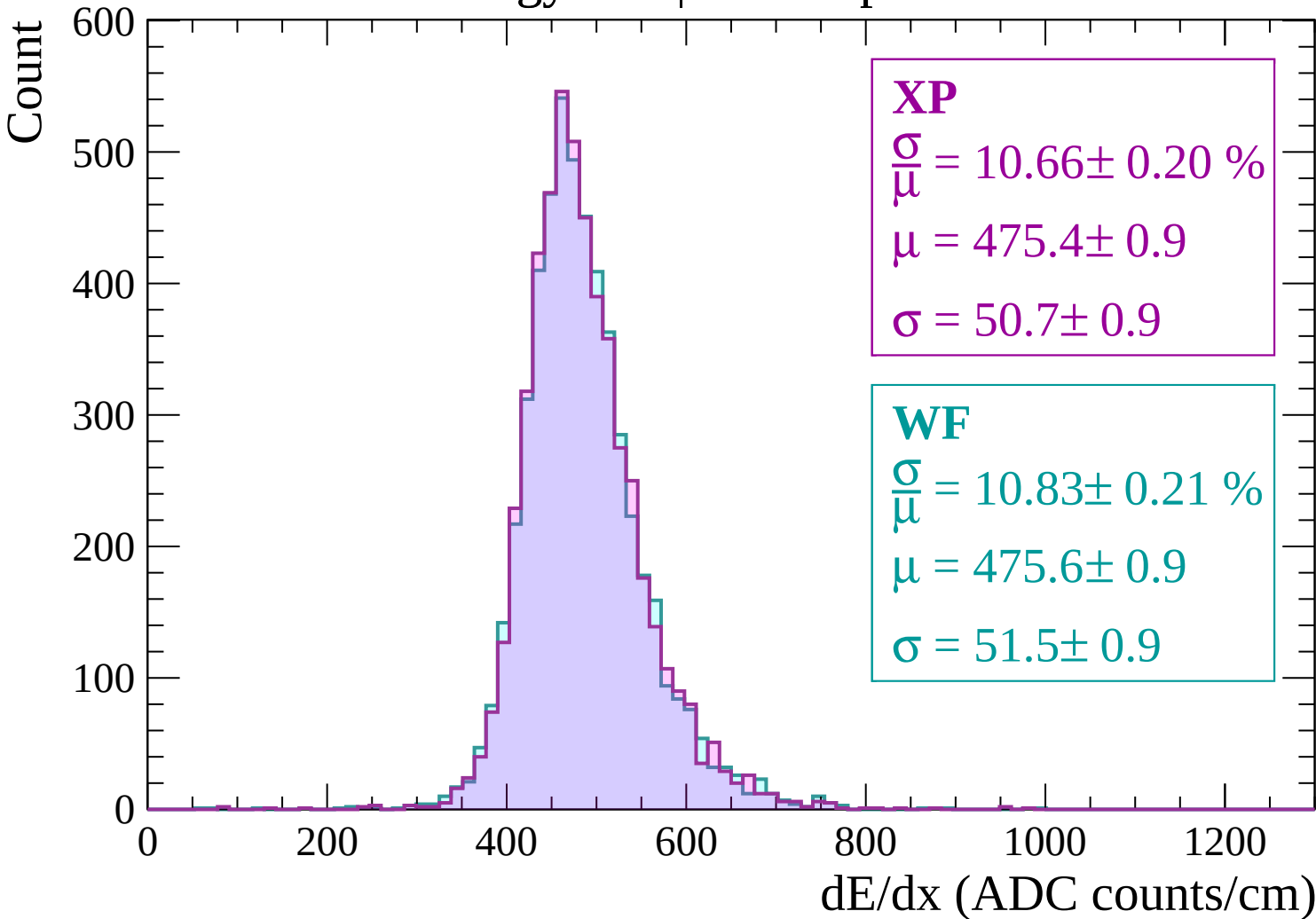
$$\frac{\sigma}{\mu} = 10.92 \pm 0.20 \%$$

$$\mu = 472.6 \pm 0.9$$

$$\sigma = 51.6 \pm 0.9$$

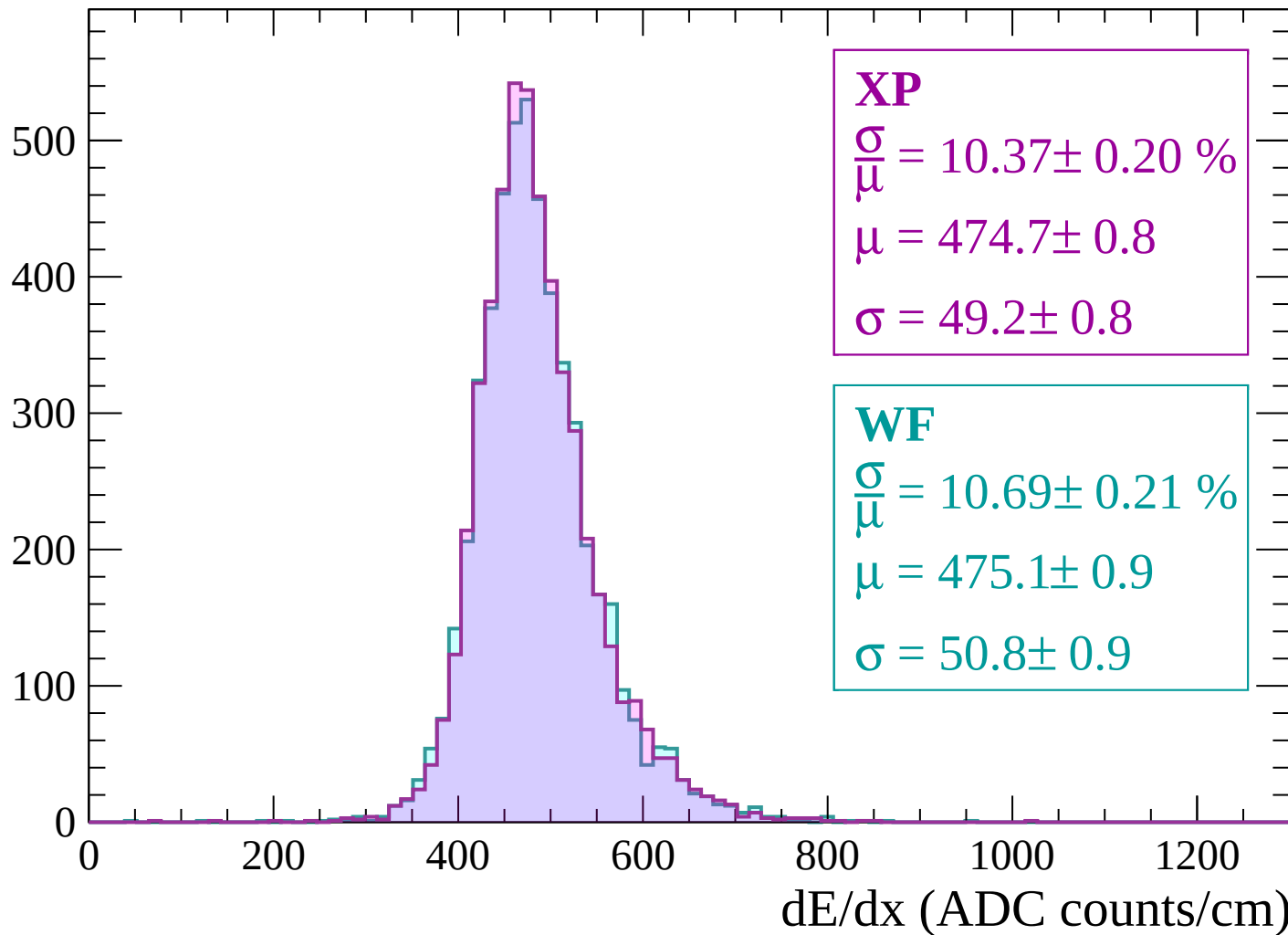
dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$



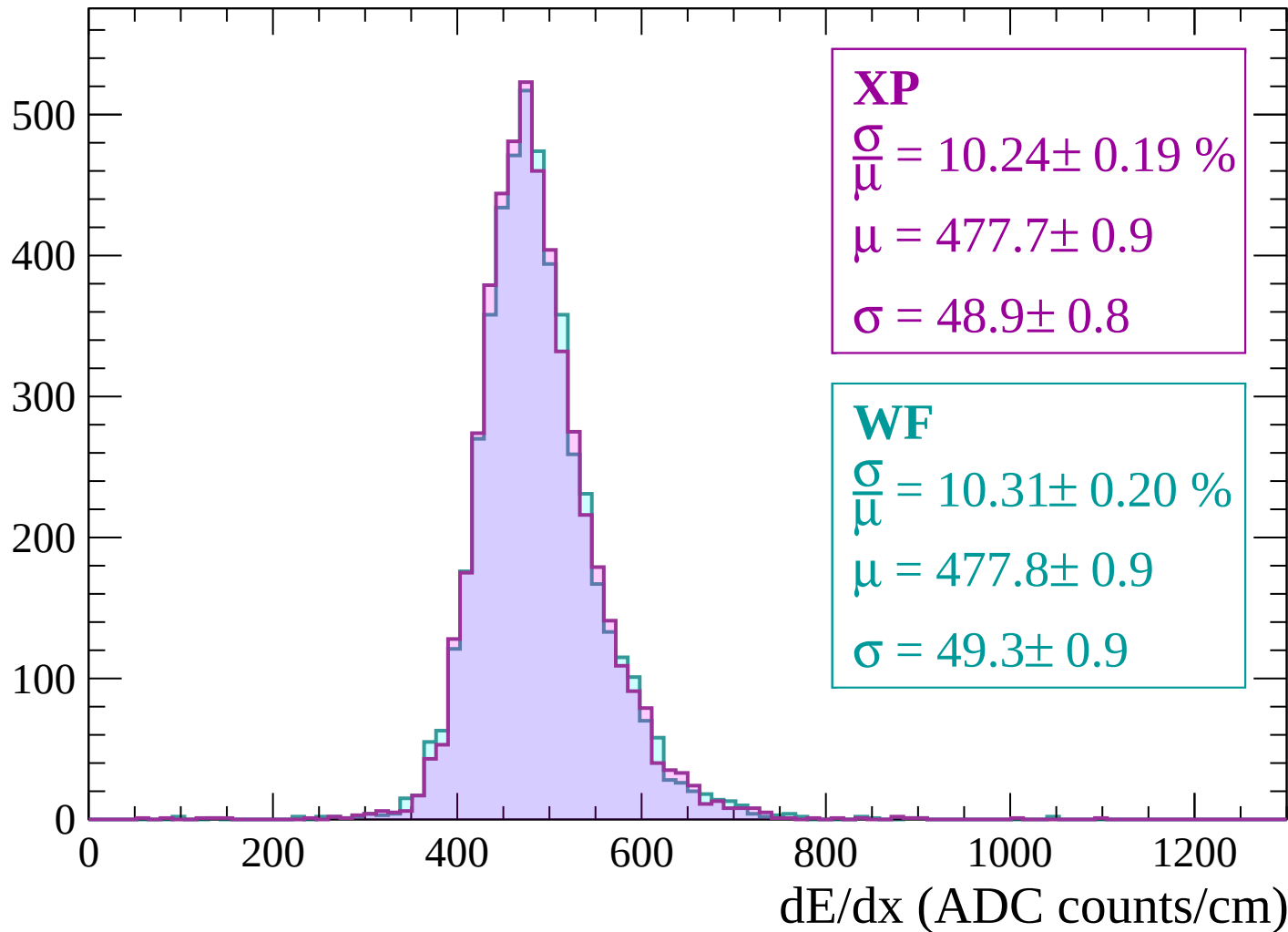
Energy loss | $1800 < p < 1860$

Count



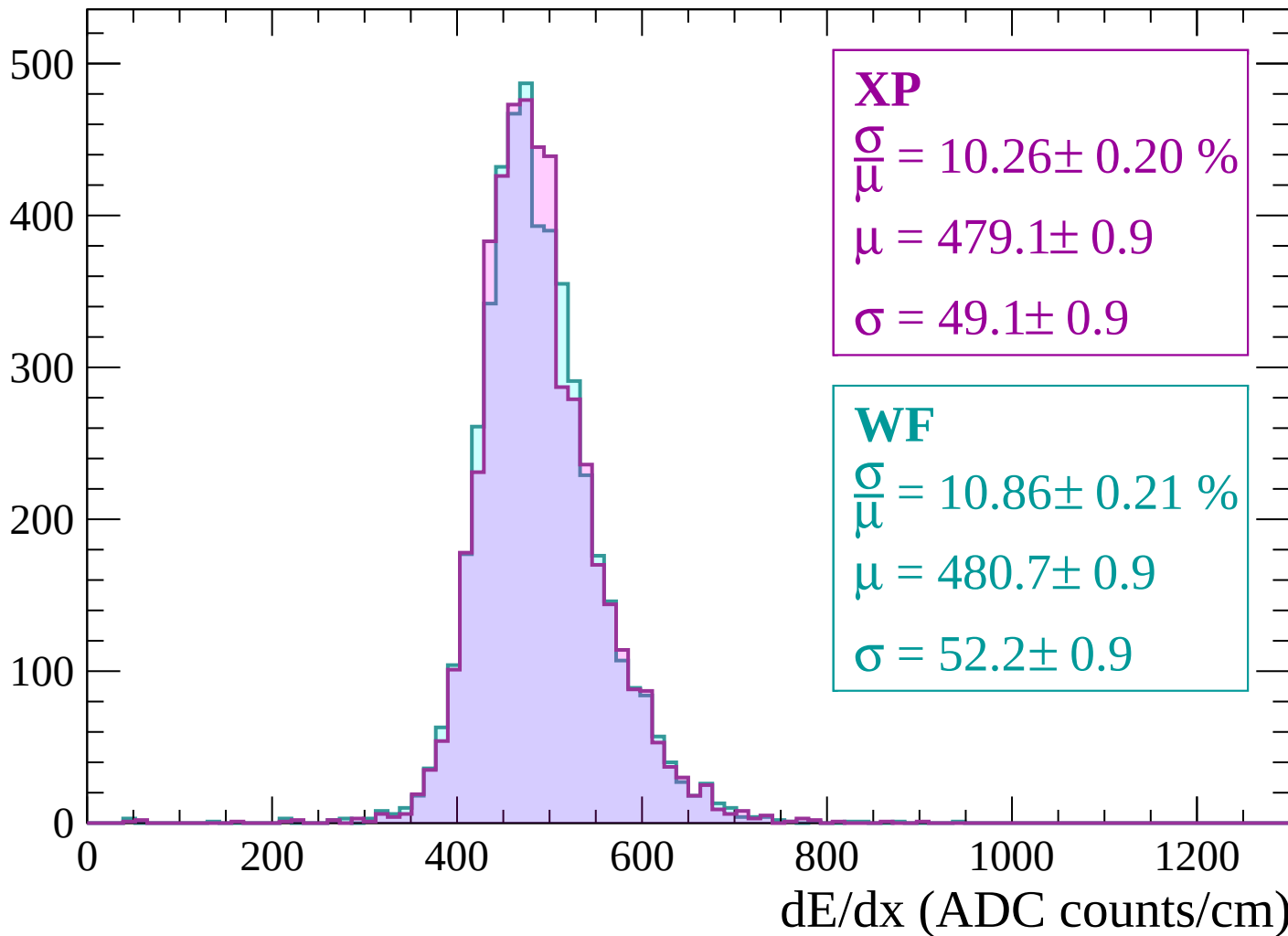
Energy loss | $1860 < p < 1920$

Count



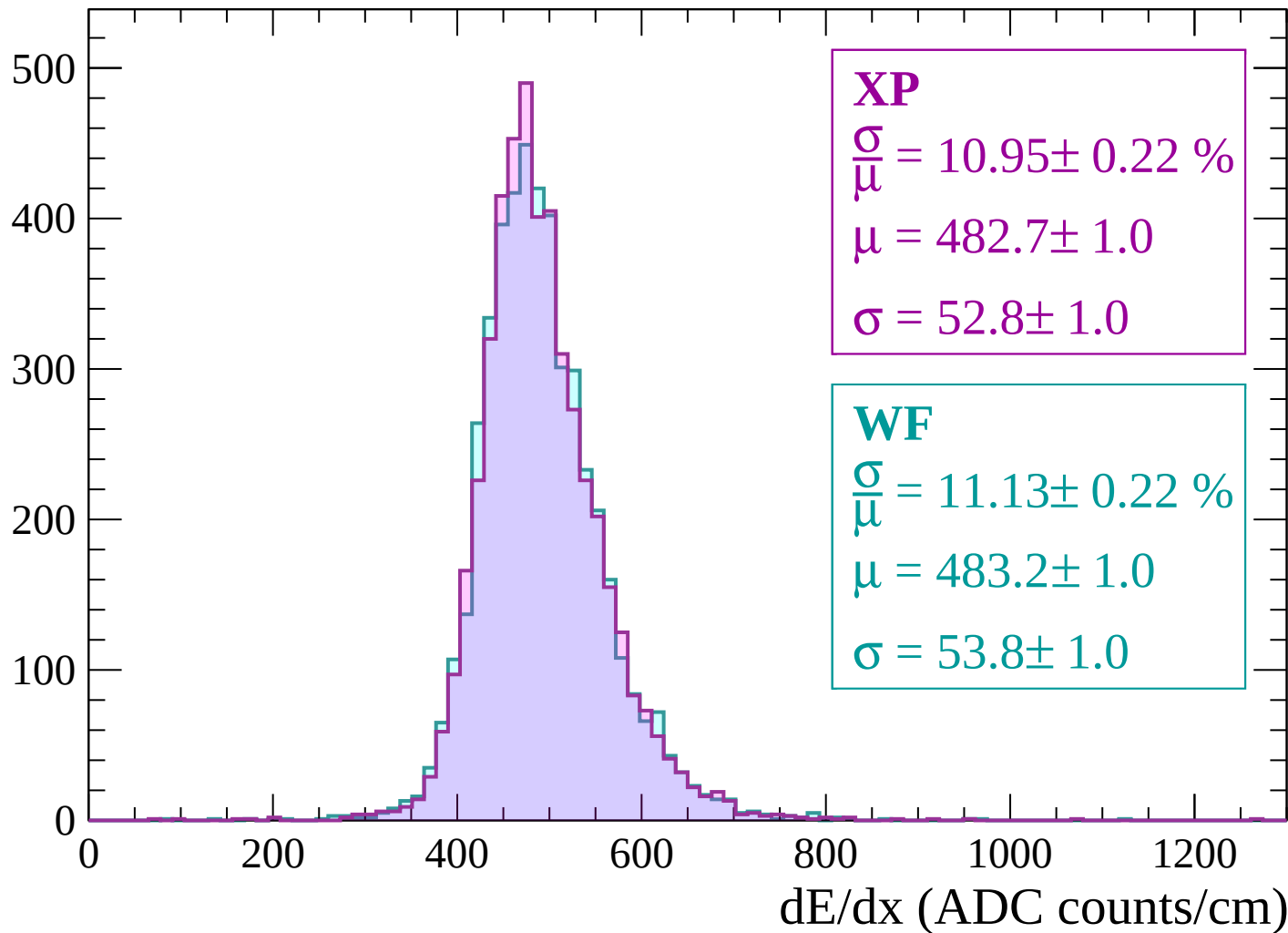
Energy loss | $1920 < p < 1980$

Count



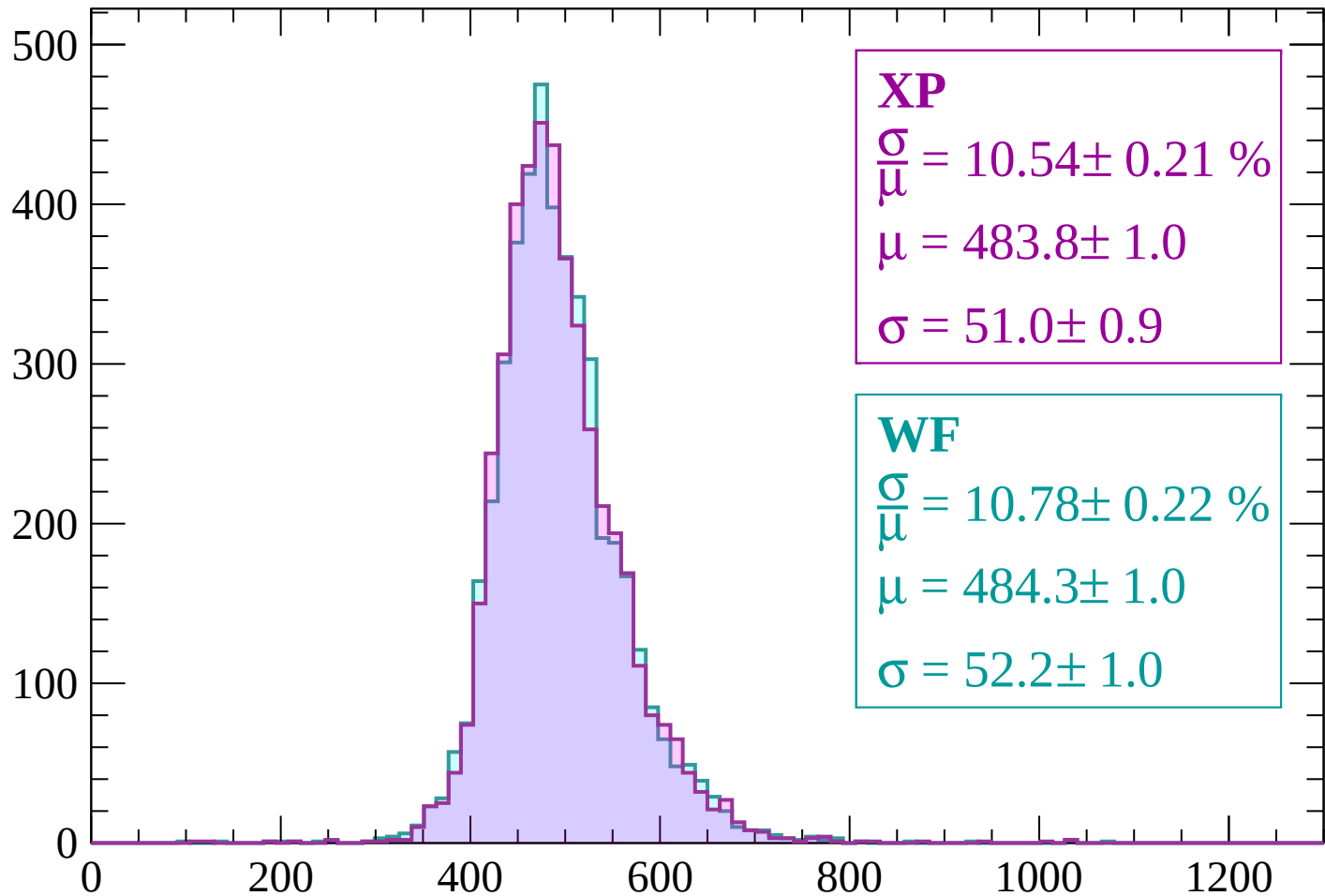
Energy loss | $1980 < p < 2040$

Count



Energy loss | $2040 < p < 2100$

Count



XP

$$\frac{\sigma}{\mu} = 10.54 \pm 0.21 \%$$

$$\mu = 483.8 \pm 1.0$$

$$\sigma = 51.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.78 \pm 0.22 \%$$

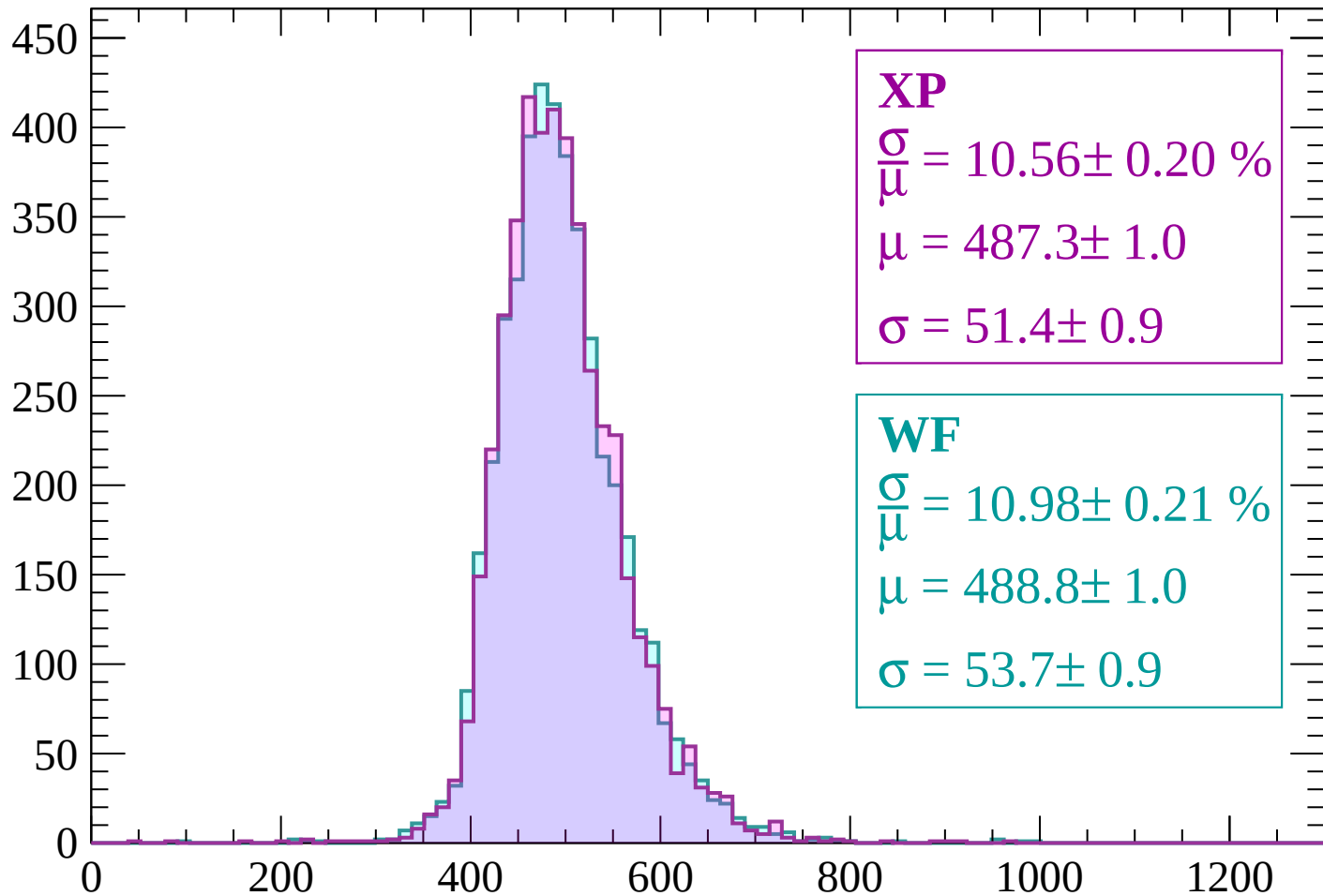
$$\mu = 484.3 \pm 1.0$$

$$\sigma = 52.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $2100 < p < 2160$

Count



XP

$$\frac{\sigma}{\mu} = 10.56 \pm 0.20 \%$$

$$\mu = 487.3 \pm 1.0$$

$$\sigma = 51.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.98 \pm 0.21 \%$$

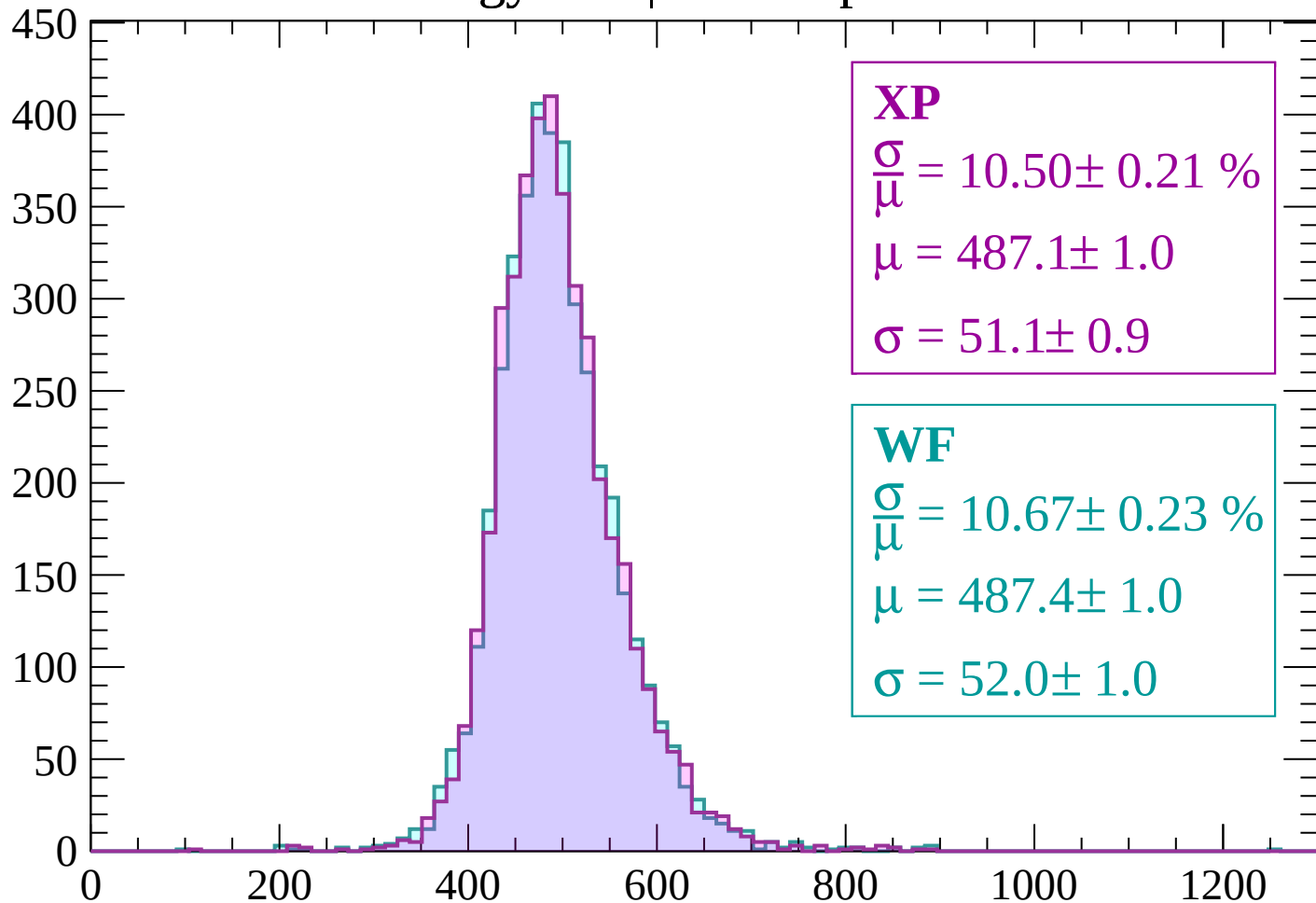
$$\mu = 488.8 \pm 1.0$$

$$\sigma = 53.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP

$$\frac{\sigma}{\mu} = 10.50 \pm 0.21 \%$$

$$\mu = 487.1 \pm 1.0$$

$$\sigma = 51.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.67 \pm 0.23 \%$$

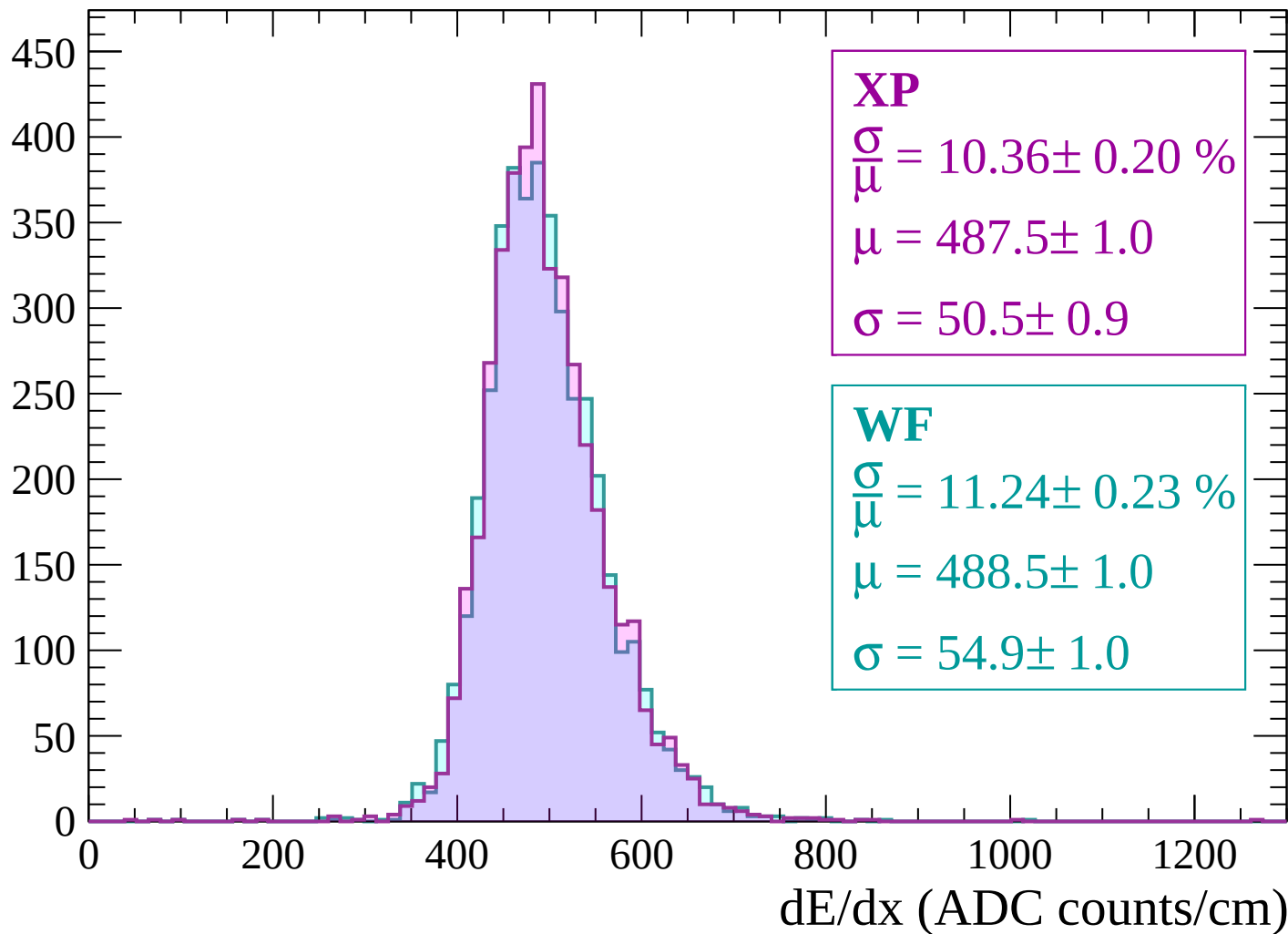
$$\mu = 487.4 \pm 1.0$$

$$\sigma = 52.0 \pm 1.0$$

dE/dx (ADC counts/cm)

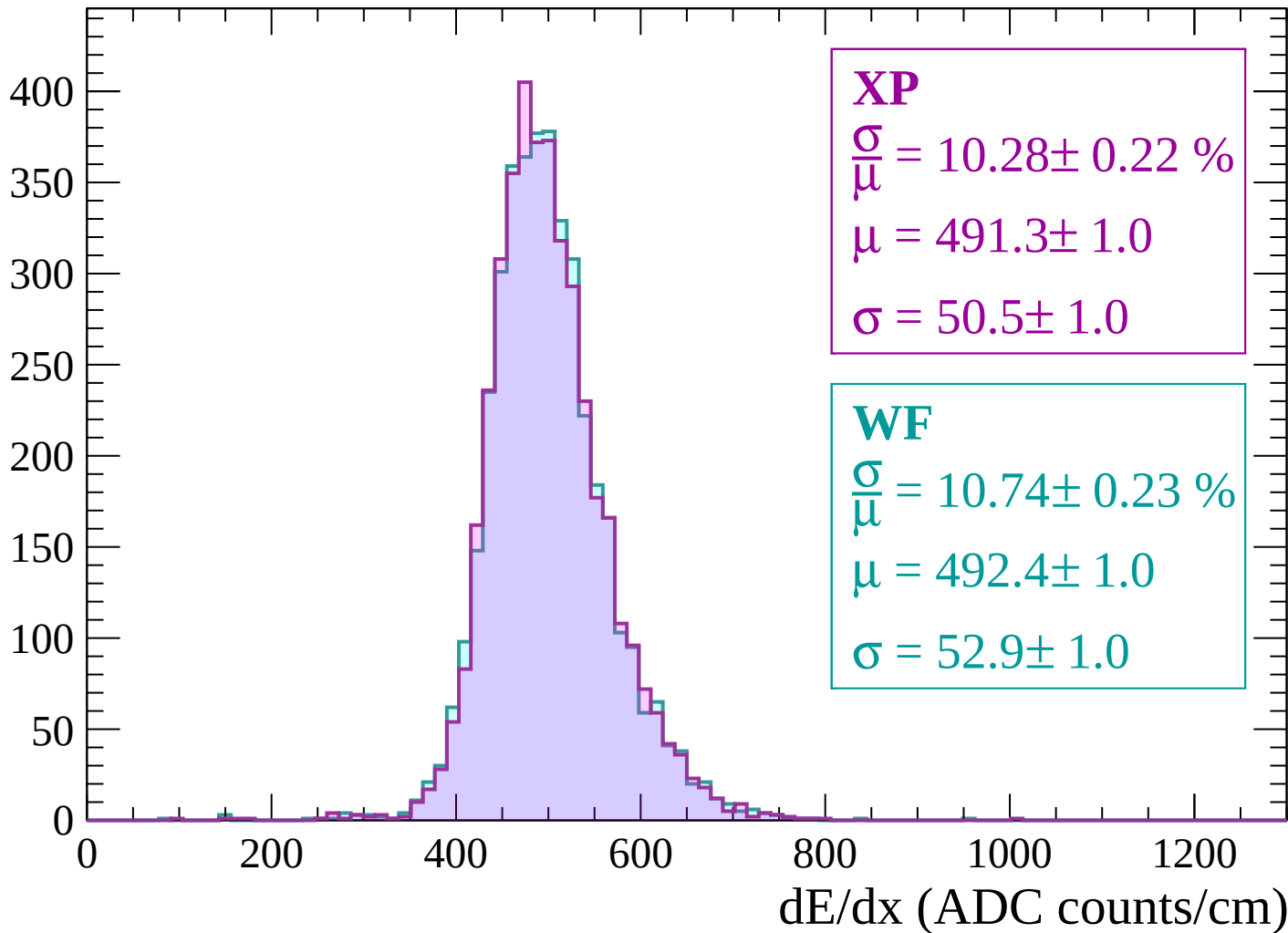
Energy loss | $2220 < p < 2280$

Count



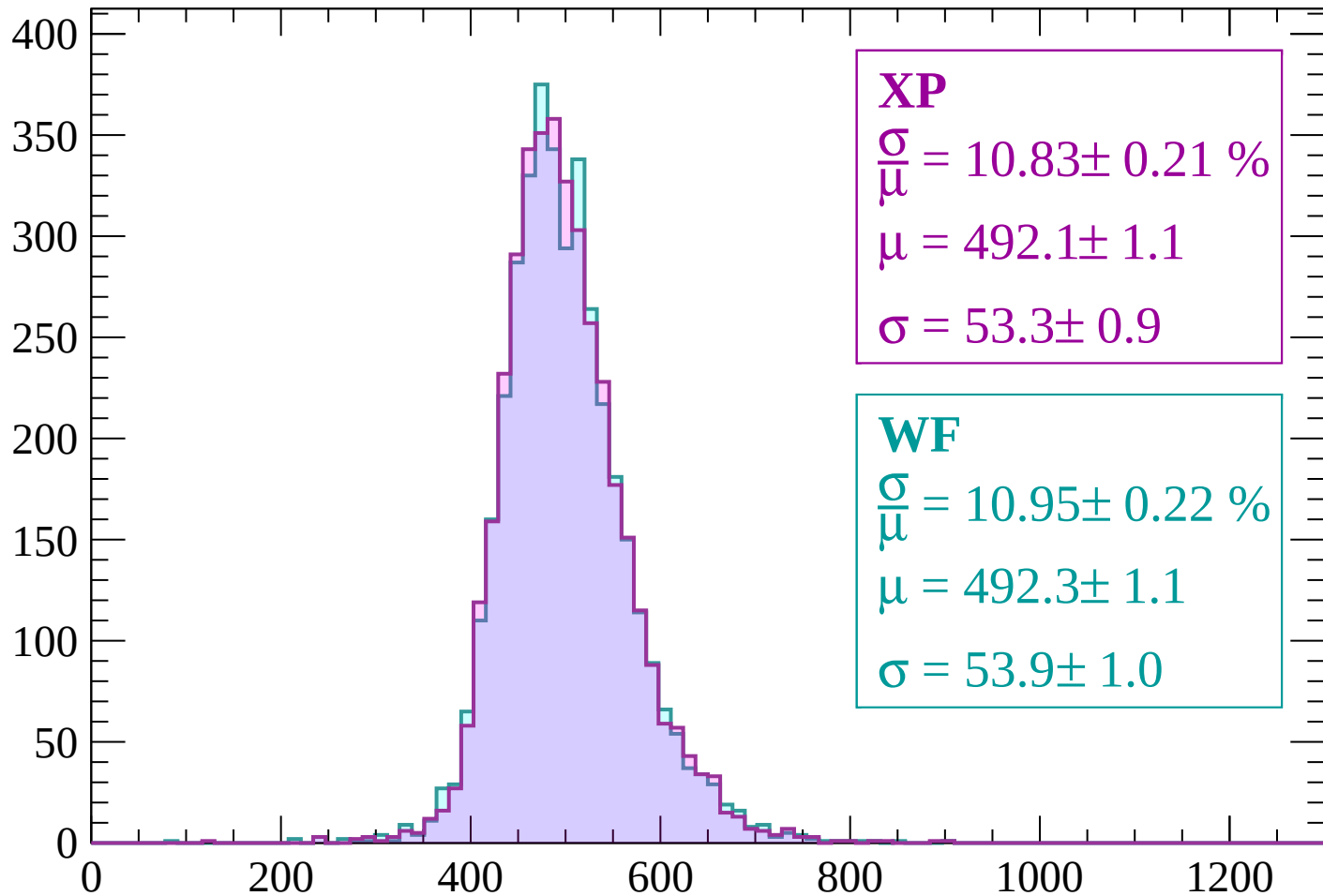
Energy loss | $2280 < p < 2340$

Count



Energy loss | $2340 < p < 2400$

Count



XP

$$\frac{\sigma}{\mu} = 10.83 \pm 0.21 \%$$

$$\mu = 492.1 \pm 1.1$$

$$\sigma = 53.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.95 \pm 0.22 \%$$

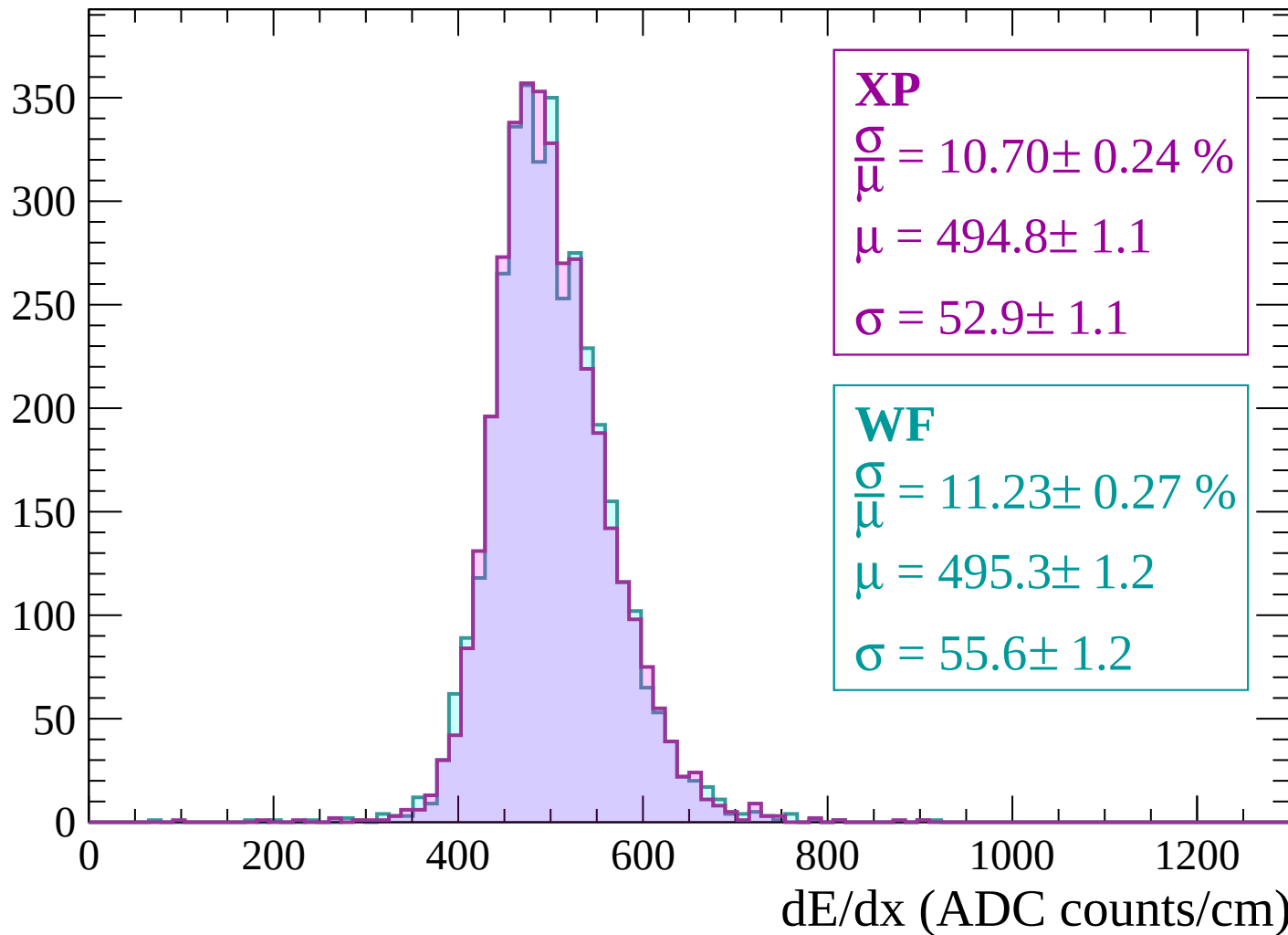
$$\mu = 492.3 \pm 1.1$$

$$\sigma = 53.9 \pm 1.0$$

dE/dx (ADC counts/cm)

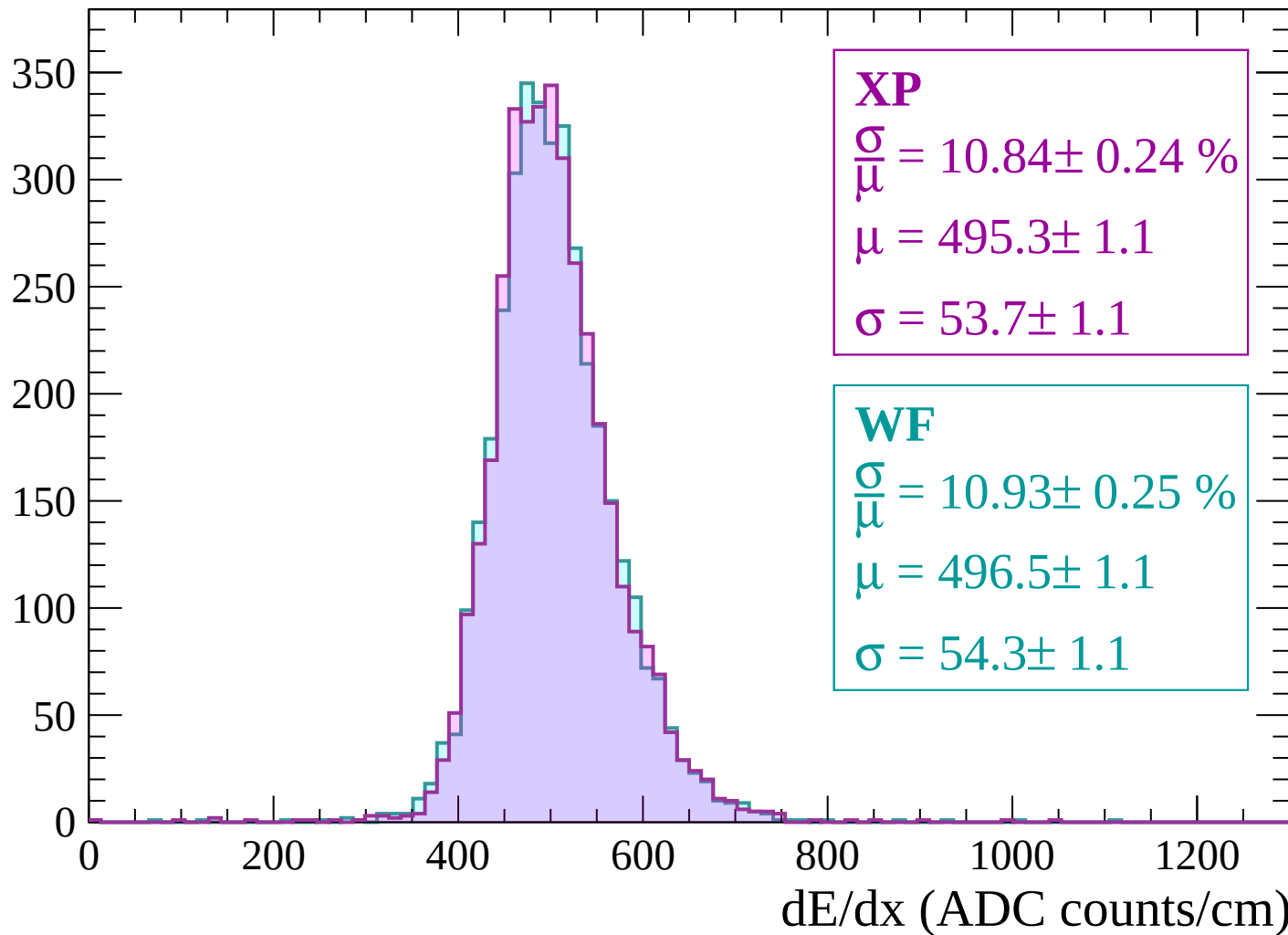
Energy loss | $2400 < p < 2460$

Count



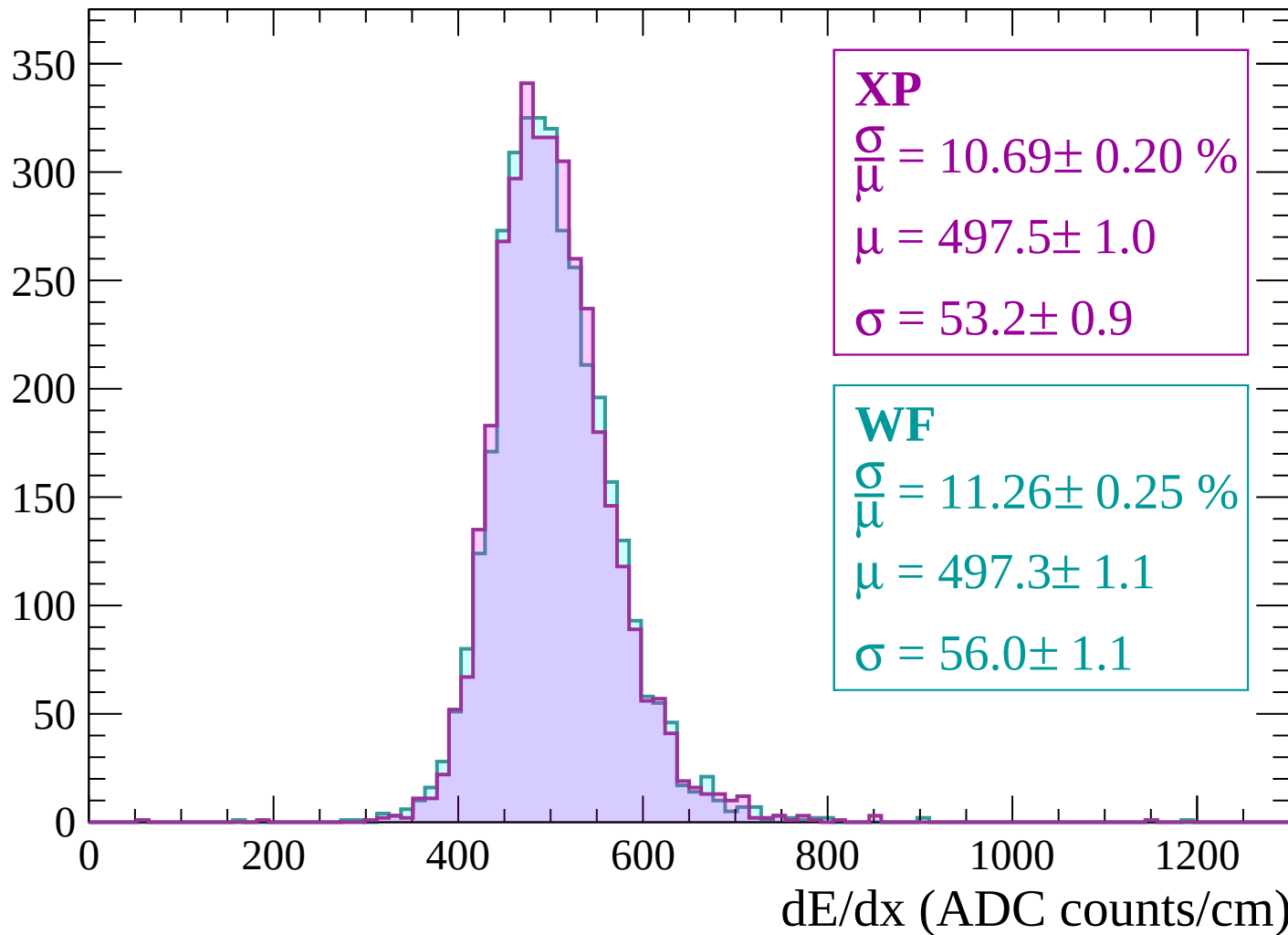
Energy loss | $2460 < p < 2520$

Count



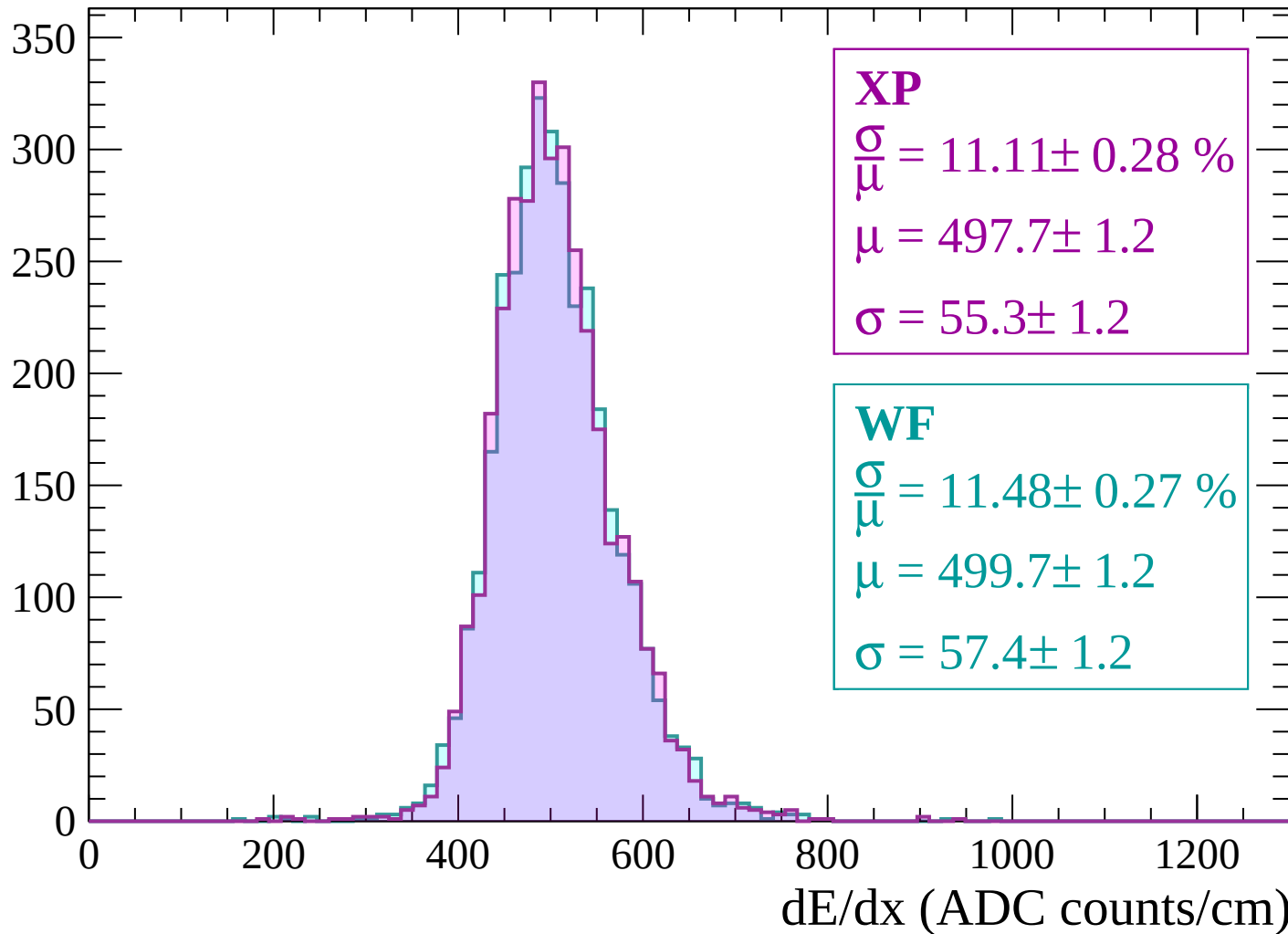
Energy loss | $2520 < p < 2580$

Count



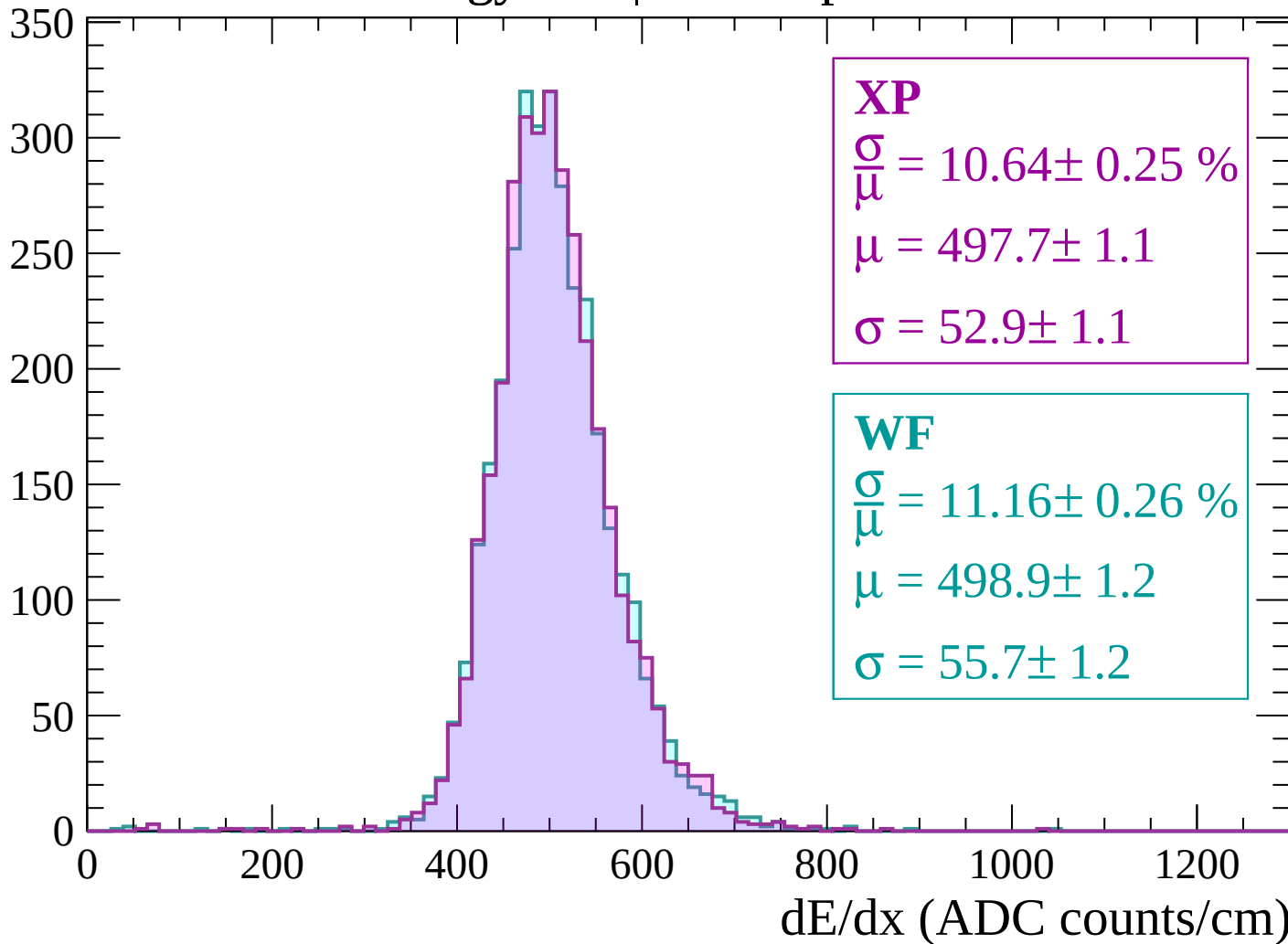
Energy loss | $2580 < p < 2640$

Count



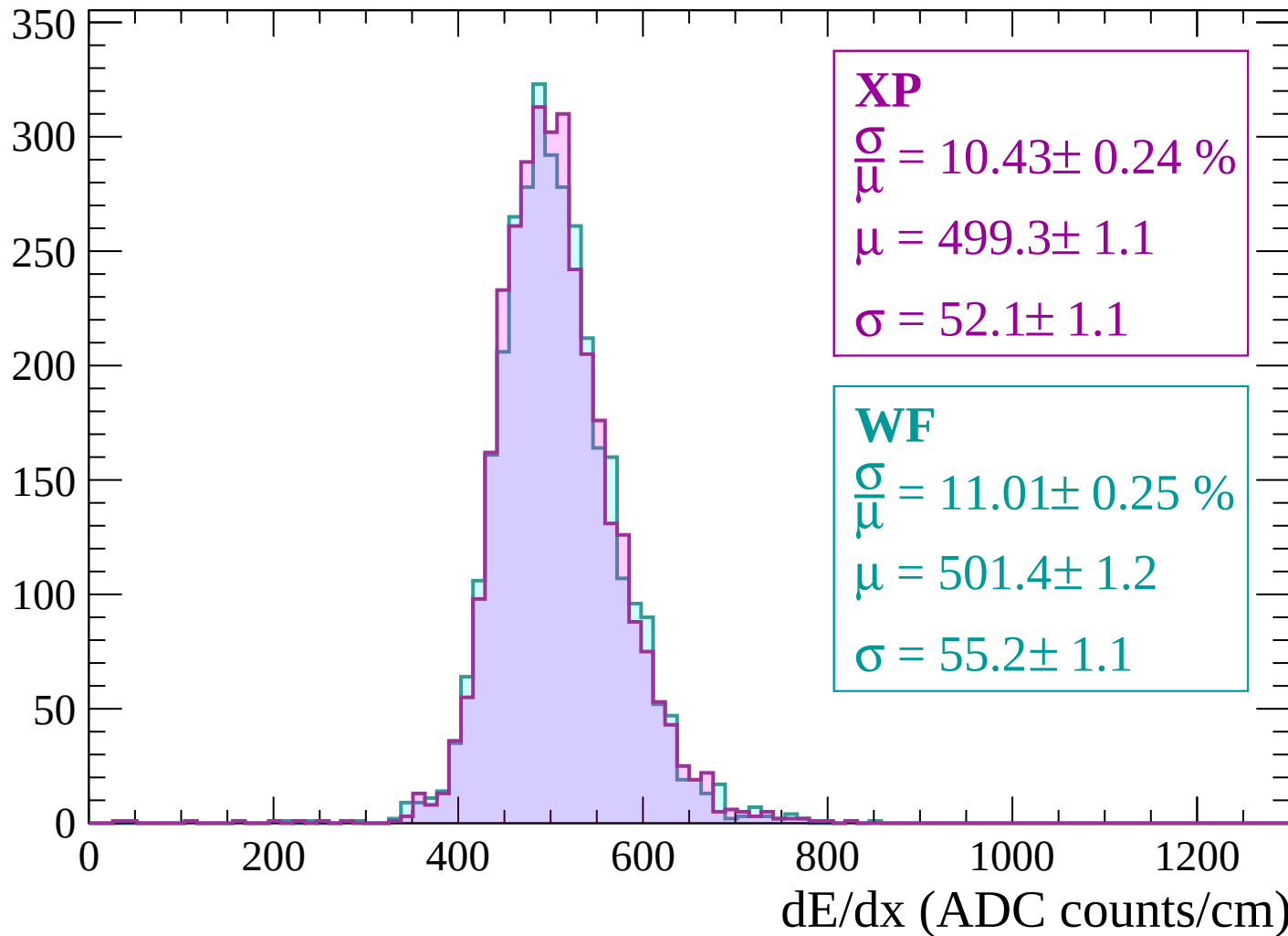
Energy loss | $2640 < p < 2700$

Count



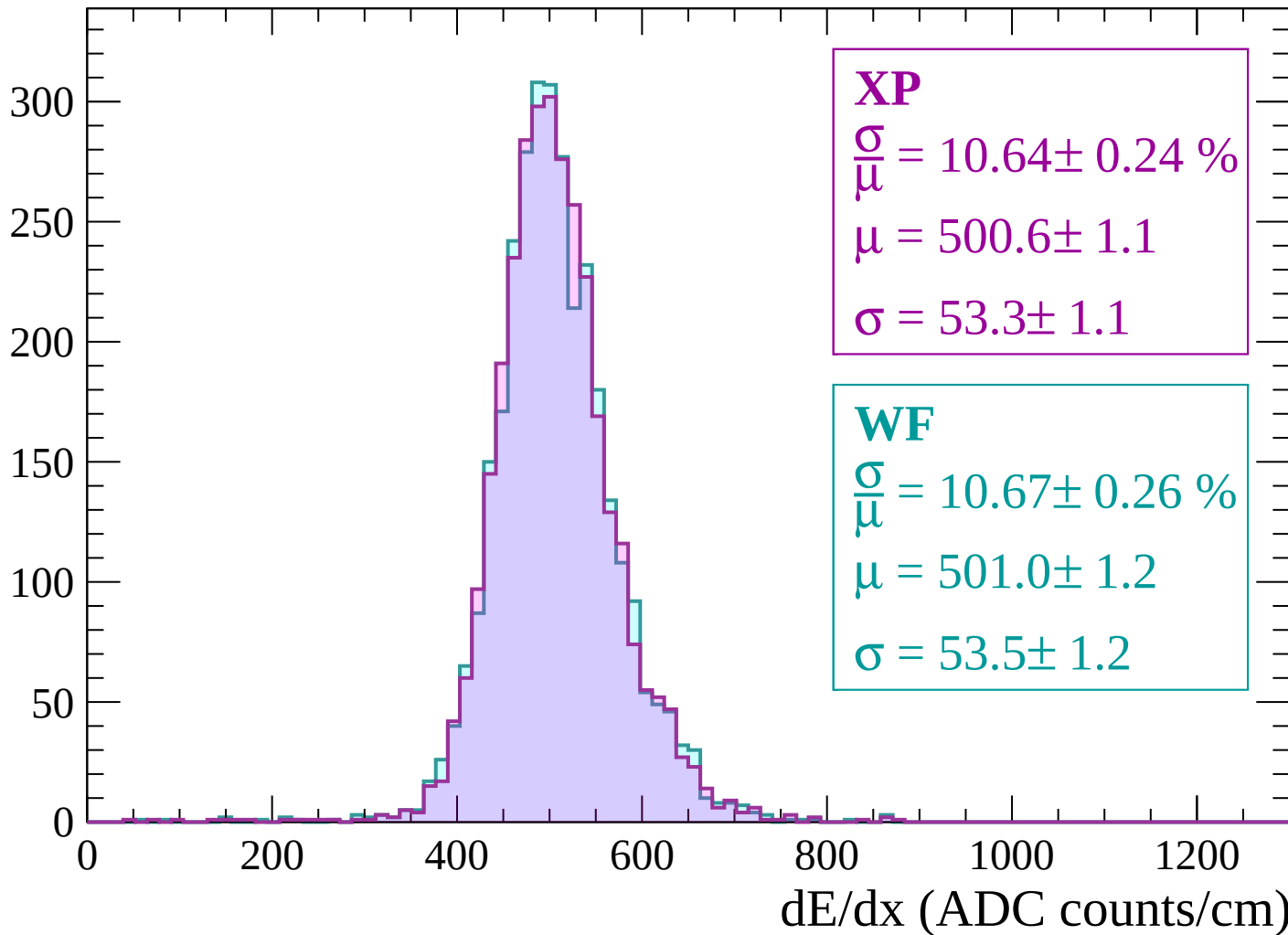
Energy loss | $2700 < p < 2760$

Count



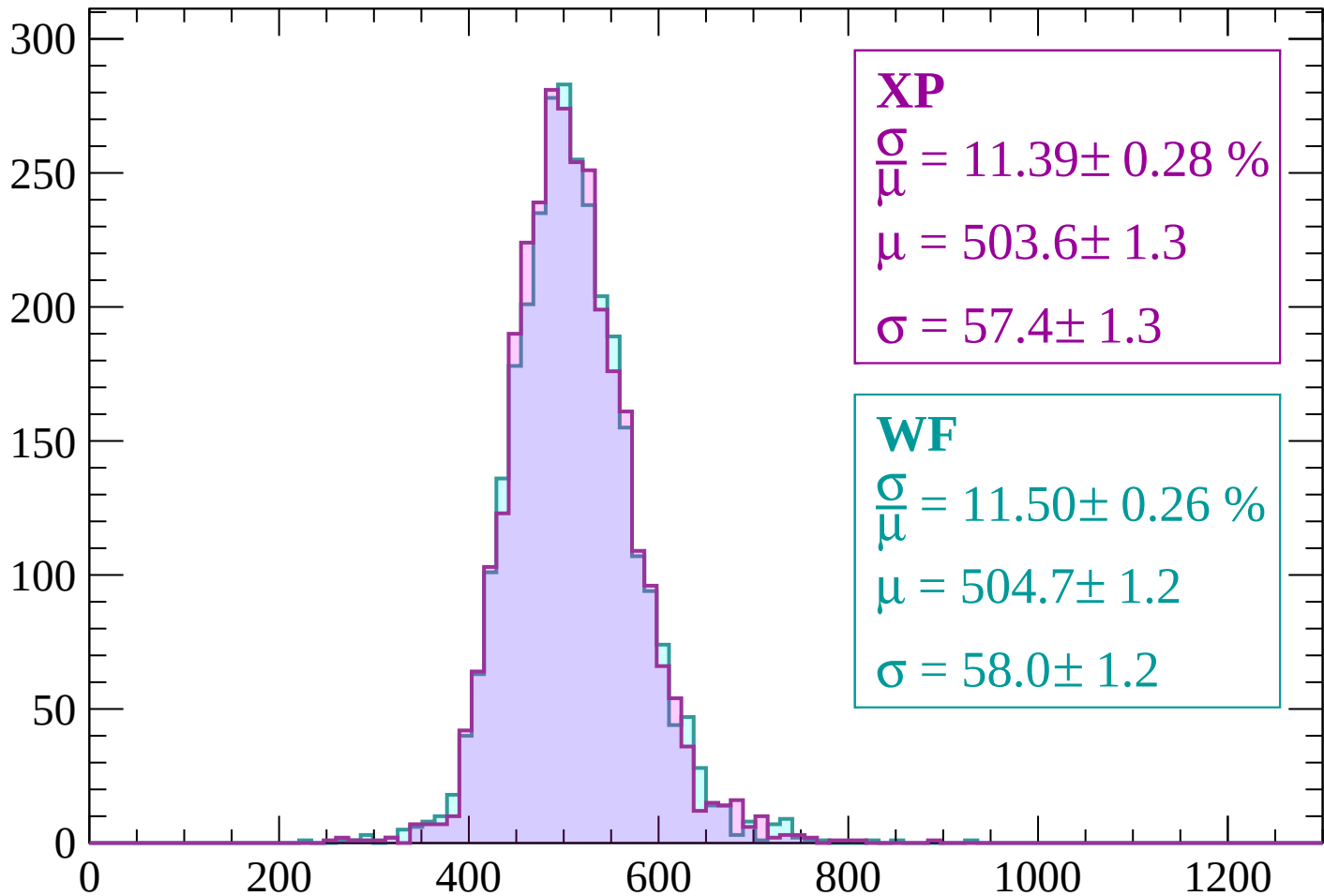
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP

$$\frac{\sigma}{\mu} = 11.39 \pm 0.28 \%$$

$$\mu = 503.6 \pm 1.3$$

$$\sigma = 57.4 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 11.50 \pm 0.26 \%$$

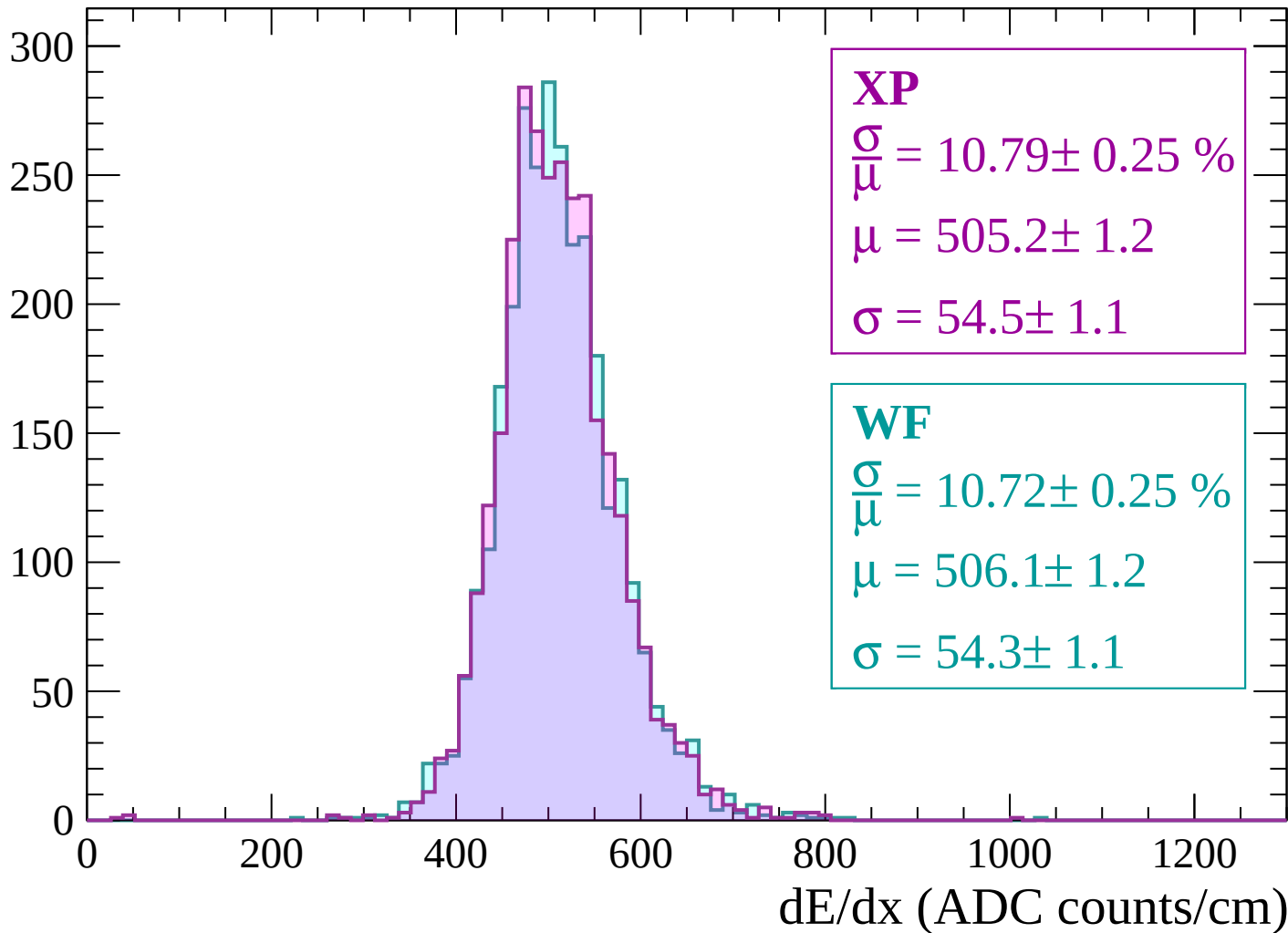
$$\mu = 504.7 \pm 1.2$$

$$\sigma = 58.0 \pm 1.2$$

dE/dx (ADC counts/cm)

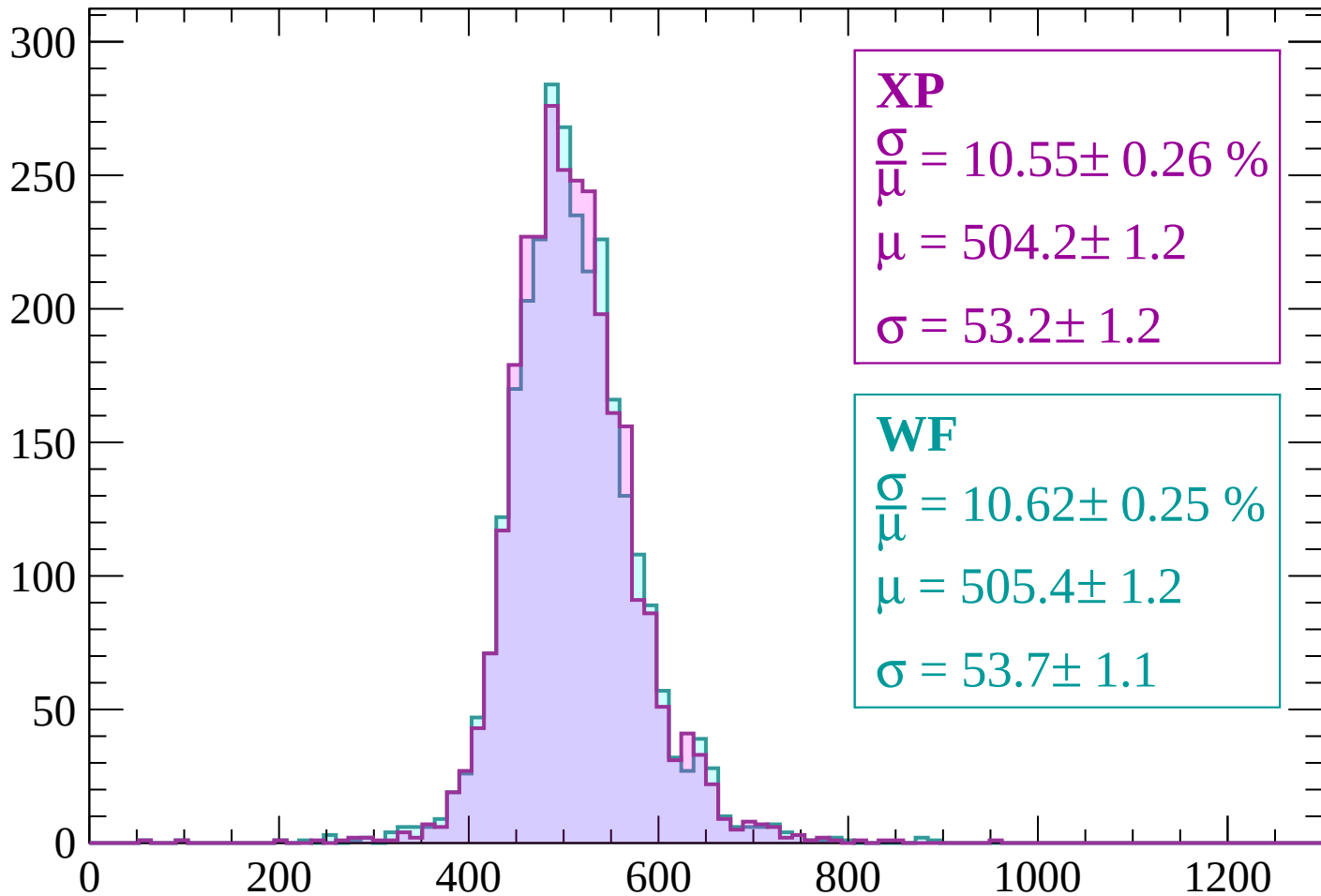
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



XP

$$\frac{\sigma}{\mu} = 10.55 \pm 0.26 \%$$

$$\mu = 504.2 \pm 1.2$$

$$\sigma = 53.2 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.62 \pm 0.25 \%$$

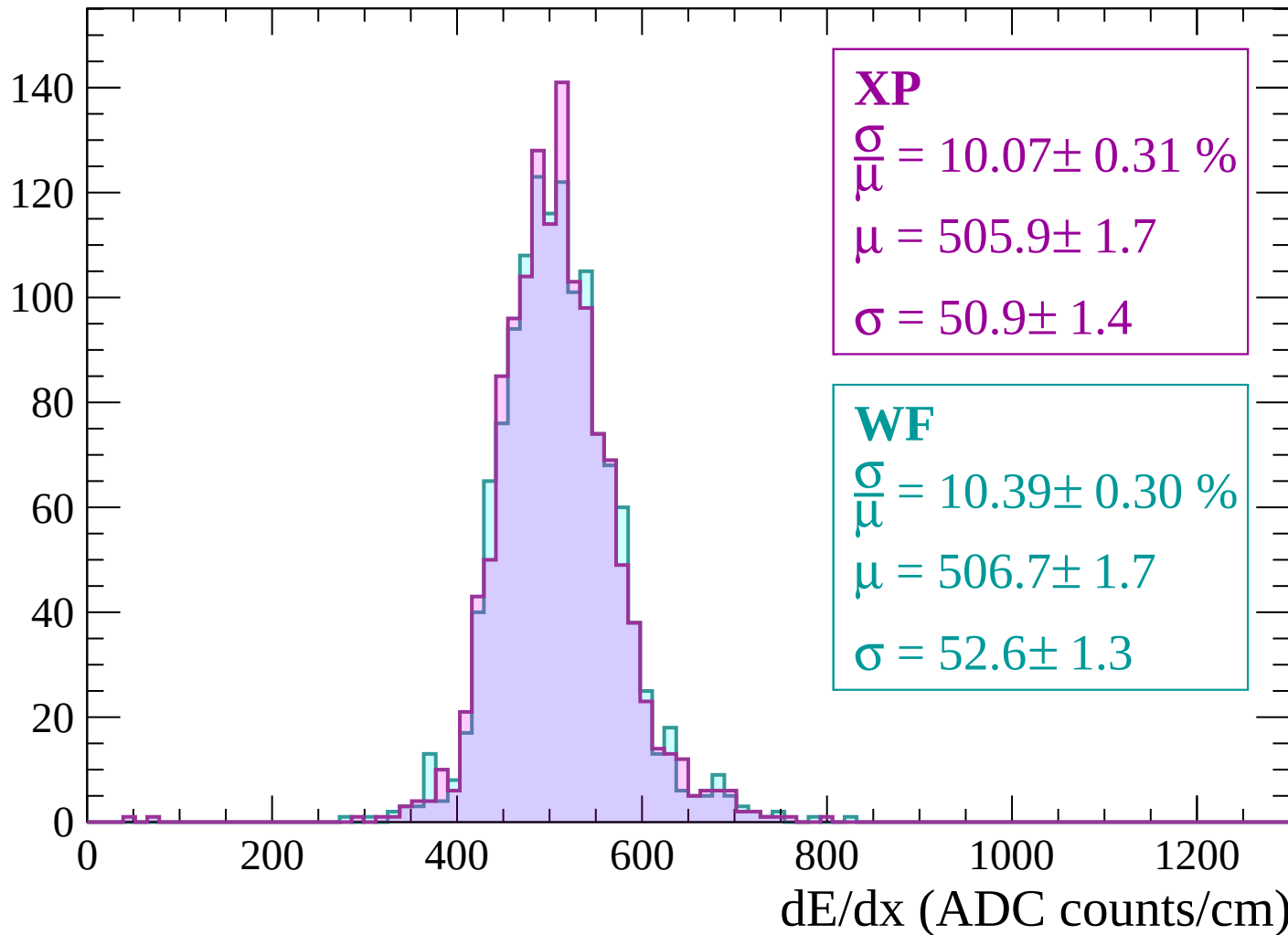
$$\mu = 505.4 \pm 1.2$$

$$\sigma = 53.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

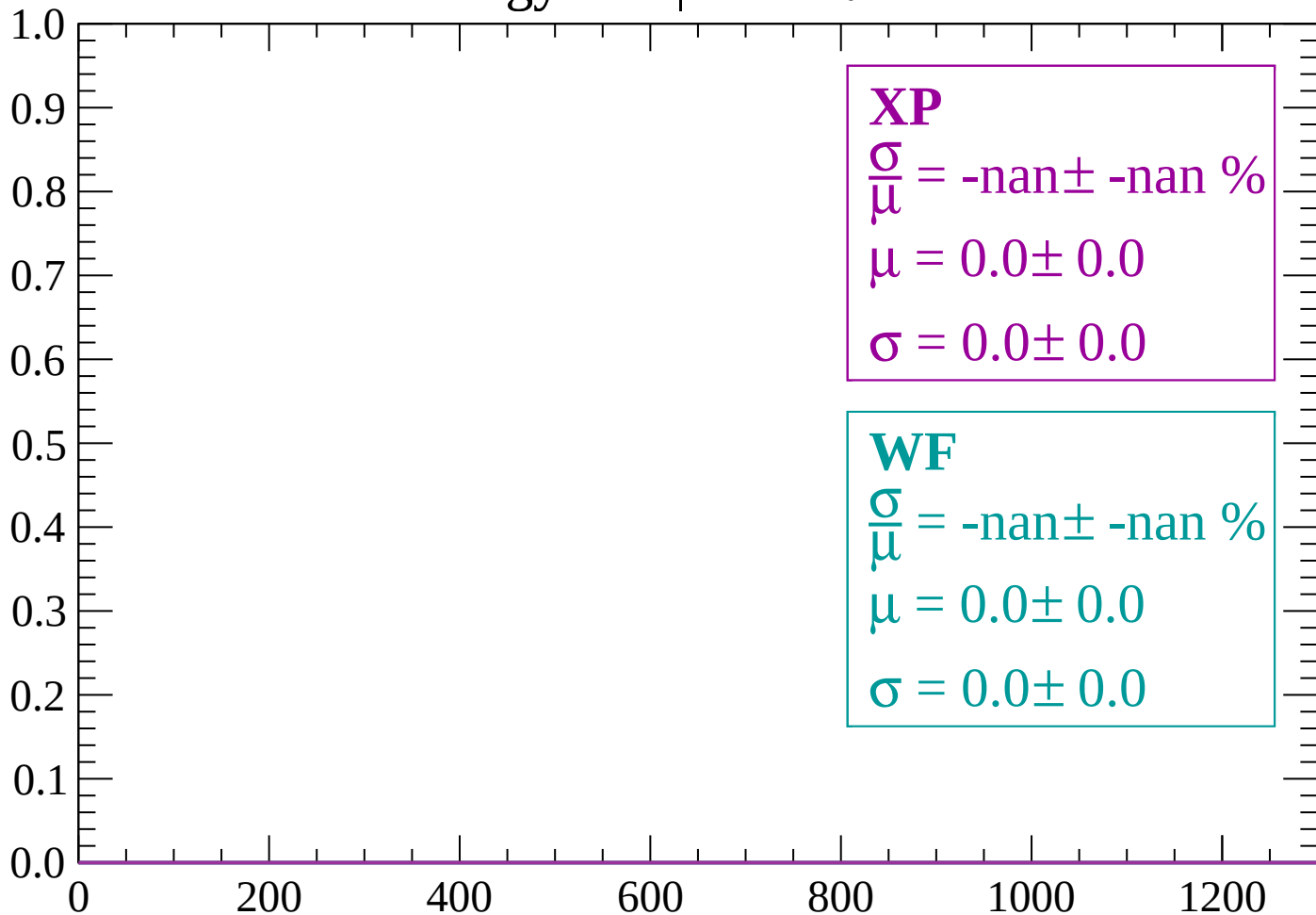
$$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

Energy loss | $-88 < \theta < -86$

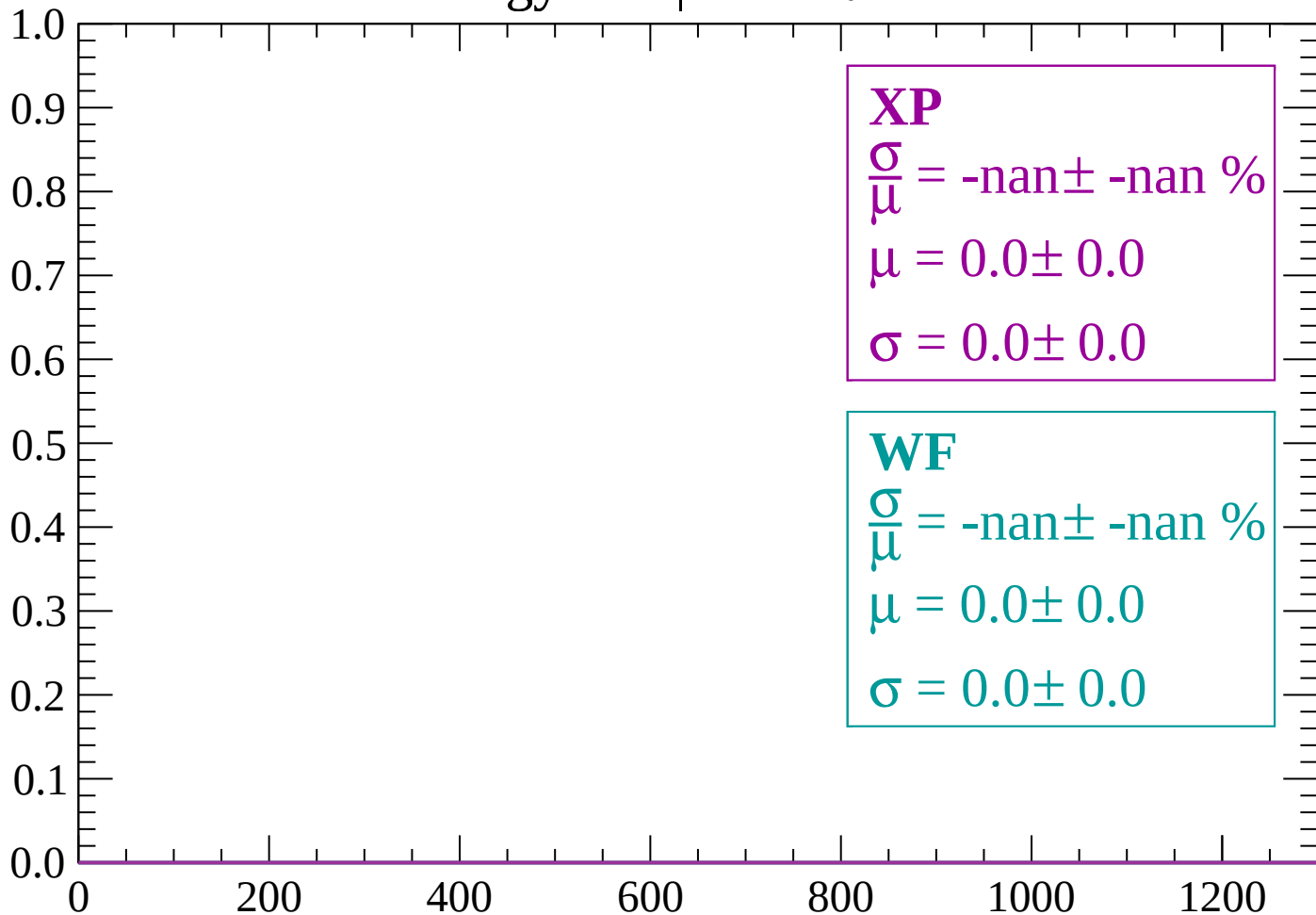
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

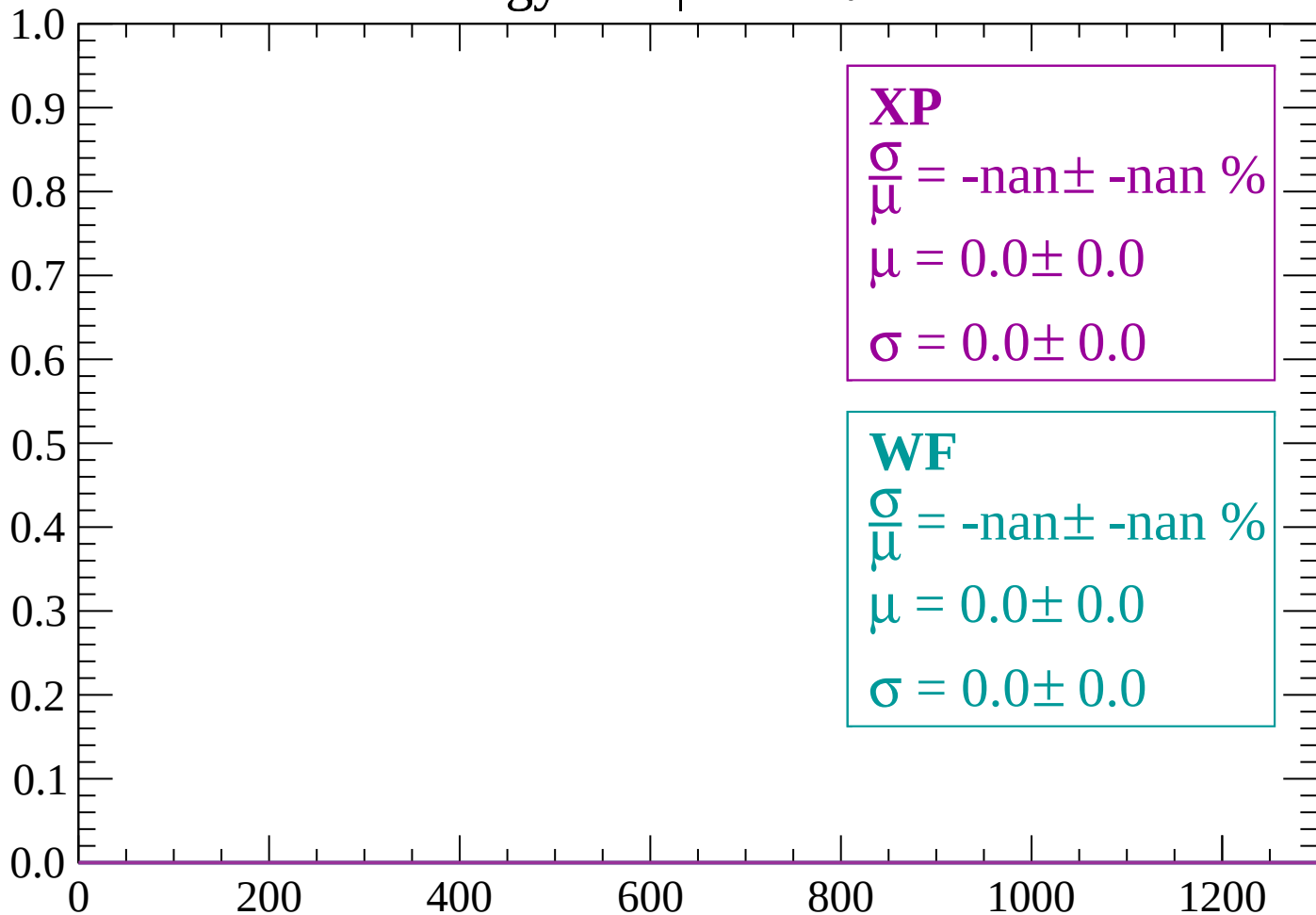
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

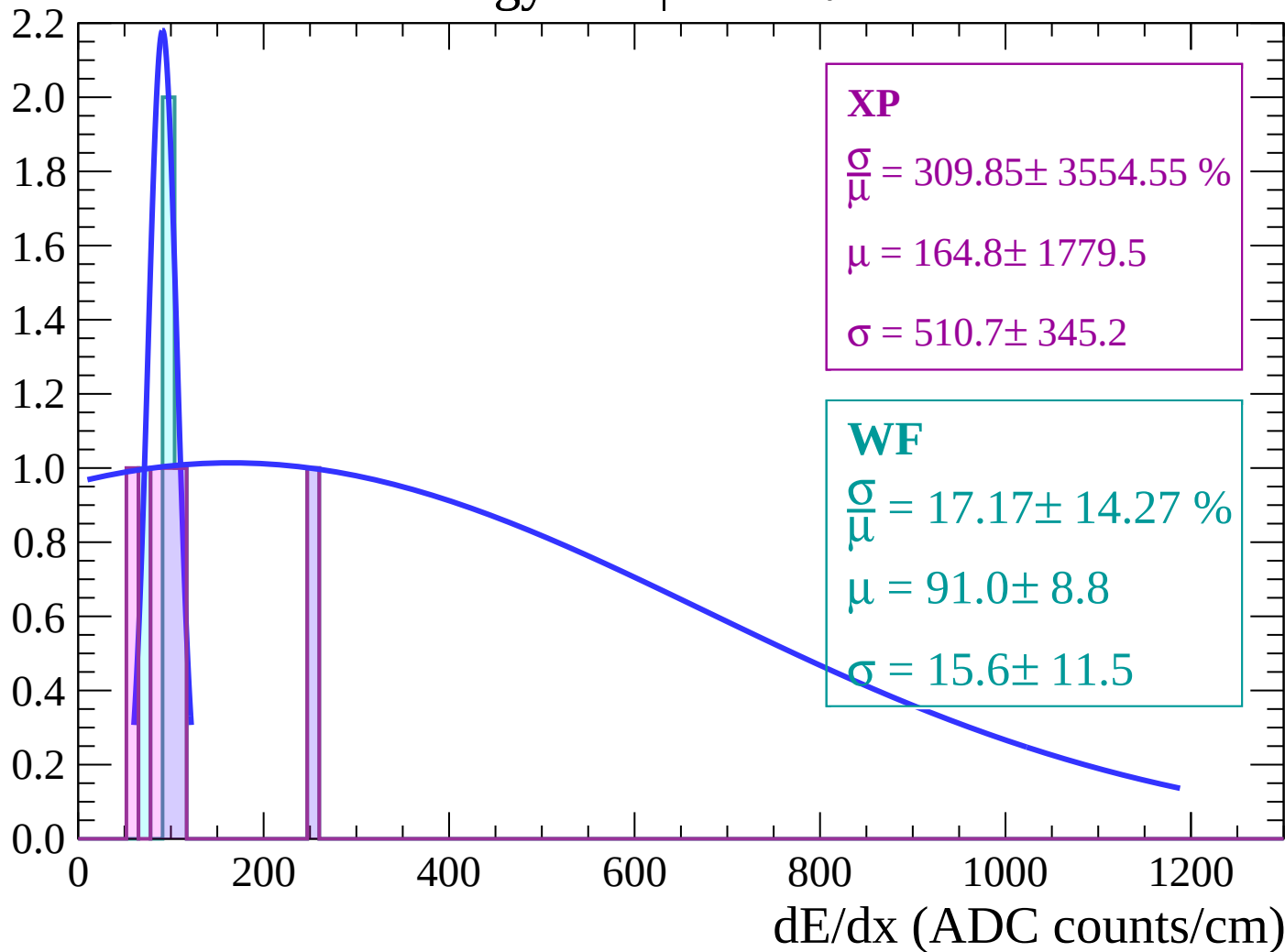
$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

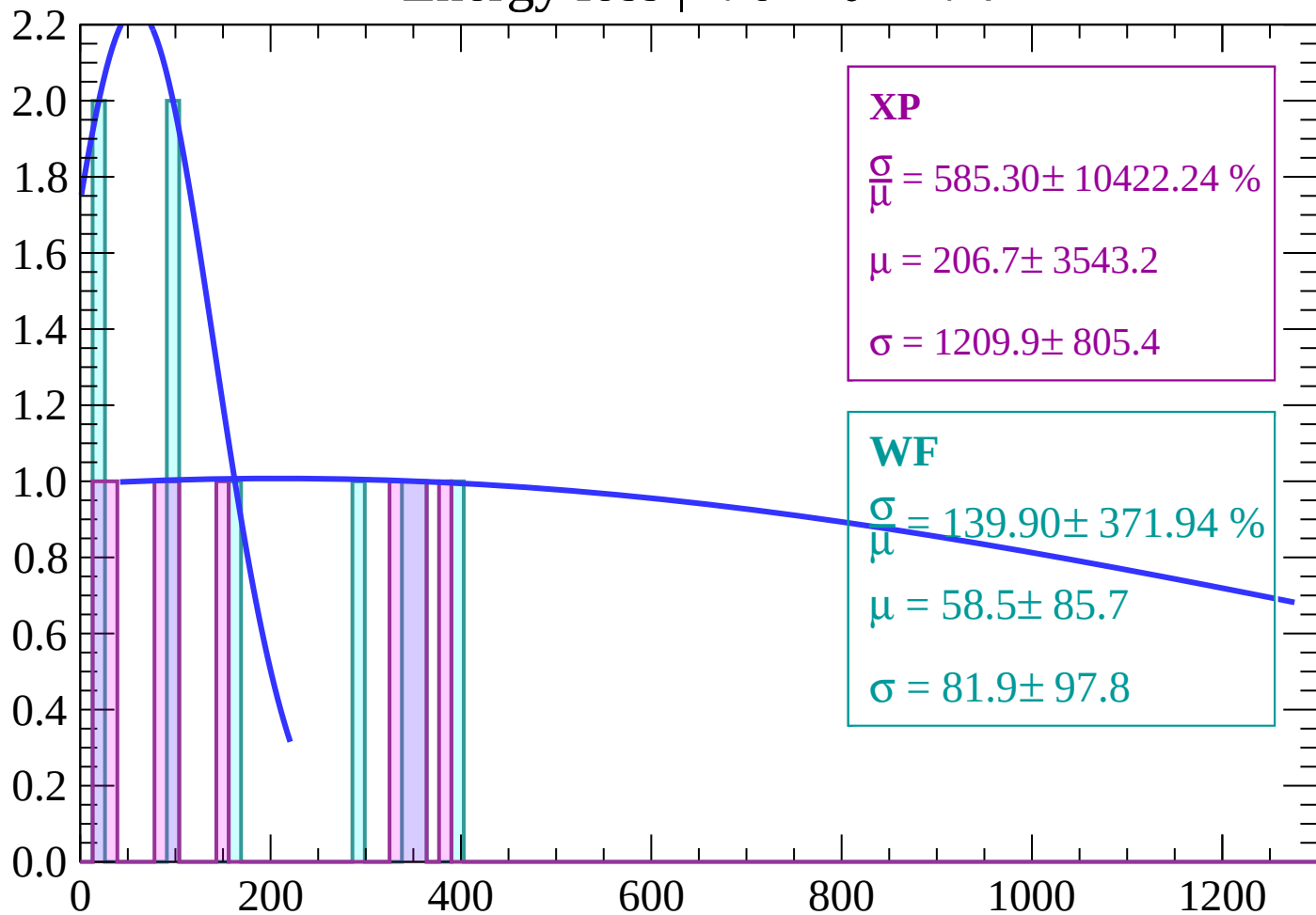
Energy loss | $-78 < \theta < -76$

Count



Energy loss | $-76 < \theta < -74$

Count



XP

$$\frac{\sigma}{\mu} = 585.30 \pm 10422.24 \%$$

$$\mu = 206.7 \pm 3543.2$$

$$\sigma = 1209.9 \pm 805.4$$

WF

$$\frac{\sigma}{\mu} = 139.90 \pm 371.94 \%$$

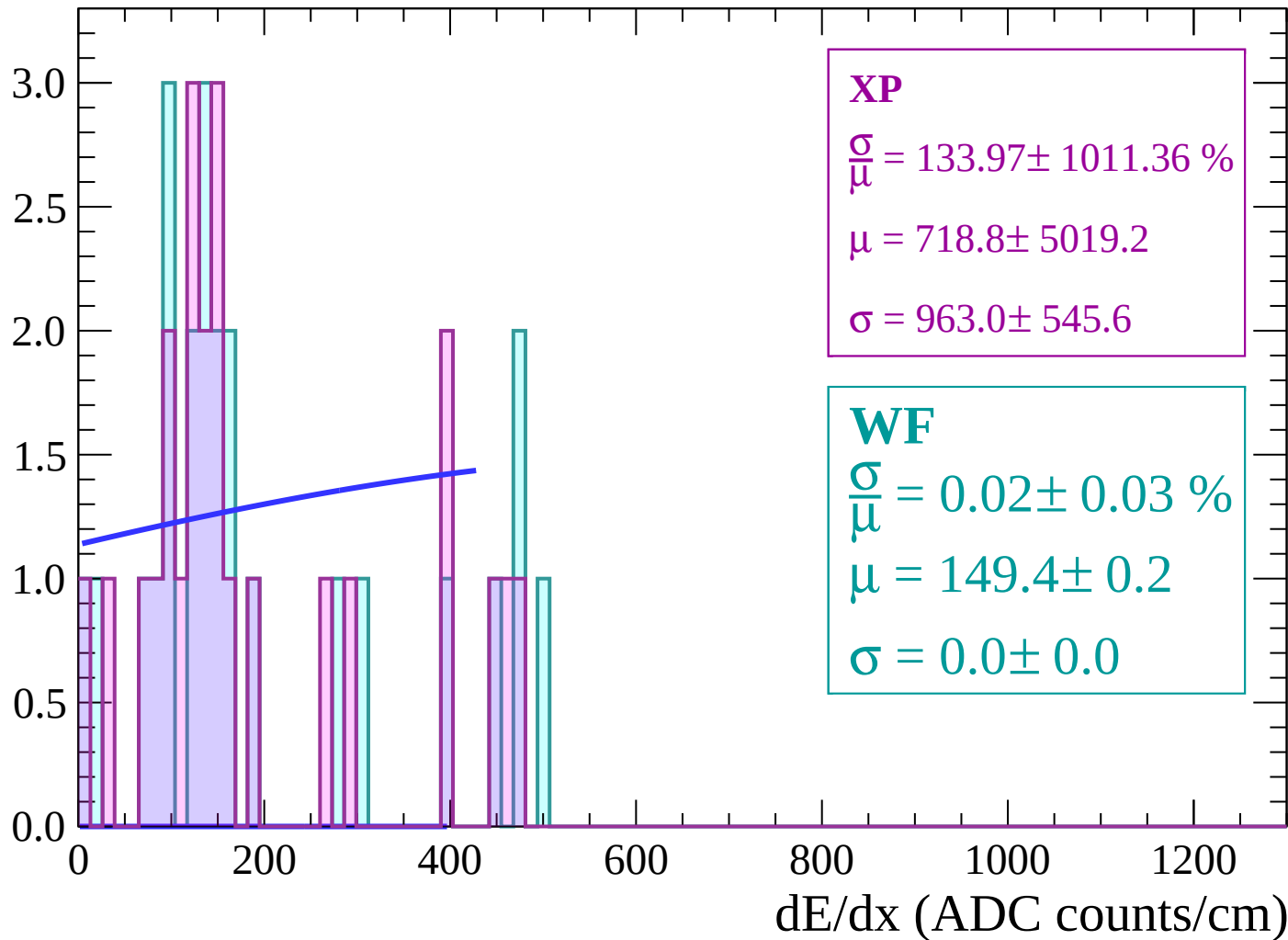
$$\mu = 58.5 \pm 85.7$$

$$\sigma = 81.9 \pm 97.8$$

dE/dx (ADC counts/cm)

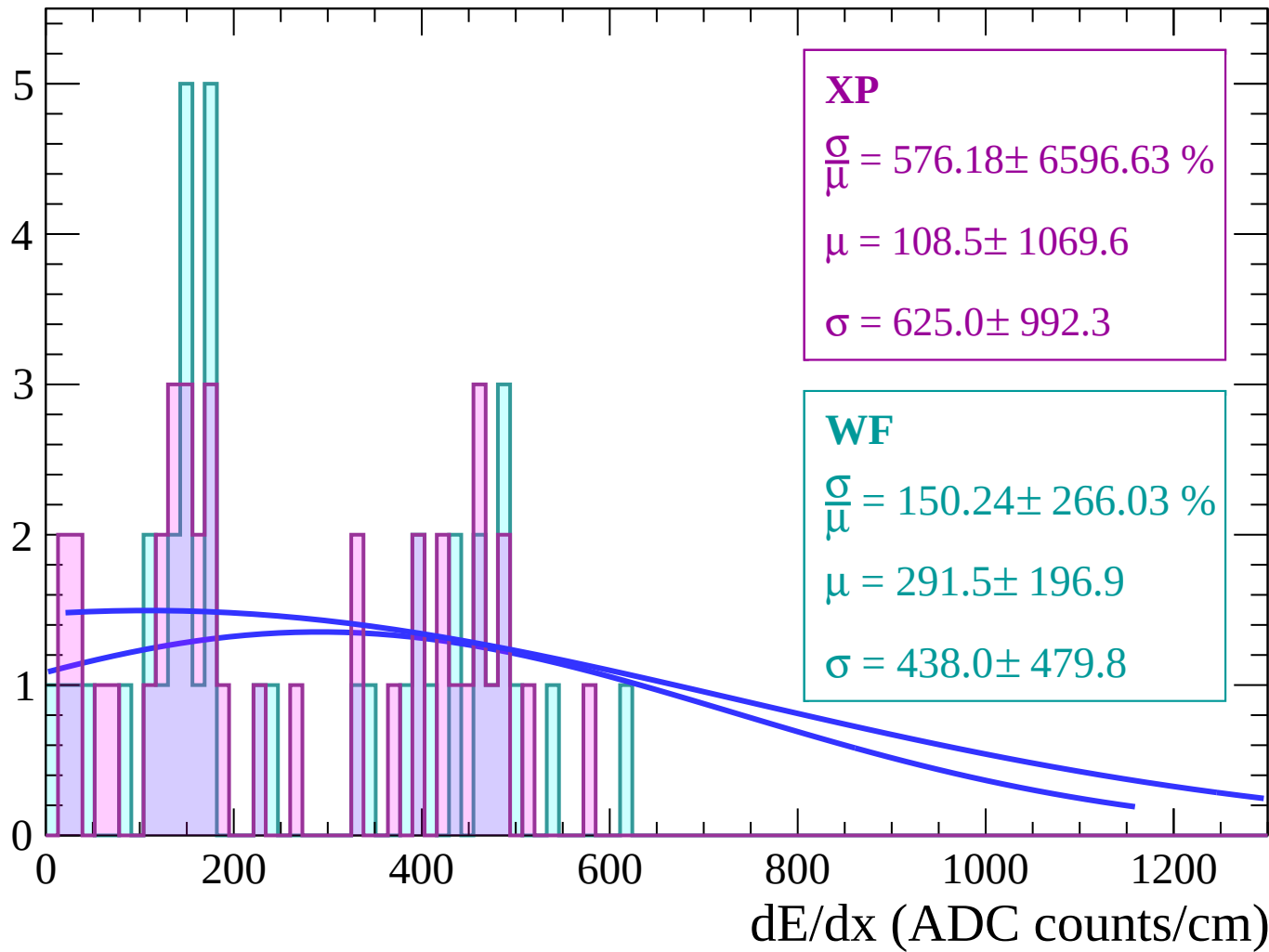
Energy loss | $-74 < \theta < -72$

Count



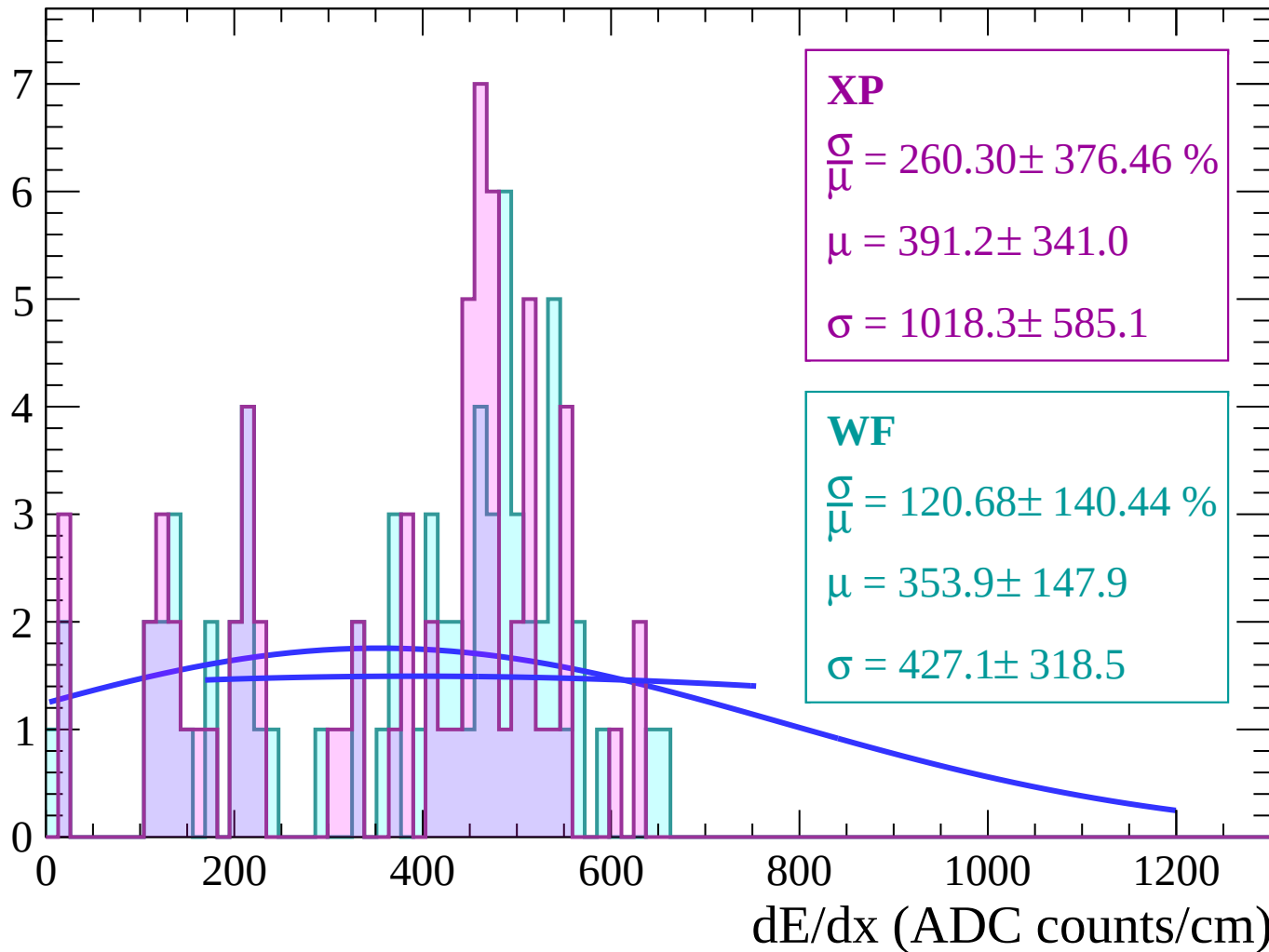
Energy loss | $-72 < \theta < -70$

Count



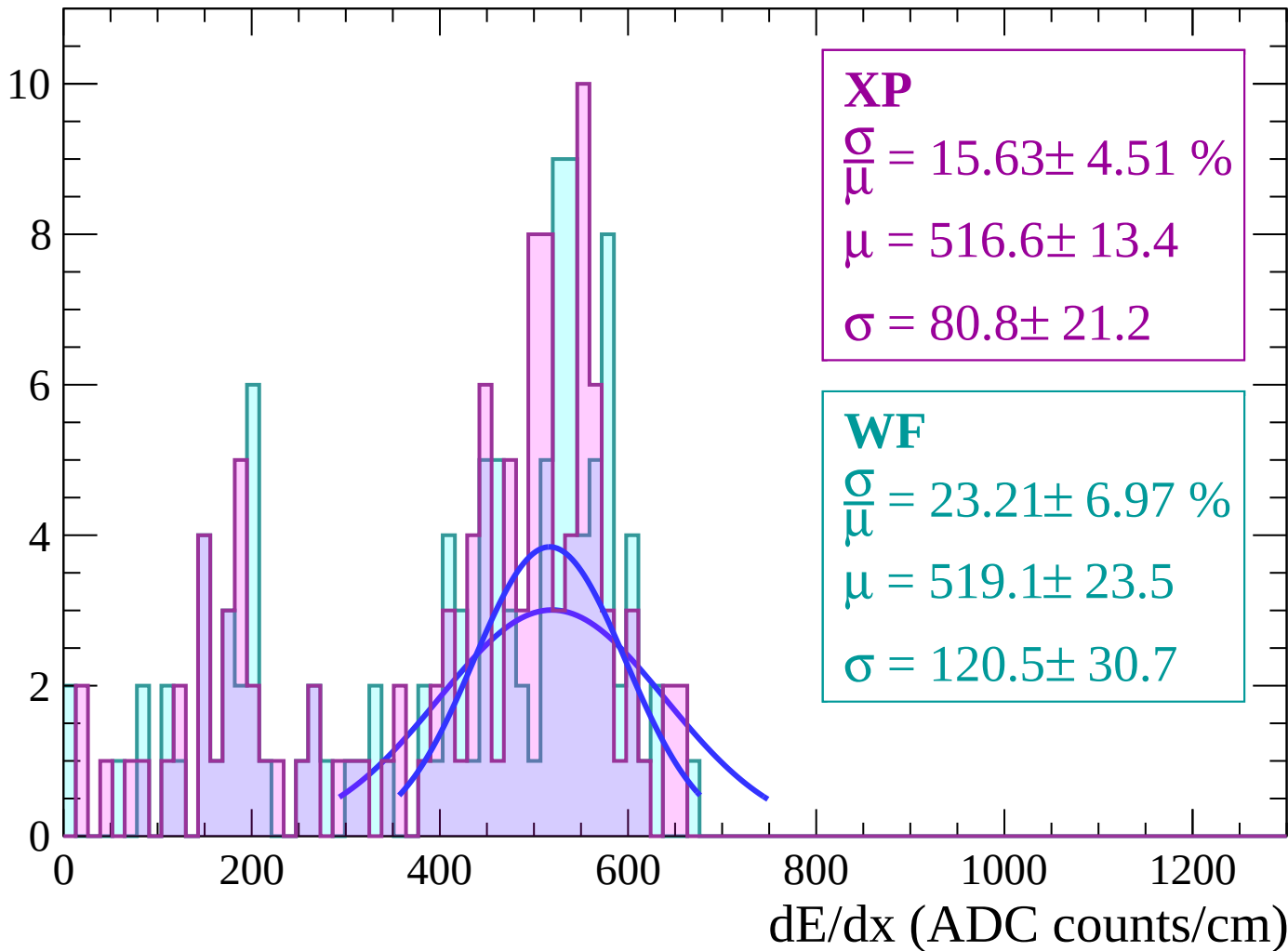
Energy loss | $-70 < \theta < -68$

Count



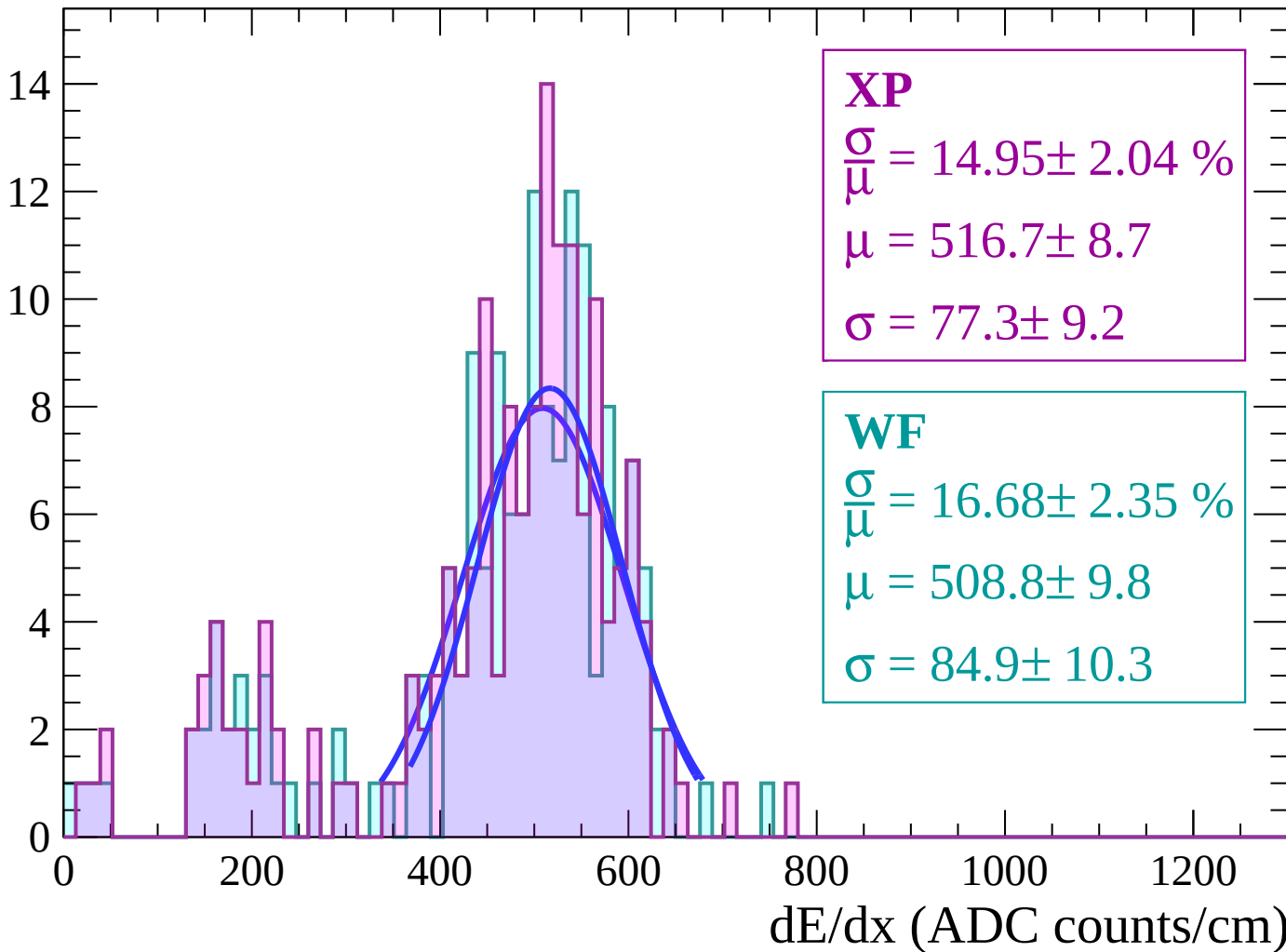
Energy loss | $-68 < \theta < -66$

Count



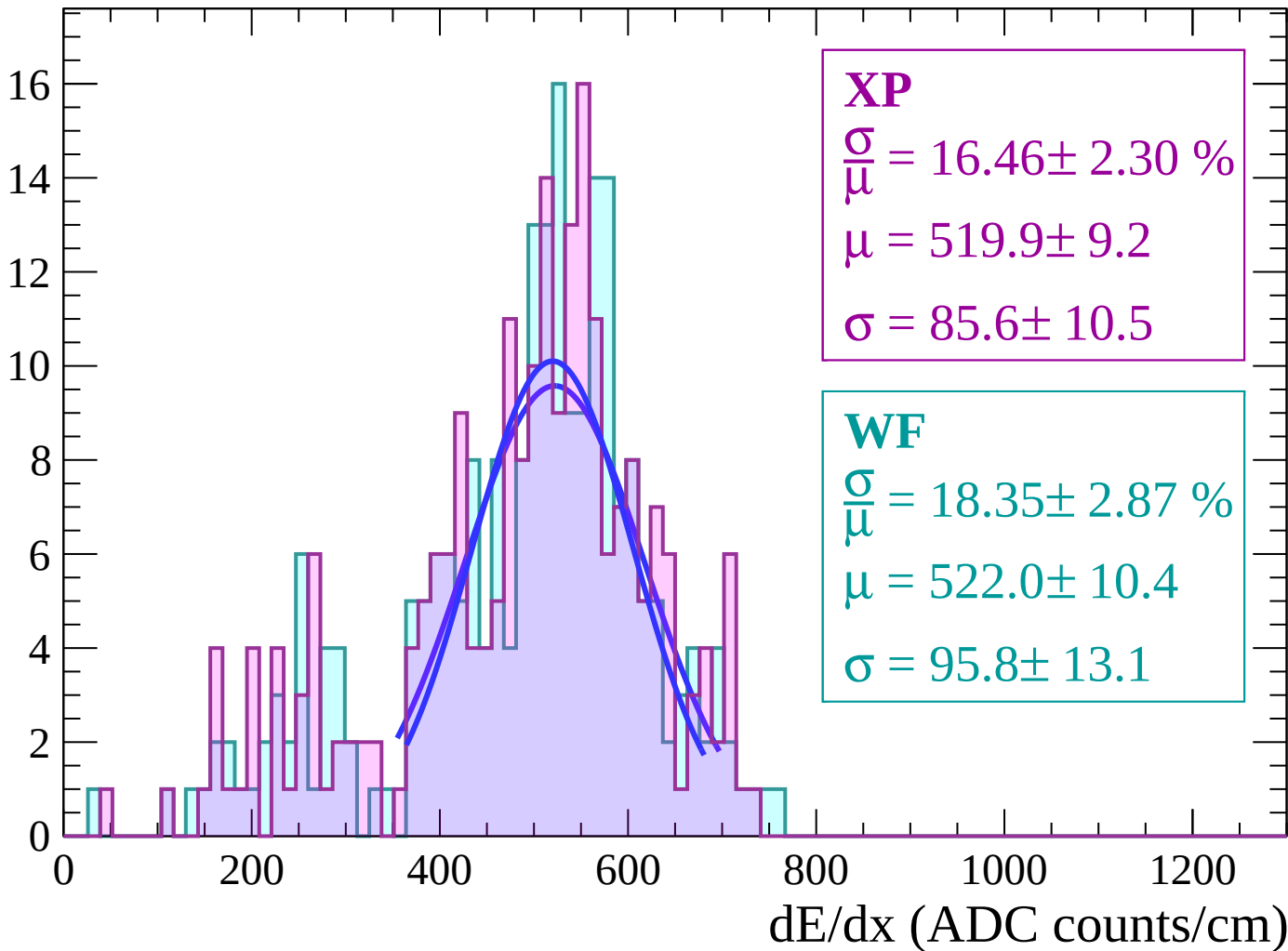
Energy loss | $-66 < \theta < -64$

Count



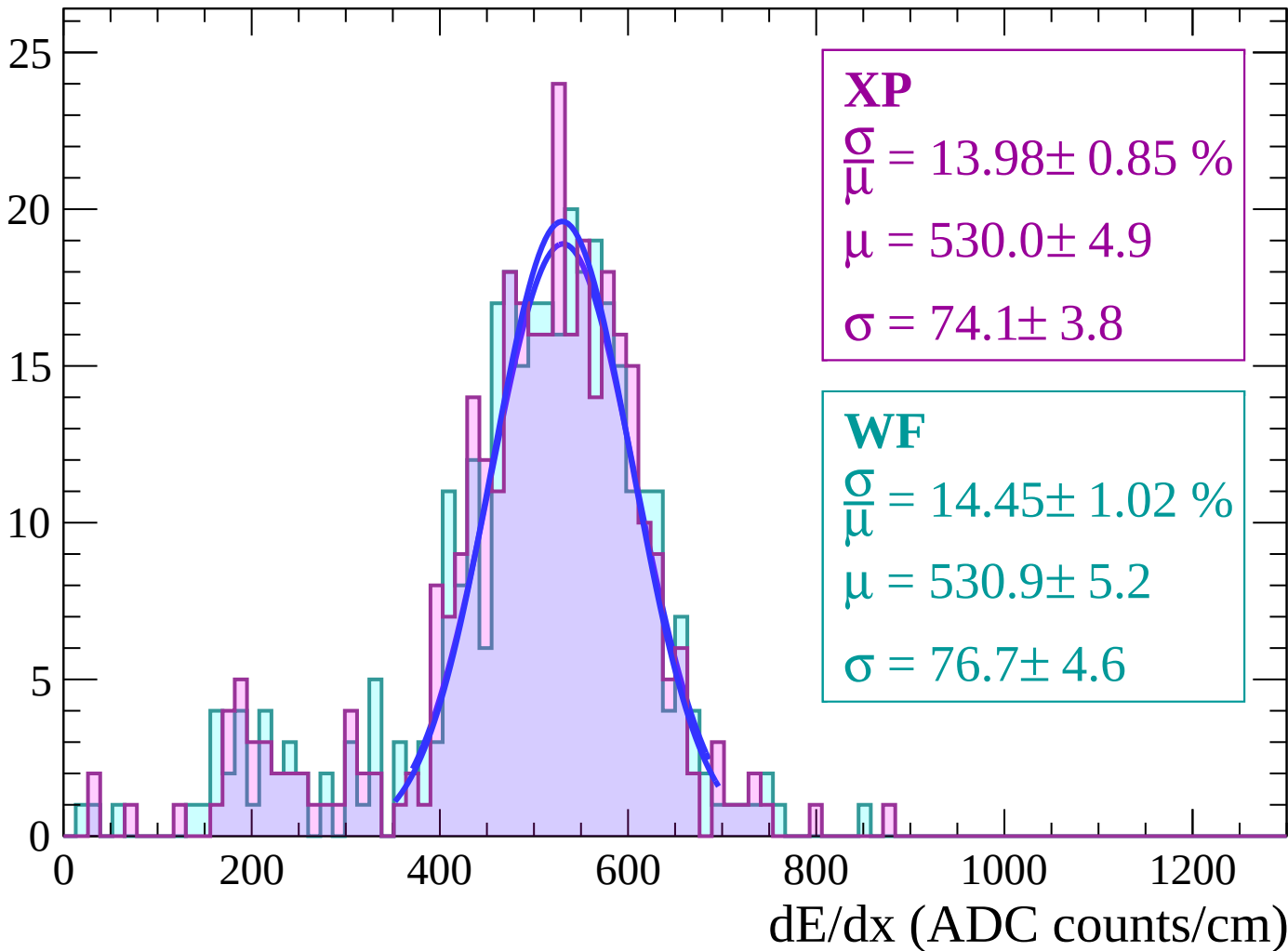
Energy loss | $-64 < \theta < -62$

Count



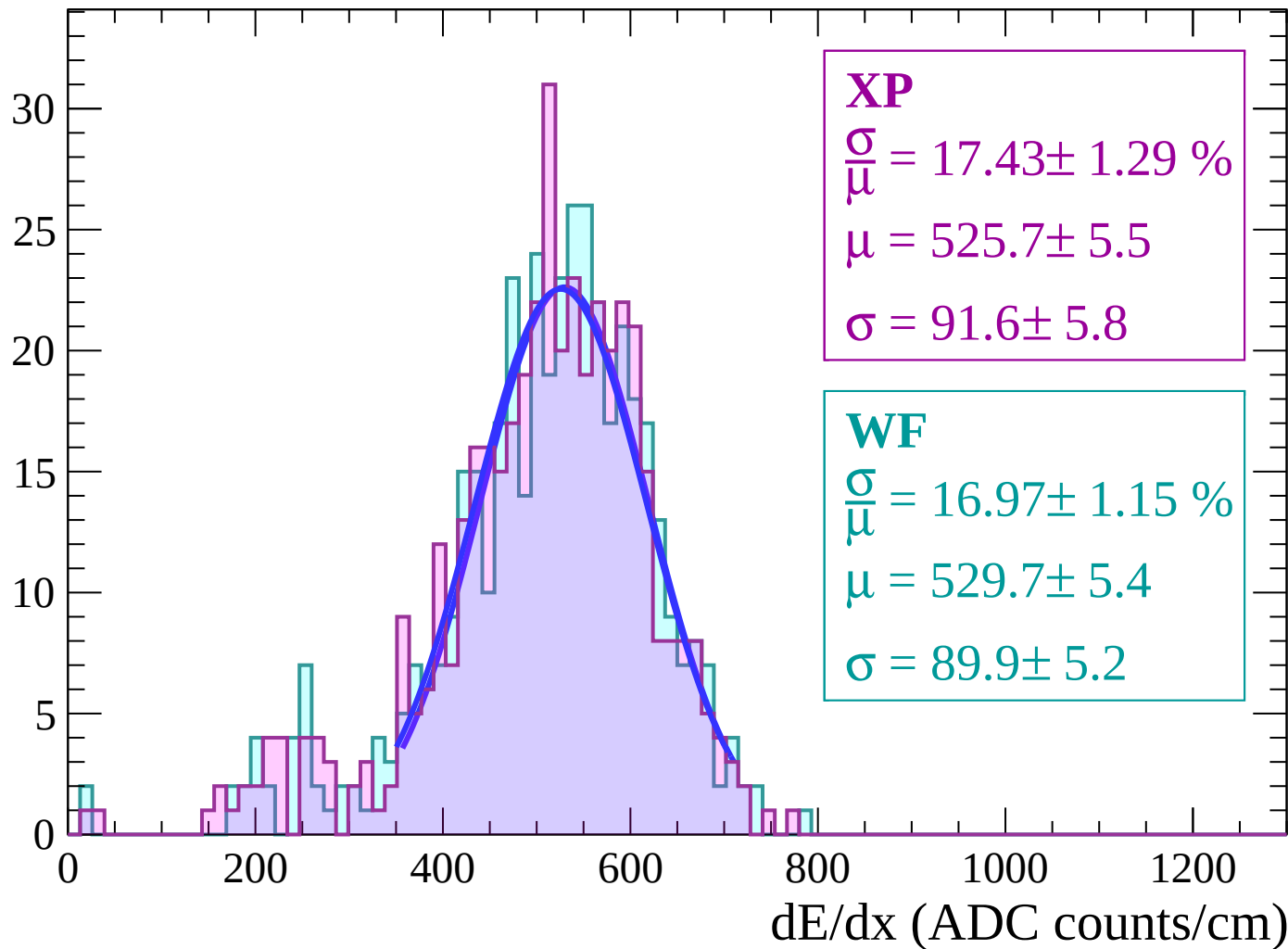
Energy loss | $-62 < \theta < -60$

Count



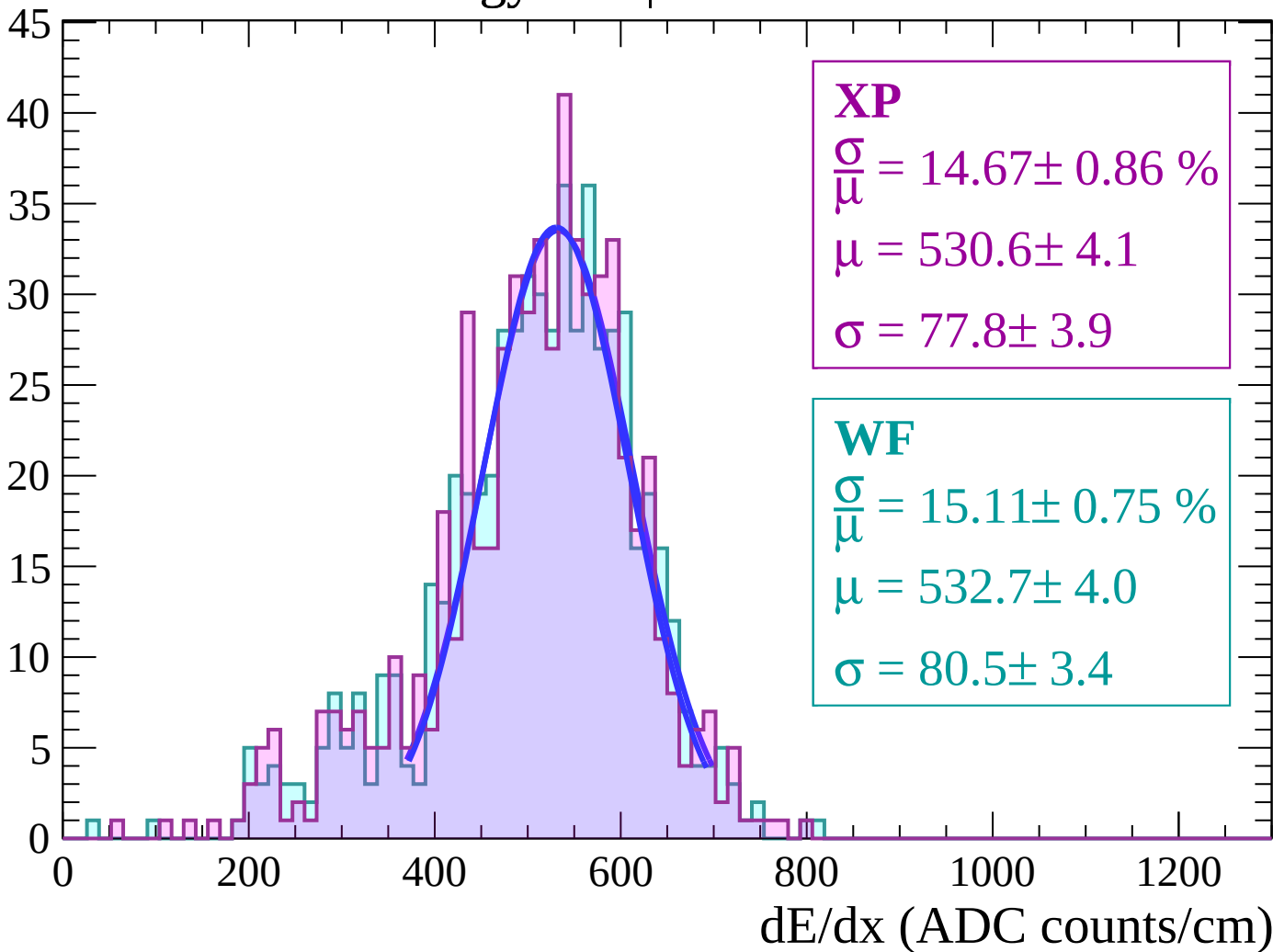
Energy loss | $-60 < \theta < -58$

Count



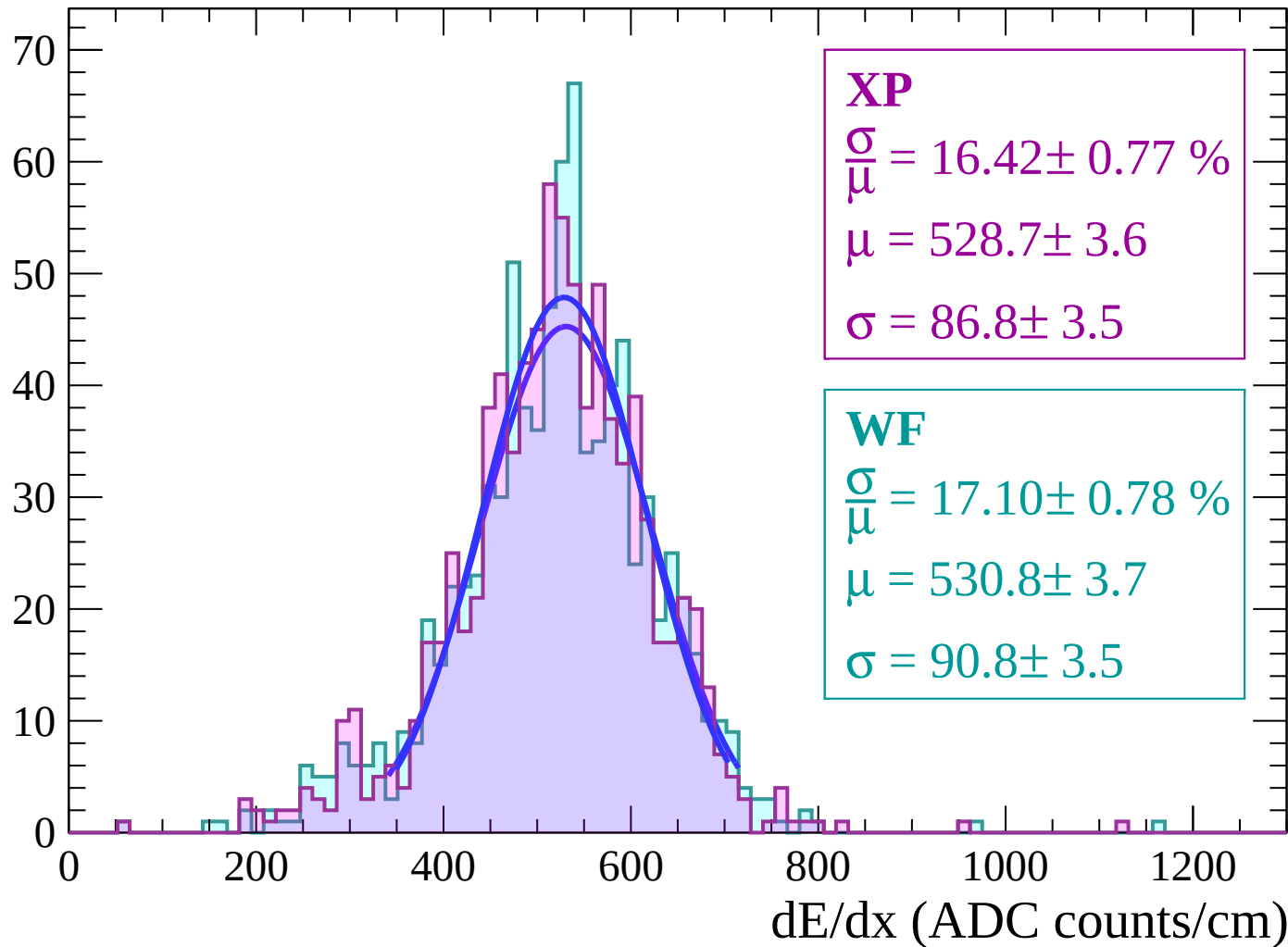
Energy loss | $-58 < \theta < -56$

Count



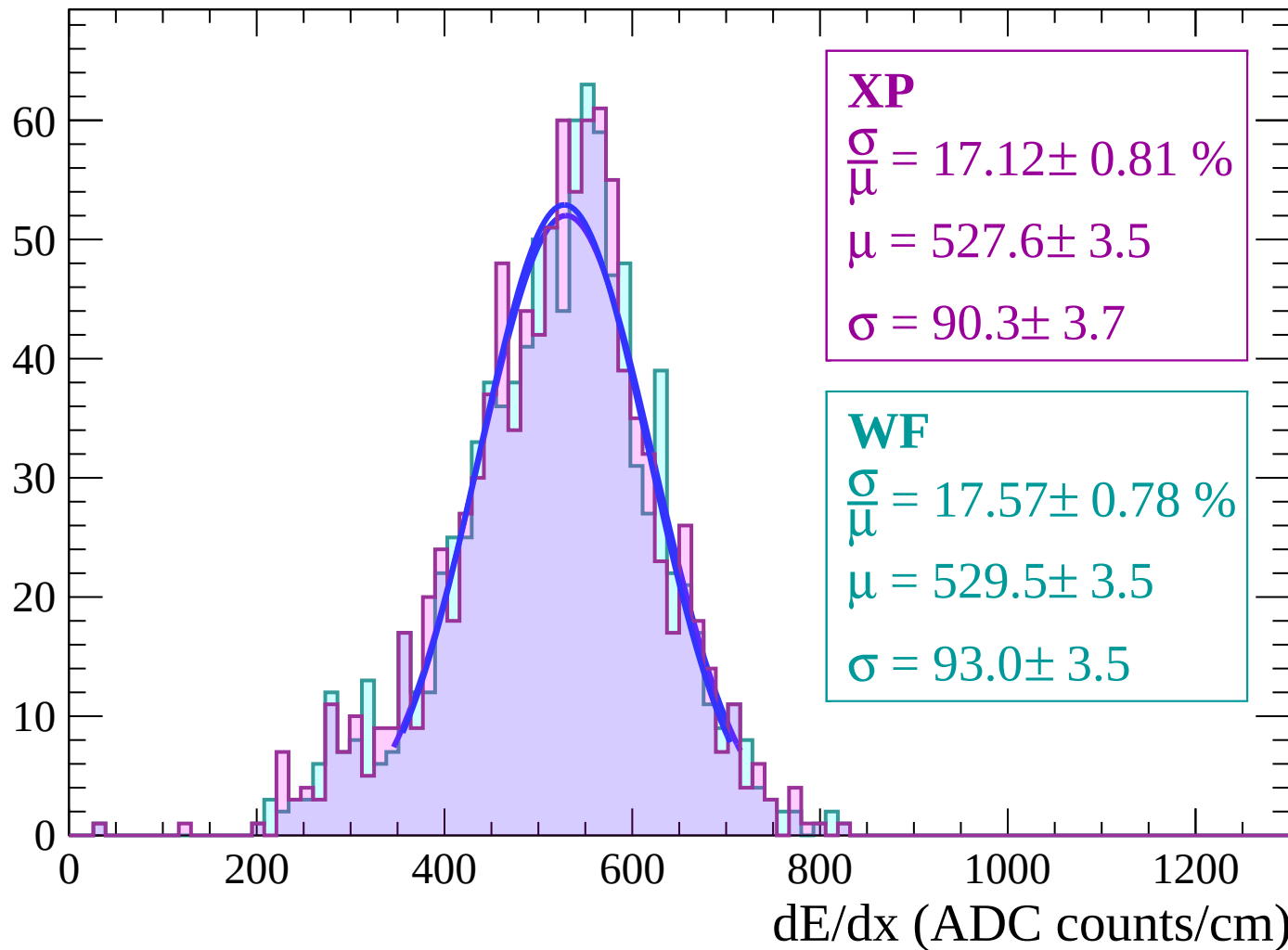
Energy loss | $-56 < \theta < -54$

Count



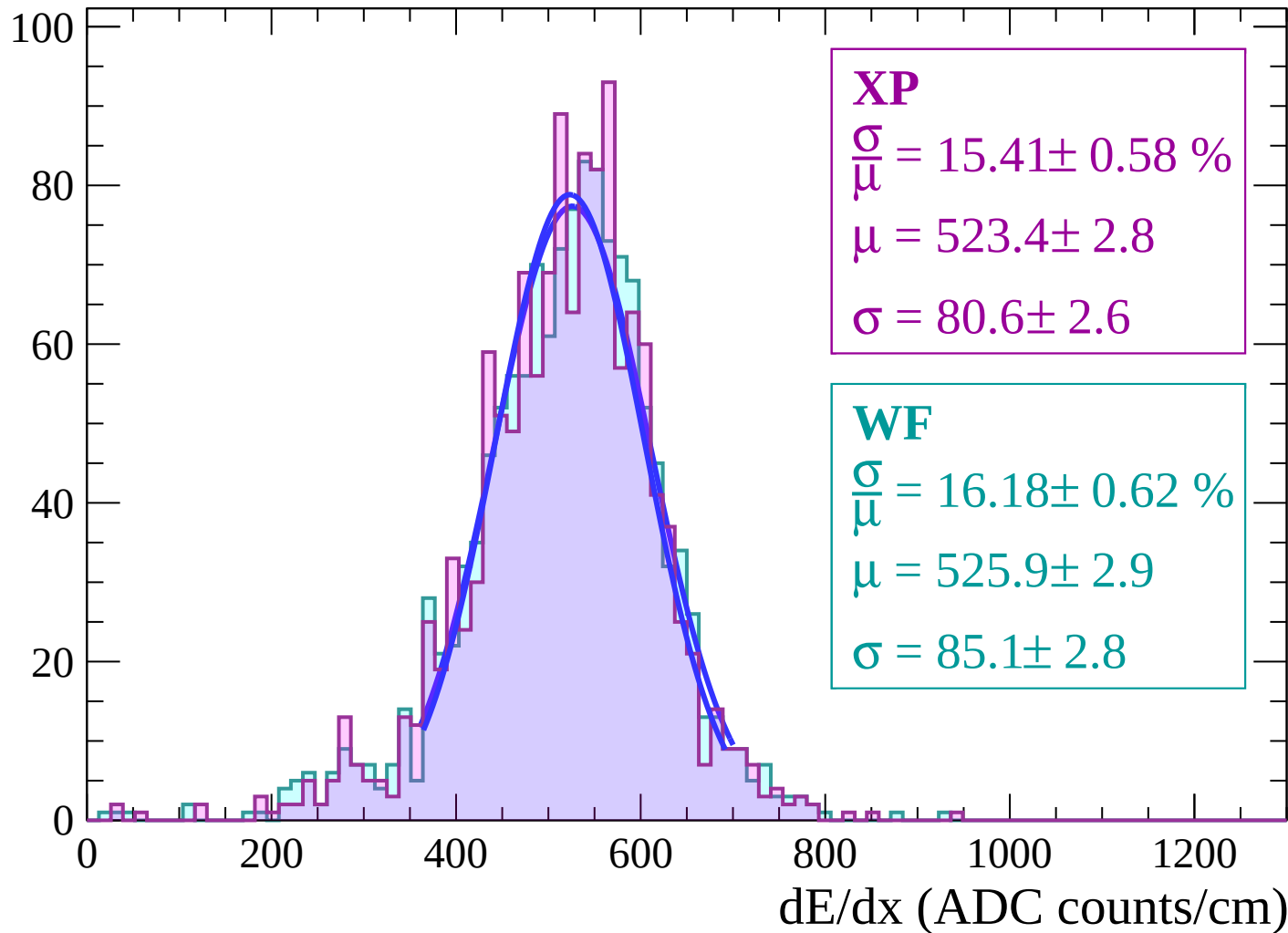
Energy loss | $-54 < \theta < -52$

Count



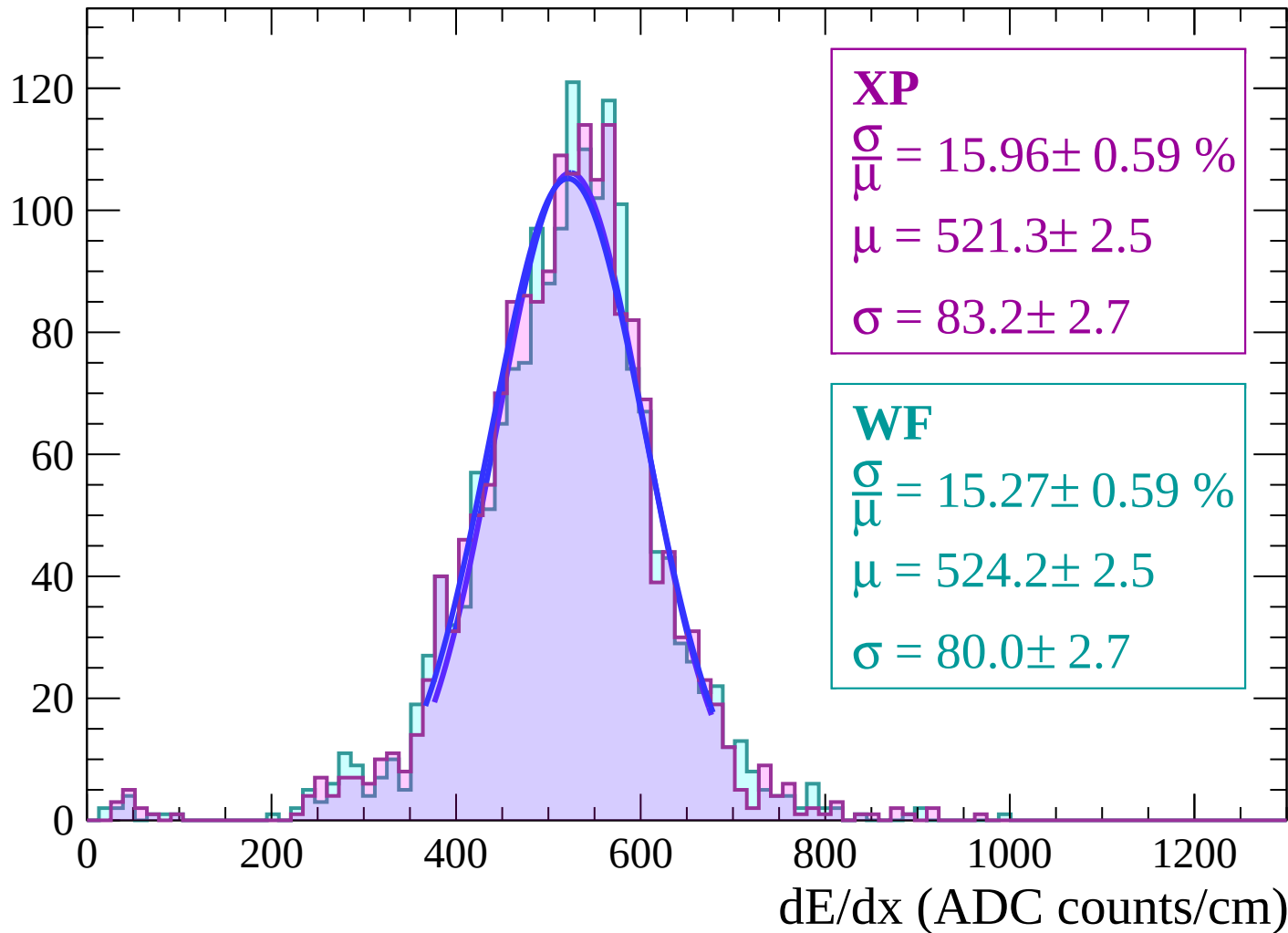
Energy loss | $-52 < \theta < -50$

Count



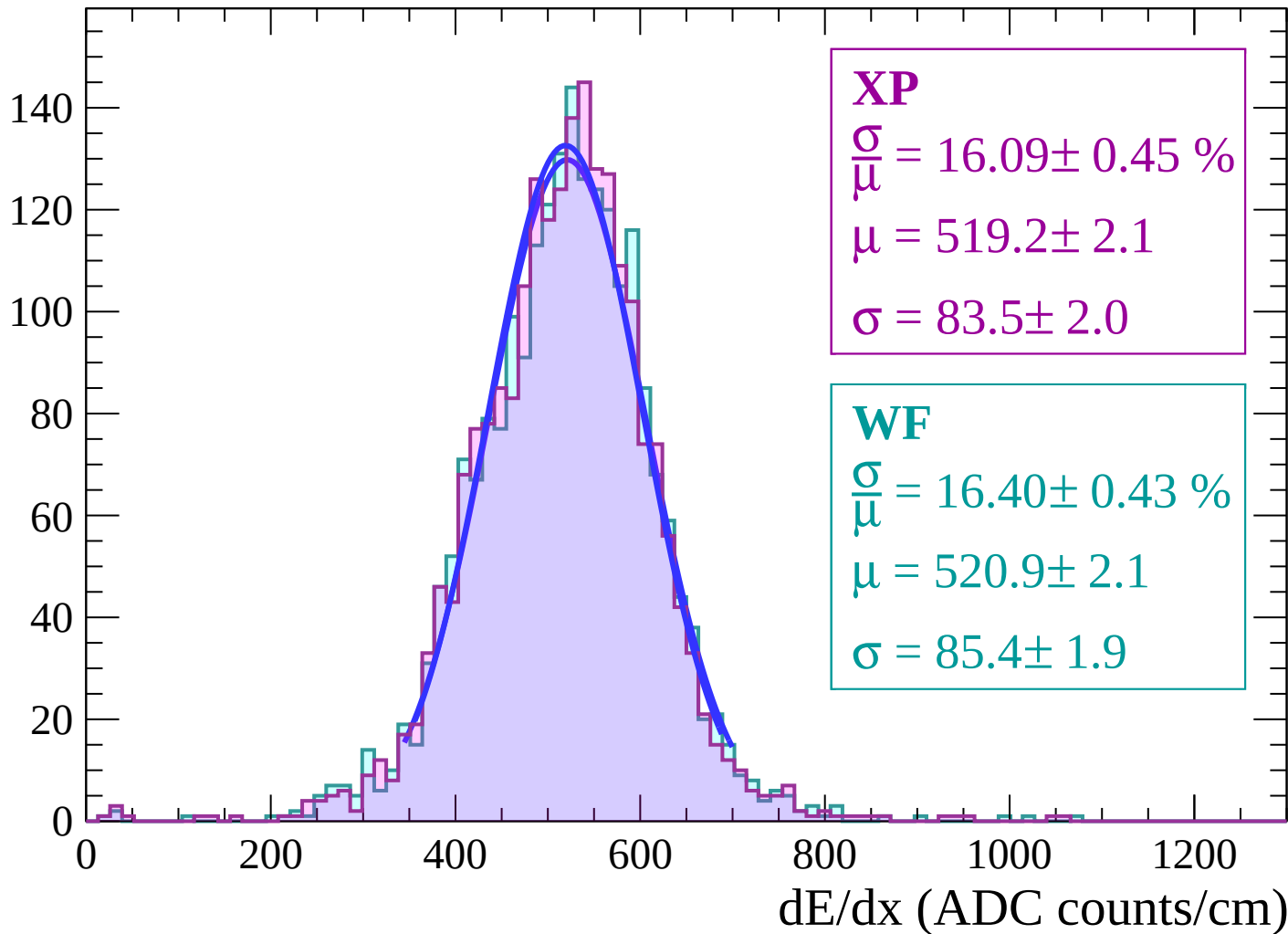
Energy loss | $-50 < \theta < -48$

Count



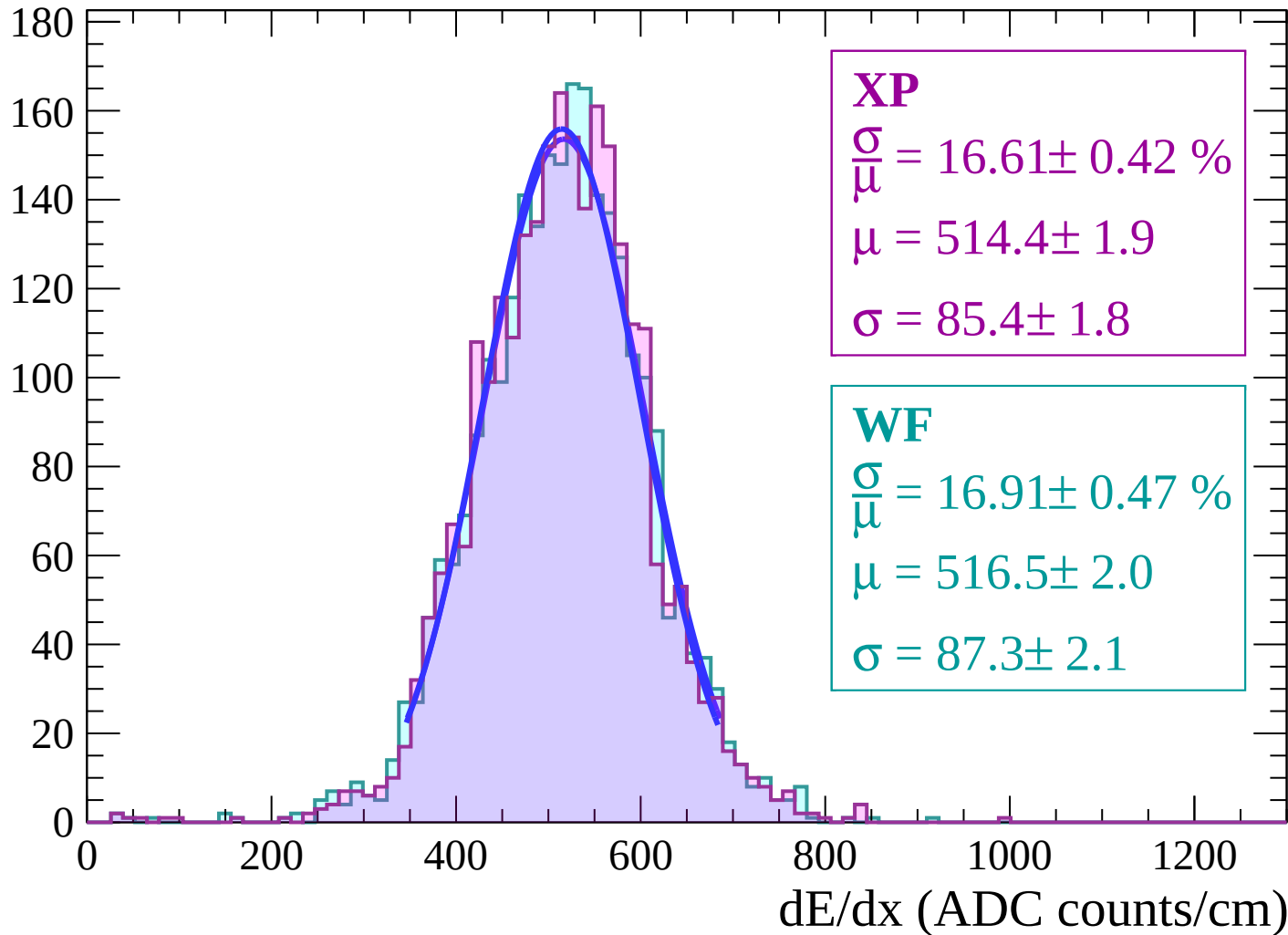
Energy loss | $-48 < \theta < -46$

Count



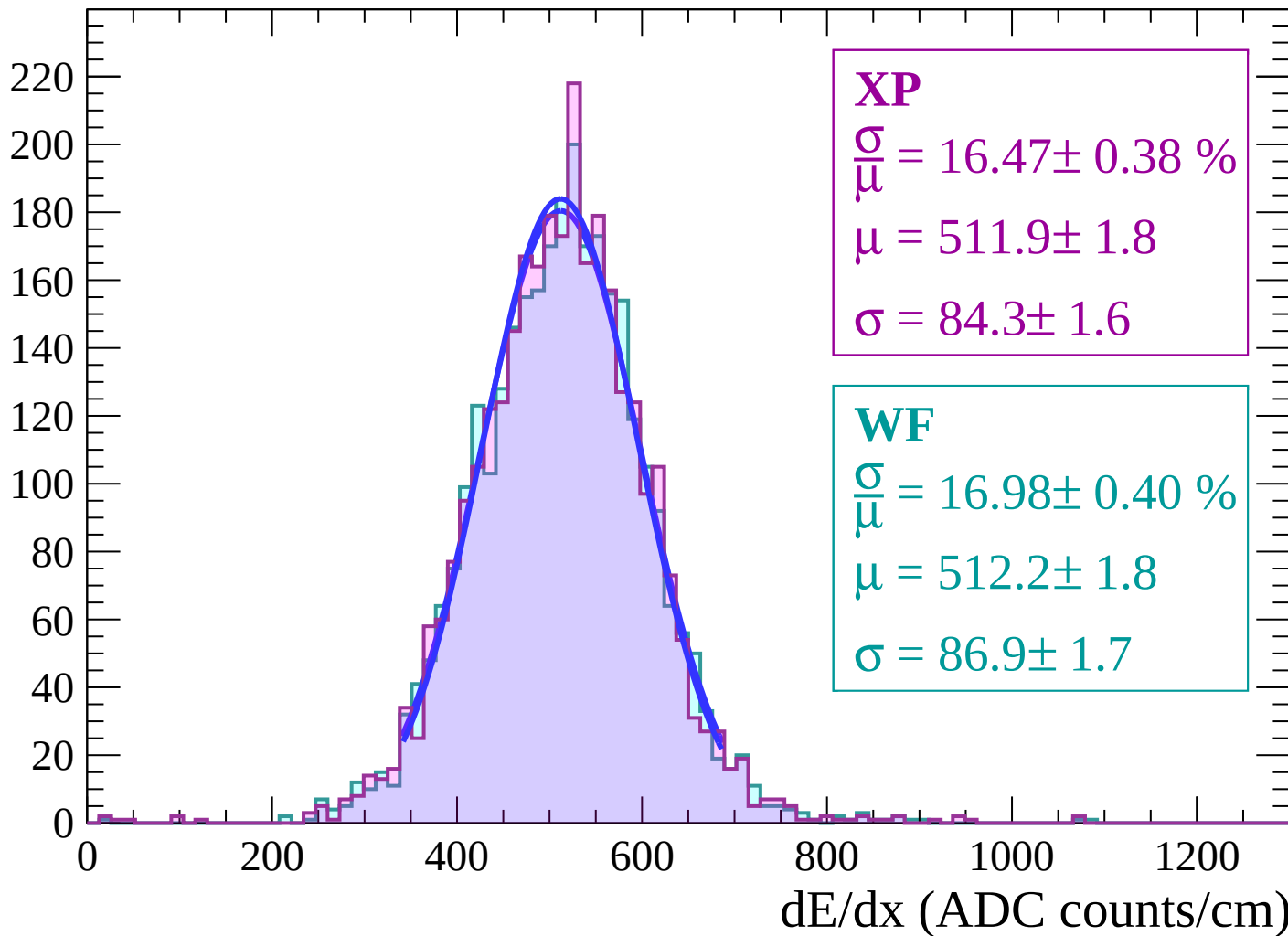
Energy loss | $-46 < \theta < -44$

Count



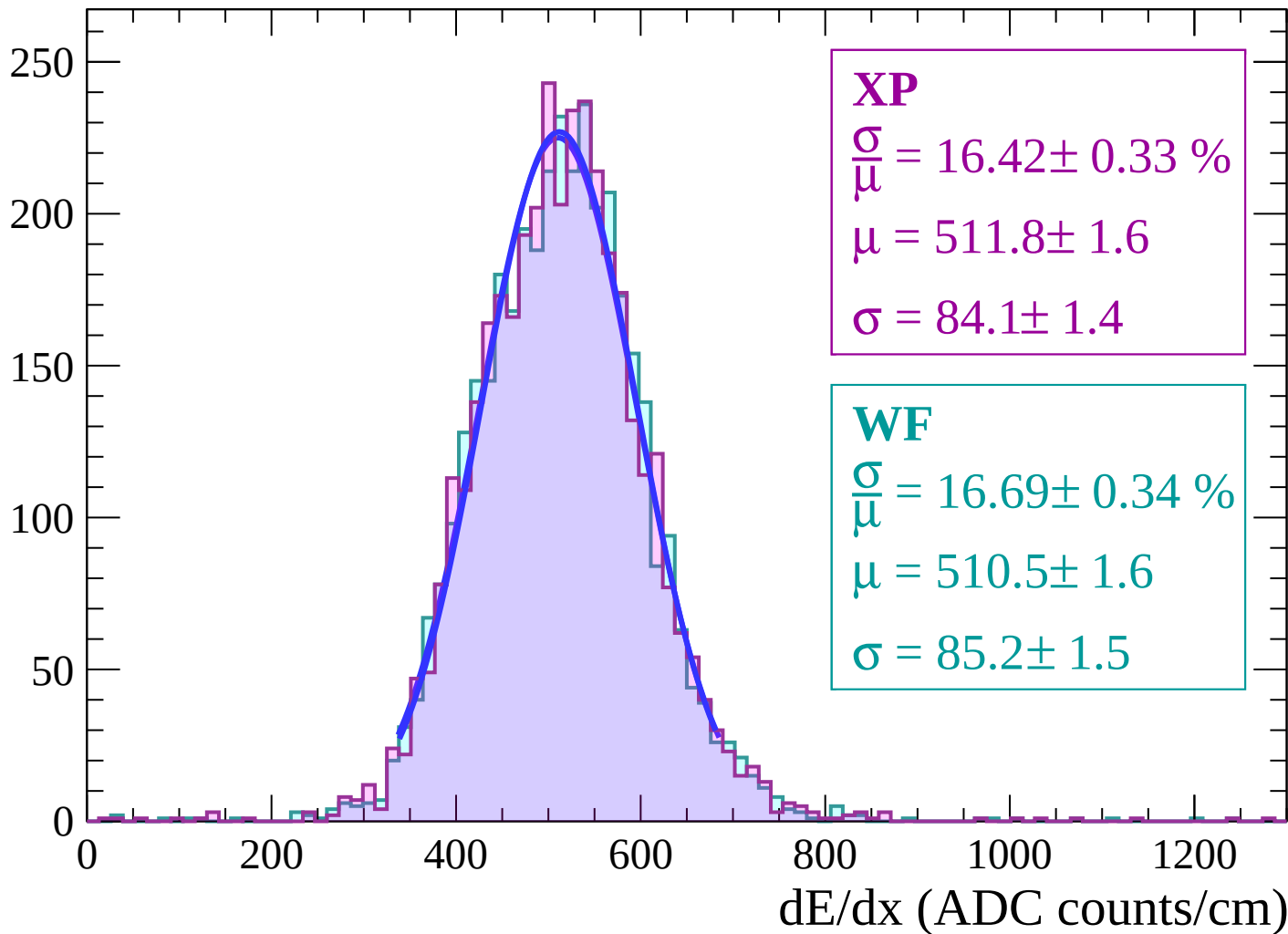
Energy loss | $-44 < \theta < -42$

Count



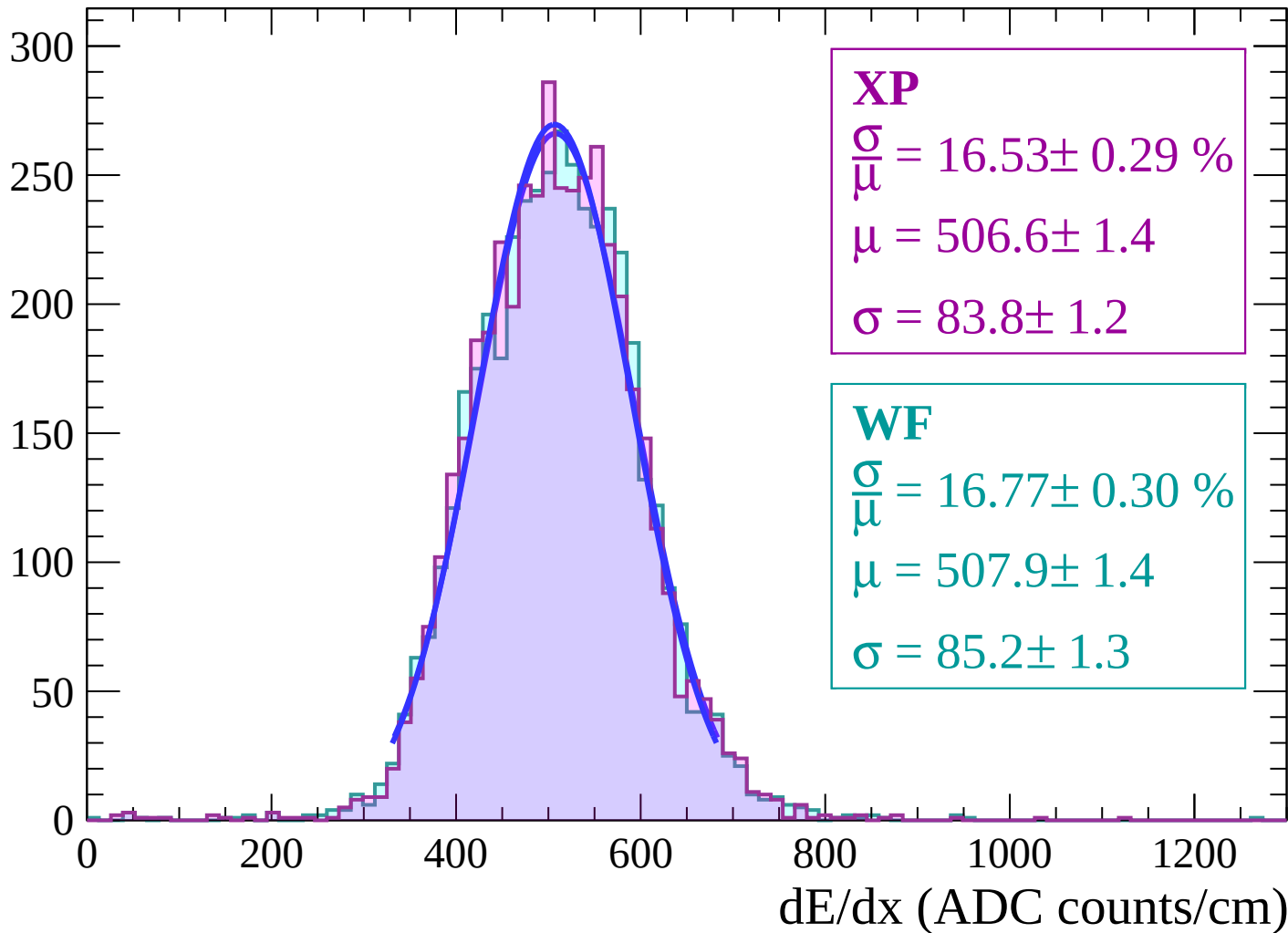
Energy loss | $-42 < \theta < -40$

Count



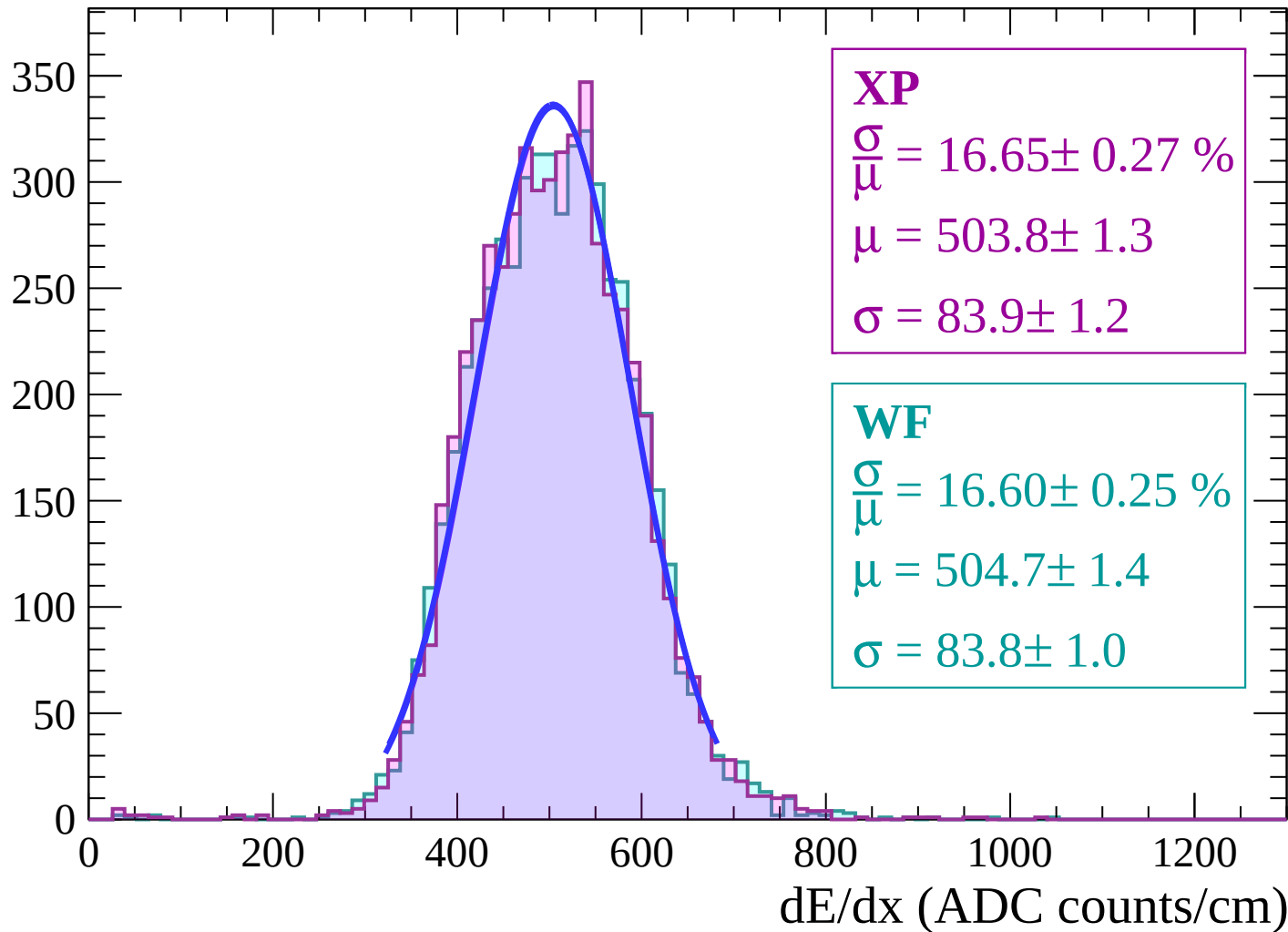
Energy loss | $-40 < \theta < -38$

Count



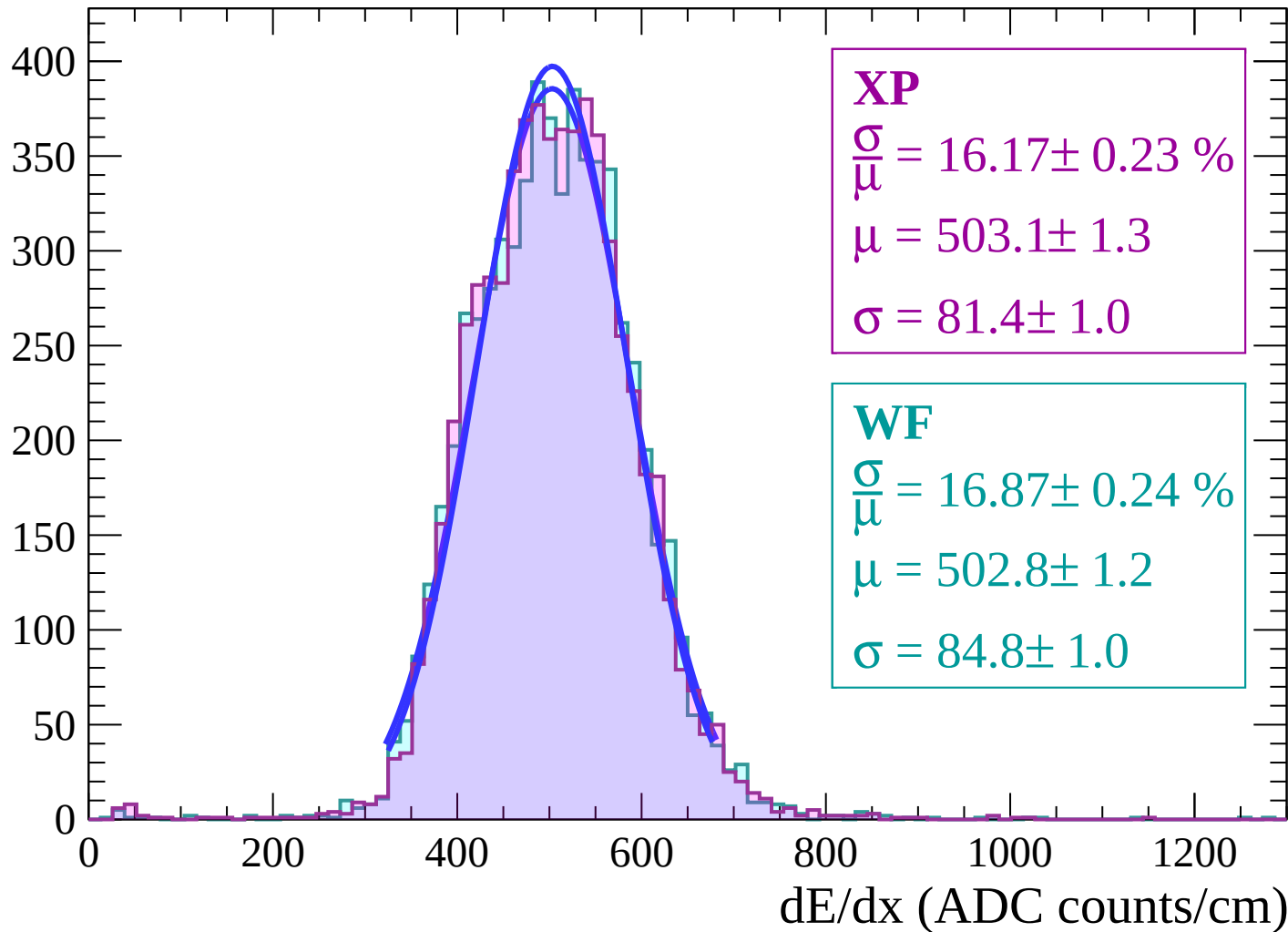
Energy loss | $-38 < \theta < -36$

Count



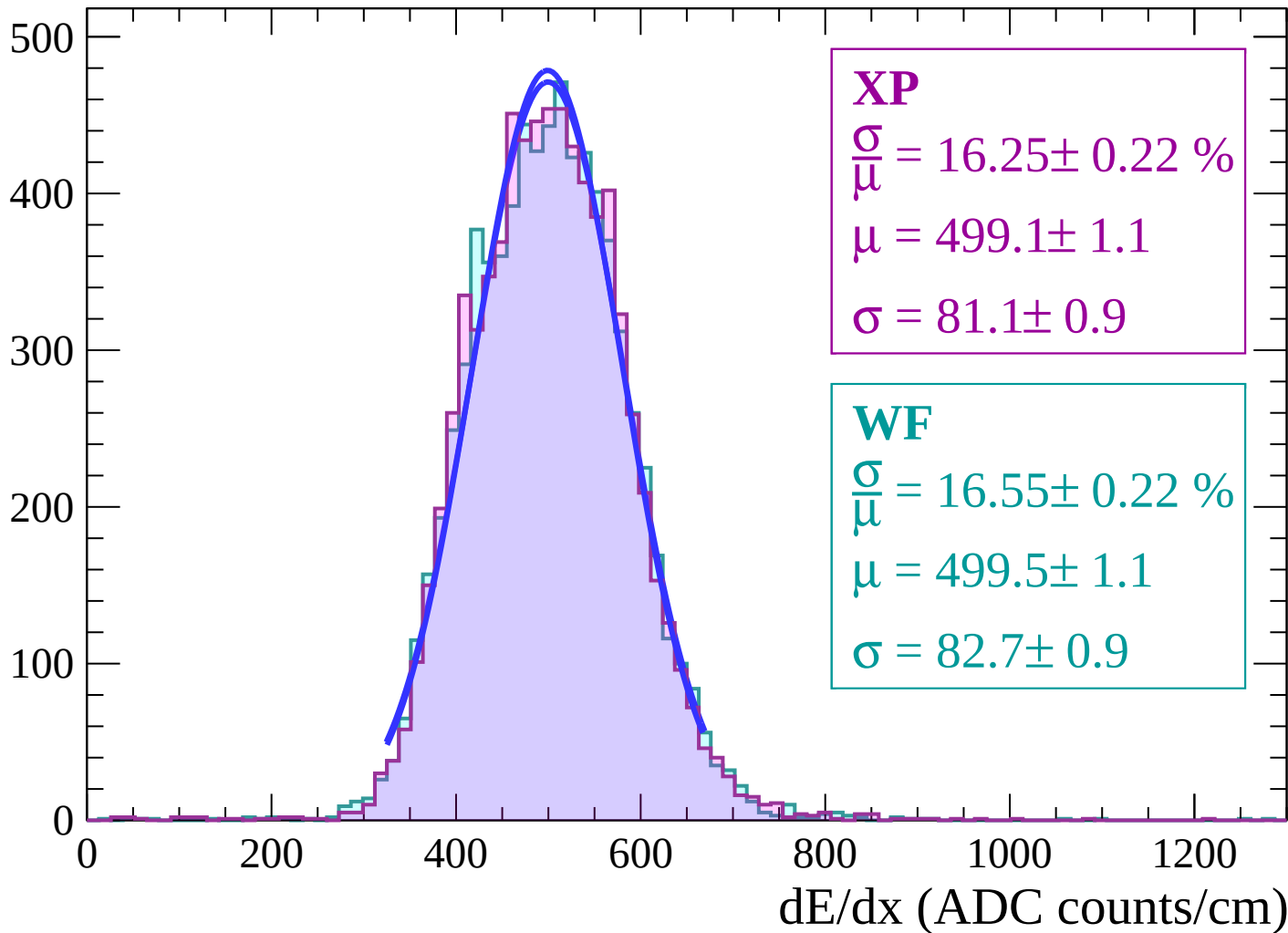
Energy loss | $-36 < \theta < -34$

Count



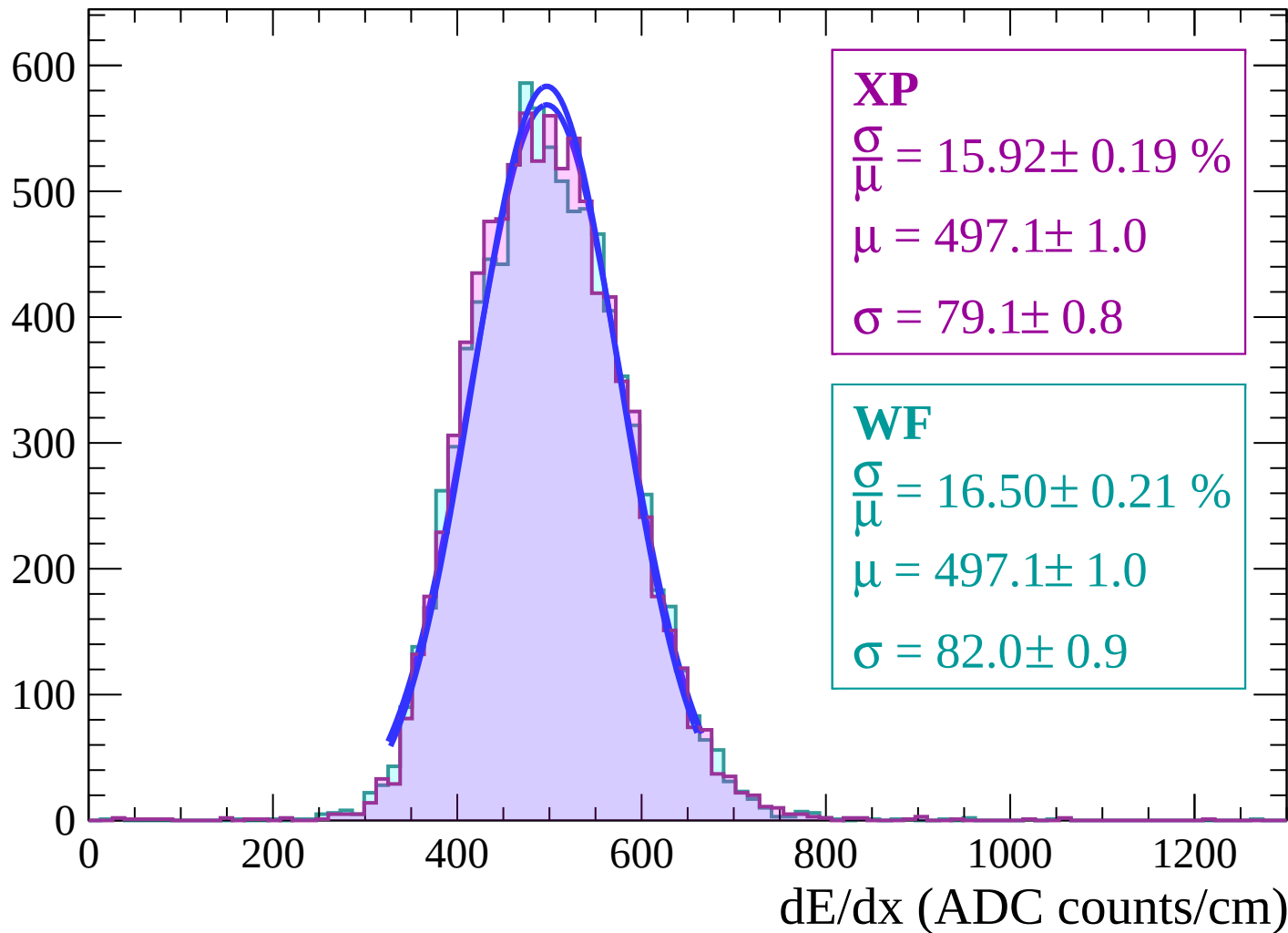
Energy loss | $-34 < \theta < -32$

Count



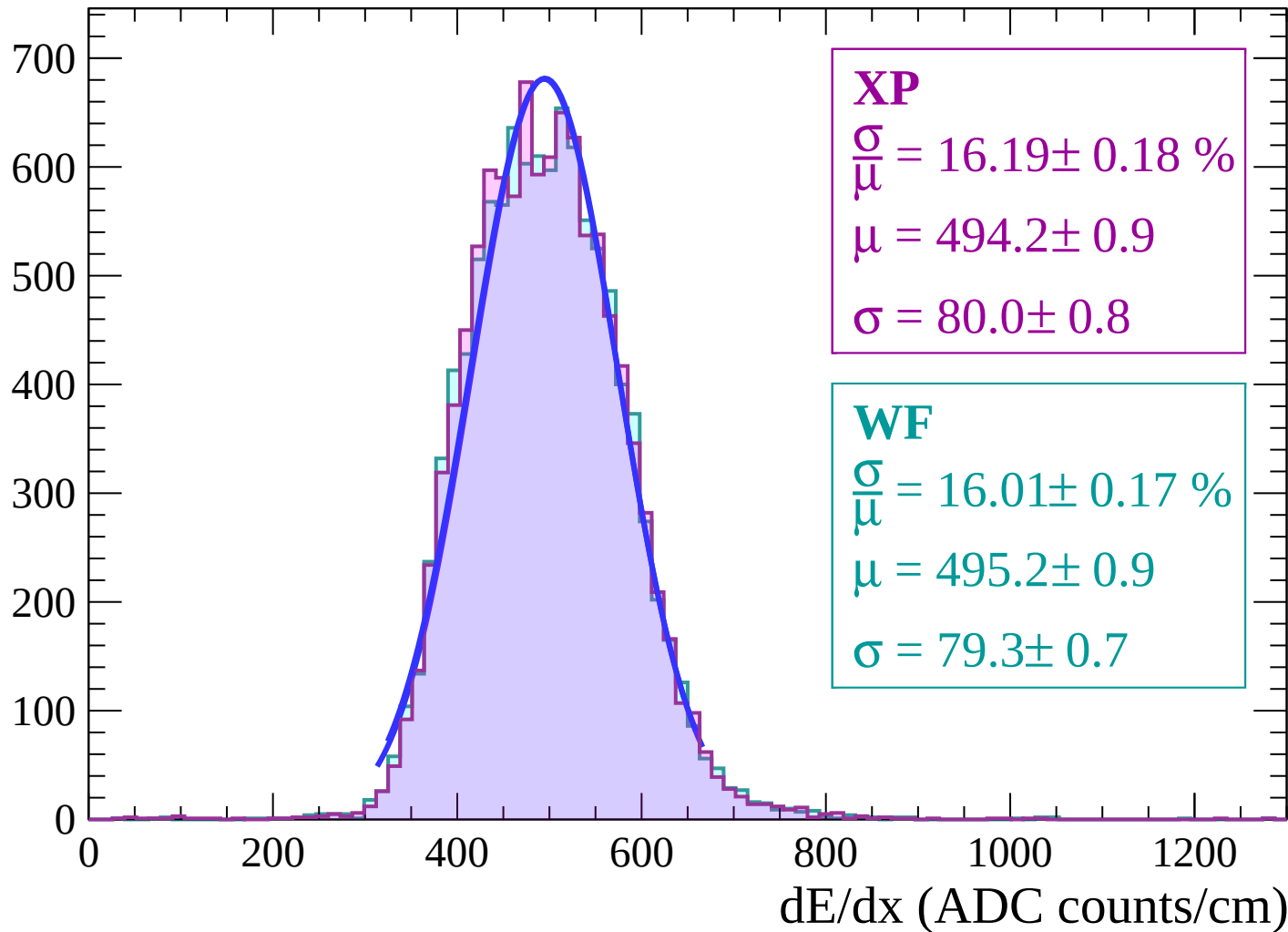
Energy loss | $-32 < \theta < -30$

Count

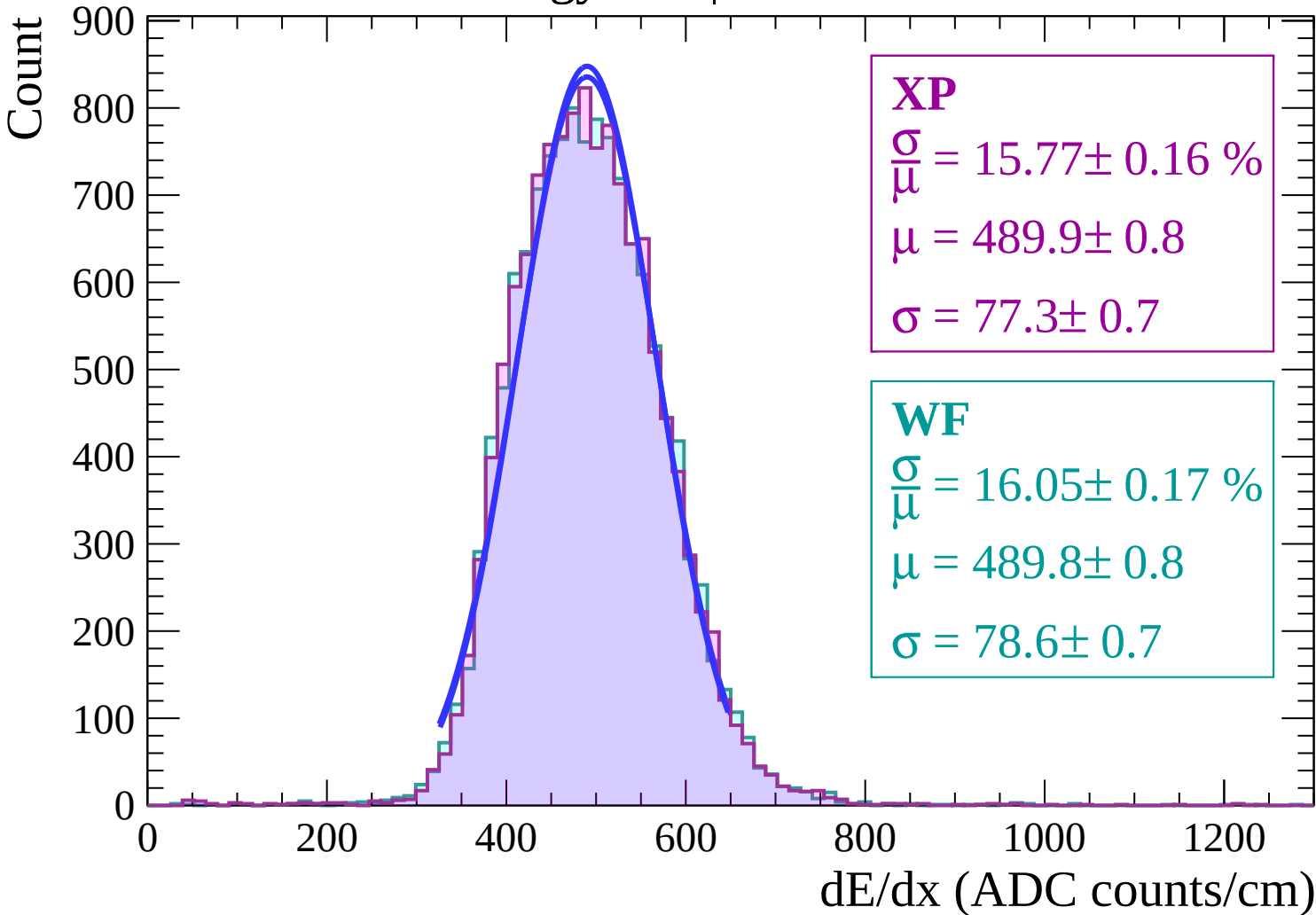


Energy loss | $-30 < \theta < -28$

Count



Energy loss | $-28 < \theta < -26$



Energy loss | $-26 < \theta < -24$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 15.55 \pm 0.15 \%$$

$$\mu = 490.9 \pm 0.7$$

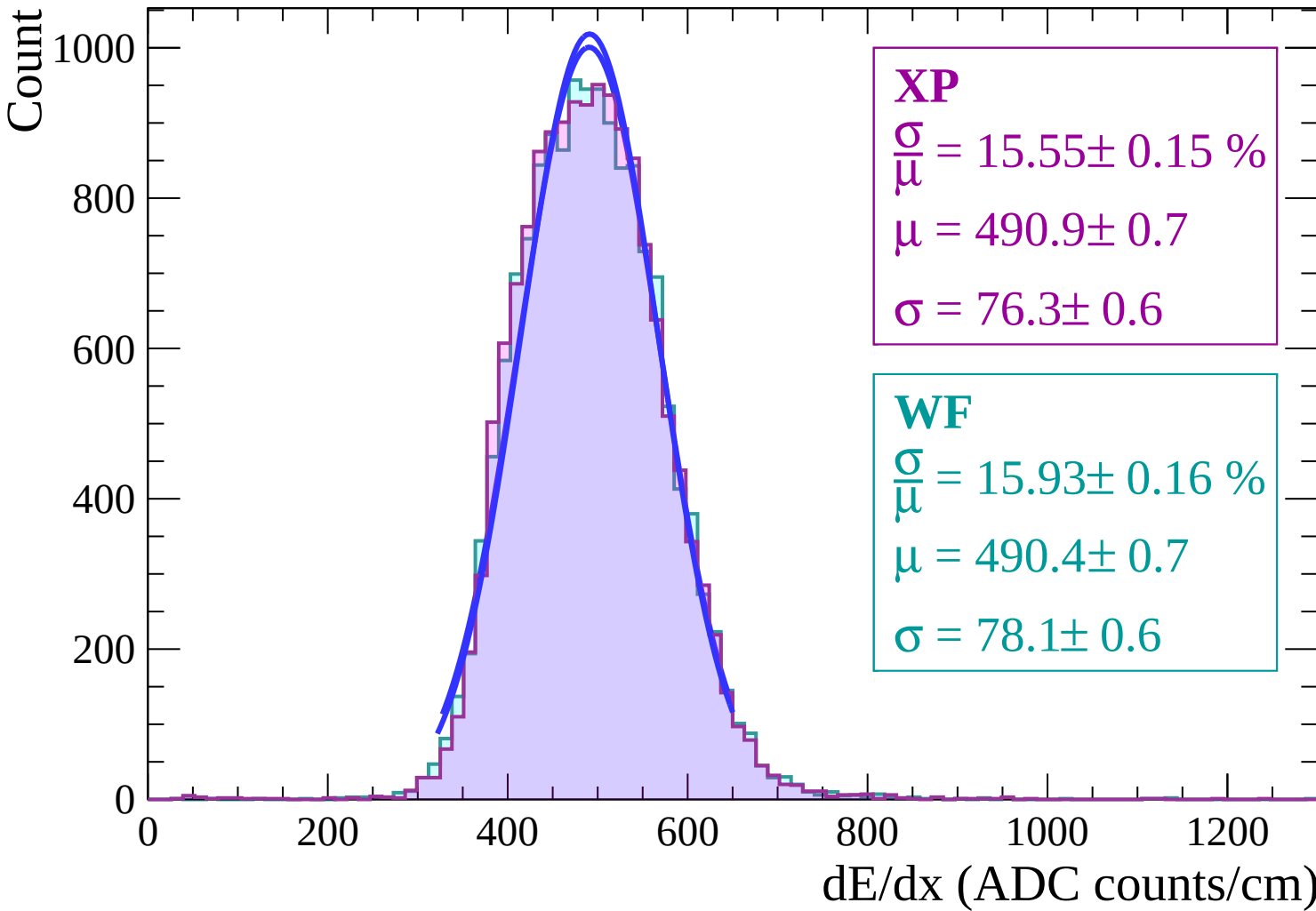
$$\sigma = 76.3 \pm 0.6$$

WF

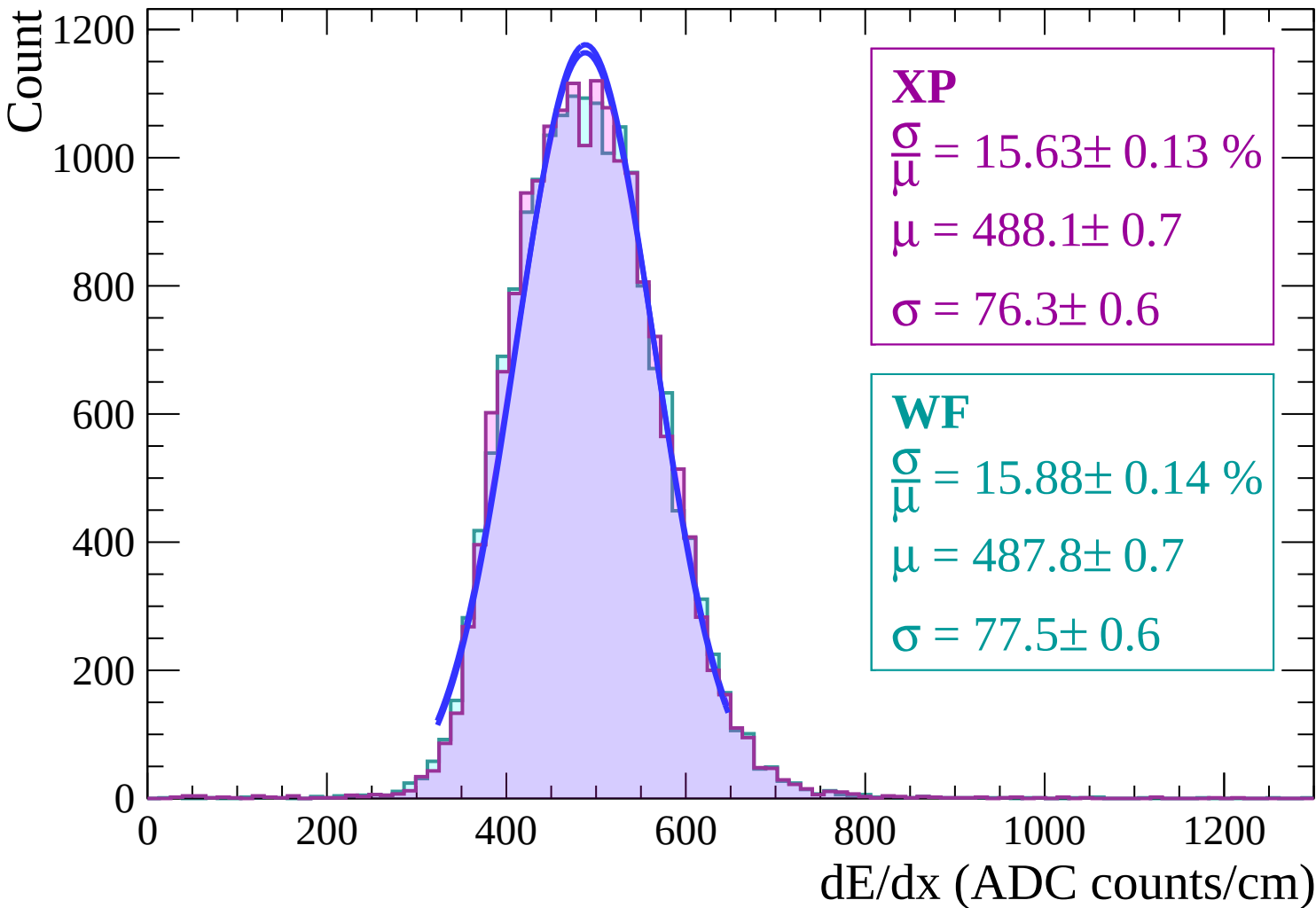
$$\frac{\sigma}{\mu} = 15.93 \pm 0.16 \%$$

$$\mu = 490.4 \pm 0.7$$

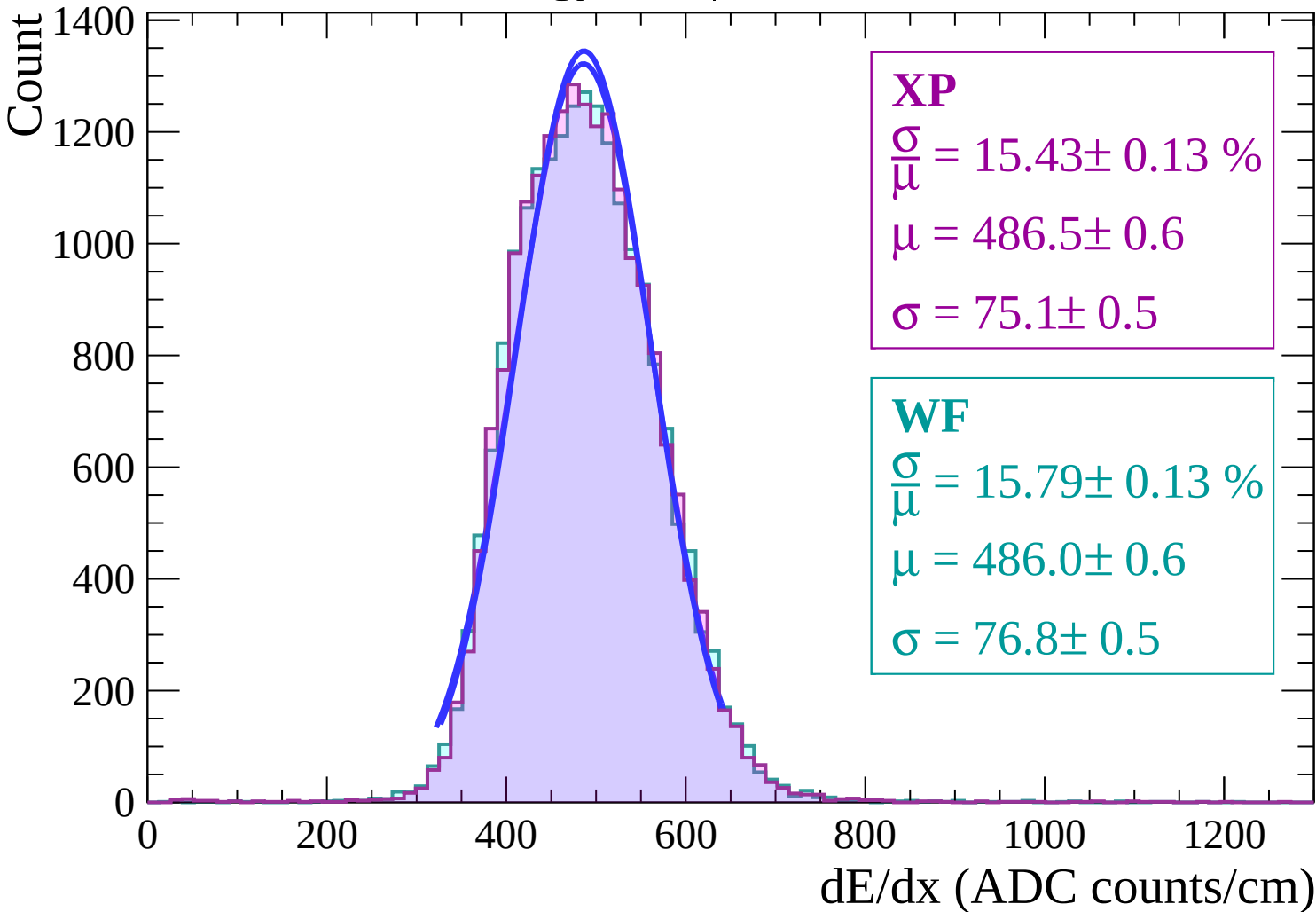
$$\sigma = 78.1 \pm 0.6$$



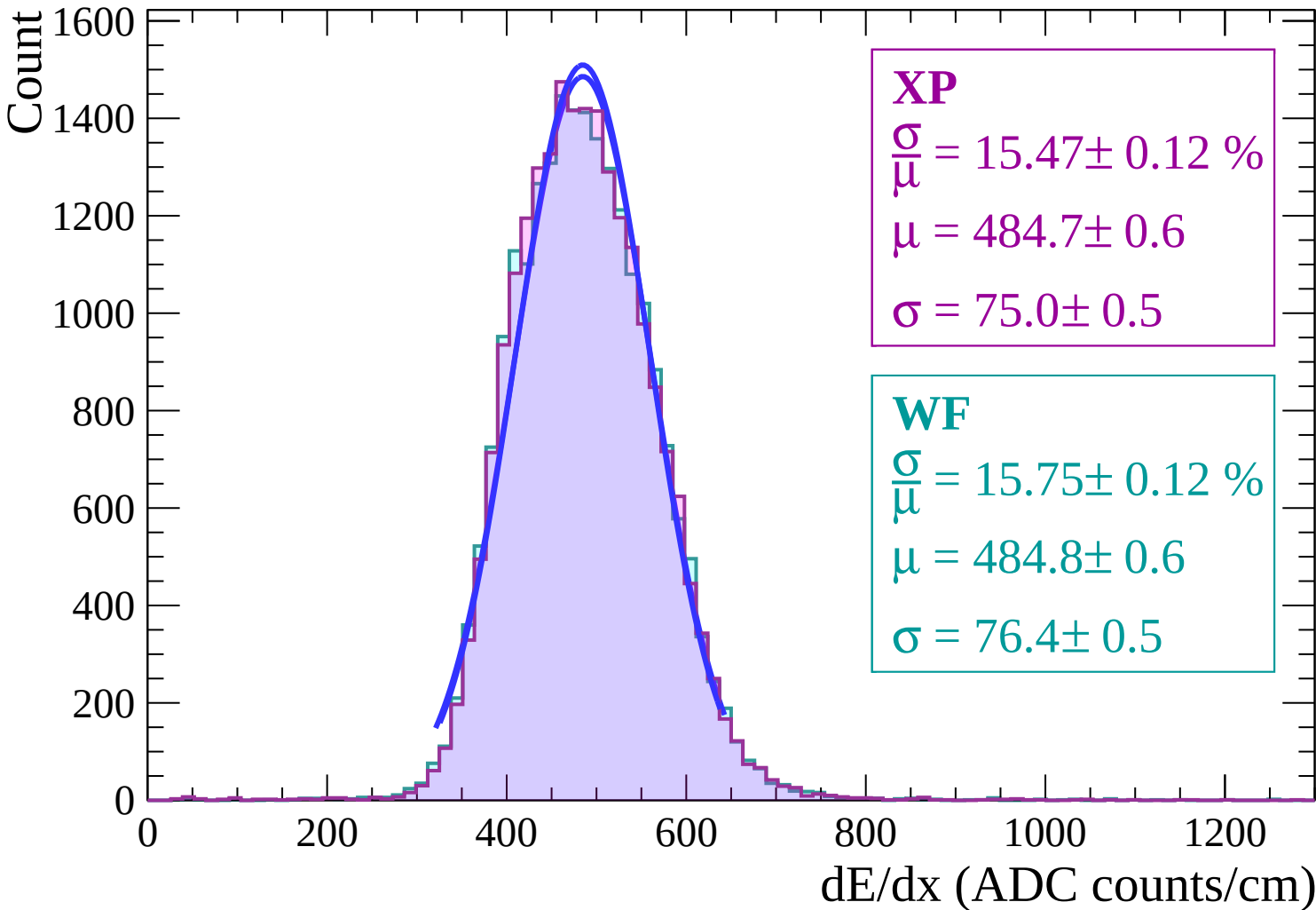
Energy loss | $-24 < \theta < -22$



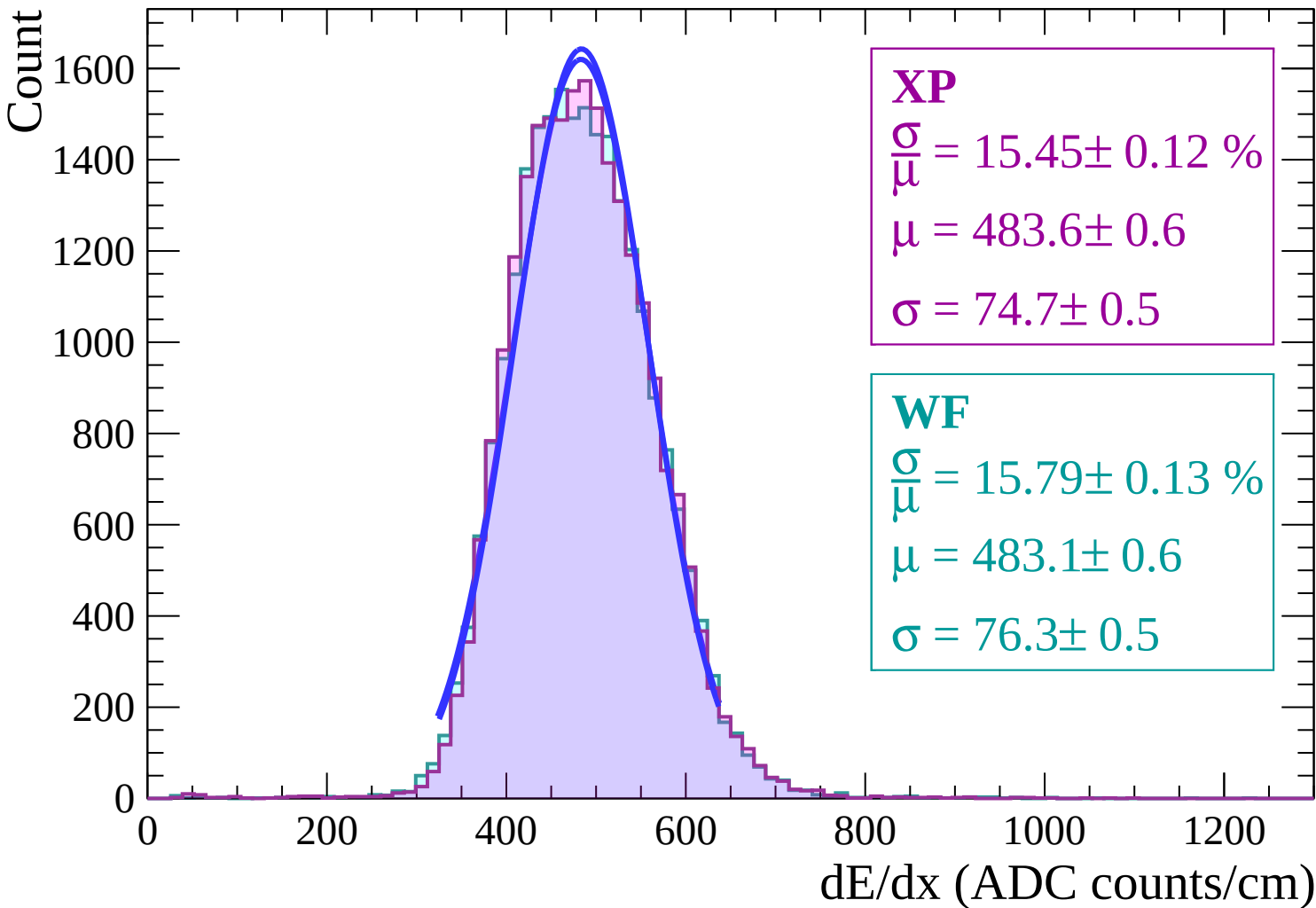
Energy loss | $-22 < \theta < -20$



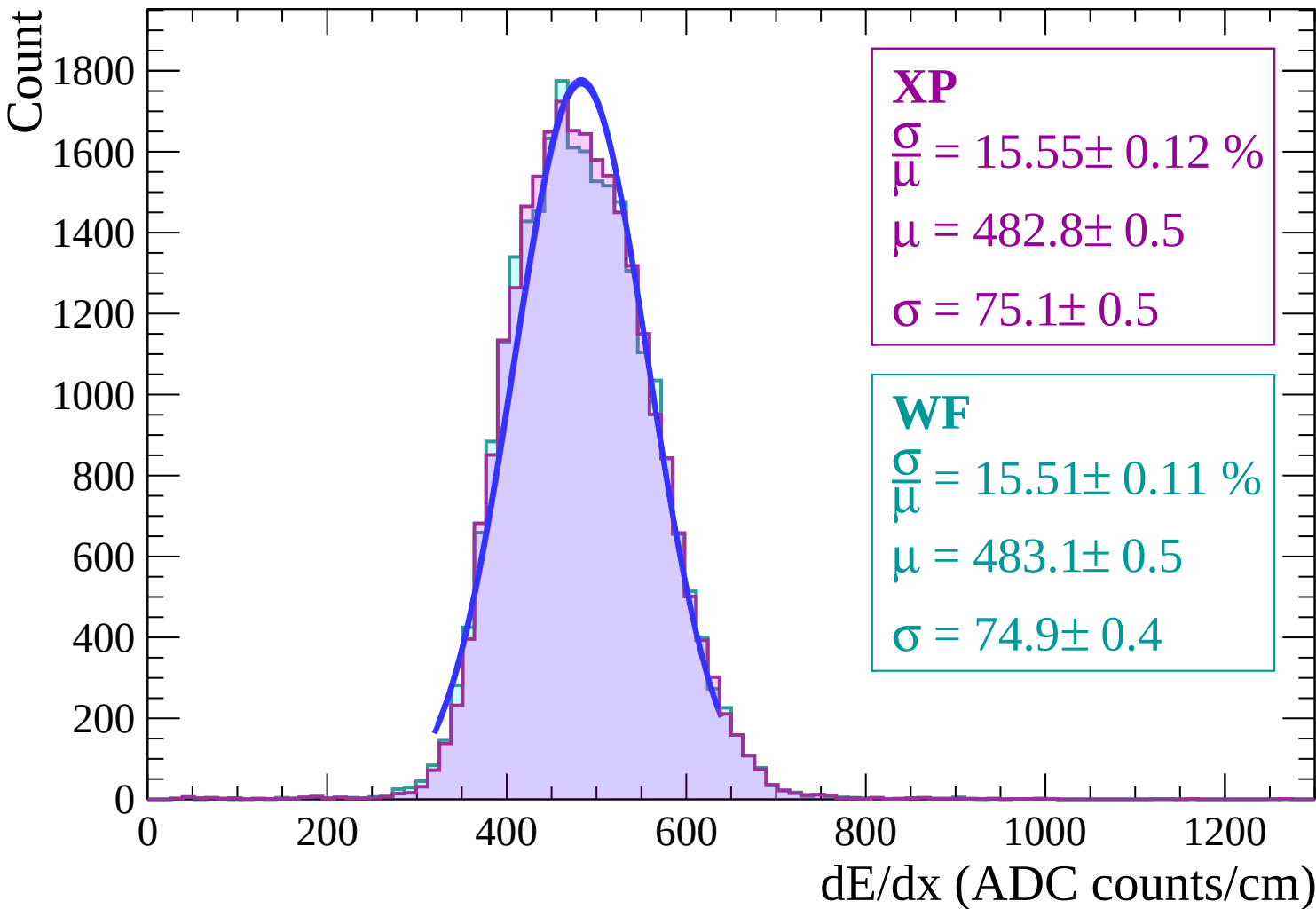
Energy loss | $-20 < \theta < -18$



Energy loss | $-18 < \theta < -16$



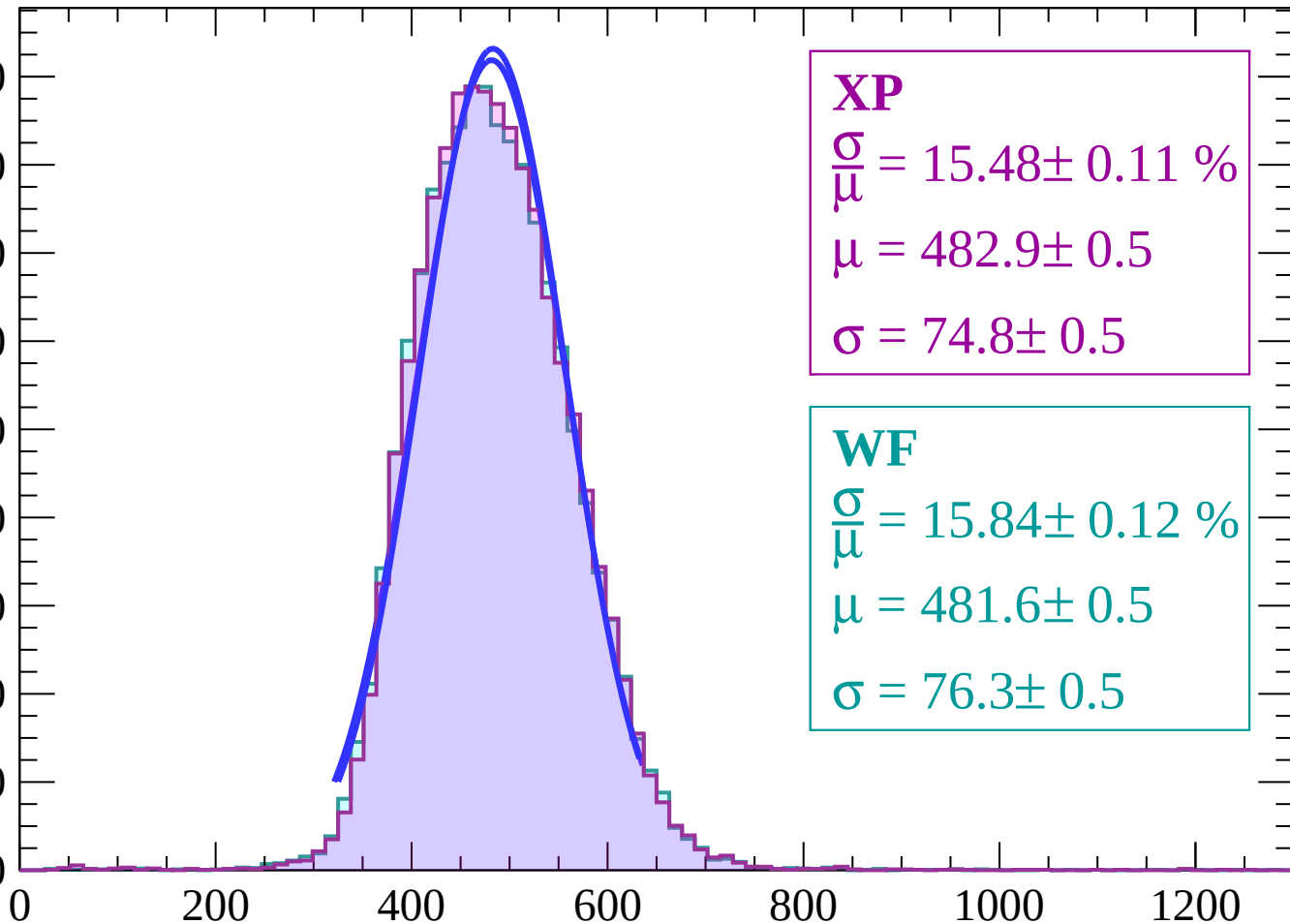
Energy loss | $-16 < \theta < -14$



Energy loss | $-14 < \theta < -12$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 15.48 \pm 0.11 \%$$

$$\mu = 482.9 \pm 0.5$$

$$\sigma = 74.8 \pm 0.5$$

WF

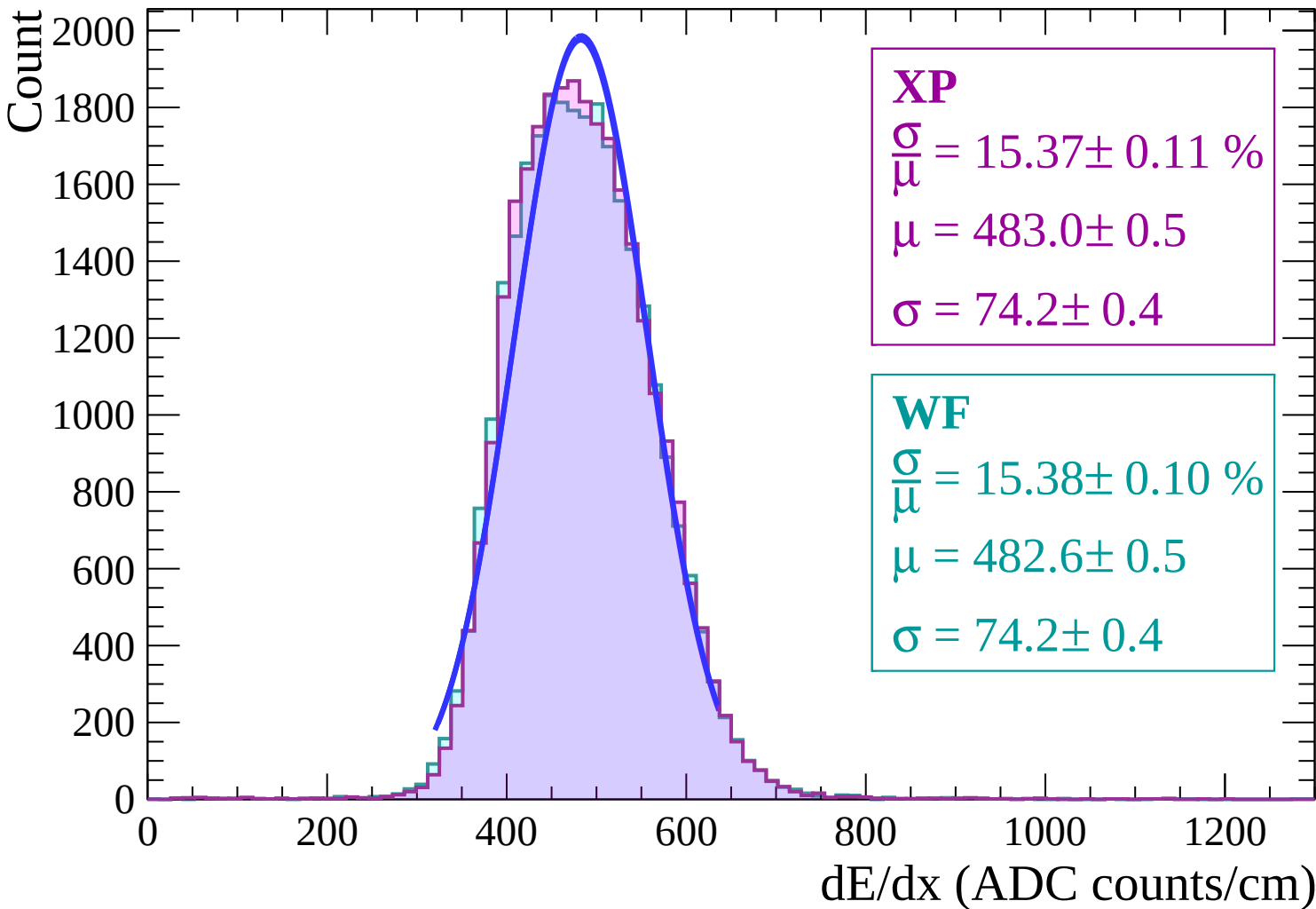
$$\frac{\sigma}{\mu} = 15.84 \pm 0.12 \%$$

$$\mu = 481.6 \pm 0.5$$

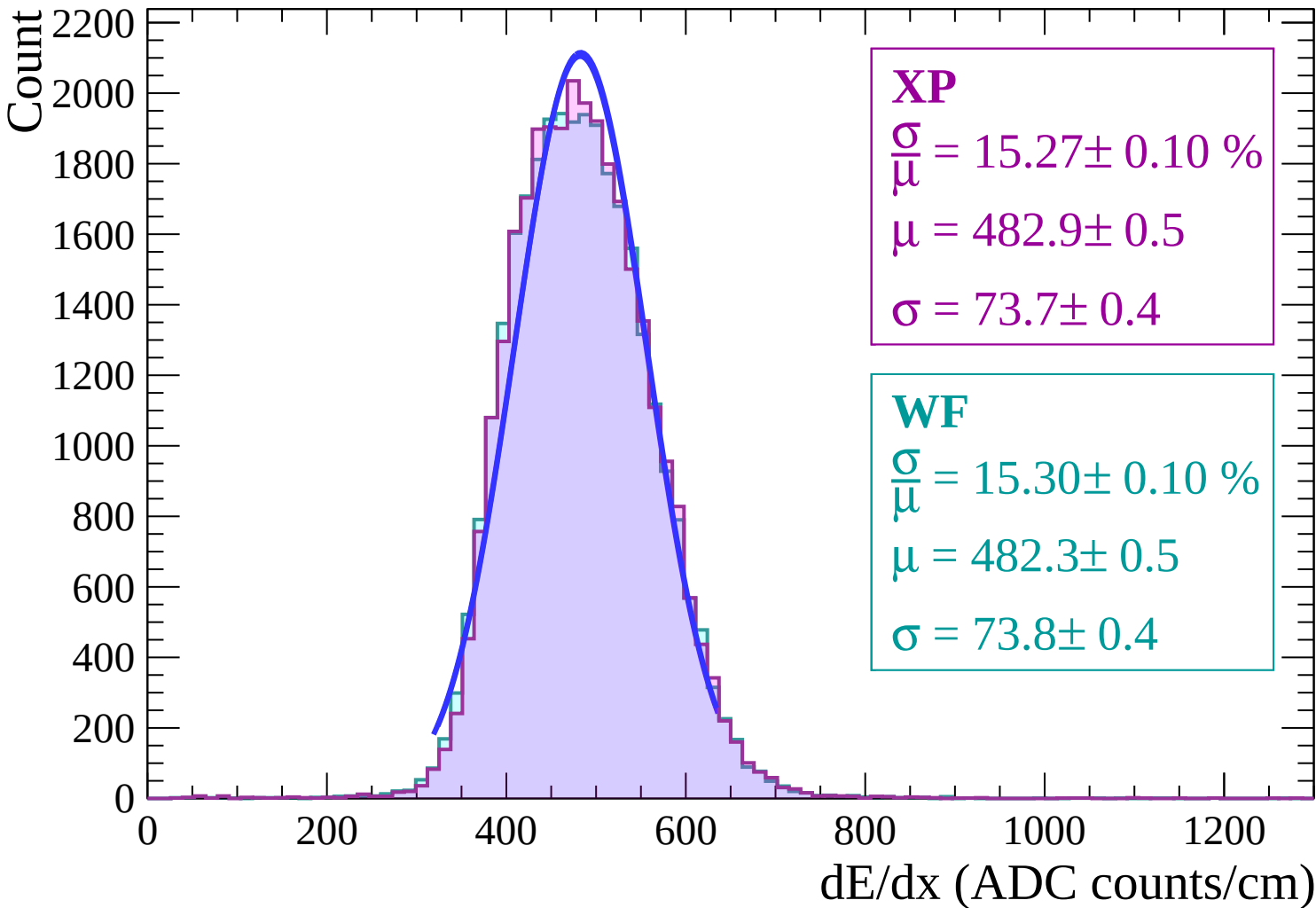
$$\sigma = 76.3 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-12 < \theta < -10$

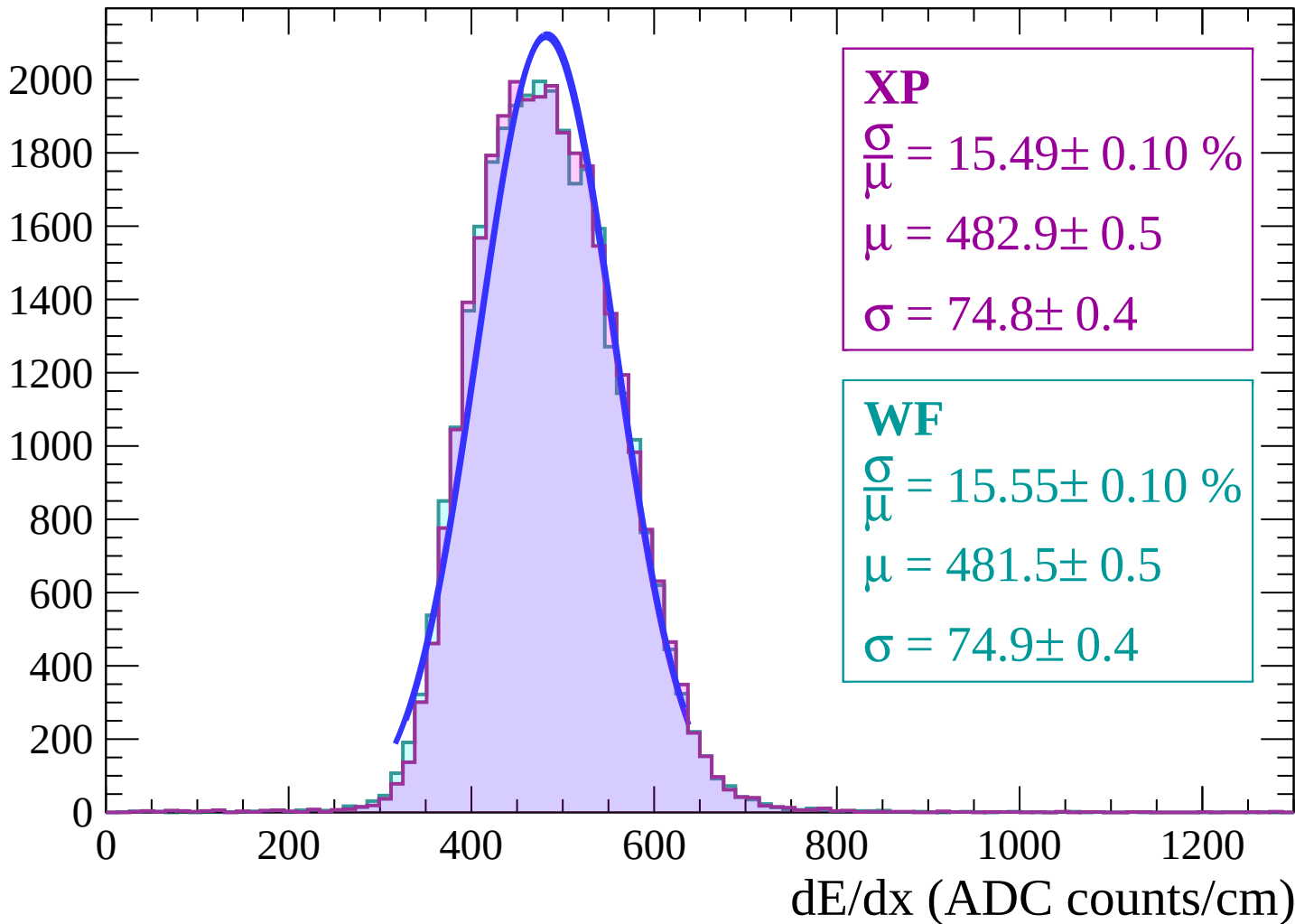


Energy loss | $-10 < \theta < -8$

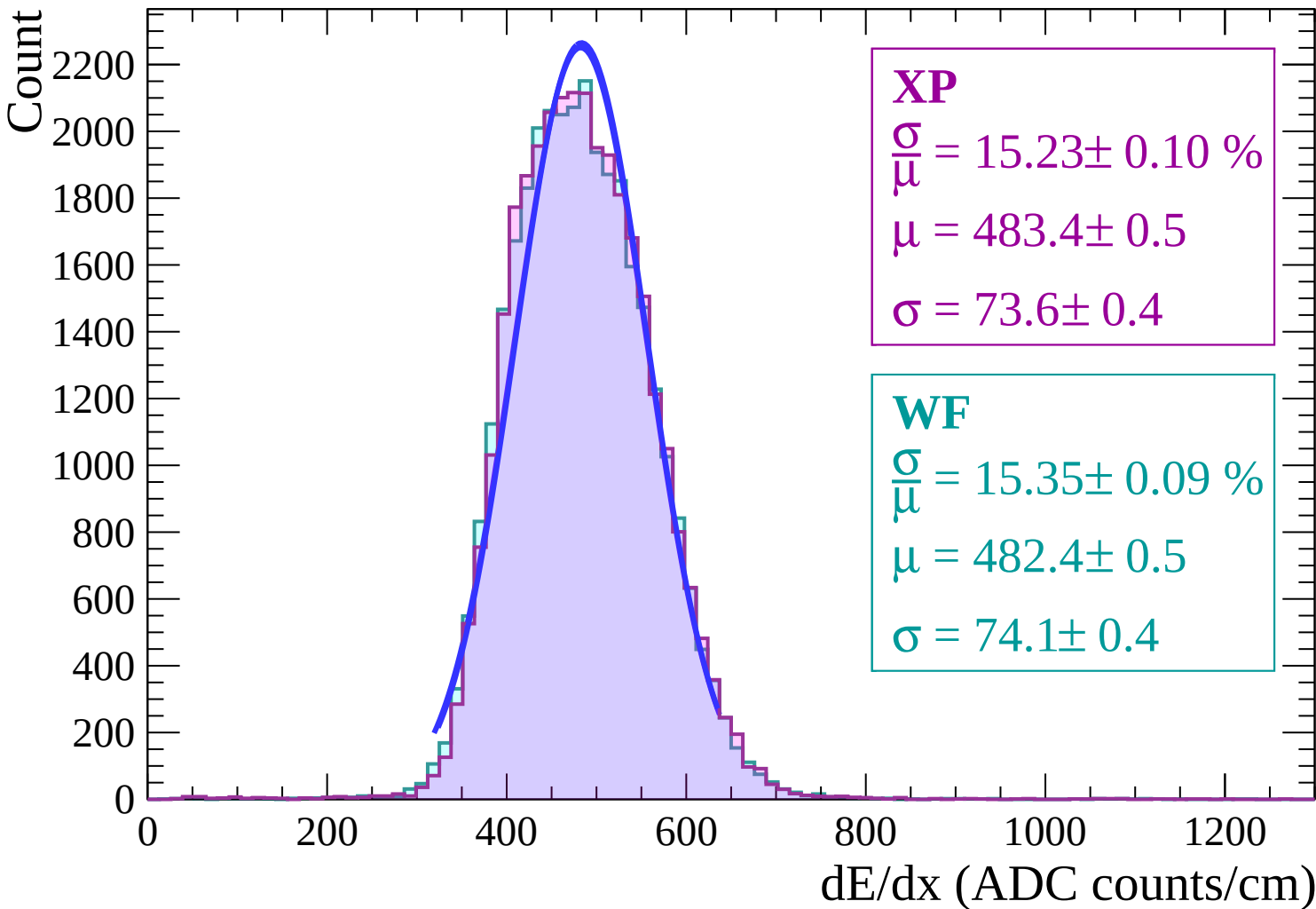


Energy loss | $-8 < \theta < -6$

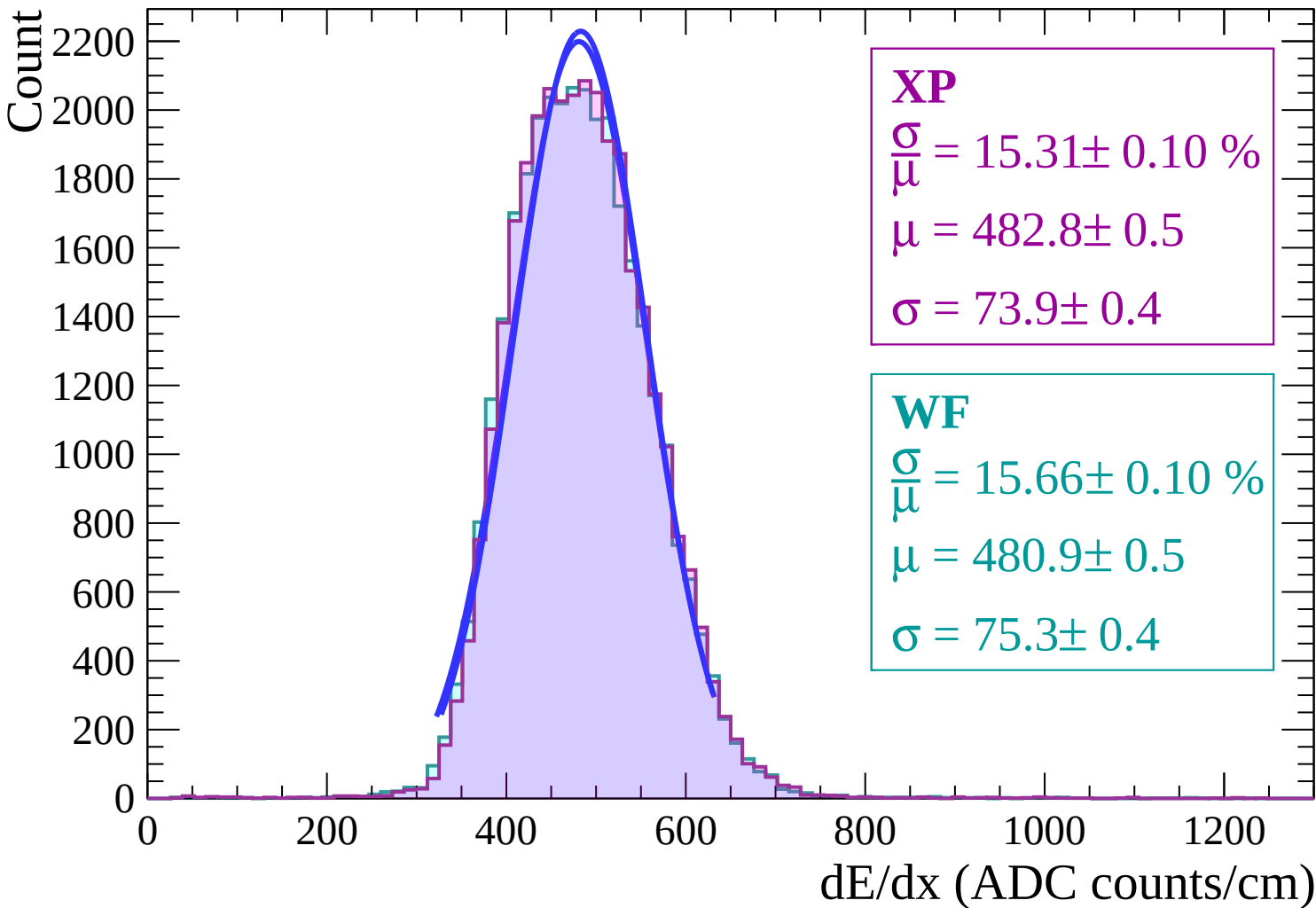
Count



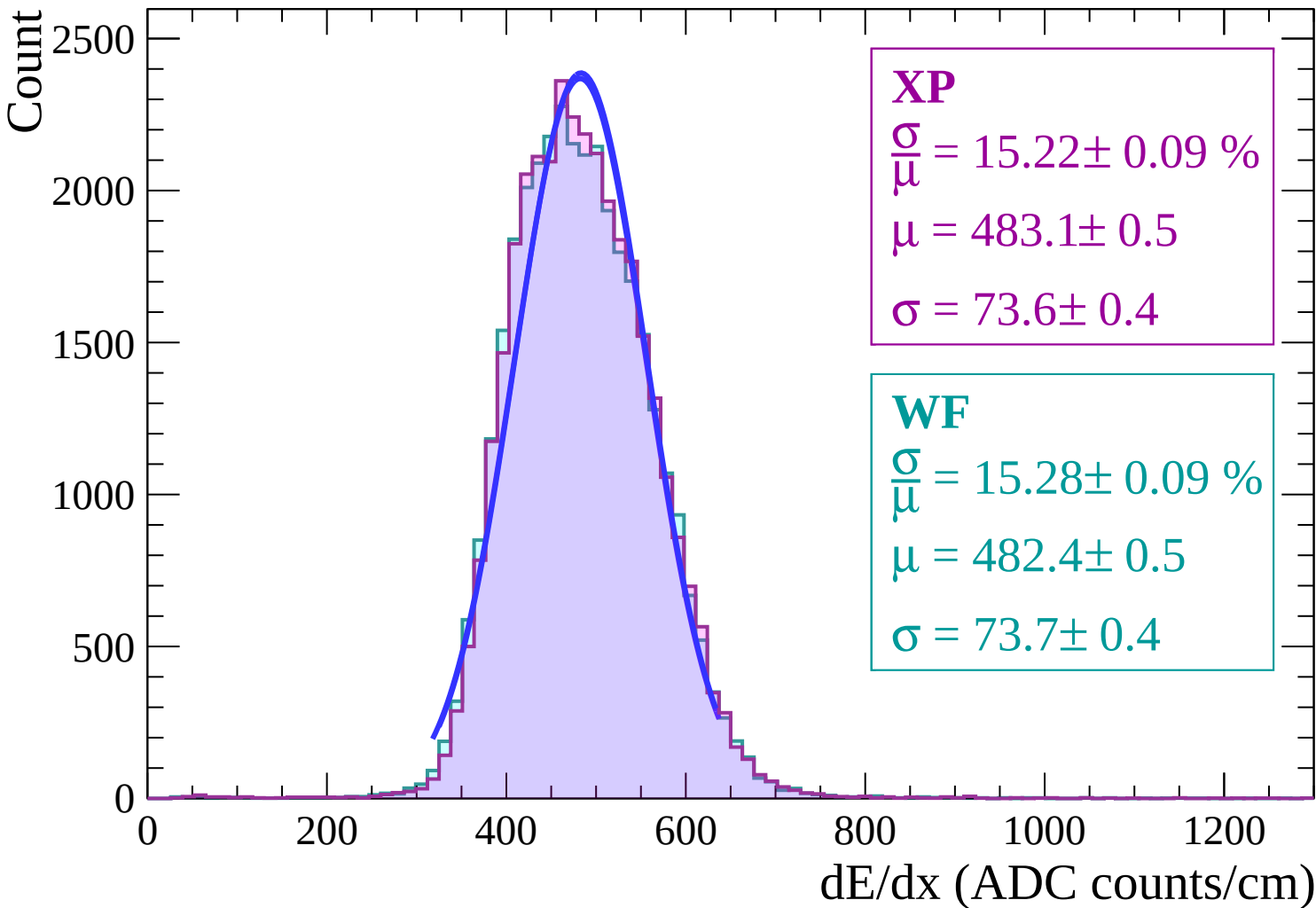
Energy loss | $-6 < \theta < -4$



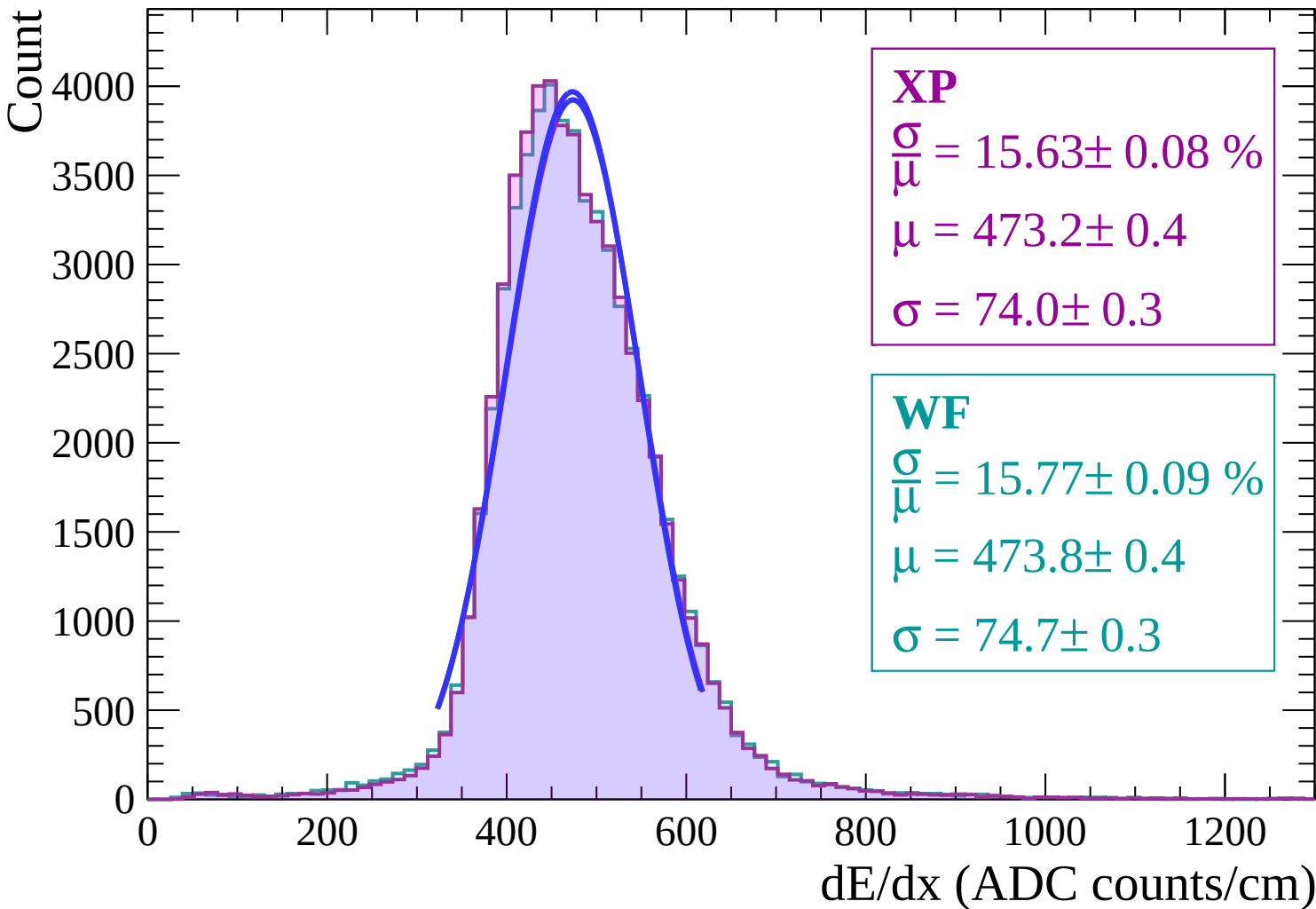
Energy loss | $-4 < \theta < -2$



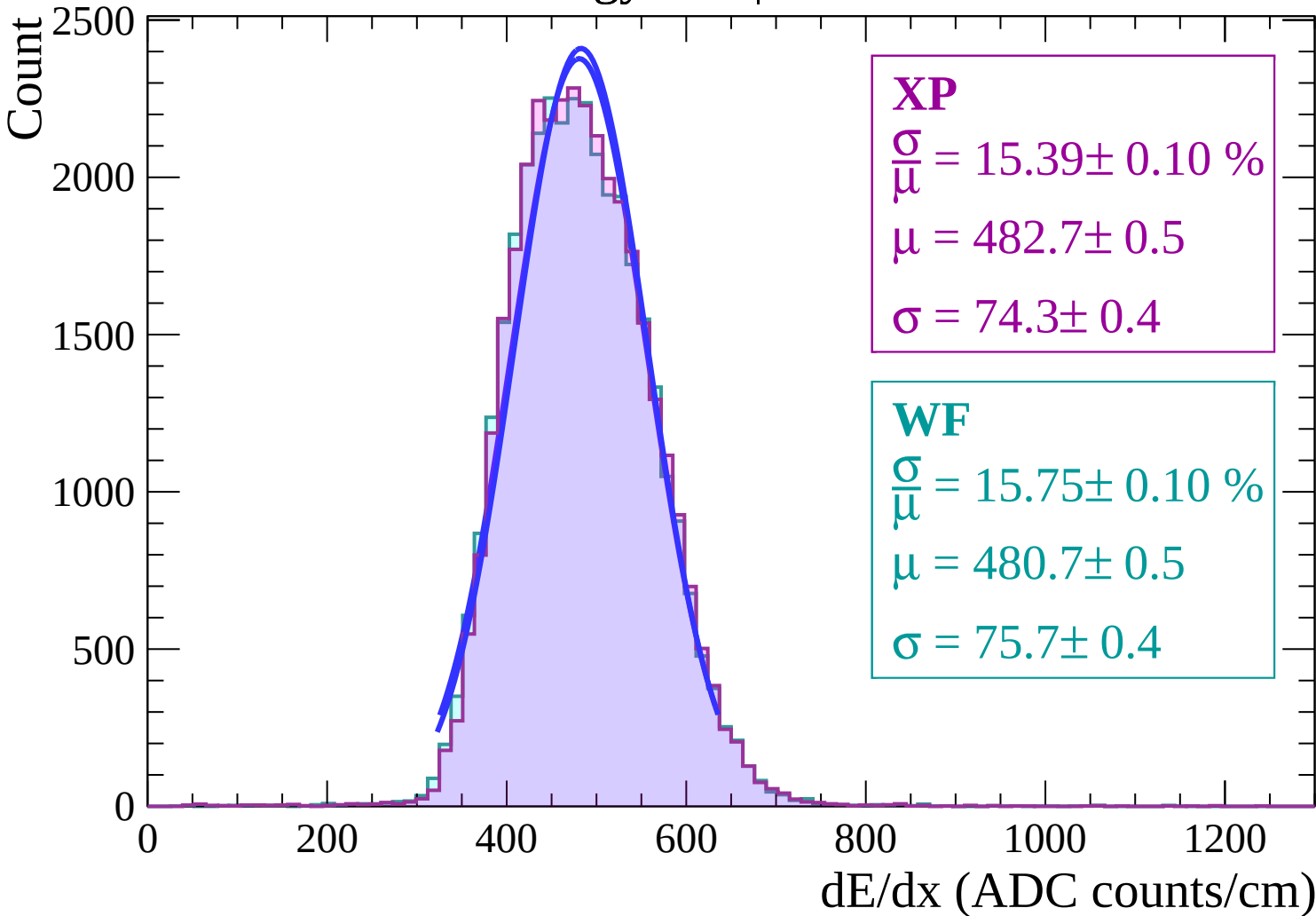
Energy loss | $-2 < \theta < 0$



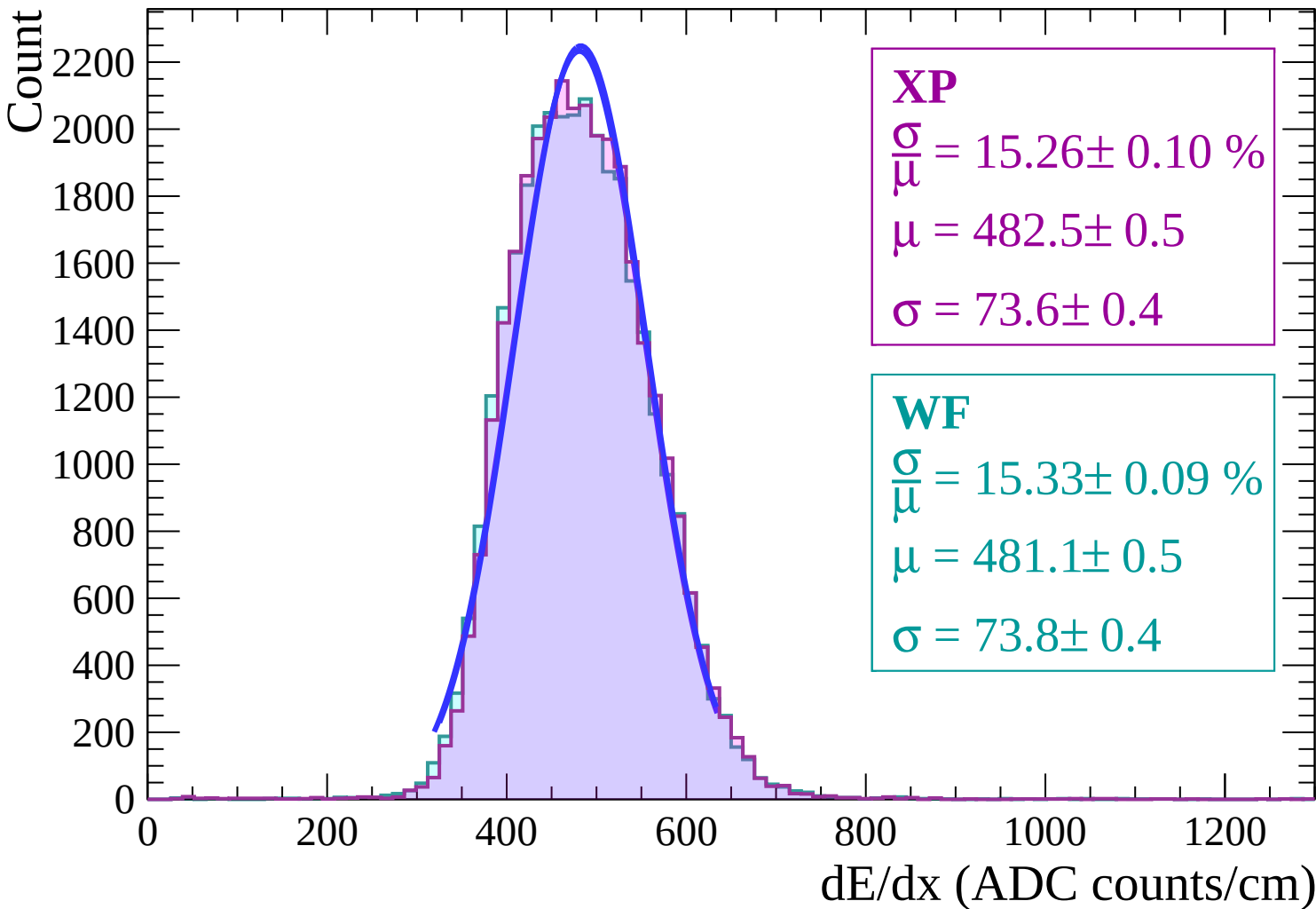
Energy loss | $0 < \theta < 2$



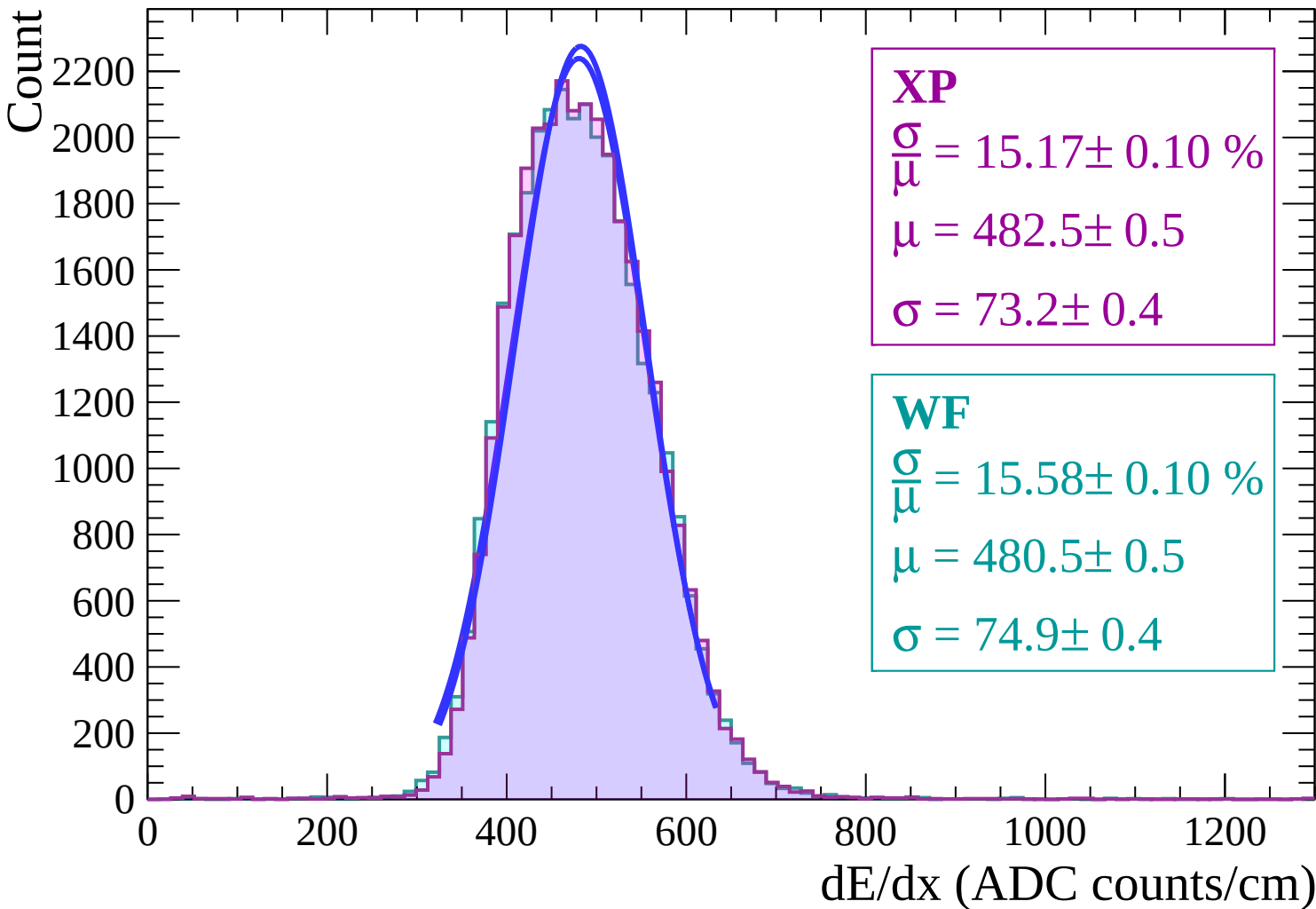
Energy loss | $2 < \theta < 4$



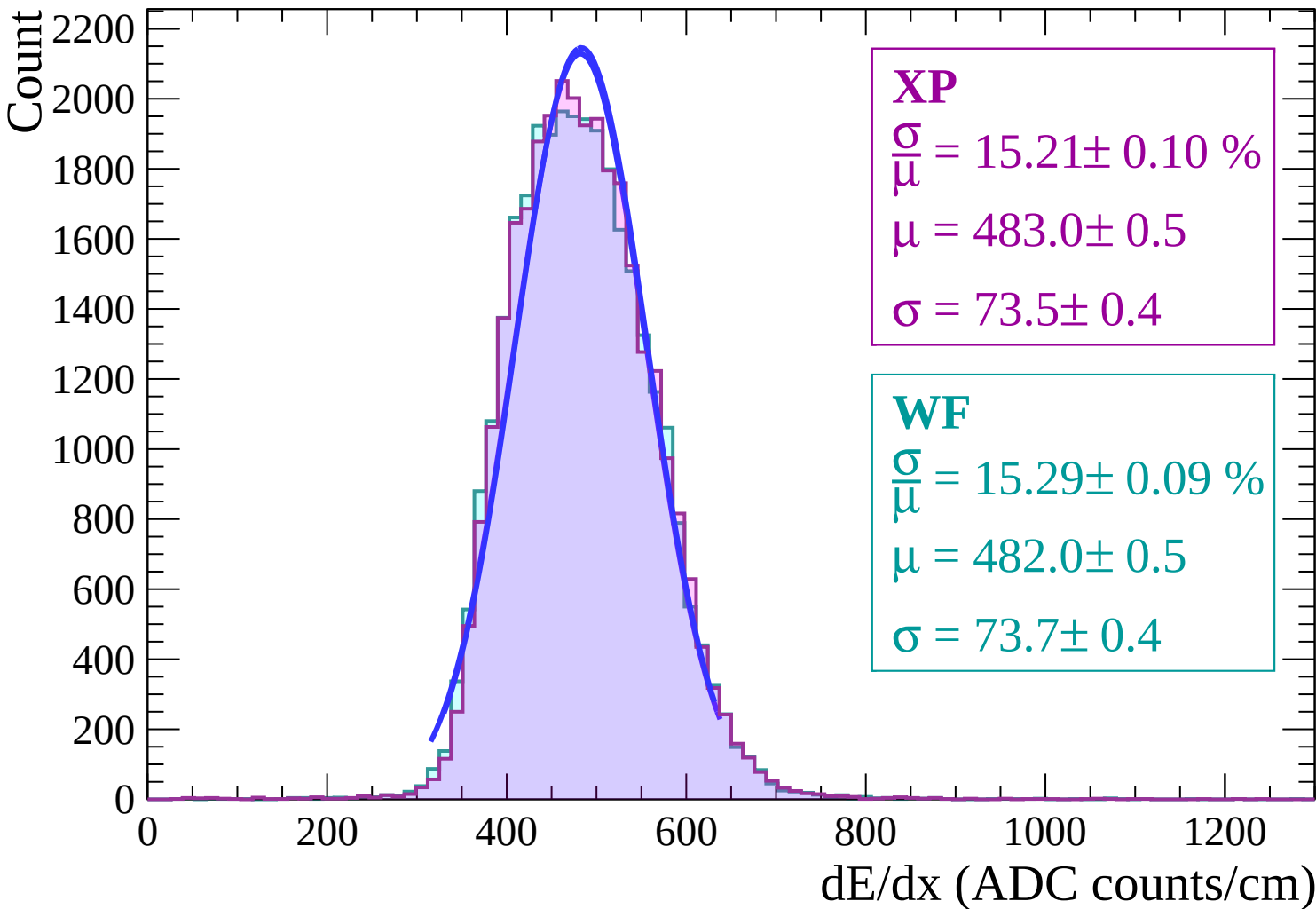
Energy loss | $4 < \theta < 6$



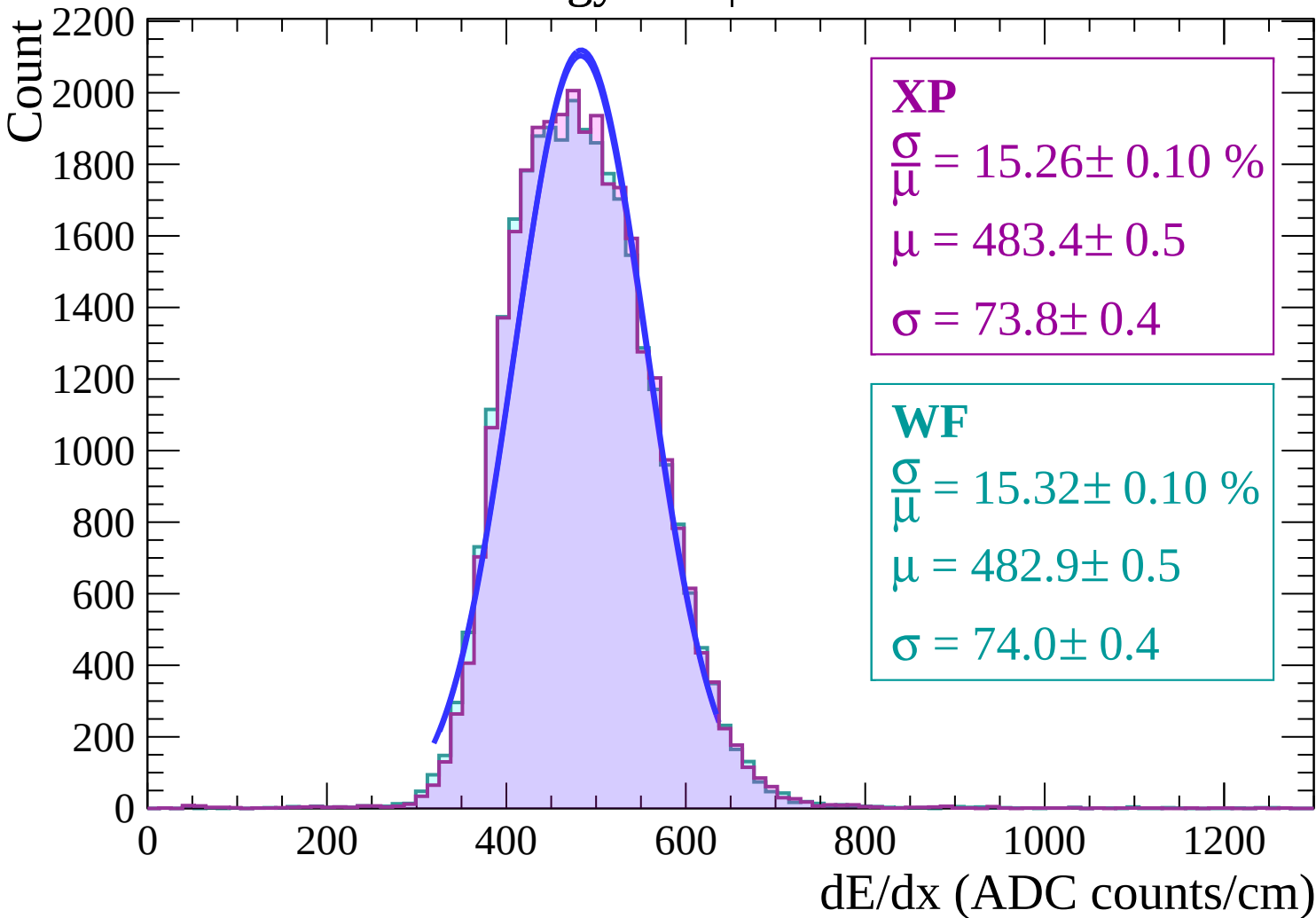
Energy loss | $6 < \theta < 8$



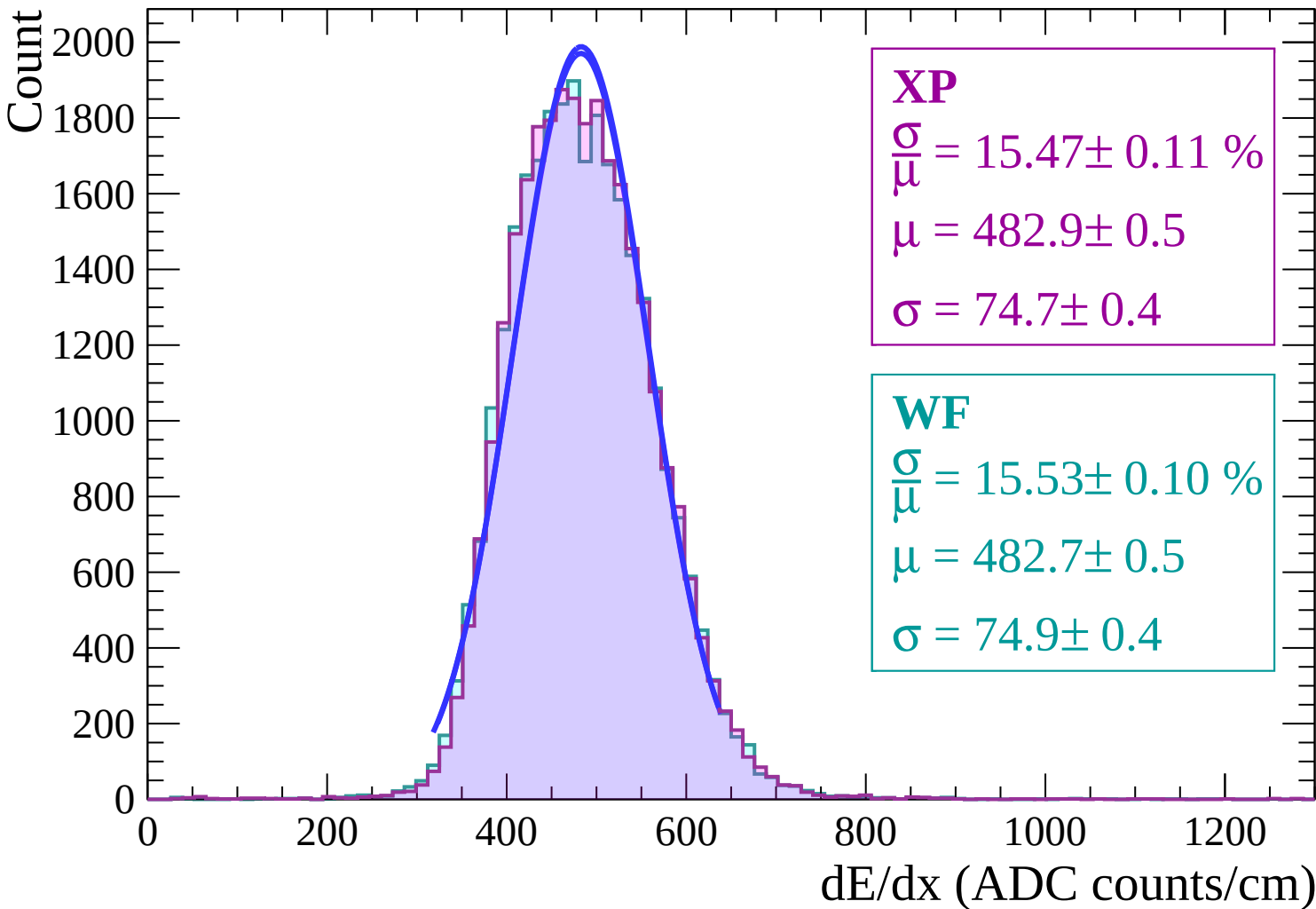
Energy loss | $8 < \theta < 10$



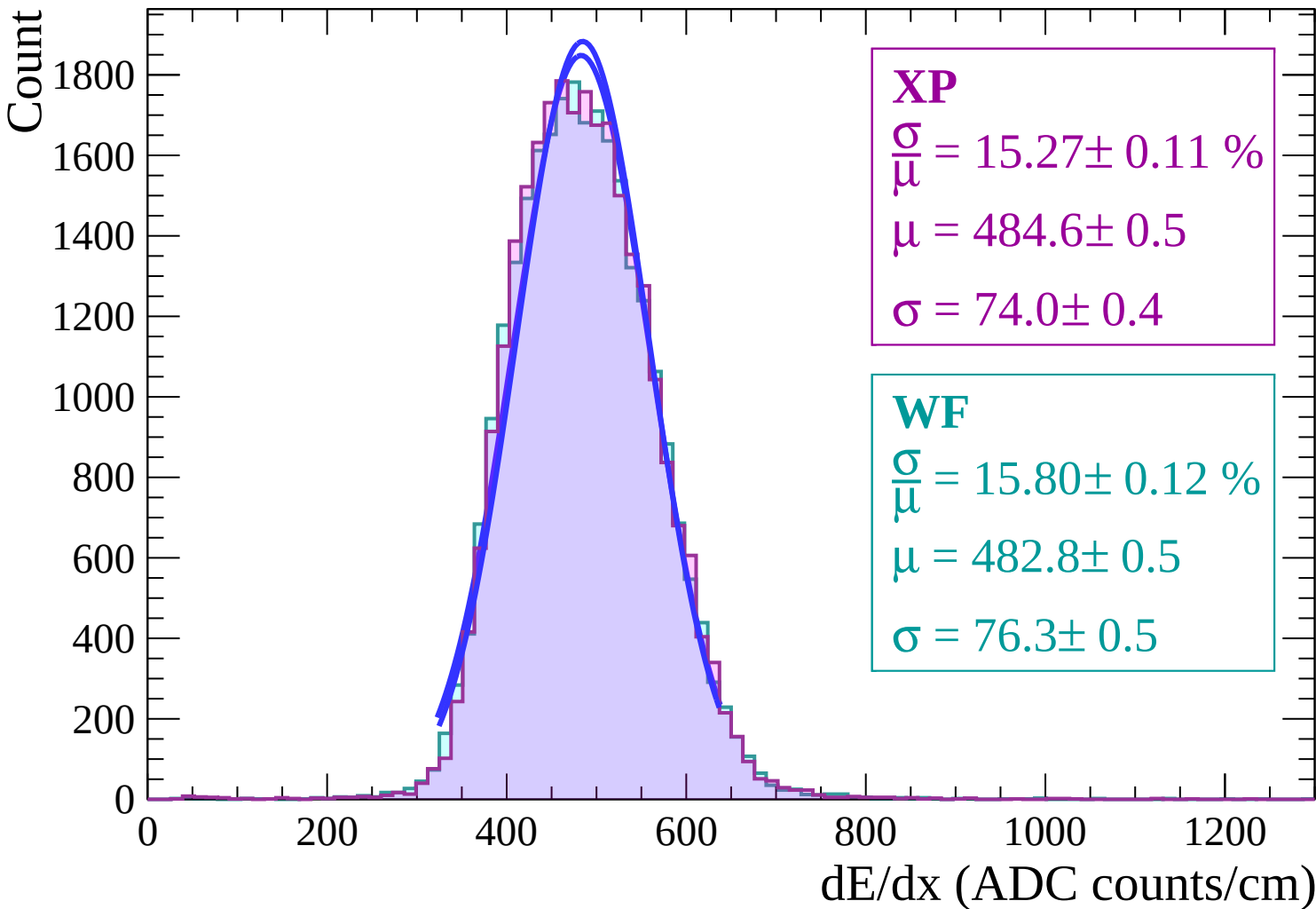
Energy loss | $10 < \theta < 12$



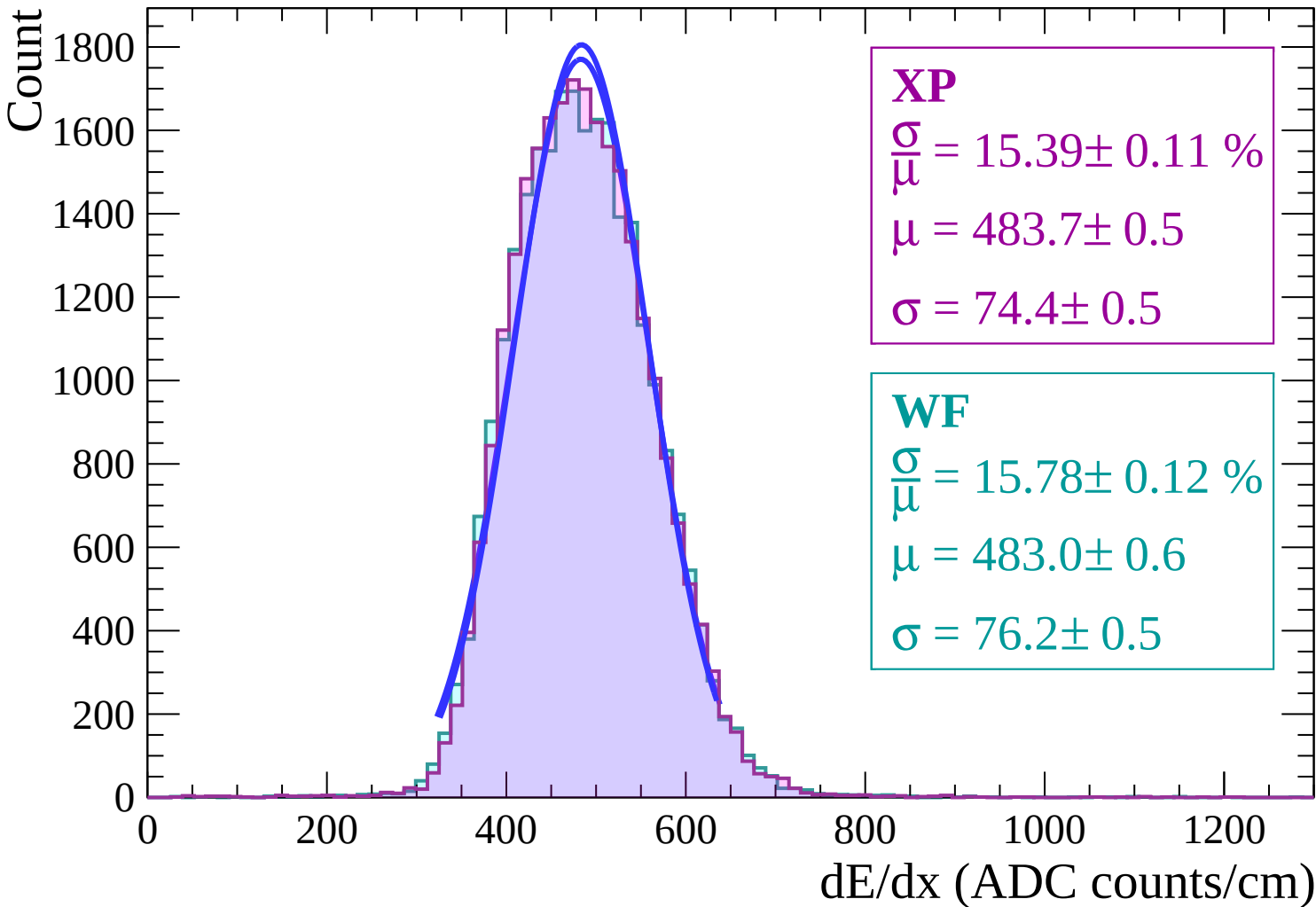
Energy loss | $12 < \theta < 14$



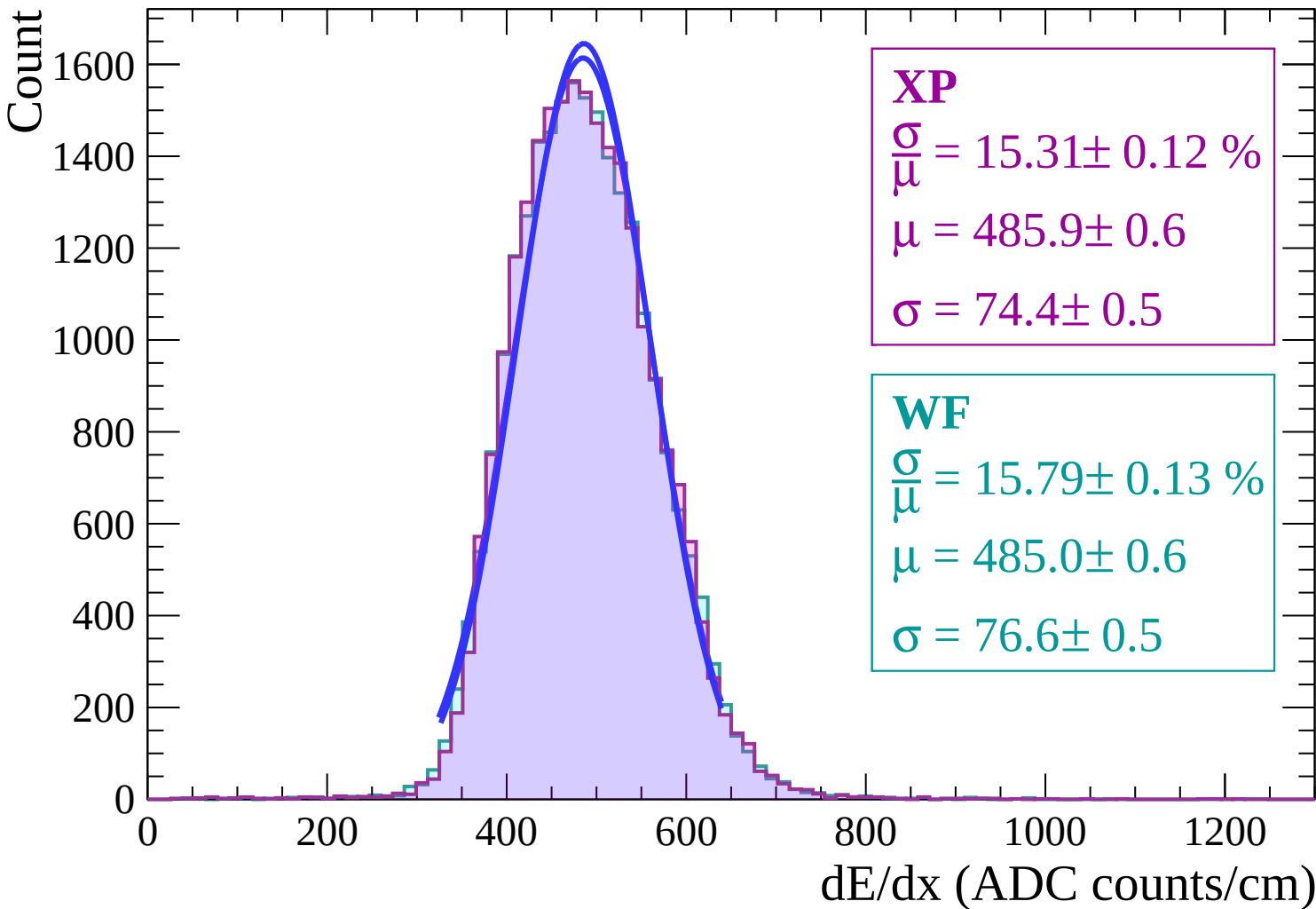
Energy loss | $14 < \theta < 16$



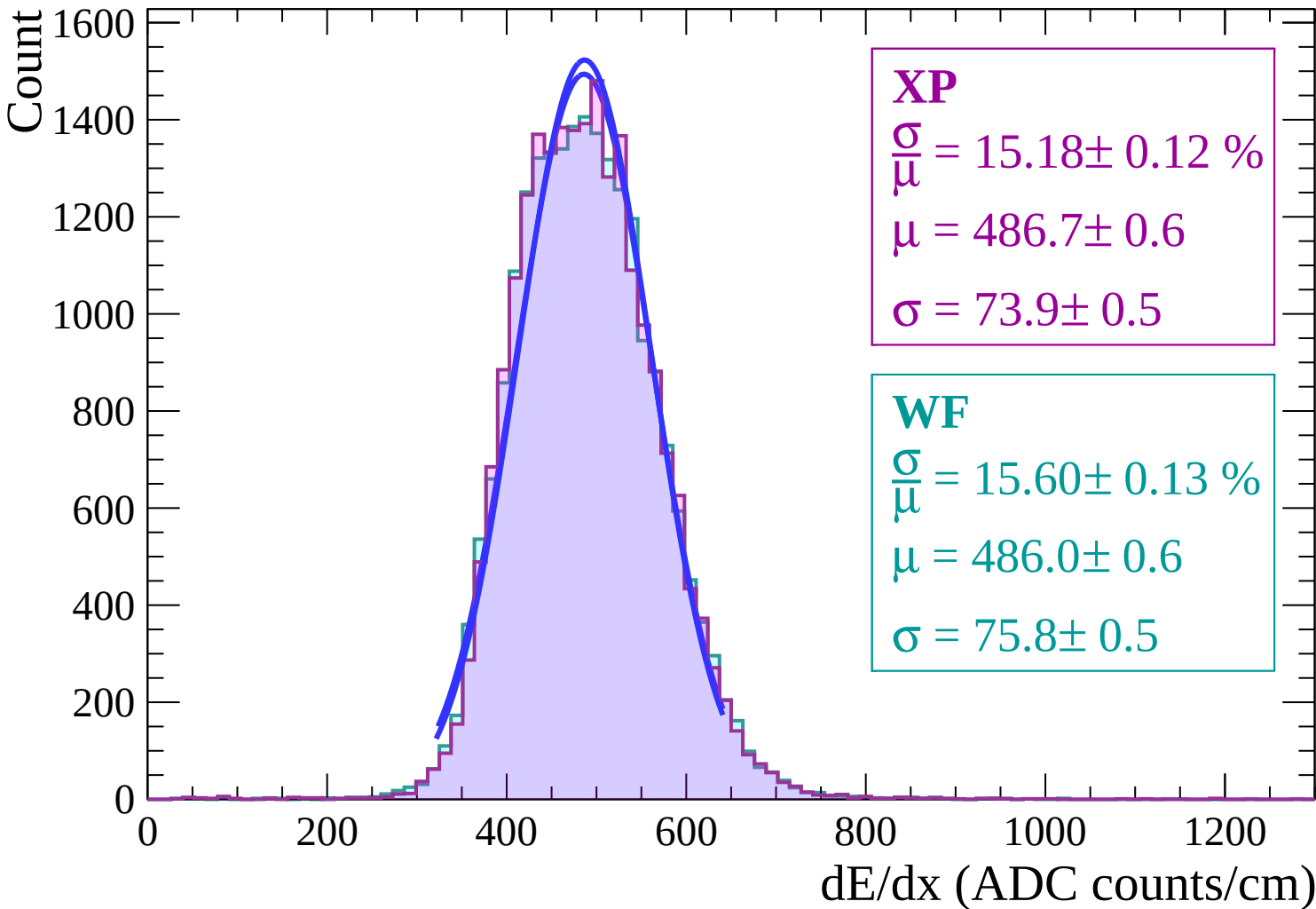
Energy loss | $16 < \theta < 18$



Energy loss | $18 < \theta < 20$

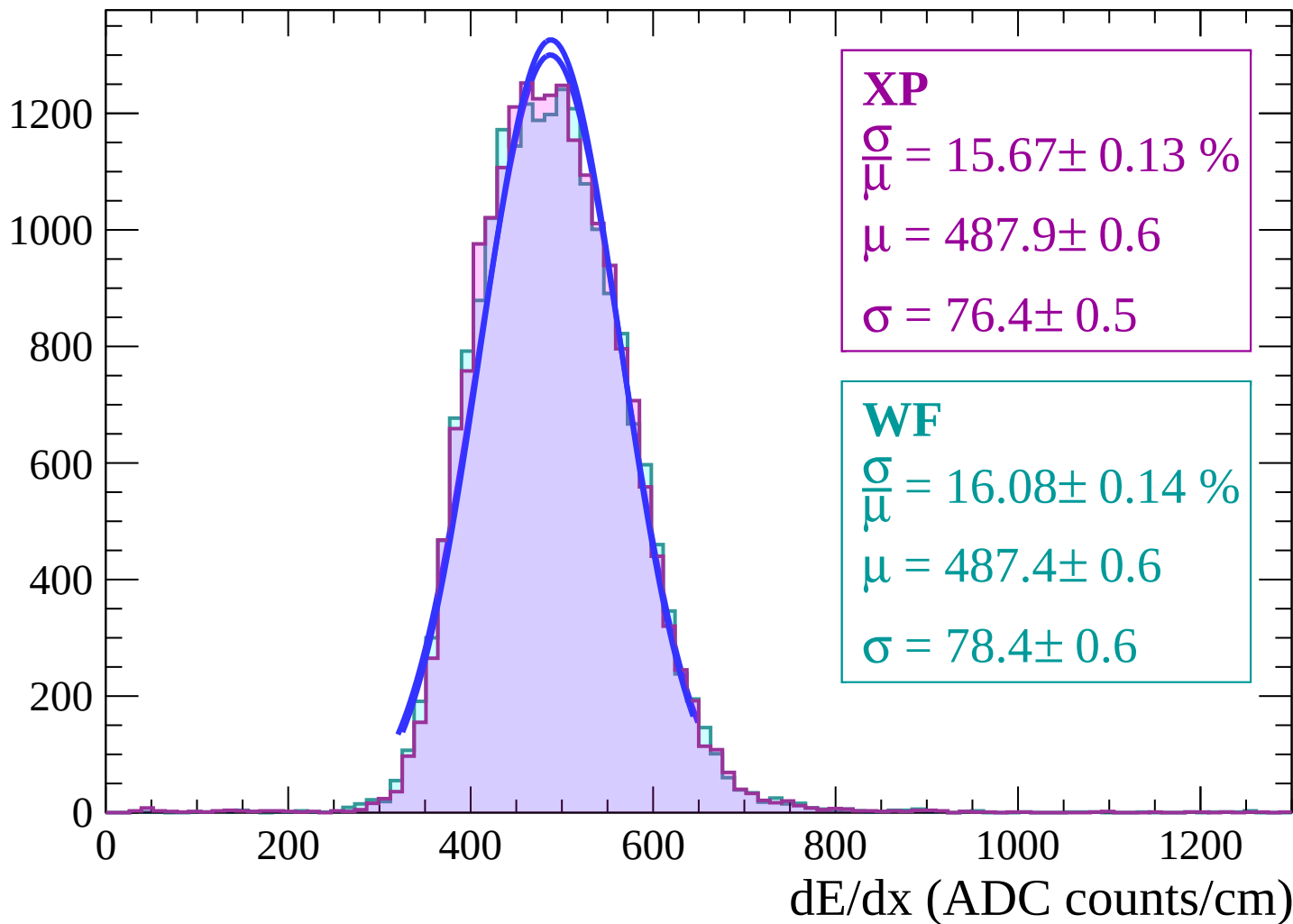


Energy loss | $20 < \theta < 22$

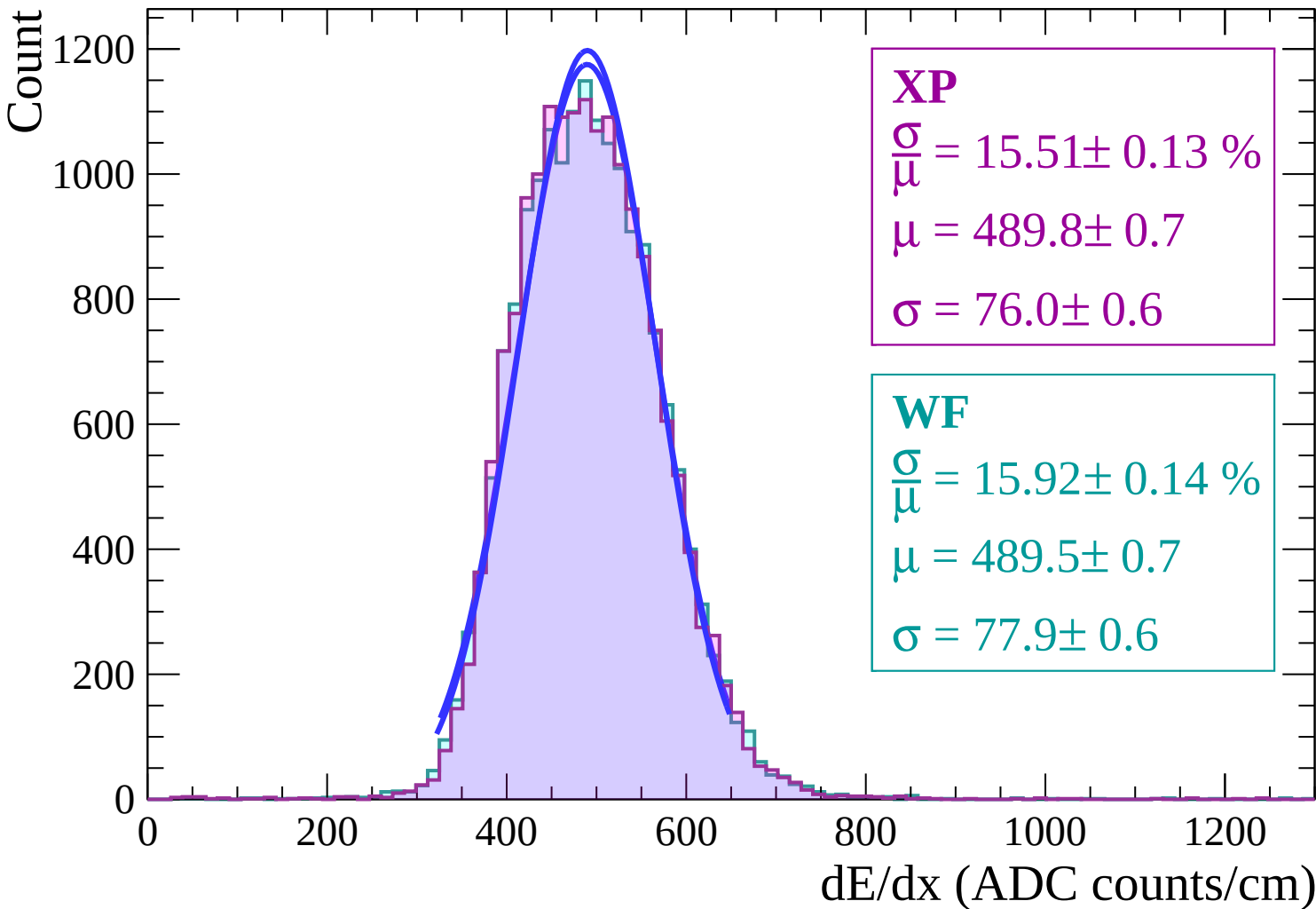


Energy loss | $22 < \theta < 24$

Count

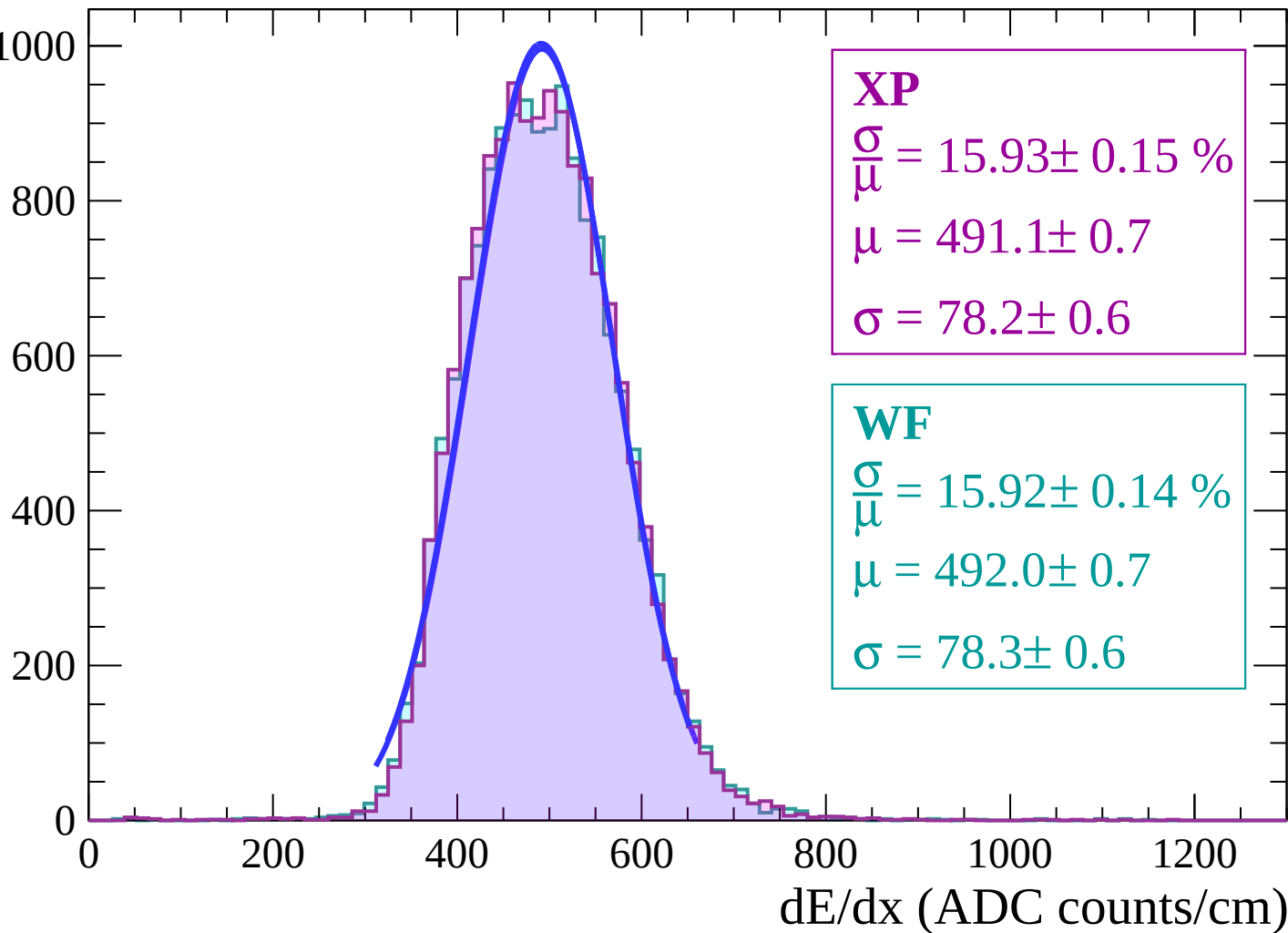


Energy loss | $24 < \theta < 26$



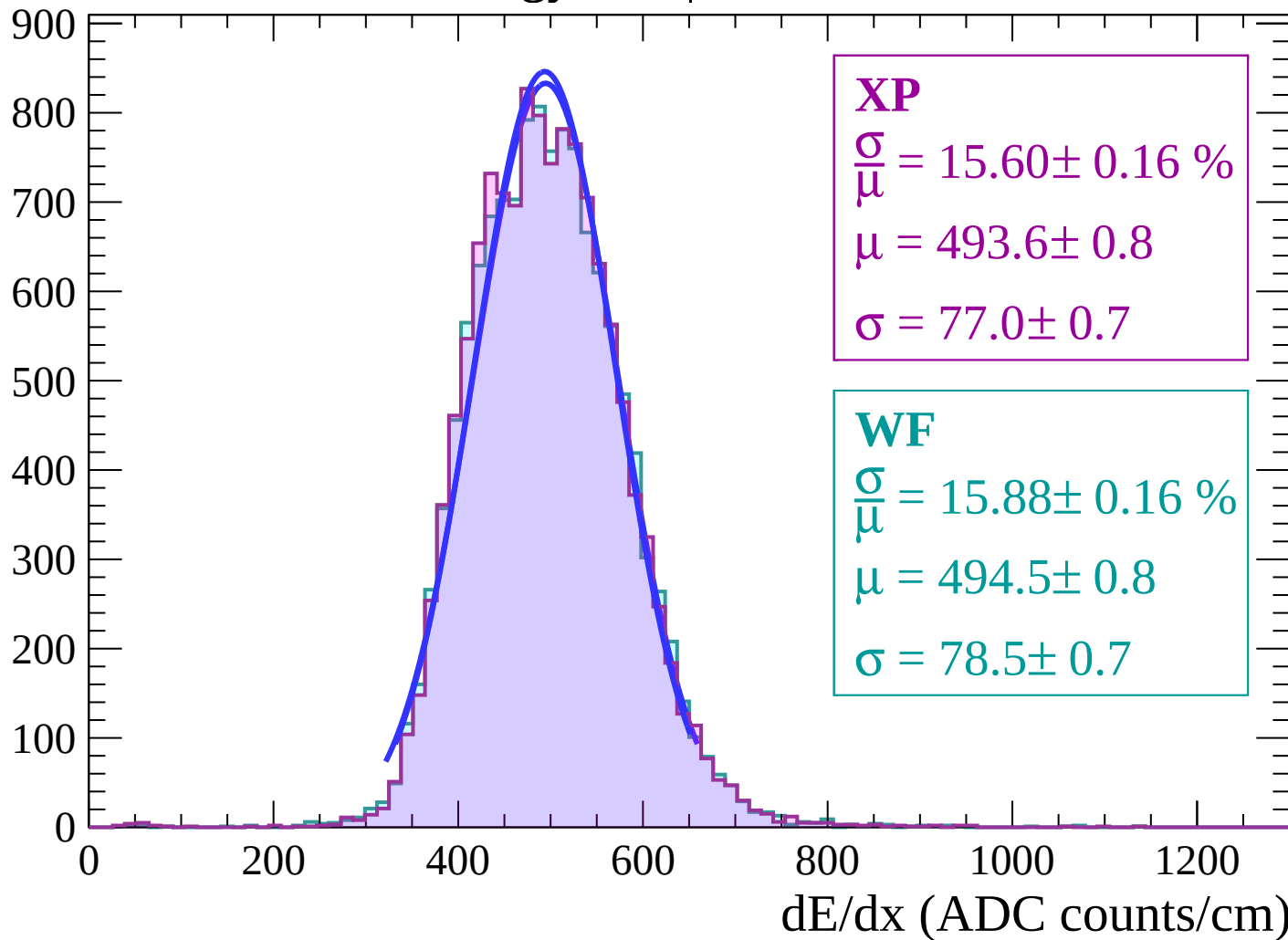
Energy loss | $26 < \theta < 28$

Count



Energy loss | $28 < \theta < 30$

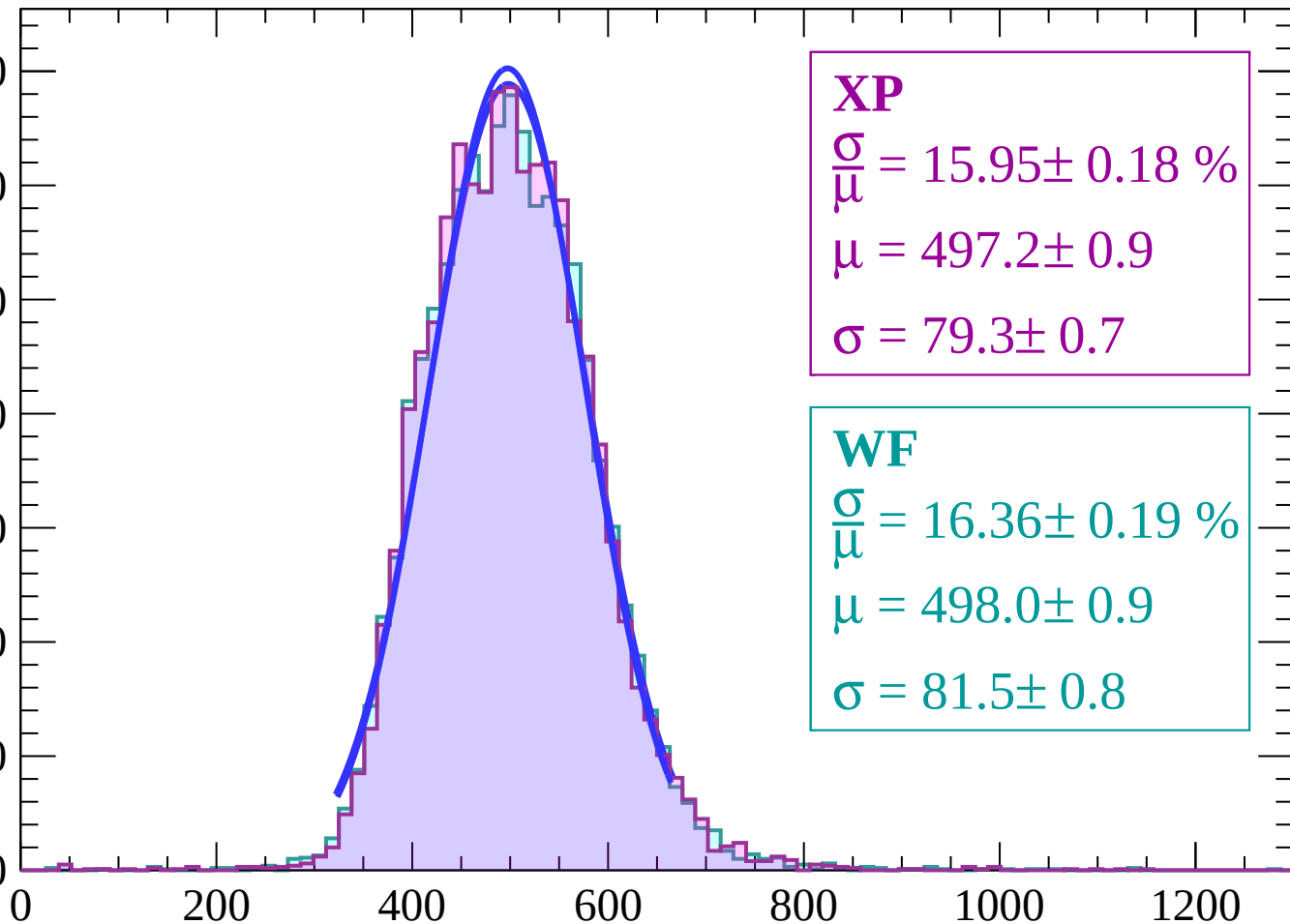
Count



Energy loss | $30 < \theta < 32$

Count

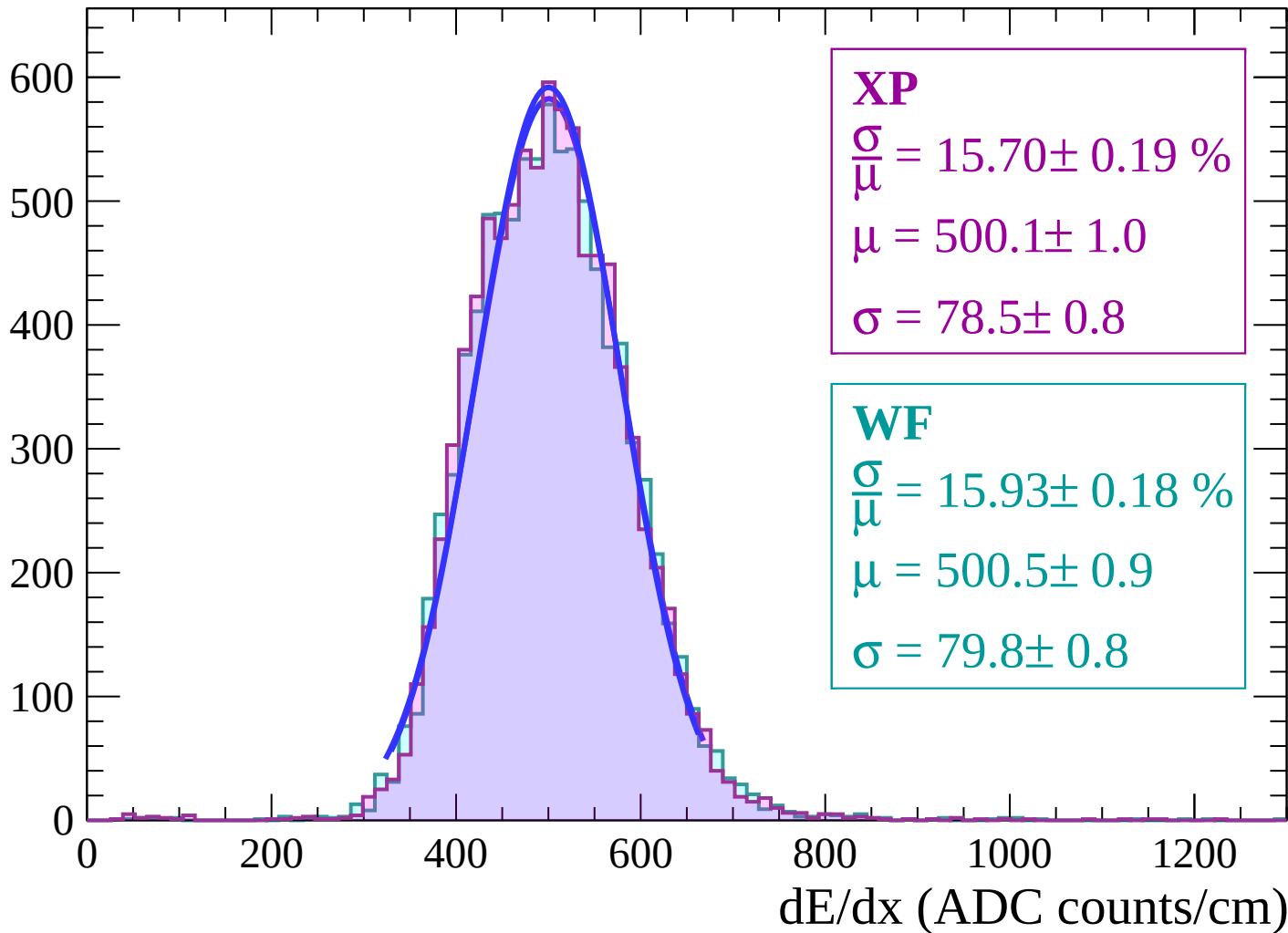
700
600
500
400
300
200
100
0



dE/dx (ADC counts/cm)

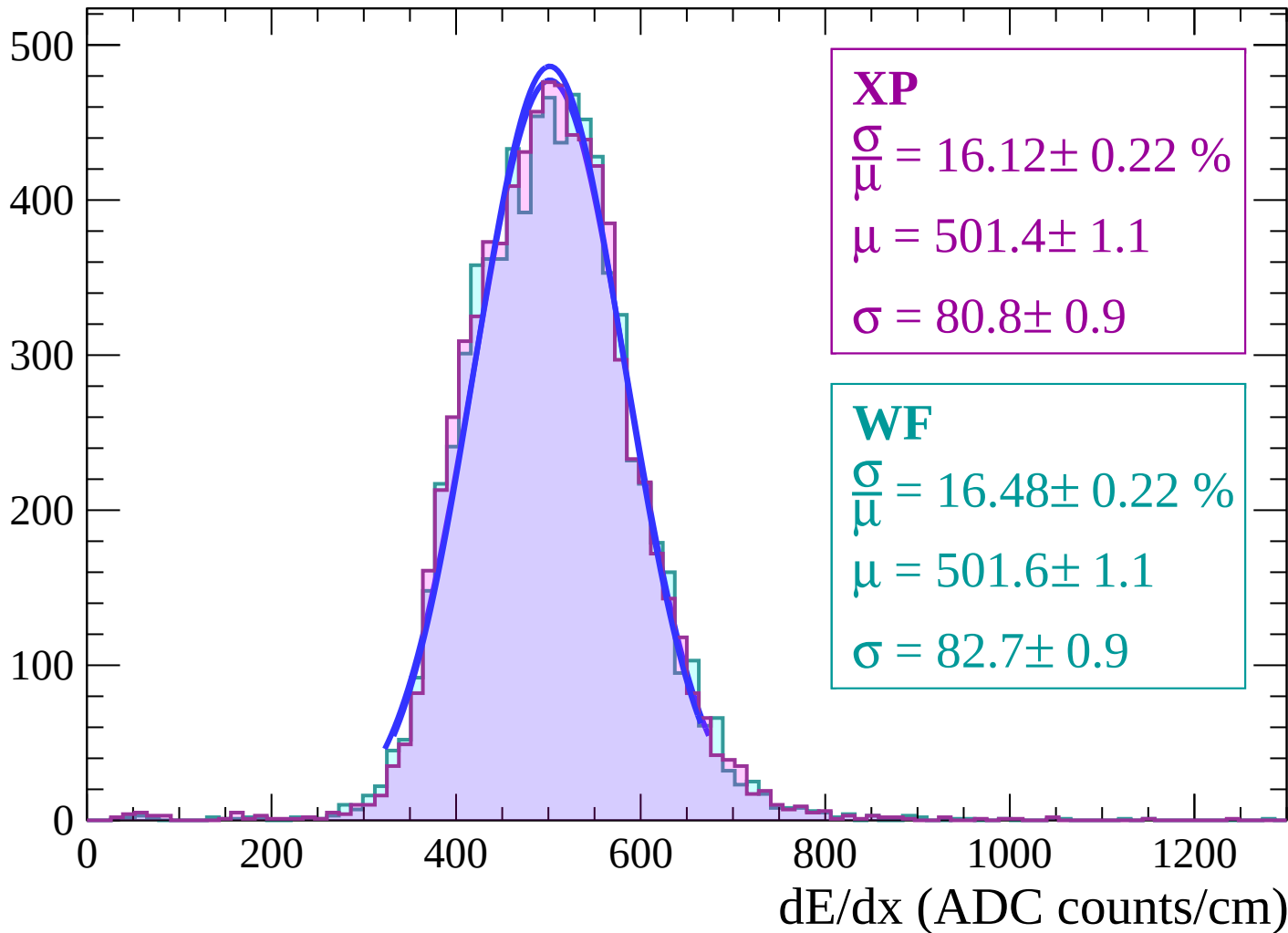
Energy loss | $32 < \theta < 34$

Count



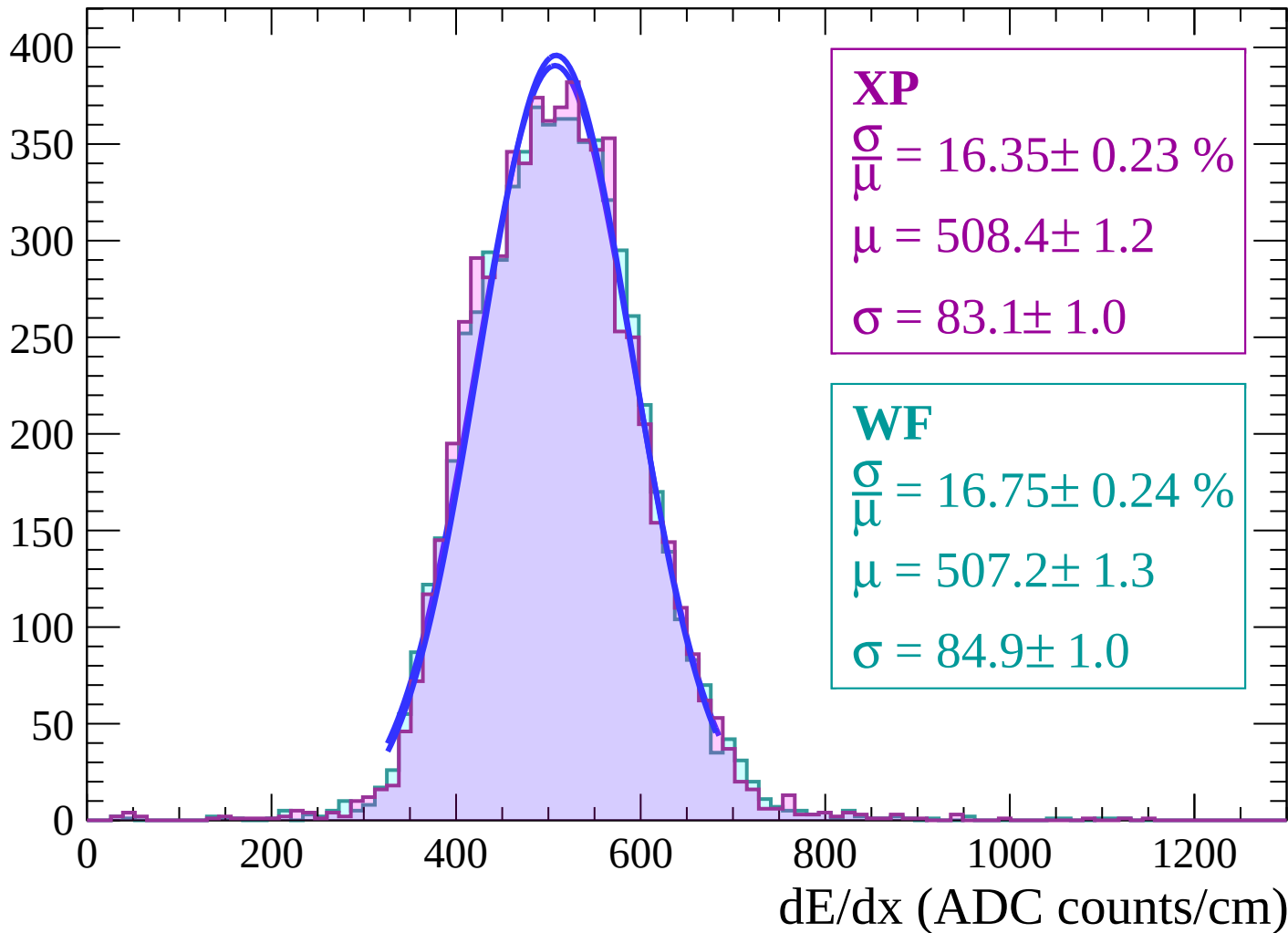
Energy loss | $34 < \theta < 36$

Count



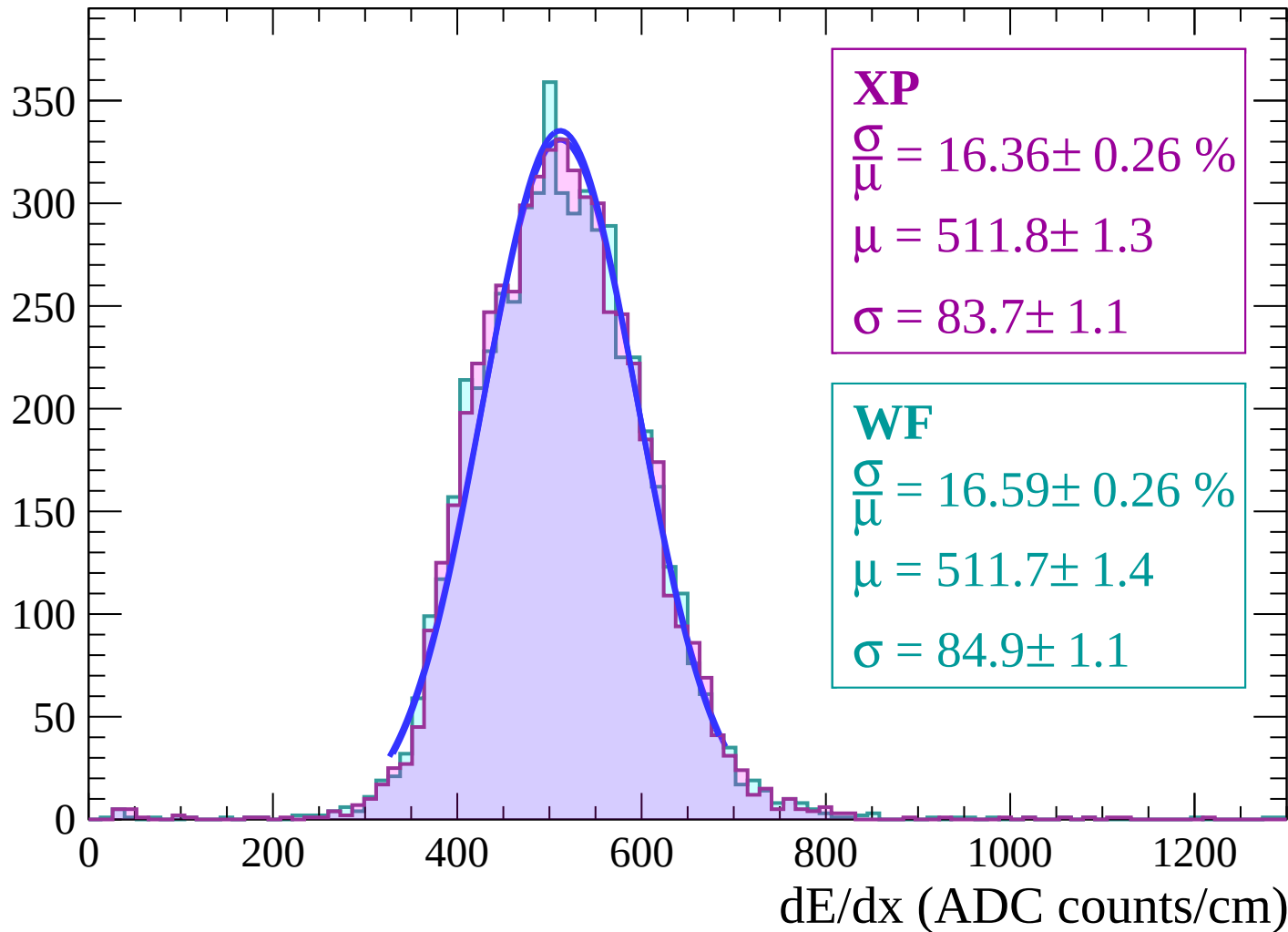
Energy loss | $36 < \theta < 38$

Count



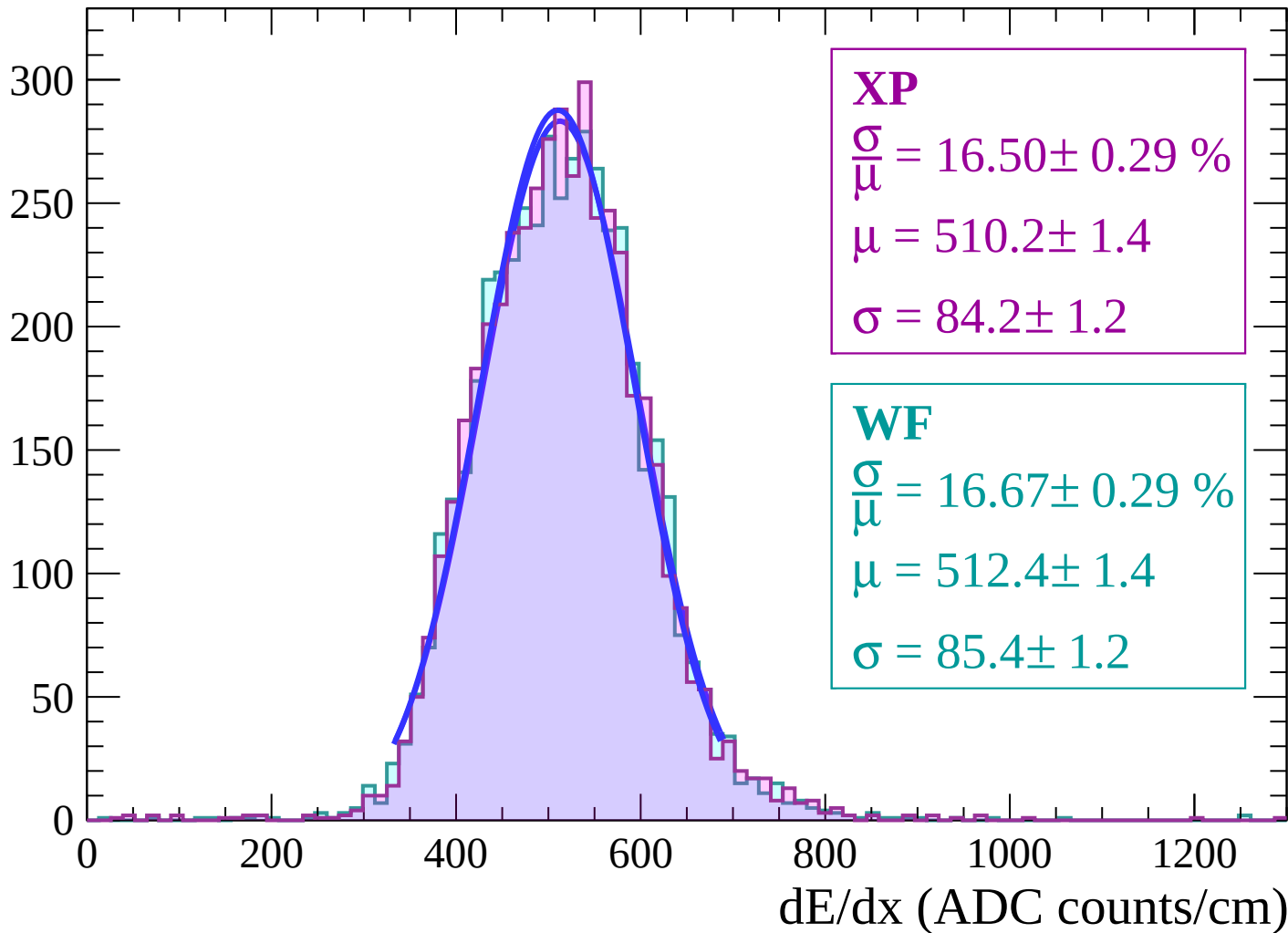
Energy loss | $38 < \theta < 40$

Count



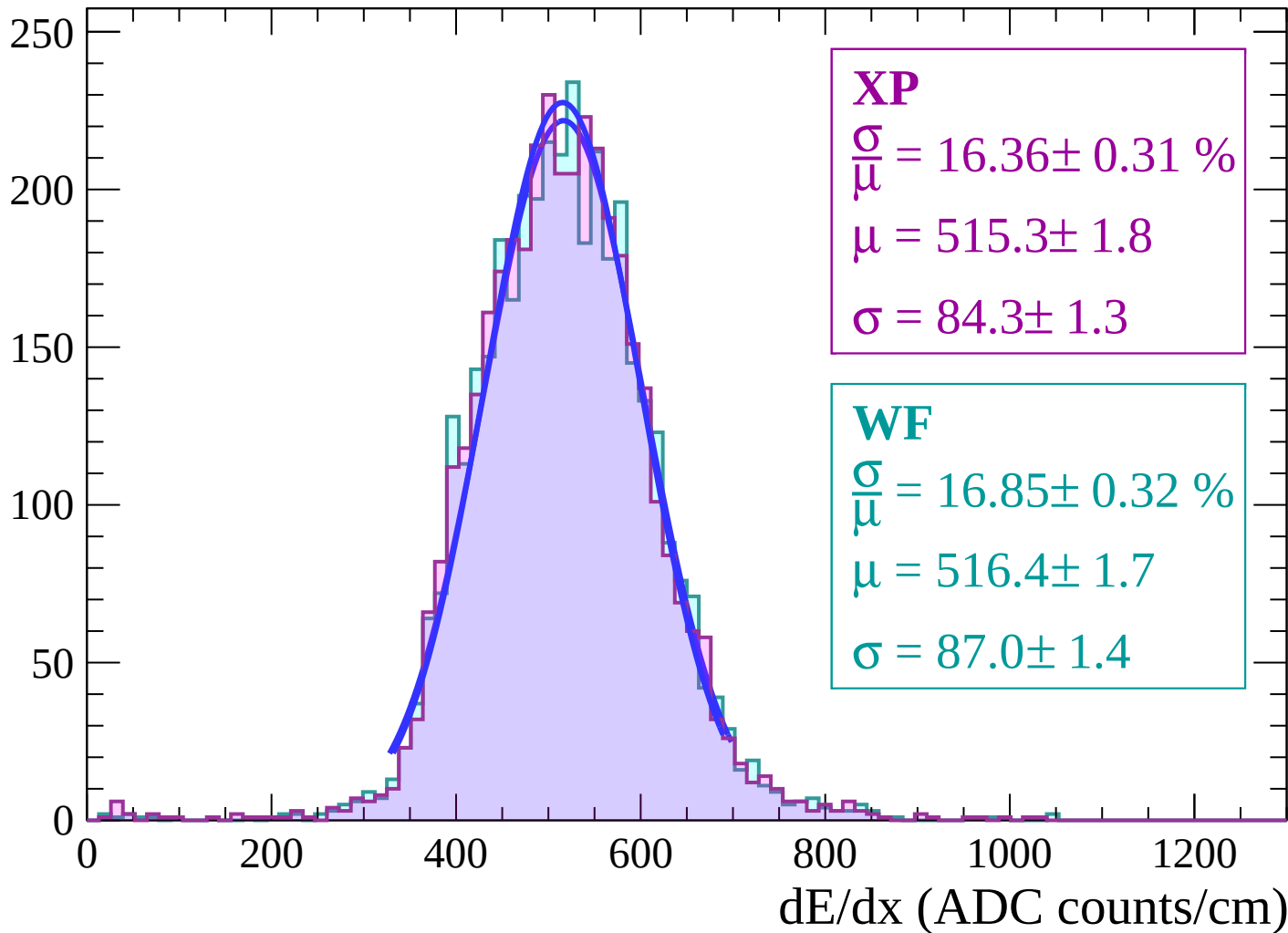
Energy loss | $40 < \theta < 42$

Count



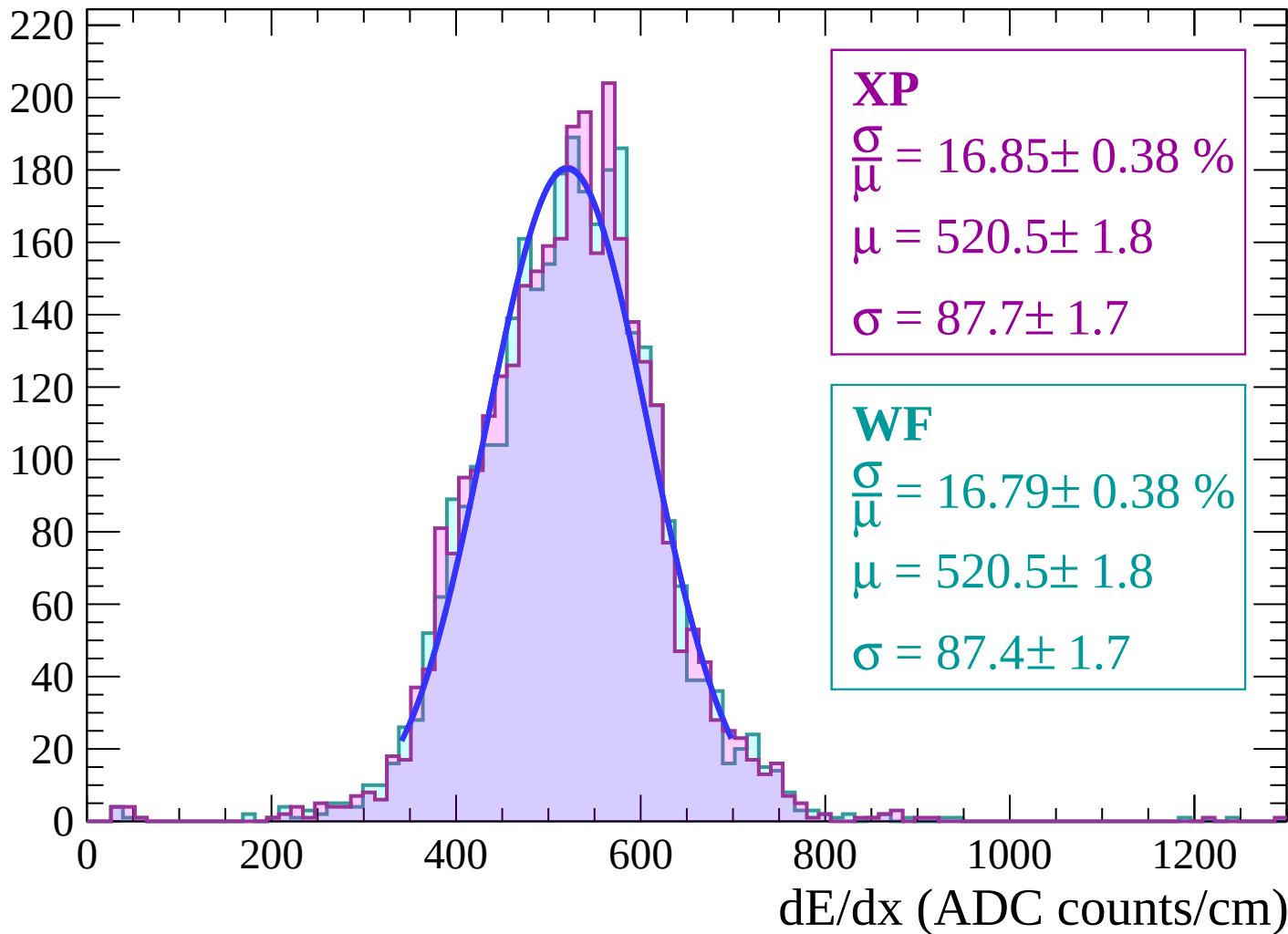
Energy loss | $42 < \theta < 44$

Count



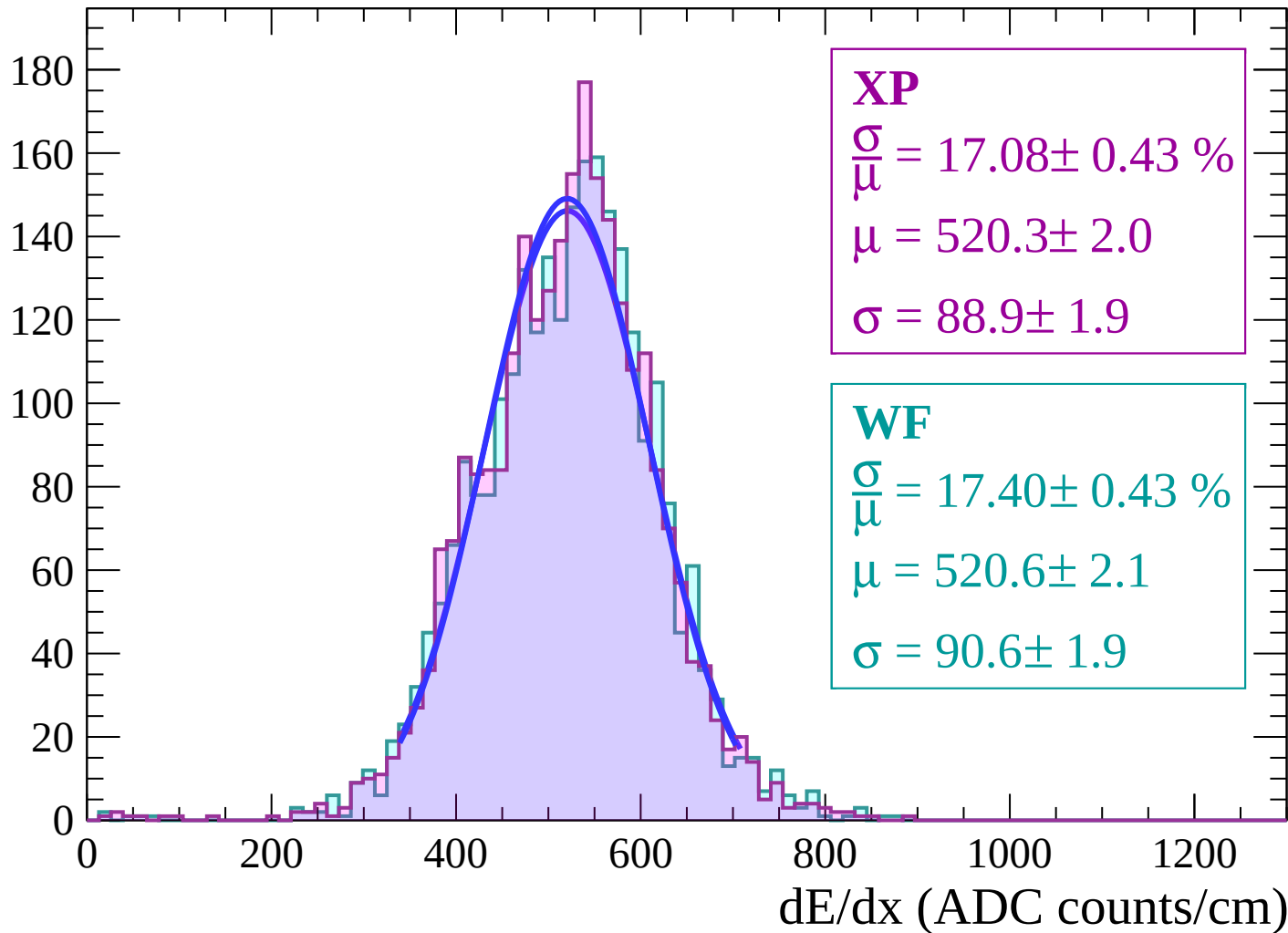
Energy loss | $44 < \theta < 46$

Count



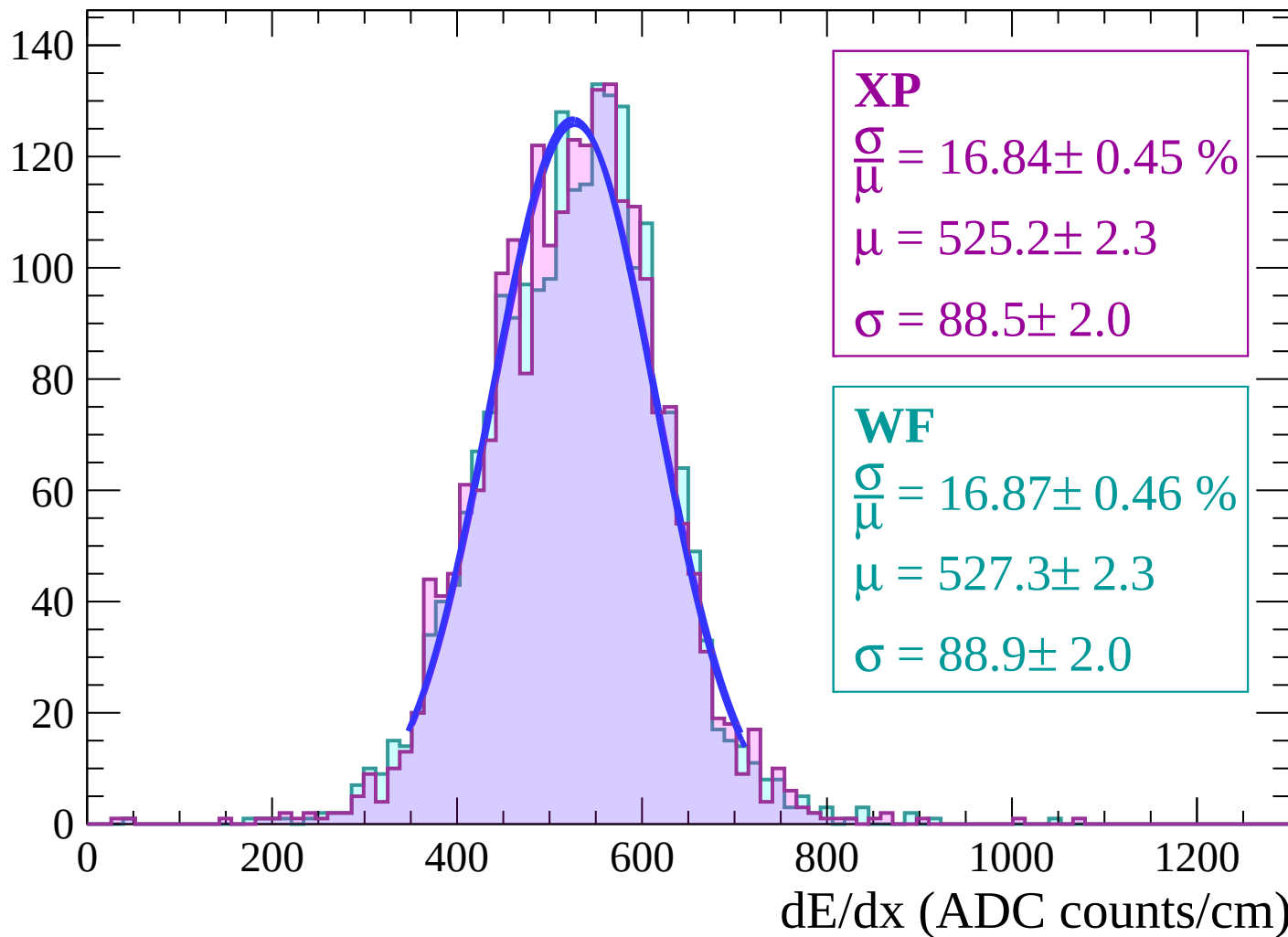
Energy loss | $46 < \theta < 48$

Count



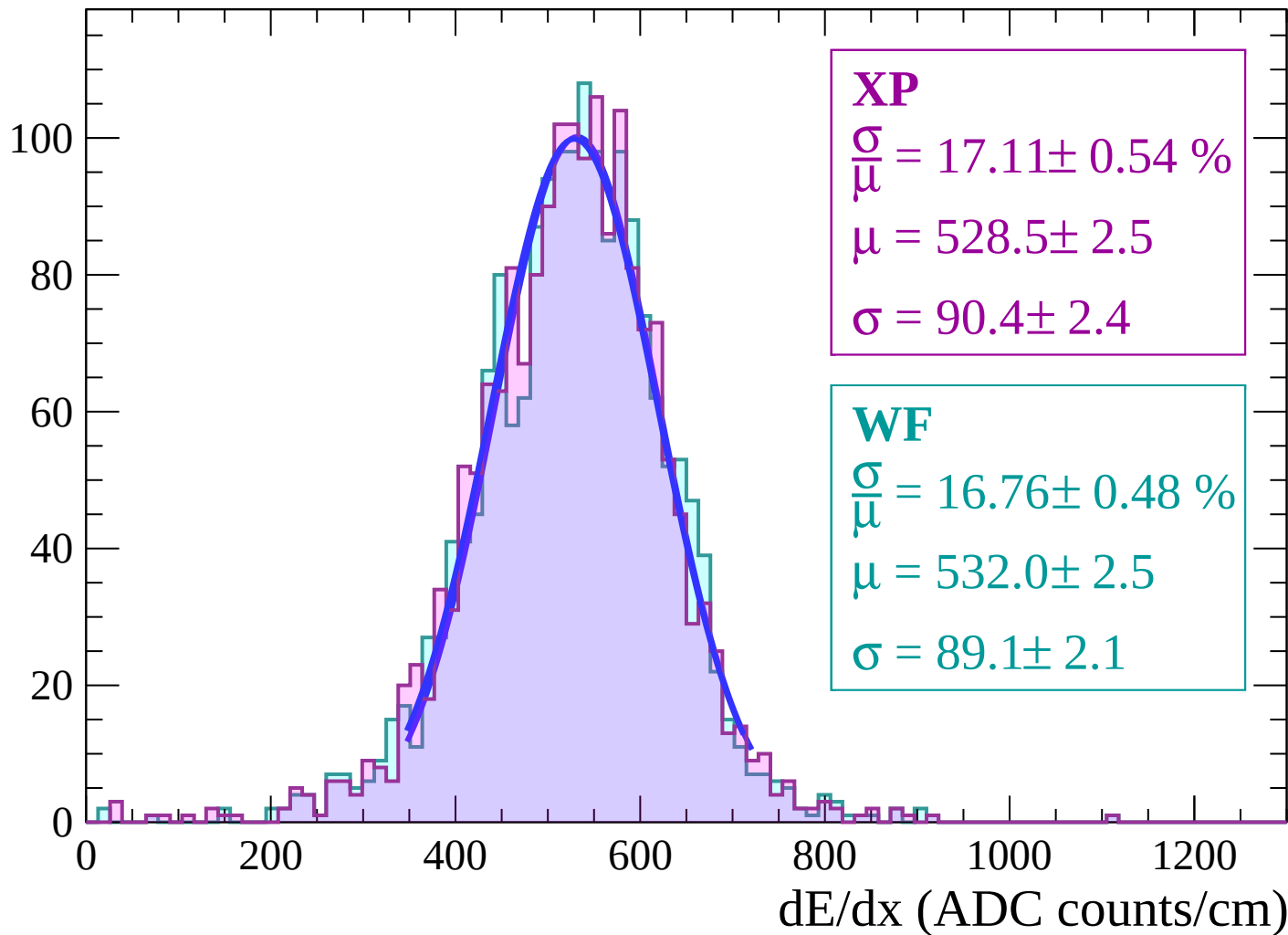
Energy loss | $48 < \theta < 50$

Count



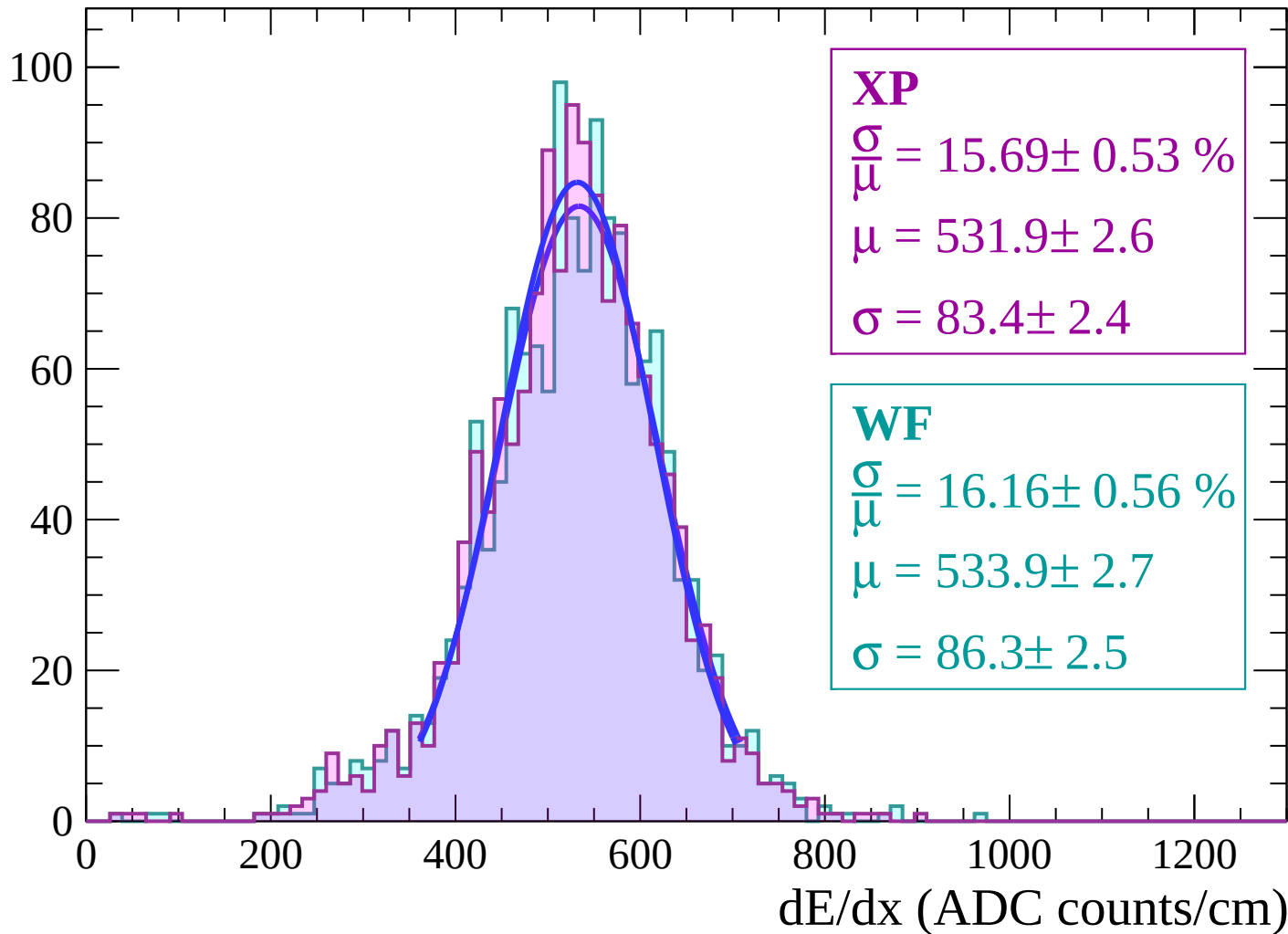
Energy loss | $50 < \theta < 52$

Count



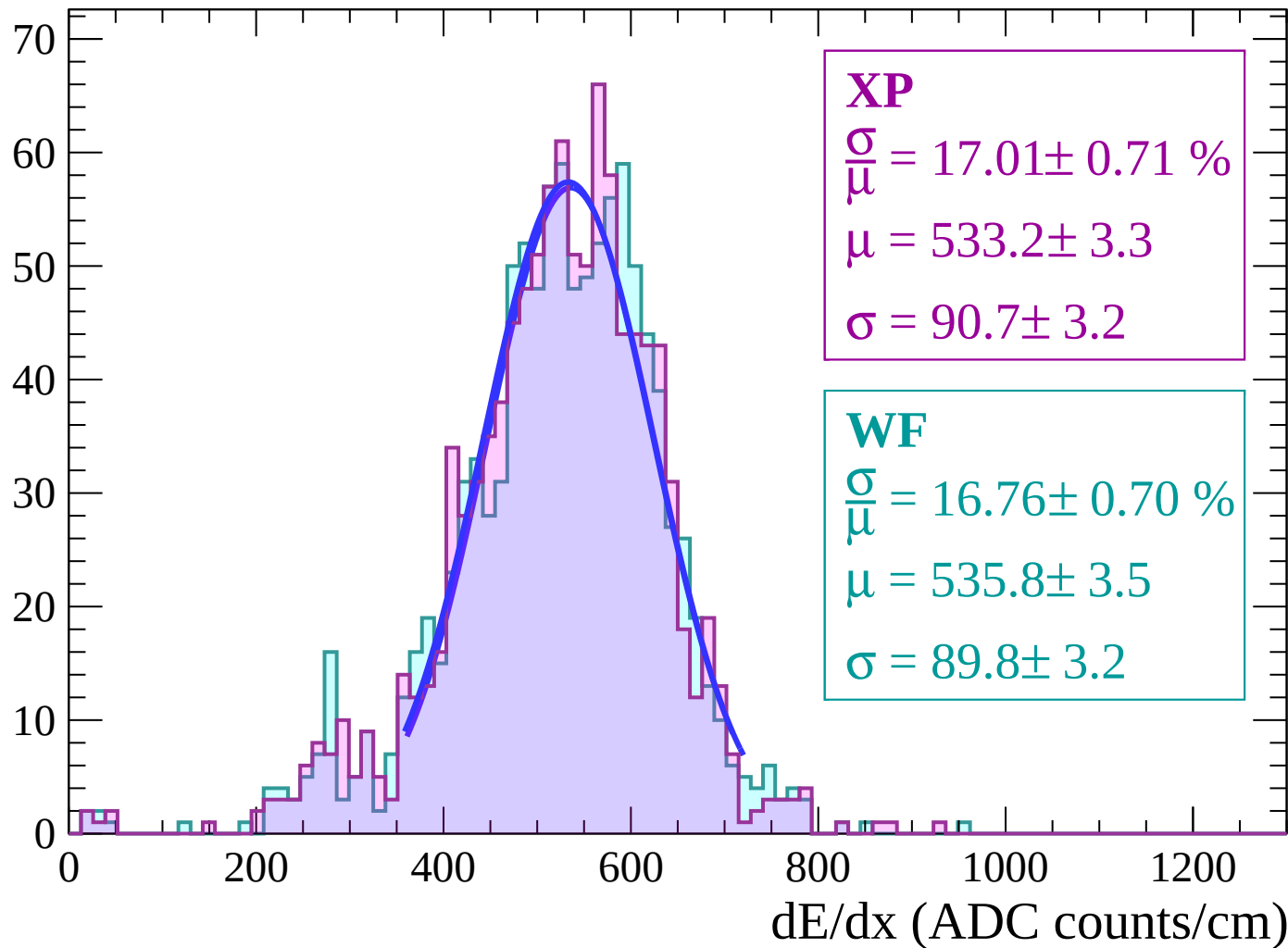
Energy loss | $52 < \theta < 54$

Count



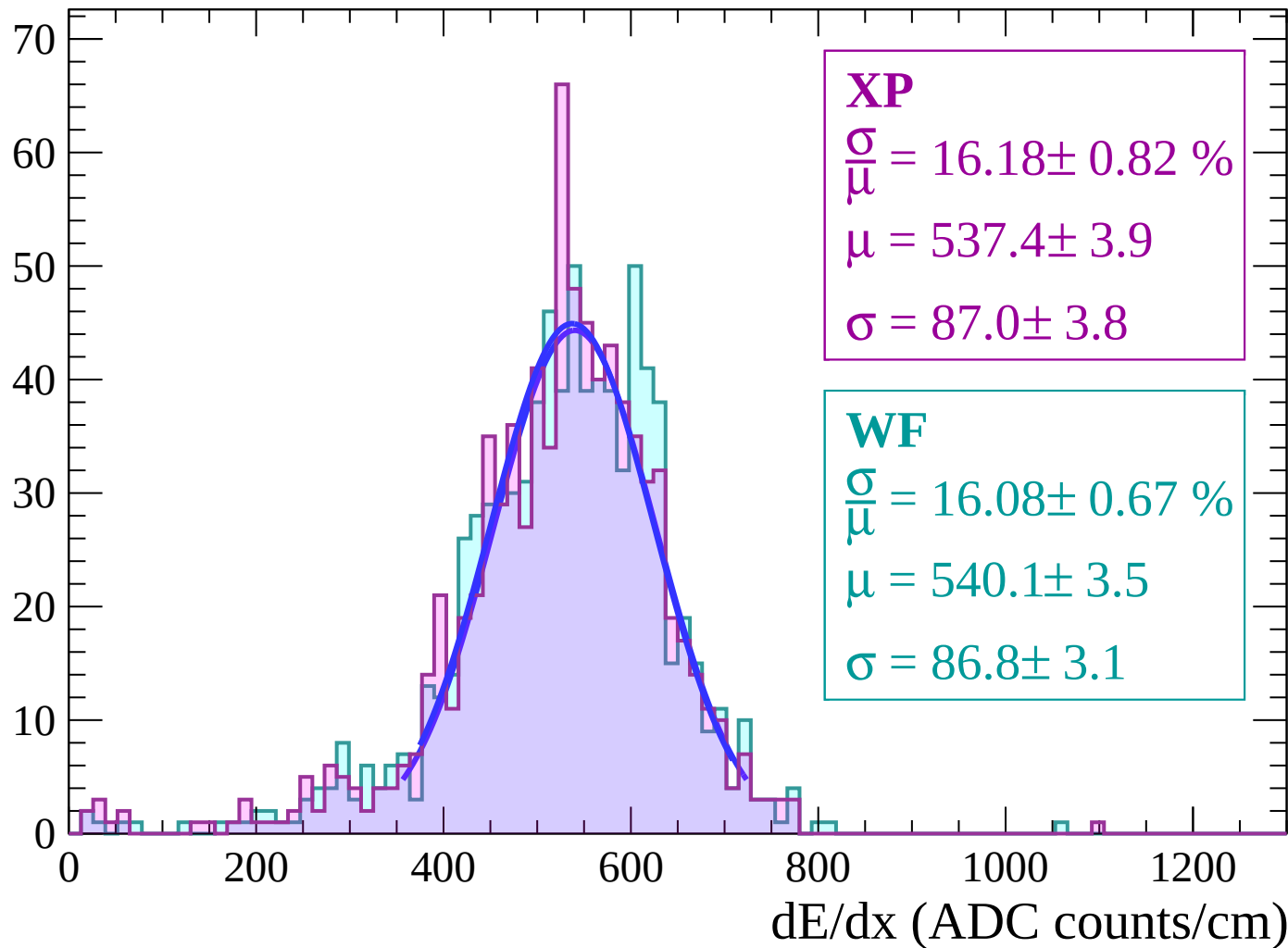
Energy loss | $54 < \theta < 56$

Count



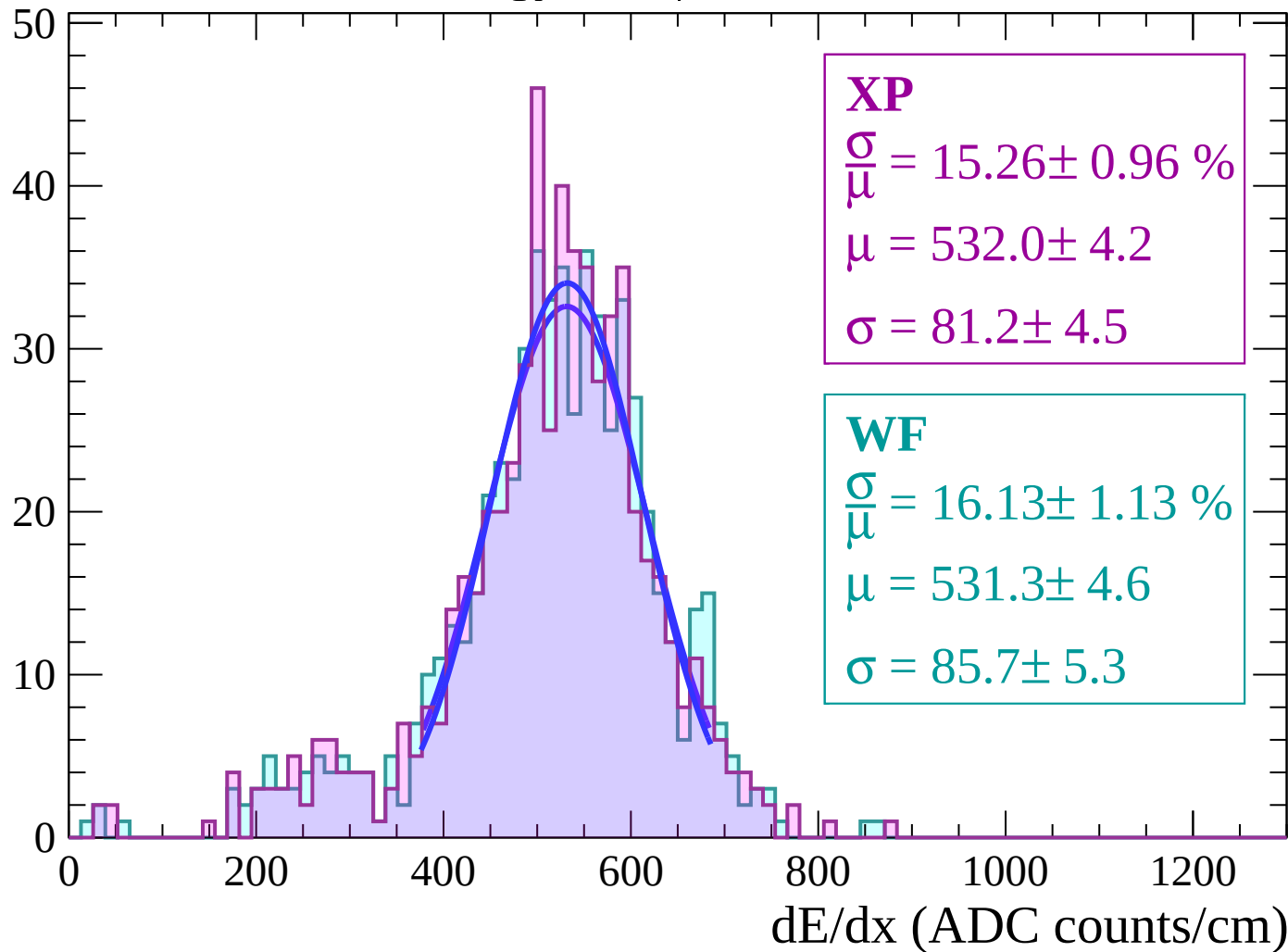
Energy loss | $56 < \theta < 58$

Count



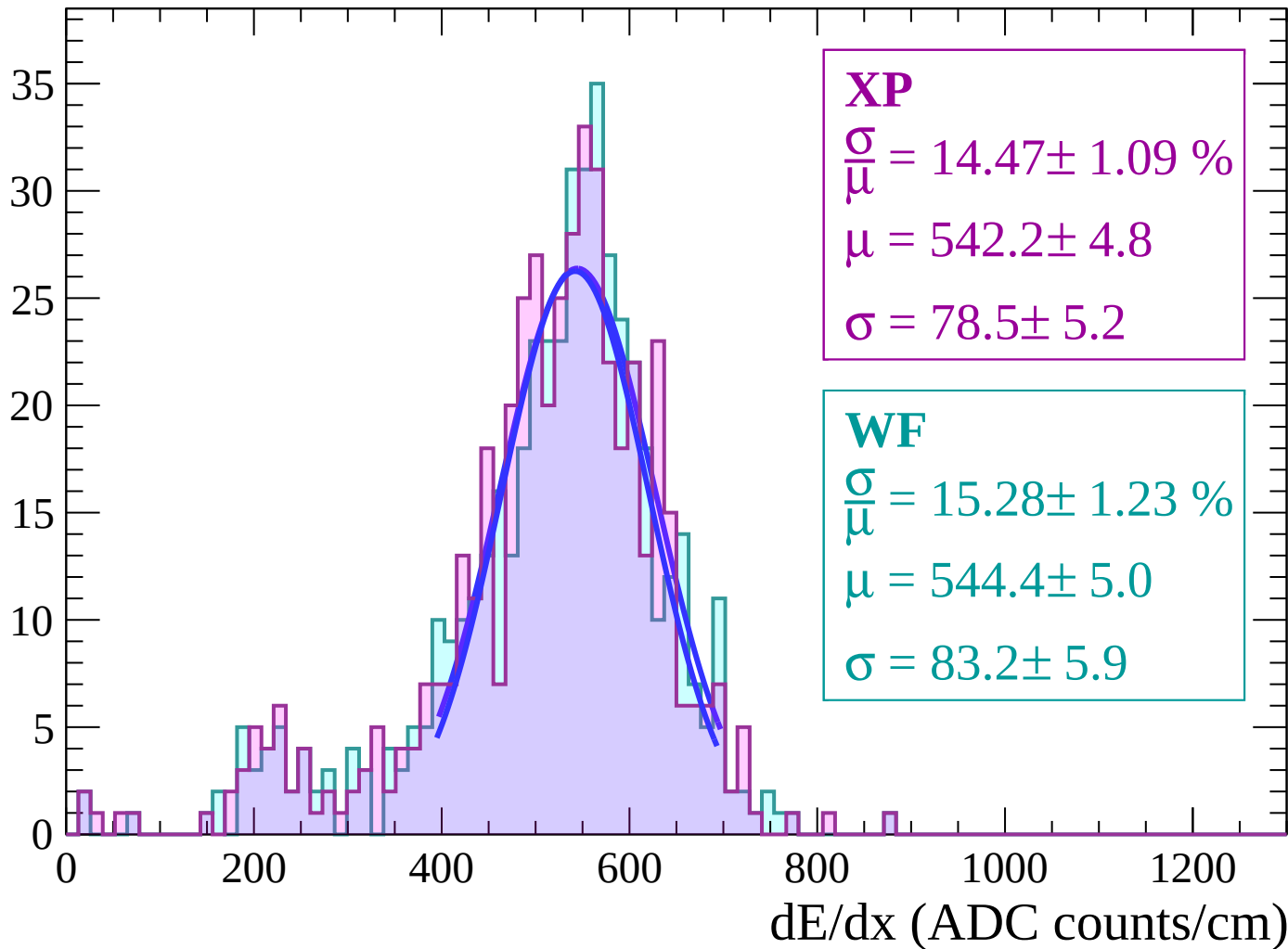
Energy loss | $58 < \theta < 60$

Count



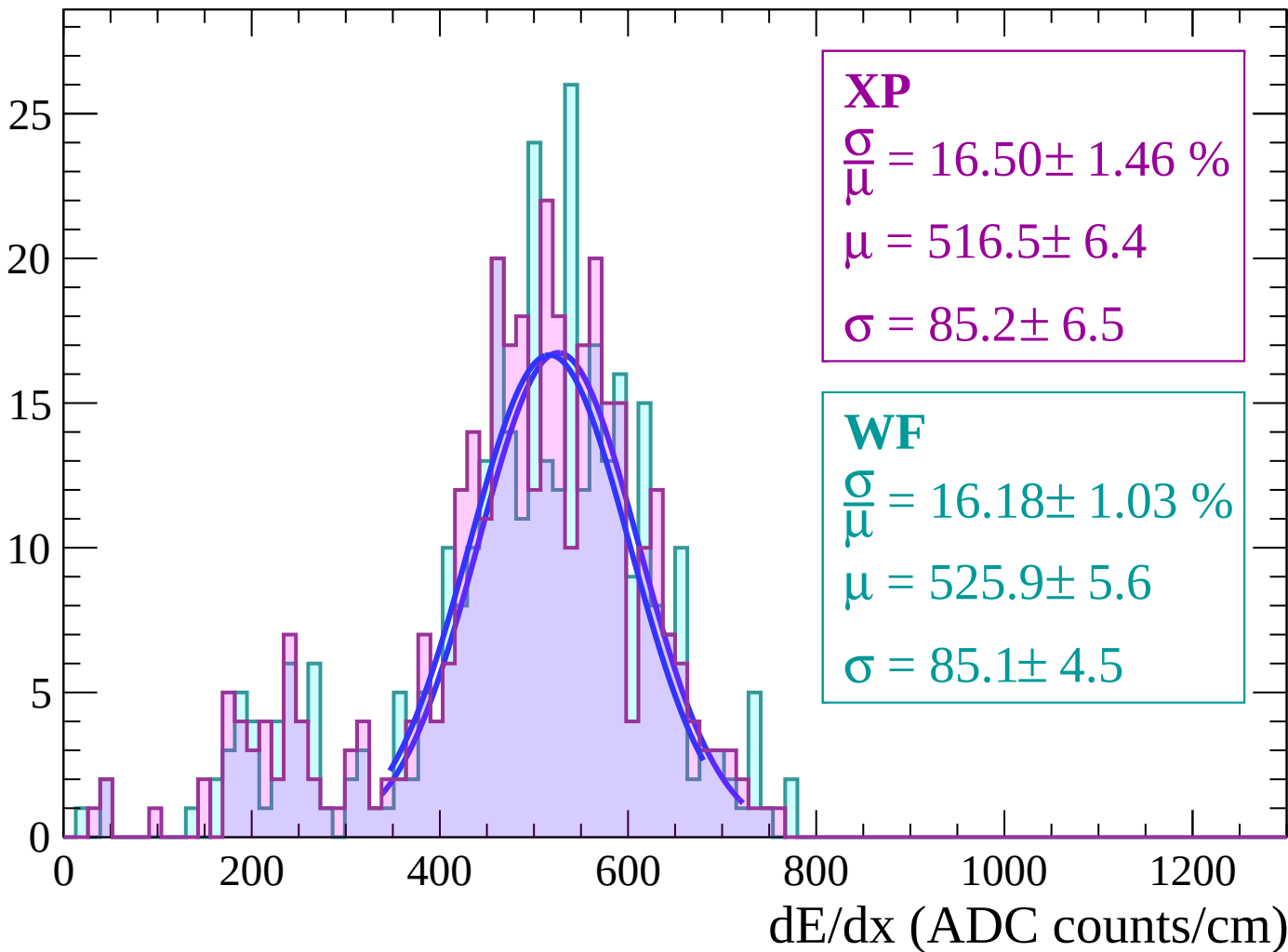
Energy loss | $60 < \theta < 62$

Count



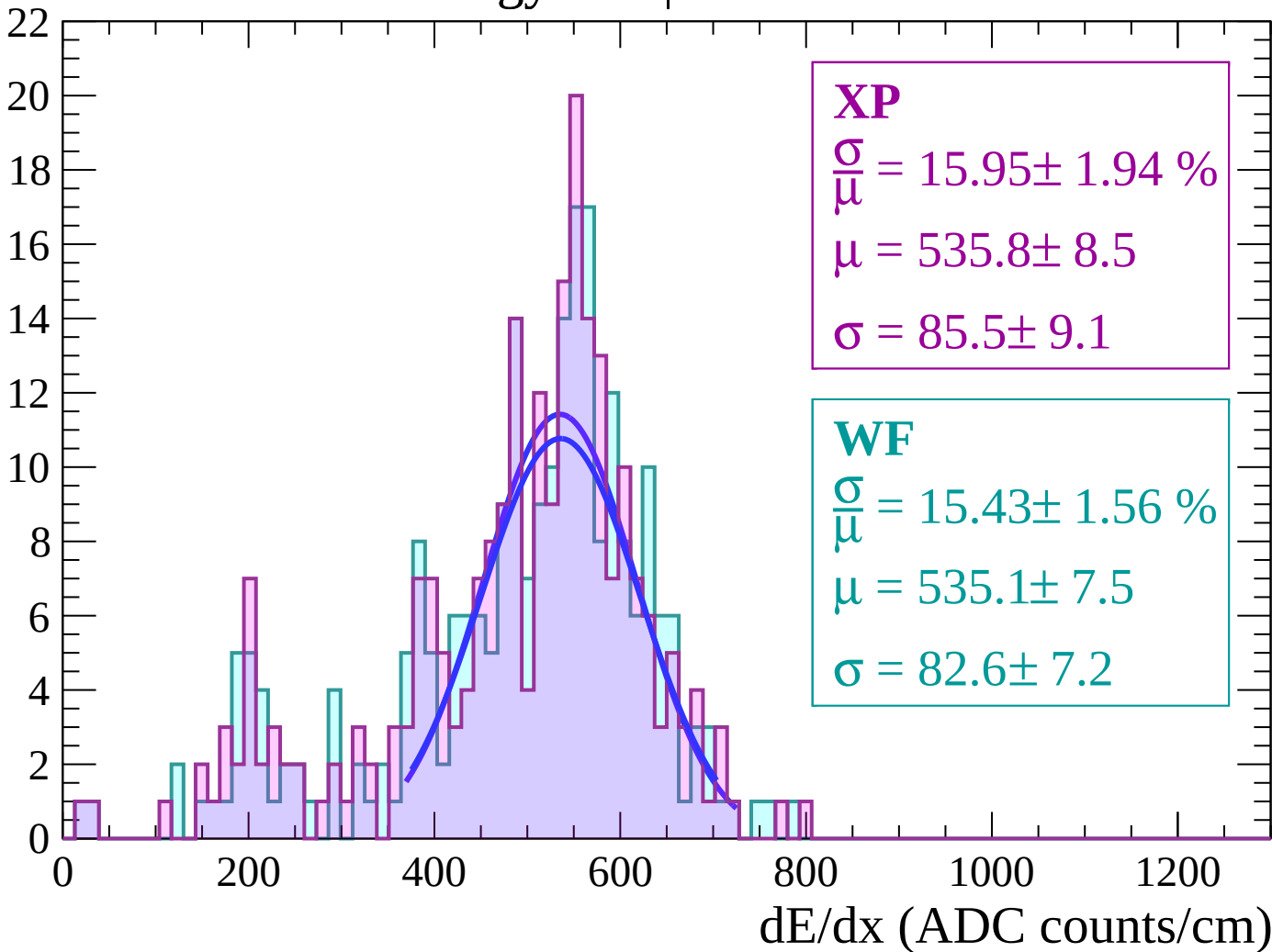
Energy loss | $62 < \theta < 64$

Count



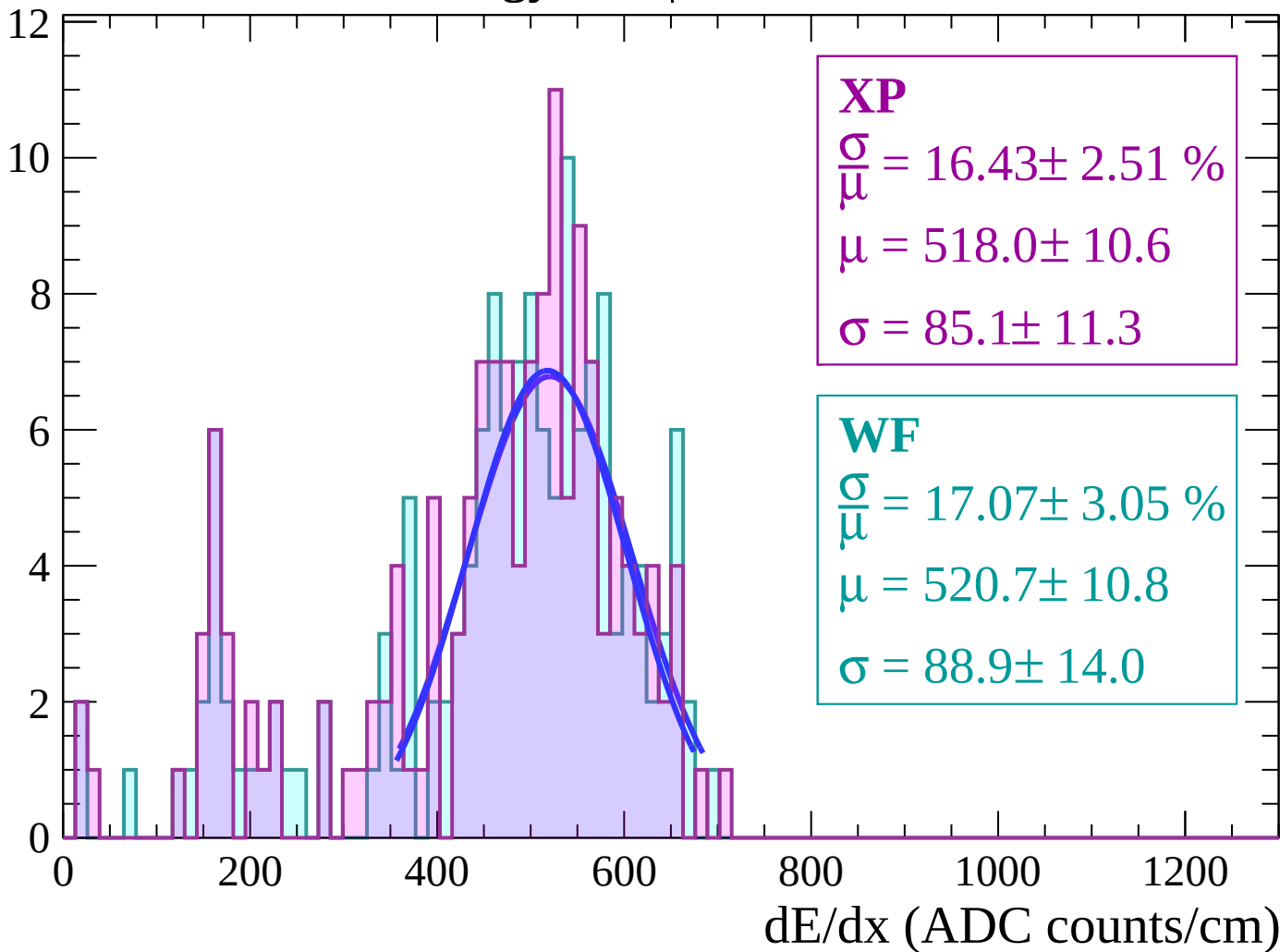
Energy loss | $64 < \theta < 66$

Count



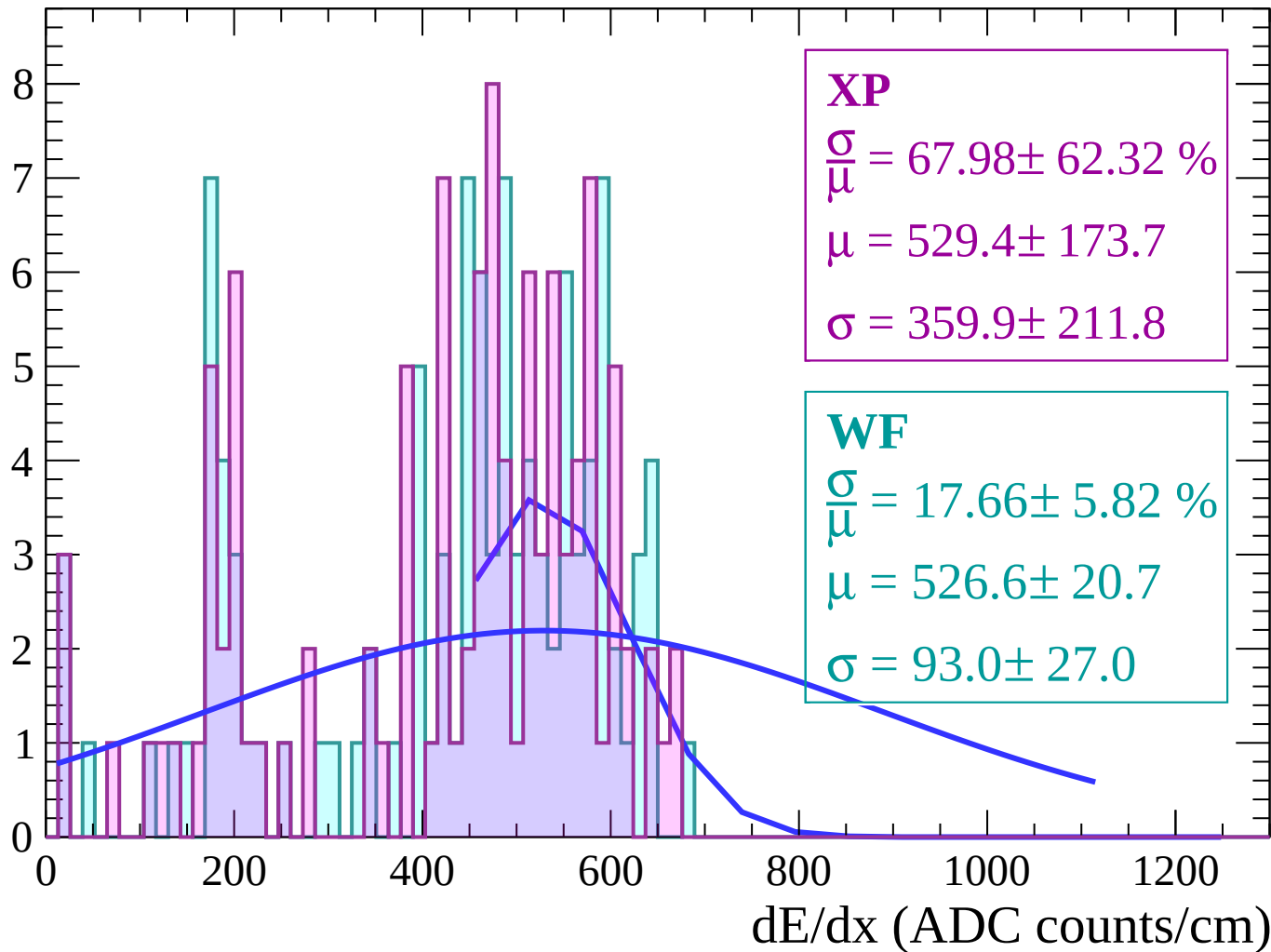
Energy loss | $66 < \theta < 68$

Count



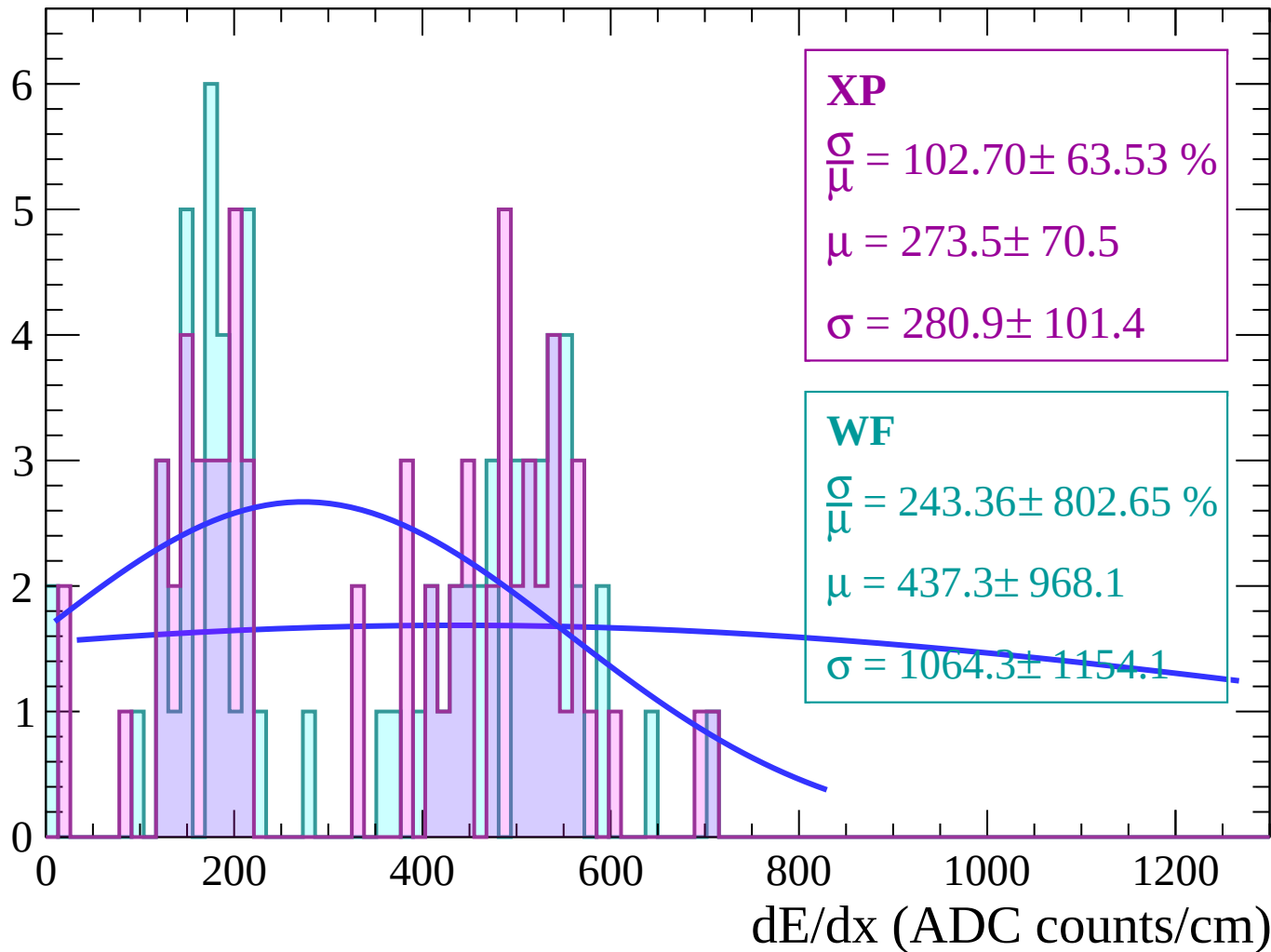
Energy loss | $68 < \theta < 70$

Count



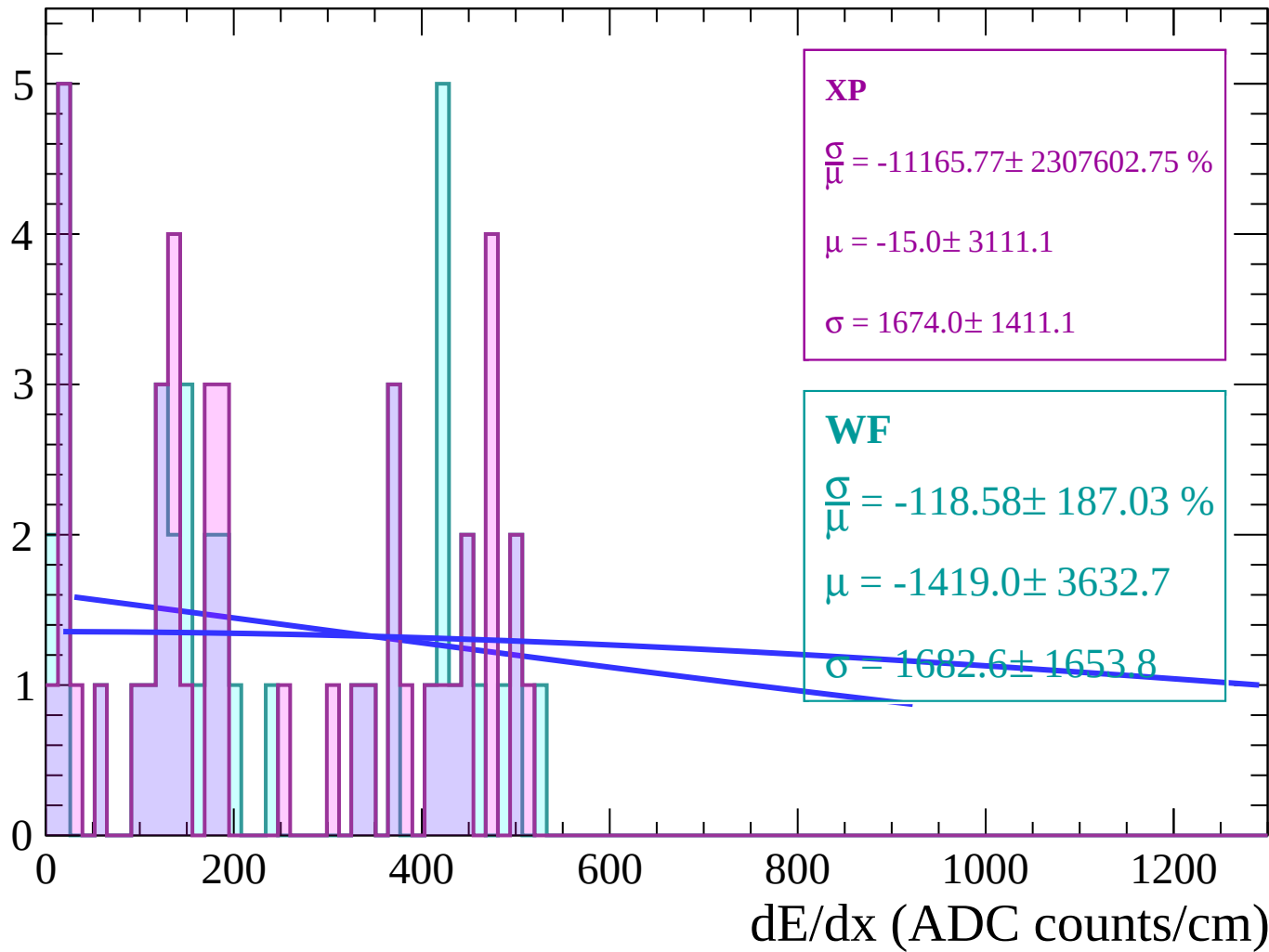
Energy loss | $70 < \theta < 72$

Count



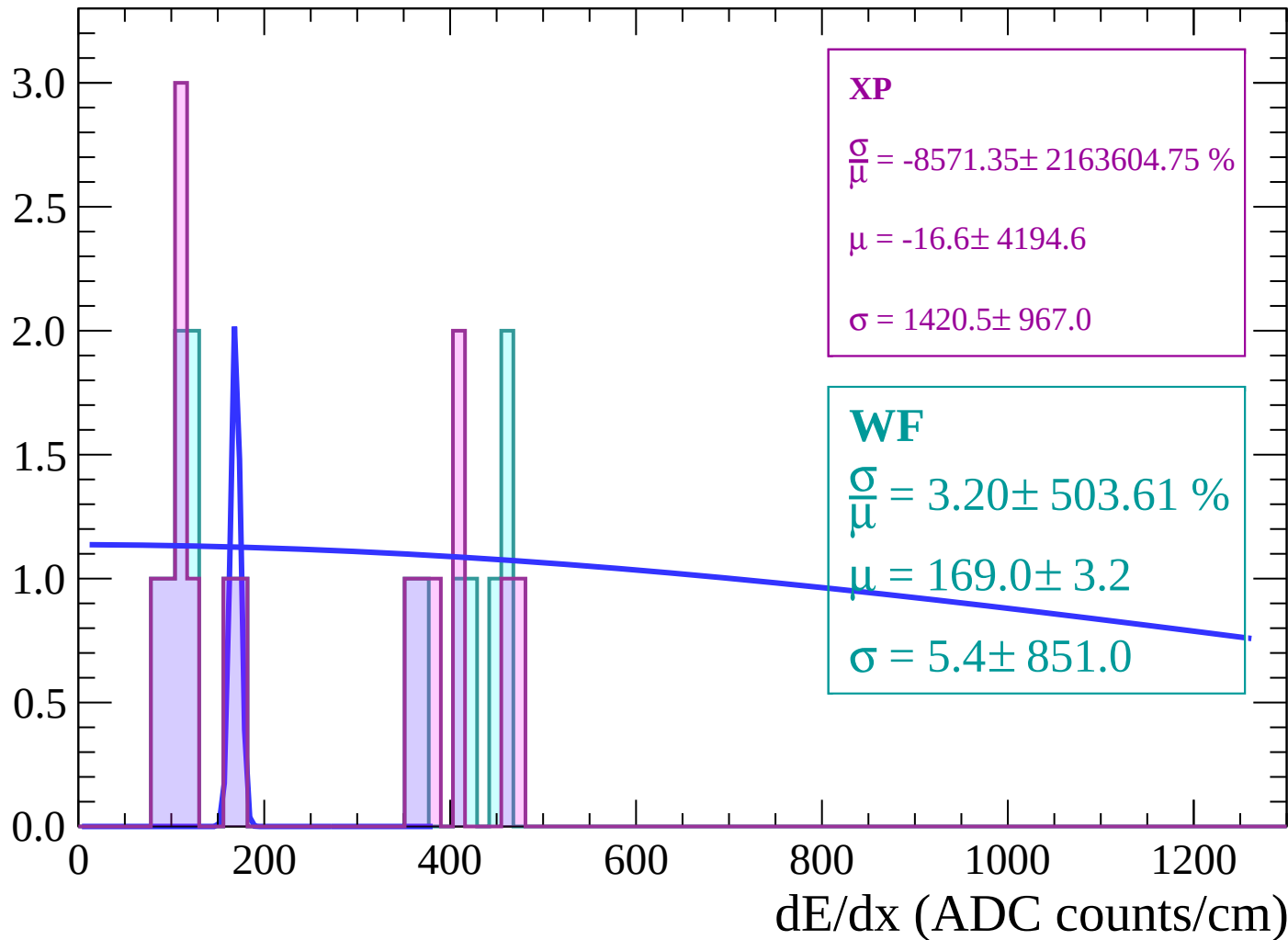
Energy loss | $72 < \theta < 74$

Count



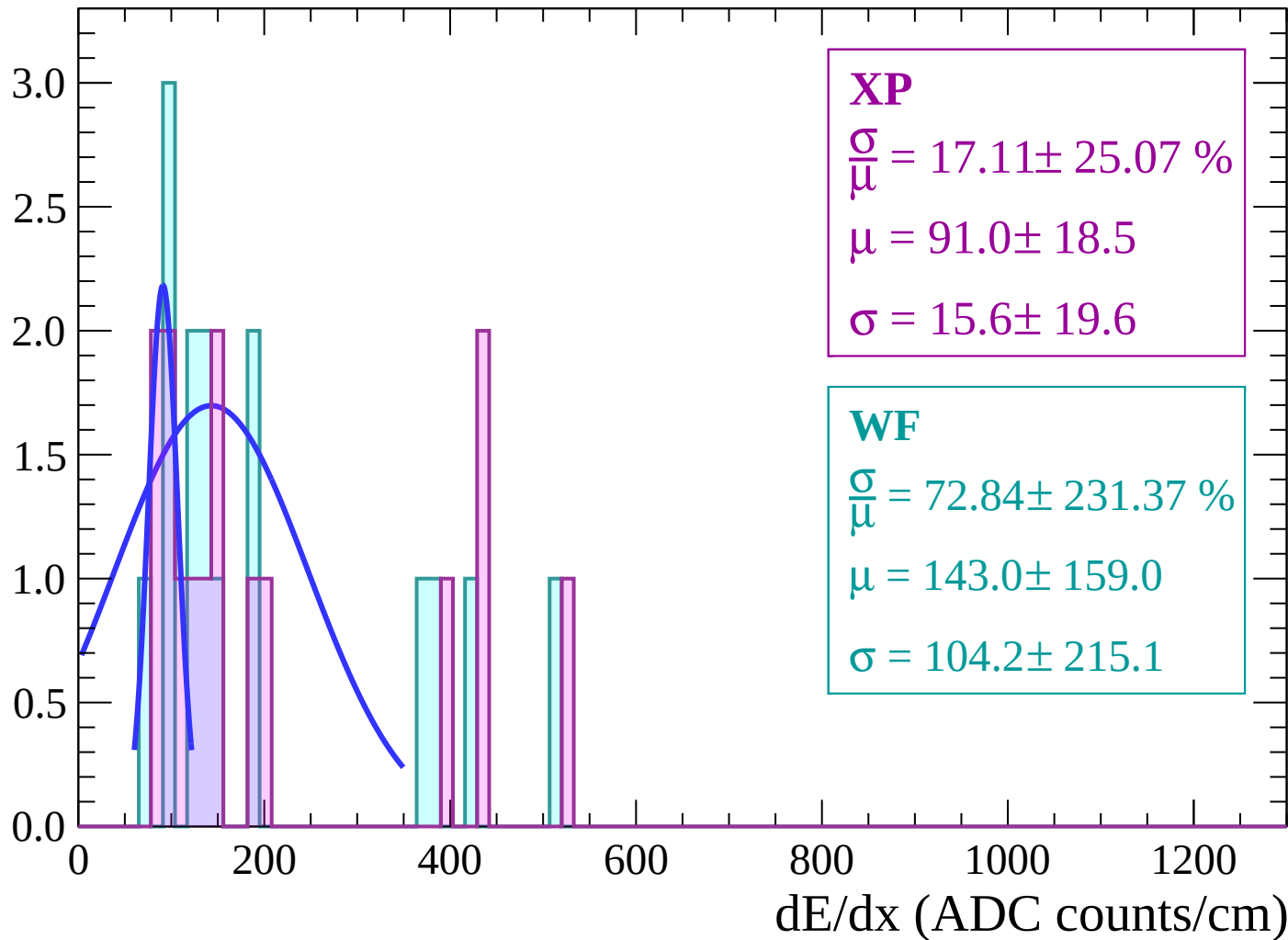
Energy loss | $74 < \theta < 76$

Count



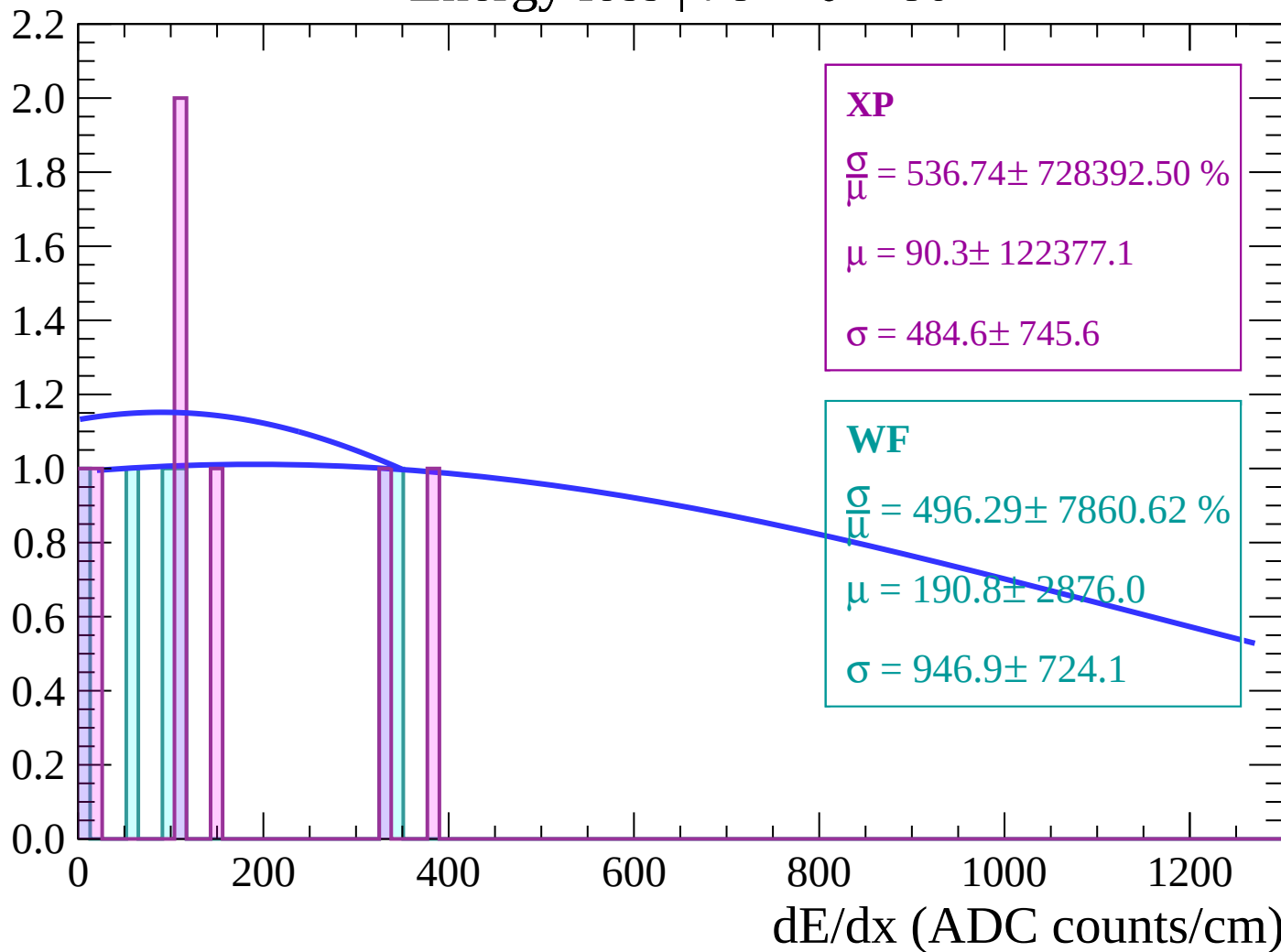
Energy loss | $76 < \theta < 78$

Count



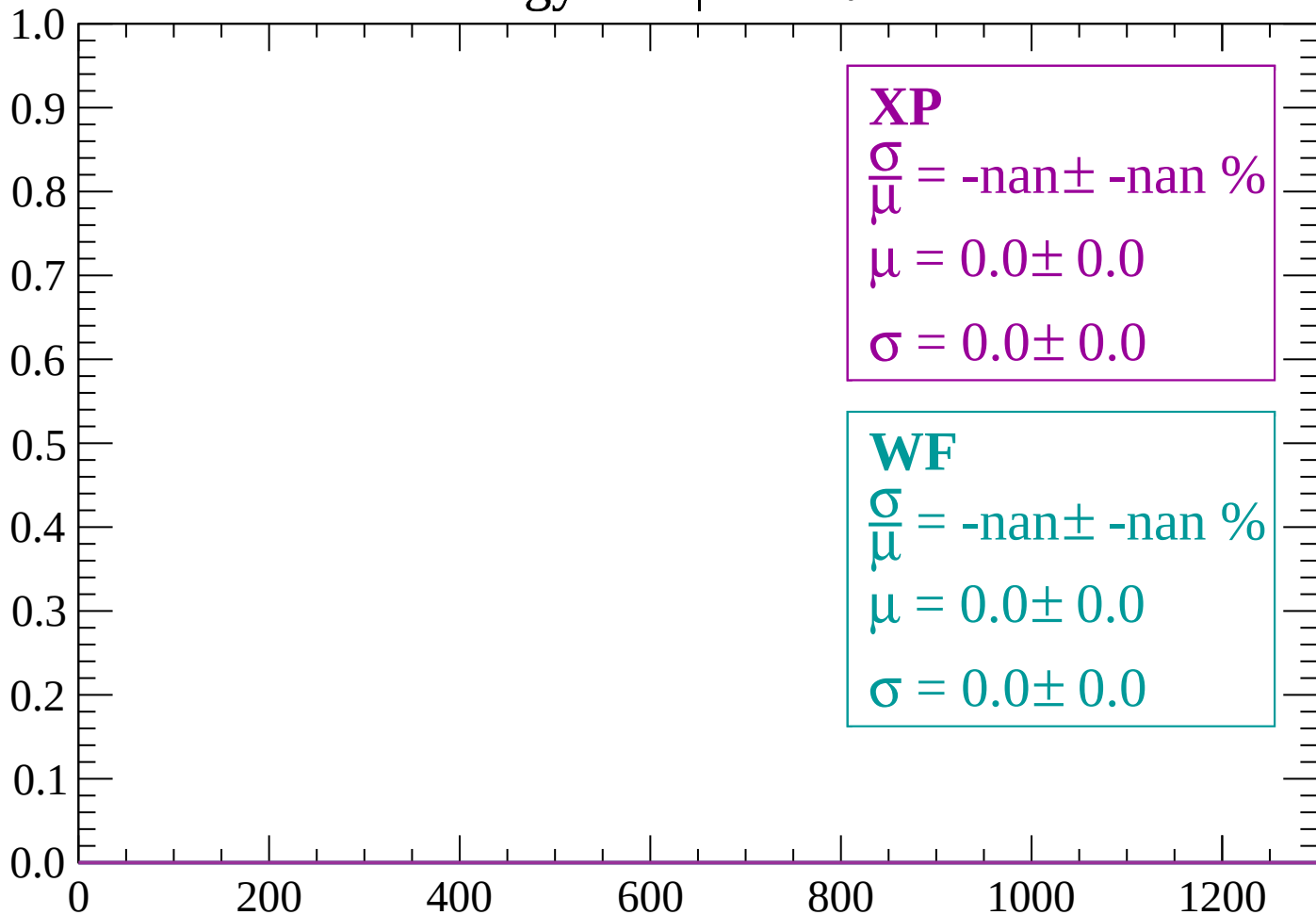
Energy loss | $78 < \theta < 80$

Count



Energy loss | $80 < \theta < 82$

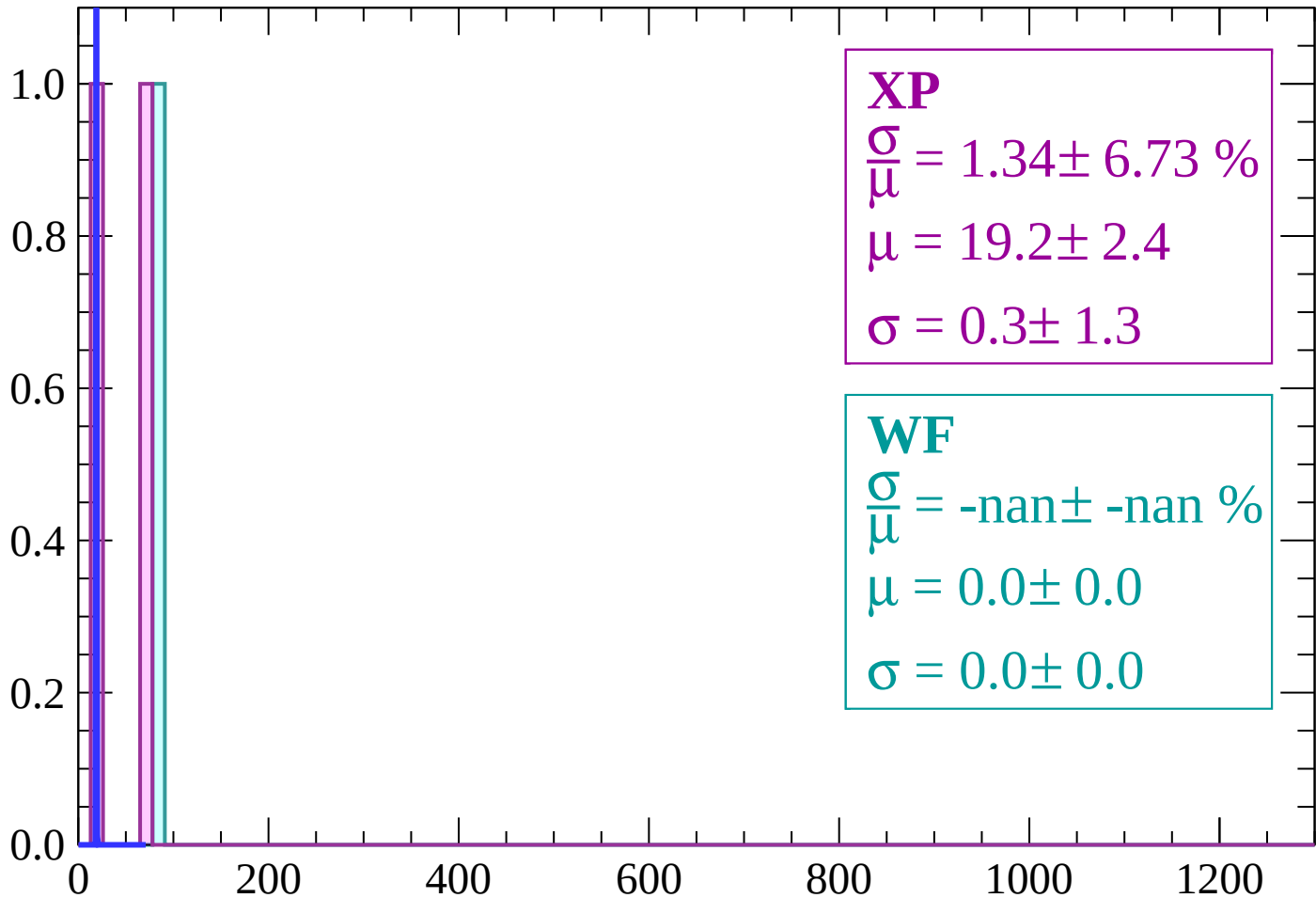
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

Count



XP

$$\frac{\sigma}{\mu} = 1.34 \pm 6.73 \%$$

$$\mu = 19.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

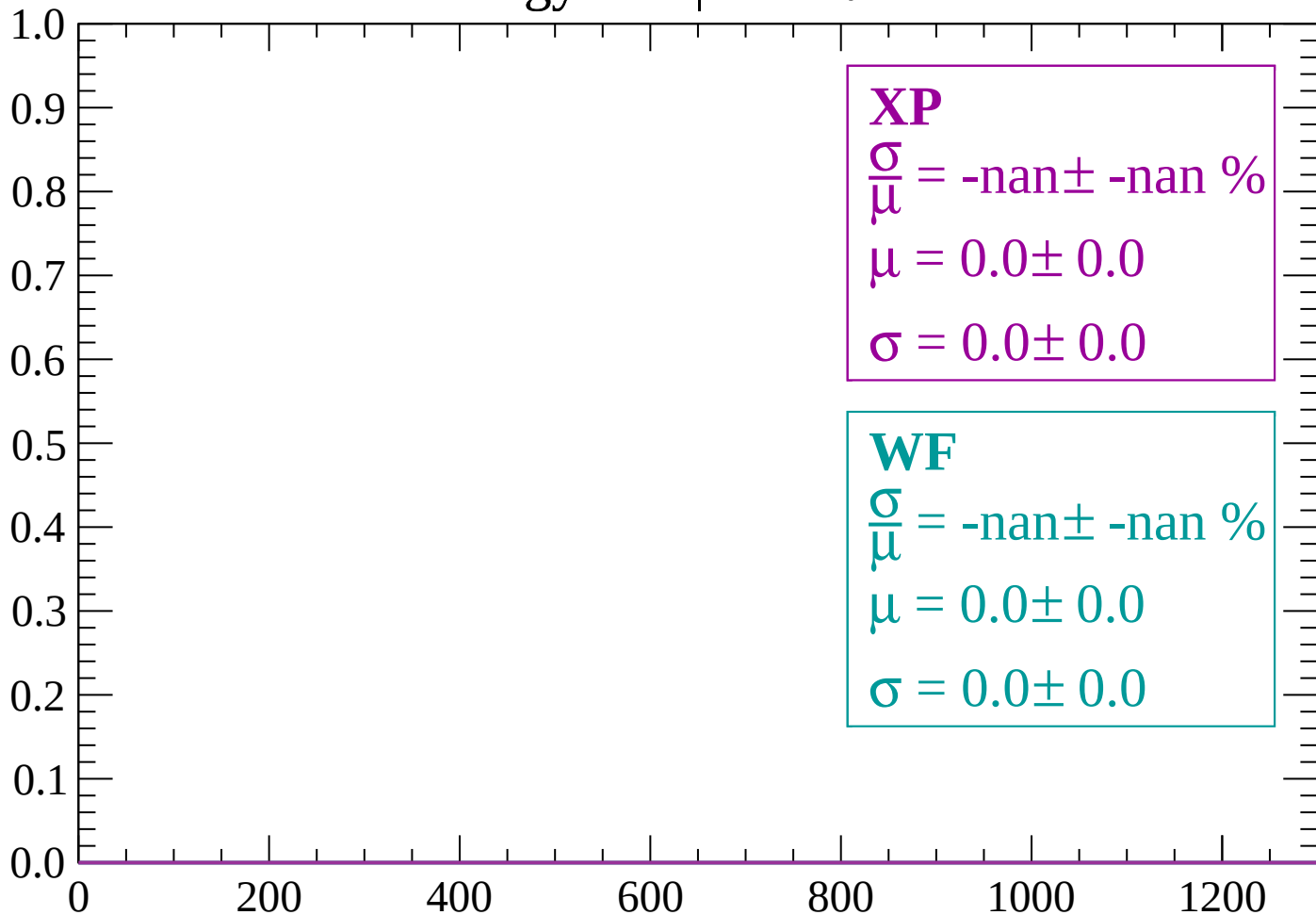
$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

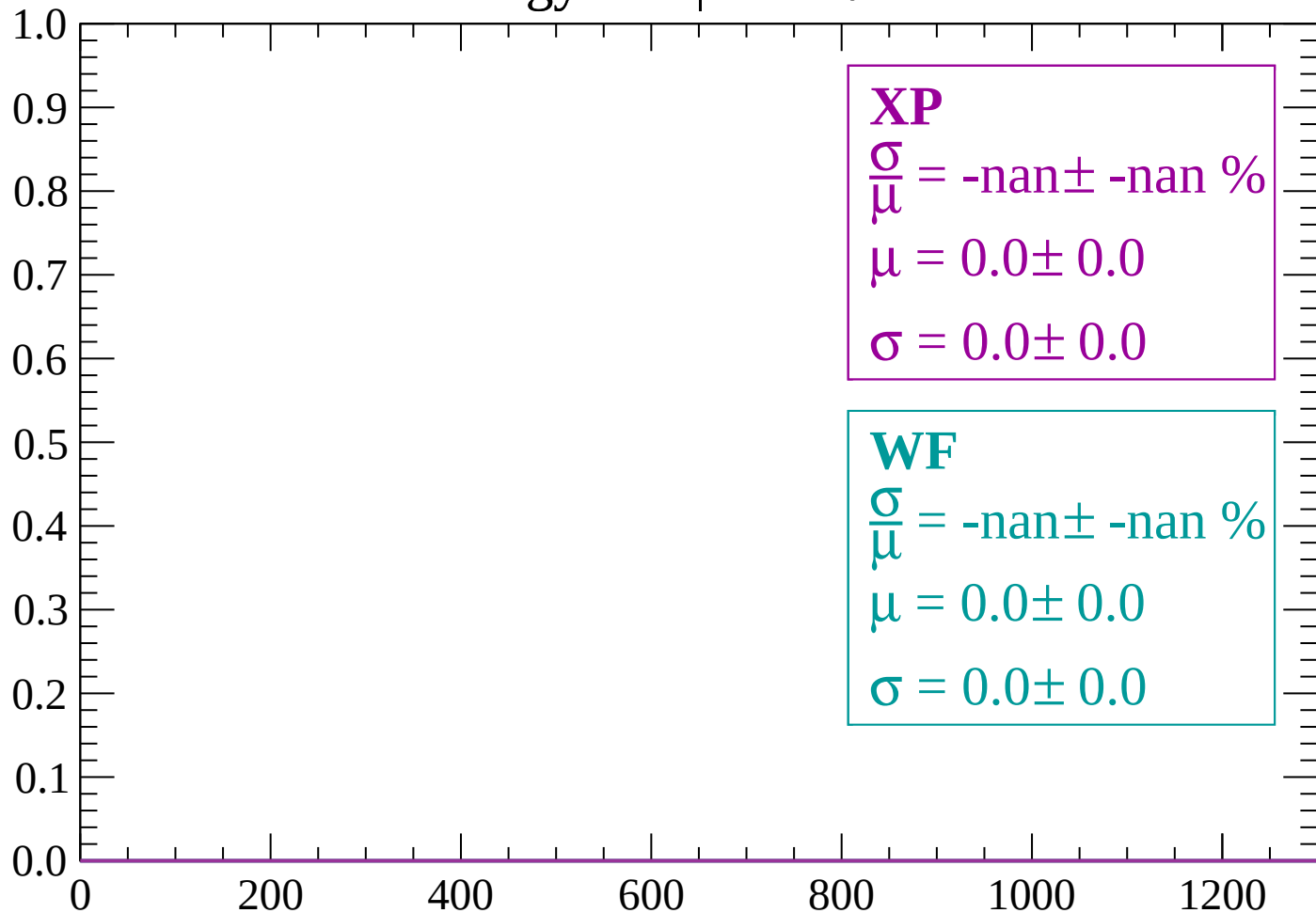
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

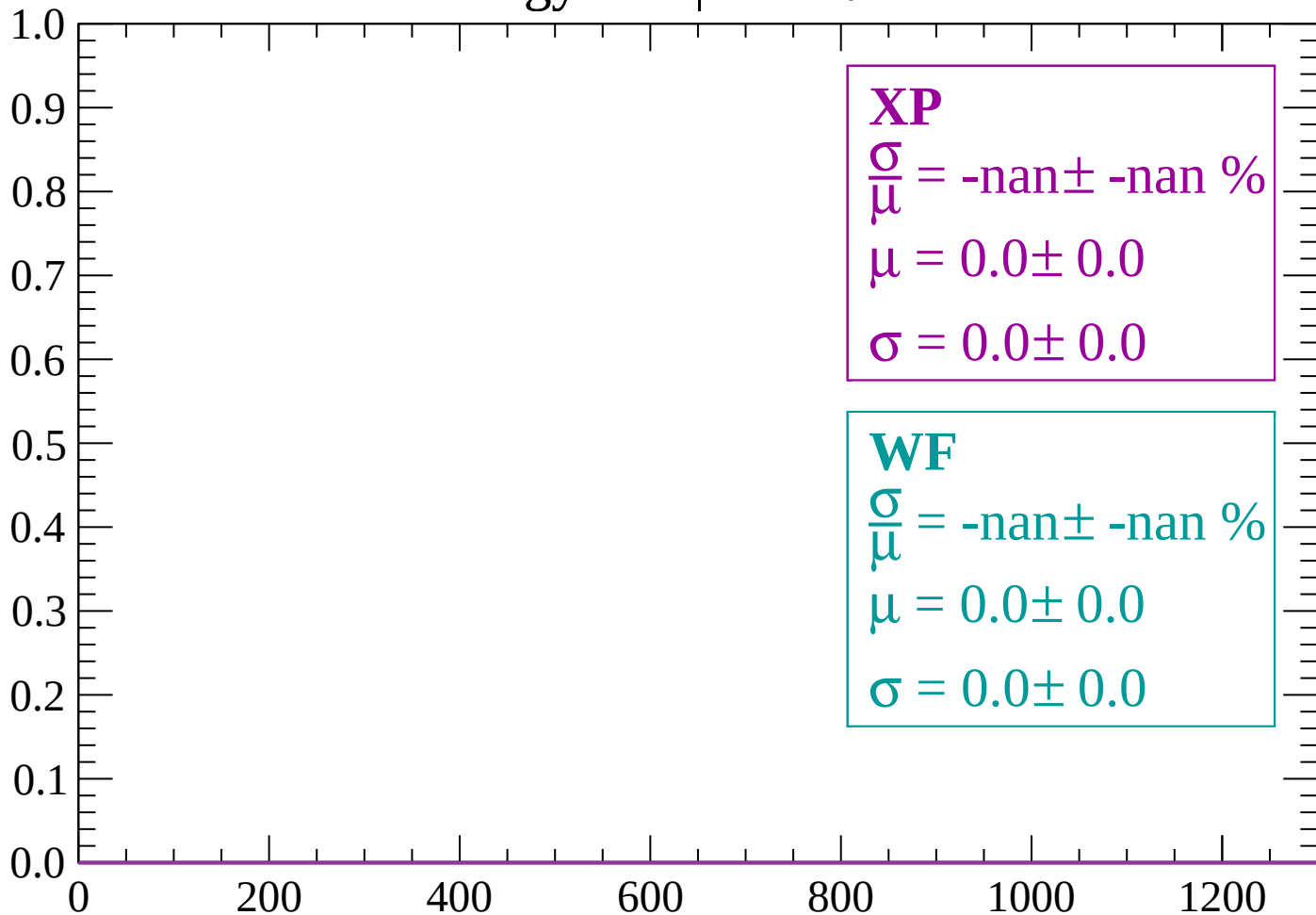
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

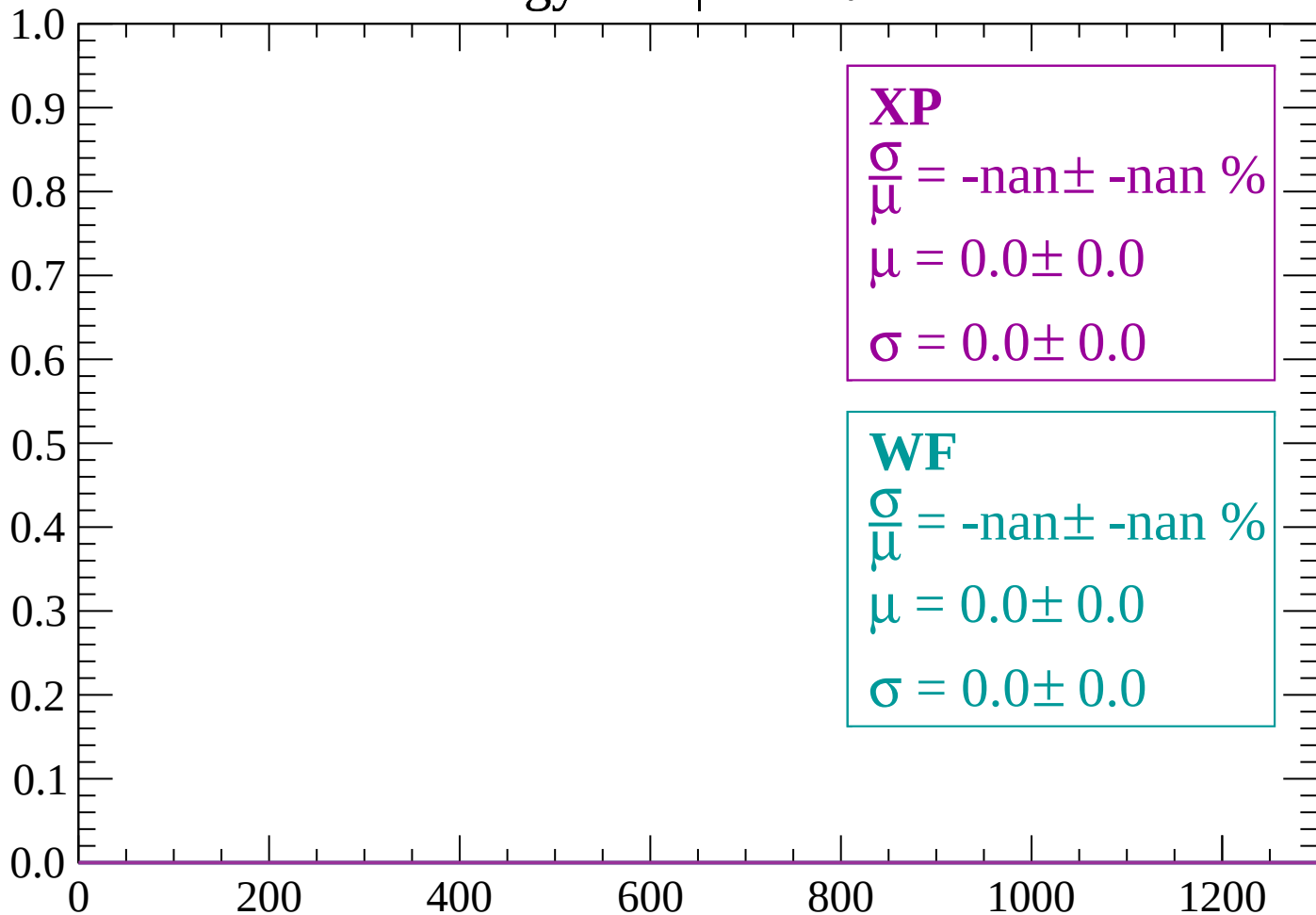
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

